

FIDES Guide 2022

-

Edition A

(July 2023)

**Reliability Methodology
for Electronic Systems**



FIDES

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Institut pour la Maîtrise des Risques

FOREWORD

The FIDES Guide 2022 is produced by a consortium co-funded by France's defence procurement agency, the DGA, and is brought to you by the *Institut pour la Maîtrise des Risques* (IMdR) via its working group "Maintenance and Development of FIDES Reliability Methodology".

SUMMARY AND PRINCIPLE OF THIS REVISION

The FIDES Guide 2022 replaces the FIDES Guide 2009 Issue A (published by the *Union Technique de l'Électricité* under the reference UTE-C 80811 January 2011). The text has been updated to include technological changes, to cover a wider range of models, and to make other improvements. The guiding principle of these modifications has been to produce a document that is as universal and practical to use as possible.

Between the FIDES Guide 2009 Issue A and the FIDES Guide 2022, many chapters of the document have been revised.

All the changes (of content rather other form) are listed in the Table of Changes.

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TABLE OF CHANGES

Page	Paragraph	Change
2 ff.	Foreword Summary and principle of this revision Copyright Page headers and footers Warning	Updated
4	Table of changes	Updated
13	Acronyms	Updated and extended
31	Mission profile and unit of time	Modified the baseline duration of all the formulae to use the total mission profile duration rather than 8760 h, which applies to the specific case of a mission profile with a duration of one year
49	Physics of failure and modelling	Rewrote the formula to make it independent from the annual duration
49	Quantification of the parameters	Updated the standards references
49	Quantification of the parameters	Added a recommendation to consider 0.01 Grms as a minimum, even in the absence of vibration
54	Data sources	Added some examples of standards and websites available for consultation
55	Method for building mission profiles	Added a methodology paragraph with recommendations on how to construct mission profiles
62	Examples of mission profiles	Modified the number of temperature cycles and durations to bring them into line with the total cycle duration per 24h period. Added sample mission profiles: automotive and space
109 ff.	III Reliability – Failure Rate Prediction Guide Calculation Sheets	Updated relative sensitivities of components Modified the $C_{\text{sensitivity}}$ paragraph Modified the $\Pi_{\text{placement}}$ paragraph Added a methodology guide on how to determine the $\Pi_{\text{placement}}$ parameter Updated criterion 1 in the evaluation of the $\Pi_{\text{application}}$ parameter
118	Thermal resistance of components	Updated values of C_{type} for integrated circuits in line with the new list of packages
123	EC: Integrated circuits	Added new categories of package Added underfill processing Updated the basic failure rates associated with the chips in integrated circuit components Added a proposed ageing model for DSM matrices Replaced t_{annual} by t_{phase}
130	EC: Discrete semiconductors	Replaced t_{annual} by t_{phase}
138	EC: Discrete power semiconductors: Silicon, MOS > 5W and IGBT	Updated the basic failure rates associated with the chips Replaced t_{annual} by t_{phase}
138	EC: Light-Emitting Diodes	Replaced t_{annual} by t_{phase}

Page	Paragraph	Change
146	EC: Resistors	Clarified the names of the different types of resistor
151	EC: Ceramic capacitors	Added classifications X5R and X7R for Type-2 ceramic capacitors Modified the classification boundaries for CV products
158	EC: Film capacitors	Added the model
177	EC: Connectors	Added details on the gauge
181	Hybrids and MCMs	Note: No update in this version of the FIDES Guide
200	RF/MW: Integrated circuits & discrete semiconductors	Modified the model
254	Counting by type	Deleted the method “Counting by family”; Note: The method “Counting by type” is not updated in this version of the FIDES Guide
256	QA _{manufacturer}	Modified the table to bring it into line with the other tables in the FIDES Guide
285	Example of a manufacturing breakdown structure (MBS)	Added details on how the MBS is integrated
287	ISO 9001 and EN	Added information on ISO 9001 and EN 9100
289	Processing the collected information	Added a detail on information gathering
294	Weighted recommendation tables	Tables (except for Ruggedizing) completely reworked, without changing the numbers of the existing recommendations
325	Detailed recommendation sheets	Sheets (except for Ruggedizing) completely reworked, without changing the numbers of the existing recommendations
494	Annexes	Added chapter

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I

Introduction to the FIDES Guide

1. Introduction

The FIDES Guide is a global methodology for reliability engineering in electronics. It consists of two parts:

- A reliability – failure rate prediction guide
- A reliability process control and audit guide.

The guide is designed, on the one hand, to enable a realistic assessment of the reliability of electronic products, including components in systems that operate in severe environments (as well as under benign conditions such as storage) and, on the other hand, to provide a concrete tool for developing and controlling their reliability.

Its key features are:

- The provision of models for electrical, electronic and electromechanical components as well as for circuit boards and certain subassemblies
- The identification and inclusion of all the technological and physical factors that play an identified role in a product's reliability
- The precise characterization of a product's mission profile
- The inclusion of accidental electrical, mechanical and thermal overstresses
- The inclusion of failures inherent in the development, production, operation and maintenance processes
- The ability to distinguish between multiple suppliers of the same component.

By identifying the factors that contribute to reliability, whether technological, physical or process-related, the FIDES Guide makes it possible to revise product definitions and intervene at any point in the product life cycle in order to control and improve reliability.

2. About the FIDES methodology

The consortium that developed the FIDES methodology is made up of companies from the defence and aerospace sectors. The consortium was created under the aegis of France's defence procurement agency, the *Direction Générale de l'Armement* (DGA).

The FIDES methodology is based on the physics of failure, supported by the analysis of test data, experience feedback and existing models. This sets it apart from earlier methods, developed mainly through the statistical analysis of field data.

This approach yields predicted reliability results that are not influenced by the industrial domains of the methodology's designers.

After fine-tuning the models, the methodology was calibrated on the basis of the experience of the consortium members, particularly as regards the process factors.

3. Terminology

3.1. Acronyms

Acronym	Definition
ADC	Analog-Digital Converter
AF	Acceleration Factor
ASIC	Application-Specific Integrated Circuit
ATA	Advanced Technology Attachment
AWG	American Wire Gauge
BGA	Ball Grid Array
BICMOS	BIpolar Complementary Metal Oxide Semiconductor
BOC	Board On Chip
BT	Bismaleimide Triazine
BTI	Bias Temperature Instability
CBGA	Ceramic Ball Grid Array
CDIP	Ceramic Dual In-line Package
CERDIP	CERamic Dual In-line Package
CGA	Column Grid Array
CLCC	Ceramic Leadless Chip Carrier
CNES	<i>Centre National d'Etudes Spatiales</i>
COCISPER	<i>COncception Circuits Intégrés Spécifiques et PErennité</i>
COTS	Commercial Off-The-Shelf
CPGA	Ceramic Pin Grid Array
CPLD	Complex Programmable Logic Device
CPU	Central Processing Unit
CQFP	Ceramic Quad FlatPack
CRT	Cathode-Ray Tube
CSP	Chip Scale Package
CV	Capacitance-Voltage
DBGA	Dimpled Ball Grid Array
DC	Direct Current
DDR	Double Data Rate
DFN	Dual-Flat No-leads
DGA	<i>Direction Générale de l'Armement</i>
DIP	Dual Inline Package
DMA	Direct Memory Access

Acronym	Definition
DPA	Destructive Physical Analysis
DPAK	Decawatt PAcKage
DPDT	Double-Pole Double-Throw
DPST	Double-Pole Single-Throw
DRAM	Dynamic Random-Access Memory
DS	Discrete Semiconductor
DSM	Deep Sub Micron
DSP	Digital Signal Processor
DT	Double Throw
Ea	Activation Energy
EEE	Electrical, Electronic, Electromechanical
EEPROM	Electrically Erasable Programmable Read-Only Memory
EIA	Electronic components Industry Association
EIDE	Enhanced Integrated Drive Electronics
EMC	Electromagnetic Compatibility
EOS	Electrical OverStress
ESD	Electro Static Discharge
ESS	Environmental Stress Screening
FC	Flip Chip
FinFET	Fin Field-Effect Transistor
FIT	Failure In Time
FLASH	Fast Low-Latency Access with Seamless Handoff
FMECA	Failure Modes, Effects and Criticality Analysis
FPGA	Field Programmable Gate Array
GaAs	Gallium Arsenide
GaN	Gallium Nitride
GPS	Global Positioning System
GRMS	G Root Mean Square
HALT	Highly Accelerated Life Test
HASS	Highly Accelerated Stress Screening
HAST	Highly Accelerated Stress Test
HQFP	Heat-sinked Quad Flat Pack
HUMS	Health and Usage Monitoring Systems
Ic	Collector current
IC	Integrated Circuit

Acronym	Definition
Id	Drain current
IEC	International Electrotechnical Commission
If	Forward current
Ig	Grid current
IGBT	Insulated-Gate Bipolar Transistor
ISO	International Organization for Standardization
ISS	In-Service Support
JEDEC	Joint Electron Device Engineering Council
JFET	Junction gate Field-Effect Transistor
JTAG	Joint Test Action Group
LAN	Local Area Network
LCD	Liquid-Crystal Display
LED	Light-Emitting Diode
LF	Lead Free
LGA	Land Grid Array
LNA	Low Noise Amplifier
LQFP	Low-profile Quad Flat Package
MBS	Manufacturing Breakdown Structure
MCM	Multi Chip Module
MELF	Metal Electrode Leadless Face
MESFET	Metal Semiconductor Field Effect Transistor
MIL HDBK	MILitary HanDBook
MMIC	Monolithic Microwave Integrated Circuit
MSL	Moisture Sensitivity Level
MSOP	Micro Small Outline Package
MTBF	Mean Time Between Failures
MTTF	Mean Time To Failure
NAND	Not AND
NDA	Non-Disclosure Agreement
NO	Normally Open
NOR	Not OR
OL	Operating Life
PAL	Programmable Array Logic
PBGA	Plastic Ball Grid Array
PCB	Printed Circuit Board

Acronym	Definition
PDIP	Plastic Dual In-line Package
PHEMT	Pseudomorphic High Electron Mobility Transistor
PLCC	Plastic-Leaded Chip Carrier
PM	Part Manufacturing
PPGA	Plastic Pin Grid Array
PQFP	Plastic Quad Flat Pack
PrPM	Periodicity of Preventive Maintenance
PSD	Power Spectral Density
PTFE	PolyTetraFluoroEthylene
PTH	Plated ThroughHole
QA	Quality Assurance
QFN	Quad Flat No-leads
QFP	Quad Flat Package
QSOP	Quarter-size Small-Outline Package
RA	Reliability Assurance
RAMS	Reliability Availability Maintainability Safety
RF	RadioFrequency
RH	Relative Humidity
ROHS	Restriction Of Hazardous Substances
ROR	Recommended Operating Ratings
RQFP	Power Quad Flat Pack
SAS	Serial Attached Small computer system interface
SATA	Serial Advanced Technology Attachment
SBGA	Super Ball Grid Array
SCSI	Small Computer System Interface
SDRAM	Synchronous Dynamic Random Access Memory
SGRAM	Synchronous Graphics Random Access Memory
SiGe	Silicon Germanium
SIL	Single In-Line
SIP	System In a Package
SMD	Surface Mounted Device
SOIC	Small Outline Integrated Circuit
SOJ	Small-Outline J-leaded
SOL	Small OutLine
SOP	Small Outline Package

Acronym	Definition
SPC	Statistical Process Control
SPDT	Single-Pole Double-Throw
SPST	Single-Pole Single-Throw
SQFP	Shrink Quad Flat Package
SRAM	Static Random Access Memory
SSOP	Shrink Small Outline Package
ST	Single Throw
STANAG	STANdardization AGreement
STN	Super-Twisted Nematic
TBGA	Tape Ball Grid Array
TCY	Temperature (or Thermal) CYcling
TFT	Thin-Film-Transistor
THB	Temperature Humidity Biased
TOS	Thermal OverStress
TQFP	Thin Quad Flat Pack
TSOP	Thin Small Outline Package
TSSOP	Thin-Shrink Small Outline Package
TVS	Transient Voltage Suppression
TVSOP	Thin Very Small Outline Package
USB	Universal Serial Bus
UTE	<i>Union Technique de l'Électricité</i>
UV	Ultra-Violet
Vce	Collector-to-Emitter Voltage
Vds	Drain-to-Source Voltage
VGA	Video Graphics Array
Vge	Gate-to-Emitter Voltage
Vgs	Gate-to-Source Voltage
VIP	Very Important Person
VQFP	Very small Quad Flat Package
VSOP	Very Small Outline Package
WLP	Wafer Level Package
ZIP	Zig-zag In-line Package

3.2. Definitions

Reliability

The ability of an entity to perform a required function under given conditions for a given length of time.

Reliability is generally expressed quantitatively, through relevant characteristics. One of these characteristics, in certain applications, is that this ability is expressed in the form of a probability, also called “a reliability”.

Failure mechanism

The chain of “cause-and-effect” relationships in a physical, chemical, or other process between the root cause of the failure and the failure mode.

Failure mode

One of the possible states entered by an entity when one of its required functions fails.

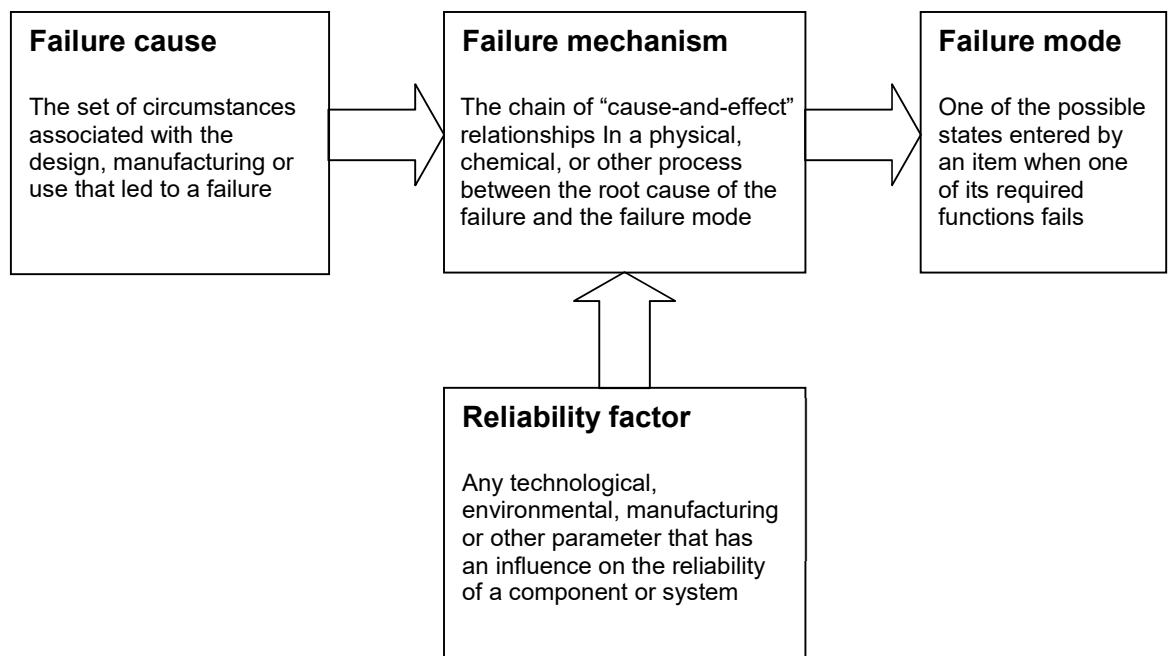
Cause of failure (or Failure cause)

The set of circumstances associated with the design, manufacturing or use that led to a failure.

Reliability factor

Any technological, environmental, manufacturing or other parameter that has an influence on the reliability of a component or system.

The logic behind the above definitions is summarized in the diagram below:



System

A set of equipment capable of performing or supporting an operational role. A complete system comprises all the equipment, hardware, software, services and personnel necessary for it to be able to function self-sufficiently in its environment of use. Examples: cars, aircraft, microcomputers.

Subsystem

A set of equipment capable of performing an operational function of a system. A subsystem is a major subdivision of a system, and is often itself called “system”. Examples: the ABS system of a car; the GPS system of an aircraft.

Equipment

Term referring to a group of items capable of performing a complete function. Examples: an ABS system computer; a GPS system monitor.

Subassembly

Term referring to an item or assembled group of items capable of performing one of the functions of a piece of equipment. Examples: circuit board of a computer; hard drive.

Electronic component

Term referring to an element designed to be assembled with others in order to perform one or more electronic functions. Examples: transistors, resistors. This definition also covers printed circuit boards (PCBs).

Product

In this guide, “product” refers to the assembled entity of which the reliability is being analysed. Generally a piece of equipment.

Item

In this guide, “item” refers to a basic entity (a component or subassembly, not broken down into smaller elements) of which the reliability is amenable to analysis.

4. References

IEC 60050 (191) A1 (1999-03)

International Electrotechnical Vocabulary – Chapter 191: Dependability and quality of service

MIL-HDBK-217F (+ Notice 1 and 2)

Reliability Prediction of Electronic Equipment

UTE C 80-810

Reliability Data Handbook: RDF 2000 – A universal model for reliability prediction of Electronics components, PCBs and equipments

IEC 61709

Electric components – Reliability – Reference conditions for failure rates and stress models for conversion

IEC 62308:2006

Equipment reliability – Reliability assessment methods

SSB-1.003

EIA Engineering Bulletin – Acceleration Factors – November 1999 and September 2002

JEDEC JEP122C

Failure Mechanisms and Models for Semiconductor Devices

5. Scope

5.1. Domains of application

The FIDES methodology is applicable to every domain that makes use of electronics:

- Aviation
- Naval
- Military
- Electricity production and distribution
- Automotive
- Rail
- Space
- Industry
- Telecommunications
- Computing, home automation, electrical appliances
- Etc.

5.2. Coverage of the models

The FIDES methodology models failures due to intrinsic causes (the technology or quality of manufacturing and distribution of the items studied) as well as extrinsic causes (the specification, design, sourcing, production or integration of the equipment).

The methodology covers the following:

- Failures resulting from errors in development or manufacturing
- Overstresses (electrical, mechanical, and thermal) linked to the use of the items and not listed as such by the user (i.e. where the overstress remains undetected).

The methodology does not cover:

- Failures caused by software
- Unconfirmed failures
- Failures linked to the non-performance of preventive maintenance
- Failures linked to accidental exposure when identified or established (propagation of failures, use outside the specifications, improper handling: i.e. where overstress is known to have occurred).

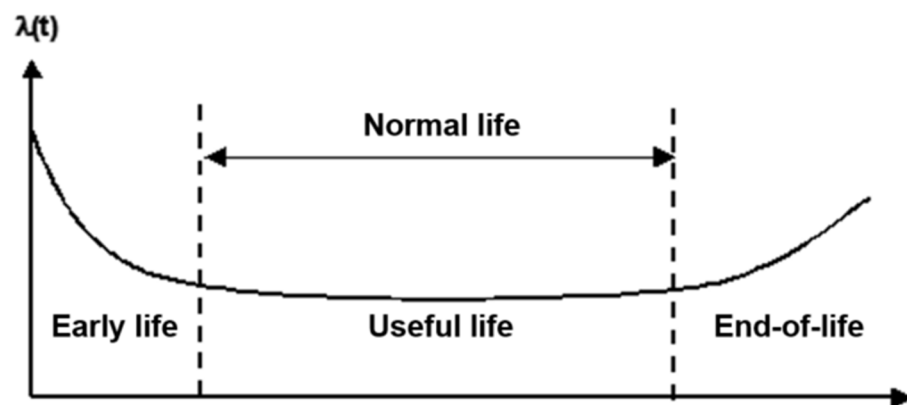
The FIDES methodology can be employed to study non-operating phases, whether dormancy periods between uses or actual storage.

5.3. Nature of reliability prediction

5.3.1. General case

The FIDES methodology produces reliability predictions in the form of failure rates, denoted with the Greek letter lambda i.e. λ .

Experimental observation shows that plotting failure rates against time usually yields the curve below, known as a “bathtub curve”.



The life of a product can be divided into three periods:

- An “early life” stage, marked by “infant mortality” failures
- A “useful life” stage, with essentially constant failure rates
- An “end-of-life” stage, marked by “wear-out” failures.

During the early-life or “burn-in” stage, the failure rate diminishes. The reliability of the product grows over time. At this stage, failures are generally due to process implementation problems and to the environmental stress screening of the design and components.

The useful life stage is represented by a constant failure rate. The failure rate is independent of the age of the product (which is why these failures are often described as “random”). This period, often non-existent for mechanical goods, is the reference period for electronics.

During the end-of-life or “wear-out” stage, reliability decreases with the number of hours of operation: the older the product, the more likely it is to fail. This type of behaviour is characteristic of items subject to wear or other forms of gradual deterioration. It is reflected in rising failure rates.

The FIDES evaluation model proposes a reliability assessment at constant failure rates (more precisely, at mean failure rates). The early-life and end-of-life stages are excluded from the prediction (though a special case is made for certain subassemblies), for the following reasons:

- Firstly, the early-life stage represents the final development of an equipment item or system. Controlling the increase in reliability during this phase is a crucial step towards achieving good reliability rapidly.
- The “end-of-life” stage is also excluded from FIDES because it is, in principle, sufficiently far removed from the useful life of the electronic systems covered by FIDES. However, it is essential, during the design phase of a product, to check that this is indeed the case. For items with a lifetime that is too short, approaches other than predictive reliability alone should be used to address this aspect, such as the definition of preventive maintenance procedures.
- At the microscopic level, of course, very few failure mechanisms have a strict “constant rate” distribution. However:
 - Many failure mechanisms, though cumulative, and therefore increasing over time, present a dispersion that makes them similar to a constant for the periods under consideration.
 - The multiplicity and diversity of components, even for a single board, make the cumulative effect close to a constant.
 - Age differences between equipment items in a single system or a pool of systems tend to result in a constant rate for an observer at system level.

For these reasons, using a constant failure rate remains the most pertinent approach for estimating the predicted reliability of a system.

The physics of failure is used in certain cases to predict probabilistic lifetime values (Time To Fail). This type of prediction is complementary to reliability prediction, but cannot replace it.

5.3.2. Ageing-related failures in subassemblies

For electronic components, in most cases, the lifetime is sufficiently long compared to its period of operational use, and its impact is therefore negligible (reminder: it is essential, during the design phase of a product, to check that this is indeed the case). But this no longer applies when, for example, wear phenomena arise due to the presence of moving mechanical parts.

In the case of certain subassemblies with a lifetime significantly shorter than the system as a whole, ageing-related failures can have a non-negligible effect on reliability. Specific modelling is proposed for these cases.

5.4. Prediction confidence level

Evaluations made using the FIDES methodology are aimed at providing realistic reliability values, similar to commonly observed average values (as opposed to pessimistic or conservative values).

One essential question following any assessment of predicted reliability is how much confidence to assign to the estimate. This question is all the more important in that users do not have confidence in the raw results supplied by earlier methodologies, and because reliability control (quantification and engineering) has become a key part of any project.

One of the objectives of the FIDES project is to build this confidence. However, accuracy of prediction is not the sole purpose of the FIDES methodology: identifying and controlling the factors that affect reliability may be seen as even more important goals.

As a general rule, an isolated predicted reliability estimate cannot be assigned a confidence interval, as can be done with a failure rate measurement based on field data. In the case of FIDES, while it might be possible to calculate a confidence interval for certain basic failure rates, it is practically impossible to estimate a confidence level for every adjustment parameter, even for well-known and widely-used laws of physical acceleration.

It is important to bear in mind that reliability belongs to the field of probability.

The representativeness of a prediction increases with the number of items analysed. Predictions are generally not applicable at the level of a single item; it is recommended that they be used at equipment level (e.g. for a set of circuit boards) or above.

Note: The presence in the models of numerical values with several significant digits does not say anything about the expected accuracy of the results.

Comparison between predicted reliability and reliability based on field data is always a delicate matter, as there are also uncertainties involved in measuring reliability in service, due for example to:

- Changes in reliability over time
- Lack of knowledge about the real lifetime of the product
- Distinguishing between failures attributable to the product and those that are not
- Instances of batch effects, which are hard to integrate into reliability calculations.

One prerequisite for comparing a predicted reliability value with a reliability value based on field data is always to ensure that the mission profile actually experienced by the product is sufficiently close to the mission profile used for the prediction. Otherwise, the comparison applies to the relative severity of the two mission profiles (predicted vs. actual) rather than to the reliability itself.

One of the characteristics of the FIDES method is that it considers failures to be largely due to the life situations encountered by the product. The confidence level of the predicted reliability can therefore never be greater than the confidence level of the predicted mission profile of the product.

5.5. Items covered

The FIDES methodology covers items ranging from basic electronic components to electronic subassemblies or modules with well-defined functions. Coverage of item families by FIDES is not fully exhaustive, but it is amply sufficient to enable a representative reliability assessment in most cases.

The methodology applies to COTS items (for which it was originally developed), but also to specific items, insofar as their technical characteristics correspond to those described in this guide.

The acronym COTS (Commercial Off-The-Shelf) refers to any catalogue-bought item, available on the domestic or foreign market, with a supplier partnumber, and for which the customer has no input in its design or production. The item may be modified, discontinued or desupported without the customer having any say in the matter. There may be a sole supplier or multiple suppliers for any one item.

FIDES covers the following COTS:

- Components such as integrated circuits, discrete semiconductor circuits and passive components
- Subassemblies such as hard drives and screens
- Assembled COTS boards.

II

Reliability – Failure Rate Prediction Guide

1. Introduction to the models

1.1. Origin of the reliability data

The data used to construct the models was taken from:

- Failure analysis databases from the defence and civil aviation domains
- Reliability data from manufacturers of components and subassemblies
- Existing reliability datasets, where relevant and usable.

This data was used to develop and calibrate the models, by applying three methods:

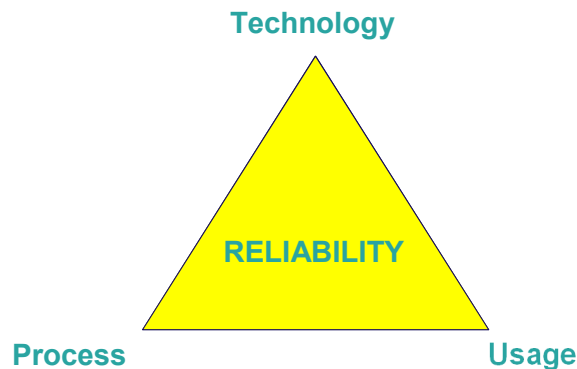
Method 1: Using operational databases (aviation and military) containing records of failure mechanisms.

Method 2: Using test data from component and subassembly manufacturers (environment tests, technological data, etc.).

Method 3: Using mixed data (manufacturer data, experience feedback, test results). This method is used mainly to build models of subassemblies.

1.2. The FIDES approach

The FIDES reliability approach is based on consideration of three dimensions: Technology, Process and Usage. These dimensions are considered for the entire life cycle, from the product specification phase through to operation and maintenance.



Technology covers the technology of the item itself as well as that of its integration into the product.

Process covers all practices and state of the art from the specification of the product through to its replacement.

Usage covers the usage constraints defined by the design of the product as well as those in operation at the end user.

The models therefore analyse a Technology with regard to its Usage constraints by adopting an approach based on failure mechanisms and associated factors; in particular, they weight the risk of failure by incorporating all of the Process factors that could activate, accelerate or mitigate these mechanisms.

1.3. Complete method and simplified method

Reliability can be assessed at different levels of detail, to adapt to the different stages of a project:

- A detailed method, the most complete one
- A simplified method of counting-based reliability estimation per type of item.

The counting-based method per type of item is derived from the complete detailed method. All the general models apply in the same way to both of the methods, the only difference being the level of information about the product under consideration.

1.4. Generic input data

The input data includes, in generic terms:

Data on the product's environments and conditions of use

Typically:

- Operating temperature
- Amplitude and frequency of temperature cycles
- Vibration amplitude
- Relative humidity
- Ambient pollution level
- Exposure to accidental overstress (depending on type of application).

The above data must be specified for each phase of the product life cycle. The granularity of the description of the product's mission profile within an operational system will determine the accuracy of the reliability assessment. This stage of the predictive analysis should therefore be conducted with great care.

Product definition data

Typically:

- Parts lists
- The technical or technological characteristics of items, as reported in the manufacturers' data sheets.

The information relating to the application needs to be assessed for each phase in the life cycle:

- Stress or load levels of the items (power dissipated, voltage stress, etc.)
- Local deteriorations (or improvements) in temperature or other environmental parameters.

In practice this data is often constant – or assumed to be constant – across all operating phases, but this is not always the case.

Product life cycle data

This data must be gathered by means of a process audit examining the reliability control system and covering the specification, design, board manufacturing, equipment integration, system integration, operation and maintenance phases, as well as the activities common to all phases. The thoroughness and depth of this audit must obviously be on a par with the required level of reliability.

Data on the suppliers of the items used in the product

This data comes from the item supplier and from the manufacturer's knowledge of the supplier.

1.5. General model

The FIDES general reliability model for an item is based on the following equation:

$$\lambda = (\Sigma_{Physical_factors}) \times (\Pi_{Process_factors})$$

Where:

- λ is the failure rate of the item.
- $\Sigma_{Physical_factors}$ is a mainly additive term, representing the physical and technological factors that contribute to reliability.
- $\Pi_{Process_factors}$ is a multiplicative term representing the impact of the development, production and operating processes on reliability.

In practice, this equation becomes:

$$\lambda = \lambda_{physical} \times \Pi_{PM} \times \Pi_{Process}$$

Where:

- $\lambda_{physical}$ represents the physical factors.
- Π_{PM} (PM for Part Manufacturing) is an expression of the quality and technical control of the item's manufacturing.
- $\Pi_{Process}$ is an expression of the quality and technical control of the development, manufacturing and operating processes of the product containing the item.

1.6. Mission profile and unit of time

The failure rates predicted by the FIDES methodology are hourly failure rates, expressed in terms of calendar hours and based on mission profile.

The failure rate for each phase is weighted by the phase duration:

$$\lambda_{\text{Physical}} = \sum_i^{\text{Phases}} \left(\frac{T_{\text{phase}_i}}{T_{\text{Total}}} \cdot \lambda_{\text{phase}_i} \right)$$

Where:

- λ_{phase_i} is the contribution of Phase i to the overall failure rate
- T_{phase_i} is the duration of Phase i
- T_{Total} is the total duration of the mission profile.

Predicted failure rates are expressed in terms of failures in time or “FITs” (1 FIT is equal to 10^{-9} failure per hour).

Remarks:

- This type of failure rate is not expressed per hour of operation, and for this reason (among others) the failure rates predicted by the FIDES methodology cannot be compared directly with results obtained using other approaches.
- If the mission profile is defined over one year, T_{Total} equals 8 760 hours.

The widespread use of the FIT rate as a unit of measurement for failure rates allows reliability managers to establish a fixed reference for comparing failure rate values. Moreover, when the product type is sufficiently well known, the failure rate also becomes an indicator of the severity of the mission profile.

Despite the greater universality of failure rates expressed in calendar-based FITs, they sometimes need to be presented in the form of an “MTBF per mission-hour”. This calculation involves allocating all failures to hours where the product is considered to be operational or “on mission”. The “on-mission” failure rate is calculated from the calendar failure rate as follows:

$$\lambda_{\text{Mission}} = \lambda_{\text{Calendar}} \times \frac{\text{Calendar duration}}{\text{Mission duration}}$$

Where:

- λ_{Mission} is the failure rate expressed in failure per mission hour
- $\lambda_{\text{Calendar}}$ is the failure rate expressed in failure per calendar hour (FIDES one)
- *Calendar duration* is the cumulative duration of life profile
- *Mission duration* is the cumulative duration of mission phases

1.7. Failure rate of electronic products

The FIDES general model can be used to calculate the failure rate of an electronic product prior to any redundancy or architecture considerations.

The overall failure rate of the electronic product (usually an equipment item) is obtained by aggregating the failure rates of all of its constituent parts.

$$\lambda_{Product} = \left(\sum_{Item} \lambda_{Item} \right)$$

Or, in another form:

$$\lambda_{Product} = \left(\begin{array}{l} \sum_{Components} \lambda_{Components} \\ + \sum_{PCBs} \lambda_{PCBs} \\ + \sum_{COTS\ boards} \lambda_{COTS\ boards} \\ + \sum_{Other_subassemblies} \lambda_{Other_subassemblies} \end{array} \right)$$

1.8. Physical and technological factors: $\lambda_{\text{Physical}}$

The physical factors can be subdivided into various subfactors as shown below:

$$\lambda_{\text{physical}} = \left[\sum_{\text{Physical factors}} (\lambda_0 \times \Pi_{\text{Acceleration}}) \right] \times \Pi_{\text{Induced}}$$

Where:

- The term between square brackets represents the contribution of nominal stresses of usage.
- Π_{Induced} represents the contribution of the induced factors (also called overstresses) inherent to a particular field of application.

1.8.1. Actual applied stresses

This element of the general model encompasses the basic failure rate assigned to the item, the influence of the characteristics of the technology employed, and acceleration factors for assigning to the item the physical stresses it undergoes in operational use.

$$\lambda_{Physical} = \left[\sum_{Physical_factors} (\lambda_0 \times \Pi_{Acceleration}) \right] \times \Pi_{Induced}$$

Where:

- λ_0 is the basic failure rate of the item
- $\Pi_{Acceleration}$ is an acceleration factor indicating sensitivity to the conditions of use.

The item's technological characteristics are taken into account:

- Either directly, in the choice of λ_0
- Or through the use of parameters in the expression of $\Pi_{Acceleration}$.

These factors, particularly the $\Pi_{Acceleration}$ factor, are defined for each type of physical stress. The term "physical stress" refers to any stress normally applied to the product during operational use, including with regard to design aspects. Physical stresses are grouped into families:

- | | |
|-------------------|--------------------|
| • Thermal | $\Pi_{Thermal}$ |
| • Electrical | $\Pi_{Electrical}$ |
| • Thermal cycling | Π_{TCy} |
| • Mechanical | $\Pi_{Mechanical}$ |
| • Humidity | Π_{RH} |
| • Chemical | $\Pi_{Chemical}$ |

The factors contributing to these physical stresses are usually additive. For some families of items, the thermal and electrical contributions are combined: $\Pi_{Thermo-electrical}$.

1.8.2. Overstresses: Π_{Induced}

The induced factors considered are of mechanical (MOS), electrical (EOS) or thermal (TOS) origin.

The Π_{Induced} factor represents the contribution of overstresses not listed as such. It is calculated for each phase “i” of the mission profile.

It takes the following form:

$$\Pi_{\text{induced}_i} = (\Pi_{\text{Placement}_i} \times \Pi_{\text{Application}_i} \times \Pi_{\text{Ruggedizing}})^{0,511 \times \text{Ln}(C_{\text{Sensitivity}})}$$

Where:

- $\Pi_{\text{Placement}}$ represents the influence of the item's position in the equipment or system. The placement here indicates the position of the item or the function of which it is a part (in particular, whether it is an interface element).
- $\Pi_{\text{Application}}$ represents the influence of the environment of use during the application of the product containing the item. For example, exposure to mechanical overstress is, in principle, greater for electronics integrated into a mobile system than into a static system. This factor varies according to the phase of the mission profile.
- $\Pi_{\text{Ruggedizing}}$ represents the influence of the policy of building overstress tolerance into product development.
- $C_{\text{Sensitivity}}$ represents the overstress sensitivity coefficient inherent to the technology of the item in question.
- i is the index of the relevant phase.

The Π_{Induced} factor theoretically ranges from 1 (best case) to 100. However, only a small part of this range is covered in practice, as extreme cases are never encountered simultaneously.

1.9. Process factors

1.9.1. The part manufacturing factor

The Π_{PM} factor (PM for Part Manufacturing) represents the quality of the item. The assessment method depends on the nature of the item considered (EEE electronic component, board assembly, or subassembly).

It takes the form:

$$\Pi_{PM} = e^{\delta_1 \times (1 - Part_Grade) - \alpha_1}$$

Where:

$$Part_Grade = \left\lceil \frac{(QA_{Manufacturer} + QA_{Item} + RA_{Component}) \times \varepsilon}{36} \right\rceil$$

The assessment method takes account of the manufacturer's quality assurance criteria ($QA_{Manufacturer}$), the quality assurance of the item (QA_{Item}), and also, where applicable, the buyer's past experience of the supplier (ε).

δ_1 and α_1 are correlating factors that determine the size of the impact of Π_{PM} on the item's reliability and provide results in the required range.

$$\begin{cases} \delta_1 = \ln(Min) \\ \alpha_1 = \ln(Max) \end{cases}$$

For active components, the Π_{PM} factor evaluation principle also includes the qualification tests and periodic reliability monitoring tests, both on the package and on the active part (component reliability assurance, $RA_{Component}$.) This data is to be found mainly in reliability reports and audit results.

The Π_{PM} factor ranges from 0.5 (supplier superior to state of the art) to 2 (worst case).

If Π_{PM} is not evaluated, a default value of 1.7 is proposed for active components and 1.6 for other components, COTS boards and the various subassemblies. Using the default value may reduce the accuracy of the final results.

1.9.2. The Π_{Process} factor

The Π_{Process} factor represents the quality and technical control of reliability in the product life cycle.

Its purpose is to provide an overall evaluation of the maturity of the manufacturer in terms of control over the reliability engineering process.

It takes the form:

$$\Pi_{\text{Process}} = e^{\delta_2 \times (1 - \text{Process_Grade})}$$

Where *Process_grade* is the score reflecting the level of process control, and δ_2 a correlating factor that determines the range of variation of the Π_{Process} factor.

The evaluation method is based on the degree to which recommendations are applied across the entire life cycle. The product life cycle is subdivided as follows:

1. Specification
2. Design
3. Board or subassembly manufacturing
4. Equipment integration (manufacturing)
5. System integration (manufacturing)
6. Operation and maintenance.

In addition to these six phases, which are consecutive in time, there is a set of transverse activities:

7. Support activities such as quality and human resources.

The recommendations make no claim to be exhaustive; they should be seen, rather, as a representative sample of good practices for improving end product reliability.

The Π_{Process} factor ranges from 1 (for the best process) to 8 (for the worst process).

If Π_{Process} is not evaluated, a default value of 4.0 is proposed. Using the default value may reduce the accuracy of the final results.

1.9.3. Remark on the supply phase

The supply of an item corresponds to a life phase between the moment the item leaves the factory and the moment it is assembled in the product (for example when it is attached to a circuit board).

In the FIDES models, there is no specific supply factor.

The influence of the supply phase on reliability is recognized as being dependent on:

- The company's procurement policy
- The item selection policy (upstream technology studies)
- The item storage, environmental stress screening, handling and inspection policy.

These points are embodied in recommendations specific to the *support*, *design*, and *board manufacturing* life cycle phases (the effect of these recommendations is taken into account in the evaluation of Π_{Process}) and in the choice of the ε factor for Π_{PM} .

The supply policy also has an indirect influence on the other parameters of Π_{PM} .

2. Mission profiles

2.1. Principles for building mission profiles

2.1.1. General

To define the mission profile for a failure rate prediction, we need to ask what is going to cause the product to fail during its life. This is a reliability engineering approach. It is crucial for reliability evaluations, because it totally determines the accuracy of the predictions.

The FIDES models have been designed to be sensitive to physical factors. When building the mission profile, choosing amplified or severe values so as to be “on the safe side” will substantially undermine the predictive value of the result.

The level of detail and precision with which the mission profile is described may be limited to the level of precision with which the lifetime of the product can be predicted.

2.1.2. General description of the mission profile

For optimal use of the mission profile, it must first be adequately described in qualitative terms.

In particular, it is important to identify:

- The precise type of platform, if the product is integrated into a system
- The location in the platform, if applicable
- The geographic region or climate zone in question
- The type of use.

A customer project manager or contractor project manager must build a system-level mission profile, as opposed to an equipment-level mission profile. An equipment-level mission profile is the expression of a system mission profile at the level of one item of equipment. The equipment mission profile must take account of local conditions internal to the equipment that are not easily generalized, such as temperature rises due the equipment itself, the damping or amplification of vibration amplitudes, any drying procedures, etc.

2.1.3. Choice of phases

The choice of phases must be sufficient to describe the different situations of use as completely as possible.

To make a complex mission profile easier to understand, it may be useful to devote a descriptive paragraph to each phase. At the very least, it is essential to give each phase a clear title.

A specific phase must be distinguished every time the environmental conditions change significantly in terms of the stresses encountered. Under this approach, the application questionnaire (which relates to the induced factor) must also be taken into account.

The situations of use must be identified first at system level and not at the level of the electronic equipment. The reason being that changes of situation at system level generally have an impact at equipment level.

In principle, the equipment-level mission profile comprises at least as many phases as the system-level mission profile. It is also possible for changes to occur at equipment level without there being any notable change at system level. In this case, the equipment-level mission profile will have more phases than the system-level mission profile.

There is no universal method for breaking mission profiles down into phases, but it is often helpful to proceed by analysing “typical days” in the use of the product.

In some cases it may be useful to distinguish between seasons (an example is provided in the VIP Helicopter mission profile).

2.2. Duration of phases

The mission profile can be built for the entire duration of the product (T_{Total}); this is the easiest way to express the characteristics of the mission profile phases. It can be instructive to build the mission profile for the operating life of the product.

It is recommended, where it can be done without losing information, that mission profiles be defined with a total duration of 1 year, i.e. 8 760 hours.

In the FIDES method, every hour is taken into account: 24 hours per day, 730 hours per month (average), 8 760 hours per year; for clarity, the term “calendar hour” is employed.

The aim is to determine failure rates expressed in calendar-based FITs (1 FIT represents one failure per 10^9 hours), with the widest possible usage. This choice is recommended rather than using failure rates expressed “per operating hour” or “per mission hour”, which can be misleading.

Phase durations must be expressed in hours.

Durations must be chosen to describe the product activity as realistically as possible.

2.3. Domains of applicability

A domain of applicability is proposed for each of the physical factors addressed by the FIDES method. The physical factors in question are:

- Temperature
- Thermal cycling
- Humidity
- Vibration.

In general, a reliability prediction is only applicable in the environmental domain for which the component is qualified. The qualification of a component for a given environment can either be guaranteed by the supplier or obtained by other means. It is, in every case, a prerequisite.

The proposed domains of applicability are defined on a theoretical basis. Using the methodology within these domains does not provide a guarantee as to the result. Even within the domain of applicability, the realism of predictions made with the FIDES method

may be diminished if the environmental conditions are very different from the reference conditions.

2.4. Temperature (thermal and thermo-electrical stress)

2.4.1. Physics of failure and modelling

The Arrhenius equation is used to model the acceleration of certain failure mechanisms due to temperature.

The acceleration factor is written:

$$AF = e^{\frac{E_a}{K_B} \times \left(\frac{1}{T_1} - \frac{1}{T_2} \right)}$$

Where:

- AF: acceleration factor
- E_a : activation energy
- K_B : Boltzmann's constant = $8.617 \cdot 10^{-5}$ eV/K
- T_1 : reference temperature
- T_2 : application temperature.

Failure mechanisms activated by electrical operation of components are often taken into account by allowing for thermal dissipation in the calculation of the application temperature (for example the junction temperature for active components) and adding the working voltage/rated voltage ratio to the model. The acceleration factor then becomes, for example:

$$AF = \left(\frac{1}{S_{Reference}} \times \frac{V_{Applied}}{V_{Rated}} \right)^p \times e^{11604 \times E_a \times \left[\frac{1}{273+T_0} - \frac{1}{(273+T_{Ambient})} \right]}$$

Where:

- $T_{ambient}$: working temperature
- T_0 : reference temperature
- $V_{applied}$: working voltage
- V_{rated} : rated voltage
- $S_{reference}$: reference level for electrical stress
- p : acceleration power for electrical stress
- The value of the activation energy E_a depends on the technology considered.

Note: 11 604 is an approximation of $\frac{1}{K_B}$.

2.4.2. Reference conditions

The reference conditions are:

- Temperature $T_0 = 20^\circ\text{C}$
- An electrical stress level defined on the basis of the technology, when necessary.

2.4.3. Domain of applicability

The reliability prediction is applicable only in the range for which the component is qualified. The theoretical temperature range for which the FIDES Guide is applicable is: $-55^{\circ}\text{C} \leq T_{\text{ambient}} \leq +125^{\circ}\text{C}$.

2.4.4. Quantification of the parameters

The following input data is required for each phase considered:

- The ambient temperature T ($^{\circ}\text{C}$)
- The state: operating or non-operating (in most cases thermal stress is ignored during non-operation).

Even if temperature is a common factor with previous methods, notably MIL-HDBK-217, it must not be estimated in the same way for FIDES as it is for MIL-HDBK-217.

For MIL-HDBK-217, temperature was the only physical factor used. Consequently, with this methodology, this factor was often used to adjust the general severity of the environment.

The FIDES physical temperature model is generally more sensitive than in previous methods. Realistic temperature values should therefore be used. The approach that involves using an artificially high fixed temperature inevitably leads to artificially exaggerated estimates.

The temperature to be used in the model is the ambient temperature of the environment. In general, the temperature to be considered for this purpose is that of the environment in which the studied item is located, whether it is powered up or not. Where necessary, the models deal explicitly with the temperature rise in the item relative to its surroundings (especially for active components, for which the model considers the junction temperature).

When the product is powered down, the immediate climatic conditions can provide the ambient temperatures to which the system is subjected.

For reliability assessments at component level, the ambient temperature to be considered is the ambient temperature around the circuit board. For example, for a board integrated into equipment, the temperature to be considered is the ambient temperature inside the equipment. During an operating phase, this temperature must include the temperature rise due to heat dissipation from the components.

Thermal simulation tools can provide detailed data on the temperatures in electronic equipment even during the preliminary phases of development. This simulation data can be further refined. In particular, when the temperature in the equipment is not uniform, the temperature can be adapted to each zone by assuming a different temperature for each board, so long as this remains consistent with the expected accuracy of the analysis. With this approach, FIDES can be used as a discriminating tool that is equally useful for designing the internal layout of electronic equipment.

For reliability assessments of hardwired boards, the same rule applies. For reliability assessments of subassemblies other than boards, the ambient temperature of the

surroundings should be used. As information on temperature rises in subassemblies is often not directly accessible, the models are designed to do without.

For a phase in which the temperature starts by changing and then stabilizes (as is usual when warming up after launch), it is generally acceptable to take the stabilized temperature as representative of the phase as a whole. For a phase in which the temperature is constantly changing and does not stabilize, the temperature to be considered is not necessarily the temporal mean (i.e. time-averaged) value, due to the non-linear influence of temperature. In this case the representative temperature is higher than the temporal mean (for an example, see the VIP Helicopter mission profile).

2.5. Thermal cycling (thermo-mechanical stress)

This stress is associated with the temperature cycles encountered by the product, whether in operating or dormant mode, considering the temperature variations relating to its operation (on/off status, in particular) and variations in its environment (day/night, for example).

2.5.1. Physics of failure and modelling

The Norris-Landzberg equation is used to model the acceleration of the fatigue mechanism due to thermal variation. It is derived from the Coffin-Manson equation commonly used for thermo-mechanical fatigue. It takes account of the fact that slower temperature cycles are more damaging, due to the phenomenon of creep (which affects soldering). The Norris-Landzberg equation has itself been modified for this guide, to convert the usual type of prediction generated by the model (a number of cycles) into an acceleration factor that can be applied to a failure rate.

The acceleration factor is written:

$$AF = \left(\frac{24}{N_0} \times \frac{N_{cy}}{t_{phase}} \right) \times \left(\frac{\min(\theta_{cy}, 2)}{\min(\theta_0, 2)} \right)^p \times \left(\frac{\Delta T_{cycling}}{\Delta T_0} \right)^m \times e^{\frac{E_a}{K_B} \times \left[\frac{1}{273+T_0+\Delta T_0} - \frac{1}{T_{max-cycling}+273} \right]}$$

Where:

- N_{cy} : Number of cycles
- N_0 : Reference number of cycles
- t_{phase} : Duration of the phase
- θ_{cy} : Cycle duration in hours
- θ_0 : Reference cycle duration
- $\Delta T_{cycling}$: Thermal amplitude of the cycle
- ΔT_0 : Reference thermal amplitude of the cycle
- E_a : Activation energy
- K_B : Boltzmann's constant ($8.617 \cdot 10^{-5}$ eV/K)
- $T_{max-cycling}$: The maximum temperature reached during the cycle
- T_0 : Reference temperature
- m : Fatigue coefficient, e.g. $m=1.9$ for the fatigue of tin-lead solder joints (value also applicable to other types of solder joints)
- $p(=1/3)$: Acceleration power of the duration factor.

In practice, the following formula is used in the rest of this document:

$$AF = \left(\frac{24}{N_0} \times \frac{N_{cy}}{t_{phase}} \right) \times \left(\frac{\min(\theta_{cy}, 2)}{\min(\theta_0, 2)} \right)^p \times \left(\frac{\Delta T_{cycling}}{\Delta T_0} \right)^m \times e^{1414 \times \left[\frac{1}{273 + T_0 + \Delta T_0} - \frac{1}{(T_{max-cycling} + 273)} \right]}$$

Note: The constant 1 414 corresponds to an activation energy of 0.122 eV.

The thermal cycling model does not deal with thermal shocks (whose transition speed is greater than 20°C / min – cf. § 12.5.3).

2.5.2. Reference conditions

The reference conditions are:

- A cycle amplitude ΔT_0 of 20°C
- A cycle frequency N_0 of 2 cycles a day
- A cycle duration θ_0 of 12 hours
- A maximum temperature $T_{max-cycling}$ (i.e. $T_0 + \Delta T_0$) of 40°C.

2.5.3. Domain of applicability

The theoretical range of the temperature cycle for which the FIDES Guide is applicable is $\Delta T_{cycling} \leq 180^\circ\text{C}$, $T_{max_cycling} \leq 125^\circ\text{C}$, thermal transition rate $\leq 20^\circ\text{C} / \text{minute}$.

2.5.4. Quantification of the parameters

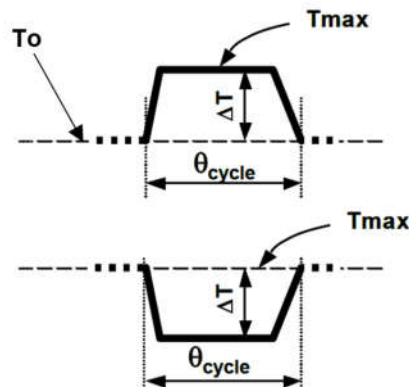
The input data is, for each phase considered:

- The amplitude of the cycle in temperature ΔT (°C)
- The associated number of cycles over one year (quantity)
- The duration of a cycle θ_{cycle} (hours)
- The maximum temperature of the cycle (°C).

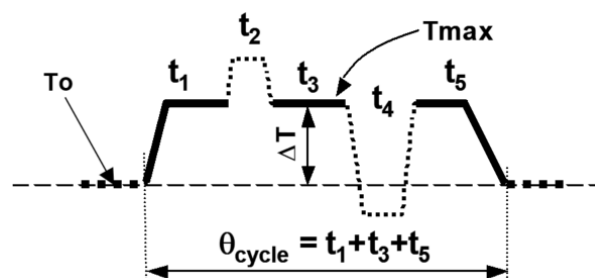
The temperatures considered for thermal cycling must be the same as those described for the temperature characteristic itself.

To ensure a representative and reproducible description of temperature cycles, the following rules should be applied:

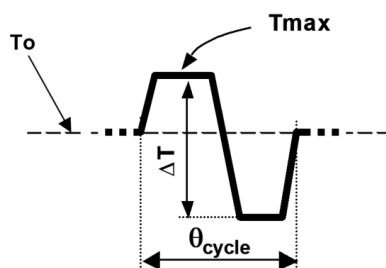
1. Cycles must be evaluated starting from an initial reference temperature on the equipment, e.g. equipment at rest (Off).
2. A cycle generally corresponds to a temperature excursion ΔT relative to the reference temperature; the cycle time θ_{cycle} extends until return to the initial temperature.



3. Other cycles may overlap with or be contained in a cycle; if so, the times of the sub-cycles must be subtracted from that of the primary cycle.



4. In some special cases, it may be necessary to analyse a cycle as a temperature variation around a mean temperature (as with day/night cycling for example).



5. In many cases, $\theta_{Cycle} = \frac{\text{Duration of phase}}{\text{Number of cycles}}$, but this is generally a simplification (for examples, see the VIP Helicopter and Military Portable Radio mission profiles).
6. A temperature cycle must correspond to an identified phenomenon that generates the stress, e.g. powering up, gaining altitude, or overheating due to a system state. Cycles must be treated as a whole, not divided up into a number of arbitrary sub-cycles that do not correspond to any real time in a profile phase.
7. A profile may comprise several identical cycles in succession. If so, the number of identical cycles must be counted.

It is especially important to comply with Rules 3 and 6 previously mentioned. Simply reading the temperature profile of the product without taking account of Rule 6 can lead to confusion. The application of Rule 6 takes priority:

- First, identify the phenomenon that causes the temperature cycle. For example:
 - Powering up – the end of the cycle must therefore correspond to powering down
 - Change of operating phase – for example switching an additional load to a power supply
 - Change of environmental temperature – e.g. day-night cycle
 - Change due to the system moving within its environment – e.g. from an air-conditioned to a non-air conditioned area, or change of altitude for an aircraft.
- When examining the temperature profile as a function of time, it is important to identify temperature cycles in full, including the temperature change phase that starts the cycle as well as the phase in which it returns to the initial temperature. For nested cycles, be careful not to associate the profile with a temperature transition cycle instead of a return cycle.
- When temperature cycles overlap, apply the counting principles in Rule 3. It is important to distinguish each temperature cycle independently, ignoring the other cycles. The plateau temperature of the first temperature cycle becomes the reference temperature of the next.

There is usually a time lag between a temperature cycle and the event that triggers it (for example, powering down does not cause an instantaneous return to ambient temperature). Depending on the specific case, this effect can sometimes be ignored.

2.6. Relative humidity

2.6.1. Physics of failure and modelling

Relative humidity (expressed as a %) is the ratio between the pressure of the water vapour contained in the air and the saturating vapour pressure (which depends on the temperature of the mass of air).

Peck's equation is used to model the acceleration exerted on certain failure mechanisms by the combination of relative humidity and temperature.

$$AF = \left(\frac{RH_{Ambient}}{RH_0} \right)^p \times e^{\frac{E_a}{K_B} \times \left[\frac{1}{273+T_0} - \frac{1}{273+T_{Ambient}} \right]}$$

Where:

- $RH_{ambient}$: Relative humidity of the environment considered
- RH_0 : Reference relative humidity
- $T_{ambient}$: Environmental temperature considered
- T_0 : Reference temperature
- E_a : Activation energy
- K_B : Boltzmann's constant ($8.617 \cdot 10^{-5}$ eV/K)
- p : Acceleration power for this stress.

In the rest of this guide, Peck's formula may be encountered in the following form:

$$AF = \left(\frac{RH_{Ambient}}{RH_0} \right)^p \times e^{11604 \times E_a \times \left[\frac{1}{273+T_0} - \frac{1}{(273+T_{Ambient})} \right]}$$

Where 11 604 is the rounding of $\frac{1}{K_B}$.

2.6.2. Reference conditions

The reference conditions are:

- A relative humidity RH_0 of 70%
- An ambient temperature T_0 of 20°C.

2.6.3. Domain of applicability

The theoretical validity range is from 0% to 100%.

Cases of condensation or frosting are not addressed.

2.6.4. Quantification of the parameters

The input data is, for each phase considered:

- The relative humidity level RH (%)
- The ambient temperature T (°C)
- The state: operating or non-operating (in most cases, humidity stress is ignored during operation).

The temperature is the same as described in the temperature section.

Like temperature, relative humidity varies depending on the climate. It is important to use a humidity level that is representative of the climate considered.

Documents such as STANAG 4370 contain tables of minimum and maximum humidity values in different parts of the world and can be used in the absence of more detailed information (cf. §2.10).

When estimating the relative humidity level, it is important to take account of the relative humidity actually experienced by components. Allowance must be made, for example, for the airtightness of the product, the possibility of moisture being trapped in a hermetically sealed case, or the effect of drying agents that can significantly reduce the humidity level experienced by components relative to that of the environment.

Drying agents are sometimes used to reduce the relative humidity. For example, if silica gel bags with a mass of about 1g are placed in a case containing less than 10L of air, the relative humidity usually drops to less than 10%. Standards document DIN 55474 suggests a method for calculating the quantity of desiccant required to bring the final relative humidity (and a number of other parameters) within tolerance. Drying agents must be renewed when their absorption capacity is depleted.

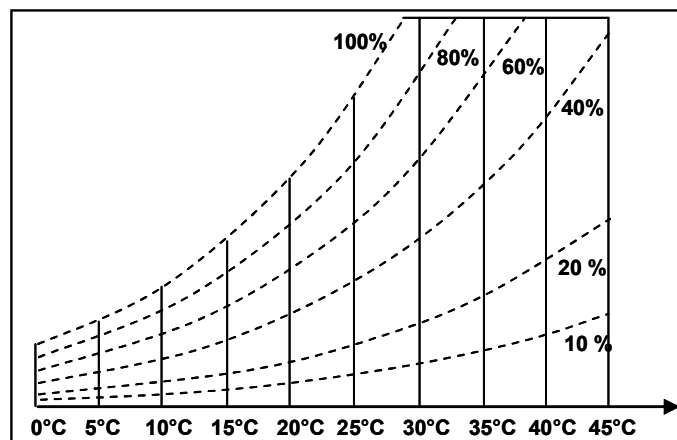
It may be necessary to take into account the possible presence of air conditioning, which often dries the air down to a relative humidity of 30% to 40% (below which the environment is less comfortable for people).

Change in humidity as a function of temperature: at constant air composition, humidity decreases as temperature increases. Air inside functioning equipment is usually warmed up, which reduces the humidity experienced by the components.

At air constant composition and in the absence of condensation, the variation in RH as a function of temperature can be calculated using the formula:

$$RH_{final} = RH_{initial} \times e^{17.2694 \times \left[\frac{T_{initial}}{238.3 + T_{initial}} - \frac{T_{final}}{238.3 + T_{final}} \right]}$$

The change in the RH as a function of temperature can also be read off from a hygrothermal diagram.



Change in humidity as a function of altitude: the humidity level varies in response to altitude. In general, as altitude increases, humidity decreases, becoming zero beyond the troposphere. However, this change is very irregular and hard to predict. In particular, the humidity level increases in cloud layers. The common simplification is to assume a mean relative humidity independent of altitude.

In mission profiles that include long storage periods, the relative humidity factor may become preponderant. As much care must then be given to determining the relative humidity as is given to determining the temperature.

Note: The influence of temperature is stronger during storage (due to humidity) than during operation.

2.7. Vibration amplitude (mechanical stress)

2.7.1. Physics of failure and modelling

The Basquin equation is used to take account of the fact that as the vibration amplitude increases, the risk of failure in circuit boards and components also rises. There is a wide variety of failure mechanisms, and not all of them are fatigue mechanisms, for which the Basquin equation is generally used. In some cases vibrations will reveal weaknesses such as, for example, dry solder joints, cracks in parts (substrate, component packages, etc.) and adherence issues at interfaces (bonding defects, delamination, etc.). Where metallic particles are present in a hermetically sealed case, vibrations will increase the risk of a short circuit by causing the particles to move around. Furthermore, if the failure mechanism is associated with mechanical fatigue, there is a wide variety of materials that can be degraded by fatigue or wear (aluminium, copper, silicon, epoxy, glass, ceramic, etc.).

$$AF = \left(\frac{G_{RMS}}{G_{RMS0}} \right)^p$$

Where:

- G_{RMS} : Root mean square vibration amplitude in the environment considered
- G_{RMS0} : Reference vibration amplitude
- p : Acceleration power for mechanical stress.

The acceleration factor derived from the Basquin equation for the FIDES model is set at $p=1.5$. This value is fairly close to the bottom of the range of “fatigue coefficients” usually encountered for the Basquin equation. This should be borne in mind when using the FIDES mechanical model in conjunction with acceleration models employed for tests.

This mechanical model does not deal with shocks. Shocks must be covered in terms of robustness when qualifying the product for its future use.

2.7.2. Reference conditions

The reference conditions are:

- A vibration amplitude G_{RMS0} of $0.5 G_{RMS}$.

2.7.3. Domain of applicability

The theoretical validity range is $G_{RMS} \leq 40 G_{RMS}$.
Shocks are not addressed.

2.7.4. Quantification of the parameters

The input data is, for each phase considered:

- The random vibration amplitude, expressed in G_{RMS} .

The vibration amplitude must be analysed in the relevant frequency range for the product considered. As the frequencies that need to be taken into account are variable, the calculation must be done on a wide frequency band (20-2000 Hz). Electronic circuit boards are generally most sensitive to vibration in the axis perpendicular to the plane of the board.

For the physics of failure at component level, the relevant vibration amplitude is that experienced by the stressed element, for example:

- The solder of a component on a board
- The pin of a through-hole component
- The solder of a strip inside a relay
- The gluing of a microcomponent within a hybrid.

But the vibration amplitude depends on many different factors:

- The vibration amplitude at the input to the circuit board
- The amplification of the board at the spot where the component is located
- The frequencies of stress relative to the natural resonant modes of the board
- The natural frequency of the stressed element (in some cases)
- etc.

As it is not practically possible to integrate all of these parameters into a predictive reliability study, the model parameter setting will be made using the input level for the product (the level experienced by the equipment or the board). One way of controlling the reliability (process aspects) is to avoid placing the most sensitive components at the most exposed locations on a board.

The level of amplitude to be used must be as close as possible to the level actually experienced in service.

In reliability studies, vibration amplitudes often have to be quantified from technical specifications, before the real levels can be measured. Caution must be exercised when using the vibration levels specified for testing. These levels often correspond either to accelerated or ruggedization tests, or to the extreme levels that the product may face. The right levels to select are the endurance vibration levels, non-accelerated and non-ruggedized.

Special care must be taken to avoid the following sources of confusion:

- Confusion between the input vibration amplitude and the amplified vibration amplitude on the components of the circuit board
- Confusion between qualification levels (generally conservative due to ruggedization) and typical in-service levels
- Confusion between accelerated test levels (e.g. for lifetime testing) and nominal test levels.

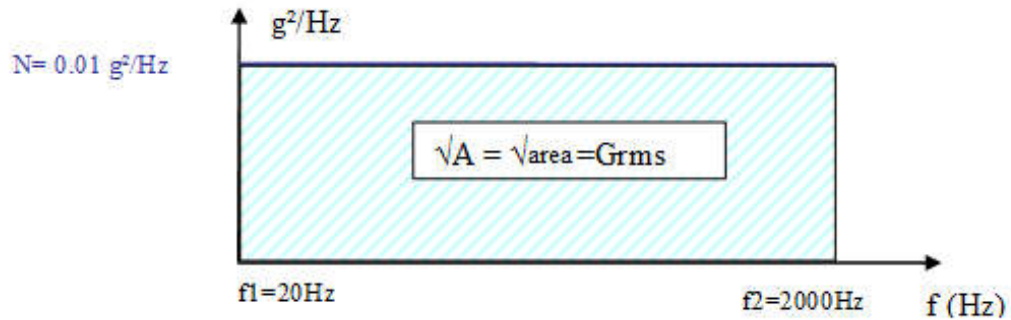
Vibration amplitudes are usually given in the form of Power Spectral Density (PSD). If so, the excitation level (in G_{RMS}) can be determined from the PSD, by taking the square root of the area under the curve of the power spectrum.

It is recommended that a value of 0.01 Grms should be kept for phases with no foreseeable random vibrations.

A calculation example is provided in the following sections.

2.7.5. Calculating the G_{RMS} (rms acceleration) from the PSD

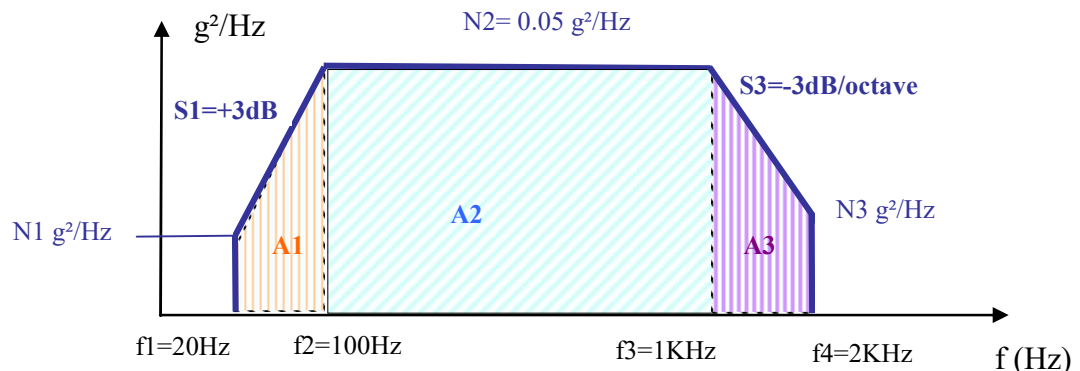
When the excitation level is constant over the entire frequency band, the area can be calculated directly.



$$G_{RMS} = \sqrt{A}$$

$$G_{RMS} = \sqrt{19.8} = 4.45$$

When the spectrum has different levels for different frequency bands, the total area must be broken down as shown below.



When the slope is other than -3dB per octave:

$$A1 = \frac{3N2}{3+S1} \left(f2 - \left(\frac{f1}{f2} \right)^{S1/3} \times f1 \right) \text{ or } A1 = \frac{3N1}{3+S1} \left(\left(\frac{f2}{f1} \right)^{S1/3} \times f2 - f1 \right)$$

$$A1 = \frac{3 \times 0.05}{3+3} \left(100 - \left(\frac{20}{100} \right)^{3/3} \times 20 \right) = 2.4$$

When the level is constant over a frequency band:

$$A2 = N2(f3 - f2)$$

$$A2 = 0.05(1000 - 100) = 45$$

When the slope is -3dB per octave:

$$A3 = f3 \times N2 \times \ln\left(\frac{f4}{f3}\right) \text{ or } A3 = -f4 \times N3 \times \ln\left(\frac{f3}{f4}\right)$$

$$A3 = 1000 \times 0.05 \times \ln\left(\frac{2000}{1000}\right) = 34$$

If slope S3 is not -3dB per octave, A3 is calculated in the same way as A1:

$$A3 = \frac{3N3}{3 + S3} \left(f4 - \left(\frac{f3}{f4} \right)^{S3/3} \times f3 \right)$$

Finally:

$$G_{RMS} = \sqrt{(A1 + A2 + A3)}$$

$$G_{RMS} = \sqrt{81.4} = 9$$

2.8. Chemical stress

2.8.1. Modelling

Chemical stress is modelled qualitatively; there is no physical model for this type of stress. In some models, chemical stress becomes an acceleration factor of other physical stresses.

The chemical contribution to product reliability is expressed through four contributing factors related to the use of the product:

- In its environment:
 - Ambient salinity (higher in coastal or marine environments)
 - Industrial or natural chemical impact on the environment (pollution).
- In its system:
 - Chemical contribution due to the placement of the product in the system or to the nature of the system (local pollution)
 - Level of product protection within the system, hermetically sealed or otherwise (note that this is not the same thing as the hermeticity of component packages).

2.8.2. Quantification of the parameters

These four criteria are described in the following tables.

Saline pollution level	Example
Low	Continental area
High	Coastal area

Environmental pollution level	Example
Low	Rural area
Moderate	Urban area
High	Urban industrial area

Application pollution level	Example
Low	Inhabited or maintained area
Moderate	Uninhabited or unmaintained area
High	Engine area

Product protection level	Example
Hermetic	Hermetic protection
Non-hermetic	Other protections

2.9. Type of application

The $\Pi_{\text{Application}}$ parameter of the induced factor is determined by means of a questionnaire.

Different criteria are used to evaluate the severity of a usage phase in terms of exposure to overstress. There are three levels per criterion. The evaluation of these levels is used to calculate the $\Pi_{\text{Application}}$ parameter. The complete method is described in detail in the calculation sheets. The criteria are as follows:

- User type: Represents professionalism, compliance with procedures, weight of operating constraints
- User qualification: Represents the user or operator's level of expertise in a given operating context
- System mobility: Represents hazards relating to potential movement of the system
- Product handling: Represents the risk of incorrect handling, shocks, falls, etc.
- Power supply type: Represents the level of electrical disturbance expected on power supplies and signals: powering up, power supply switching, connection/disconnection
- Exposure to human activity: Represents exposure to hazards relating to human activity: shocks, use for other purposes, etc.
- Exposure to machine disturbance: Represents hazards relating to the operation of machines, engines and actuators, e.g. shocks, overheating, electrical disturbance, release of aggressive pollutants
- Exposure to weather: Represents exposure to rain, hail, frost, sandstorms, lightning, dust, etc.

The complete description of a mission profile must include the answers to this questionnaire.

The main rule to follow for answering these questions is to address the question at the right level, i.e. product or system level. The following table indicates which level to address:

Criterion	Level
User type in the phase considered	Complete system
Level of user qualification in the phase considered	User of product within complete system
System mobility	Complete system
Product handling	Product
Type of electrical network in the system	System and product
Exposure of product to human activity	Product within complete system
Exposure of product to machine disturbance	Product within complete system
Exposure of product to the weather	Product within complete system

2.10. Data sources

There are many data sources for establishing a mission profile. The main sources are:

- Statement of requirements by the customer or system integrator
- Weather and climate statistics
- Standards
- Experience feedback/field data
- Test and simulation results.

For example, international standards such as STANAG 4370 supply tables of minimum and maximum values that can be used to identify the climate conditions associated with certain geographic zones around the world; standards such as NF X 50-122 can help clarify the sequencing of situations and their frequency of occurrence; websites such as <https://www.metoffice.gov.uk> can provide more accurate local climate data than international standards.

Note that documents like STANAG 4370 are intended for configuring military systems tests and should be used with care when defining mission profiles for reliability studies.

The following confusions may cause difficulty in using the input documents:

- Confusion between qualification levels (which are extreme values) and typical levels (useful for reliability)
- Confusion between test levels and specified levels
- Confusion between accelerated (lifetime) test levels and nominal test levels
- Heterogeneity (or even conflict) between the different reference standards.

2.11. Method for building mission profiles

The steps recommended for building a mission profile are as follows:

- Analyse and “translate” the mission profile supplied by the customer
- Create a flowchart of situations encountered
- Create thermographs for each typical day/situation
- Create a mission profile for each typical day/situation
- Concatenate the different typical day/situation profiles
- Group/merge similar mission profile phases
- Evaluate $\Pi_{\text{application}}$
- Simplify the mission profile.

Recommendations:

1. The various steps for building a mission profile should be set out in a file that records the input data, assumptions and intermediate steps of the process.

2.11.1. Analyse the customer mission profile

Recommendations (ctd.):

2. All the assumptions made when building the mission profile must be registered in a document.⁽¹⁾ This document should be discussed and validated with the customer. Each assumption should be translated, where possible, into a retained value. It is important for each value to be recorded; it will then be used in the formulae, ensuring the traceability of the whole process.

⁽¹⁾ The input data is generally not sufficient to create a FIDES mission profile; it needs to be supplemented by a certain number of assumptions that make it possible to build and quantify the mission profile.

2.11.2. Create a situation flowchart

Standards such as NF X50-144-2 (Annex A) can help visualize the sequence of typical situations that the product will encounter during its lifetime, as well as the number of occurrences and the previous and subsequent situations. The proposed approach also provides a useful illustration of the different branches in the life of a set of products.

2.11.3. Create thermographs

Watch points:

3. It is important to do a thermograph (for an illustrative example see §3.1.3 or §3.2.7ff.) to account for the sequencing of mission phases.
4. Storage phases need to be included in the typical days to avoid the very frequent errors encountered in the description of these phases.
5. There are two types of temperature cycle:
 - “Up-and-down” cycles, usually represented in terms of time slots, where the plateau duration predominates over the rise and fall times; these generally correspond to changes of operating mode (on/off, standby/operation, transmission/reception, etc.) or to sudden changes in the thermal environment (e.g. changes of altitude).
 - “Oscillating” cycles for which the temperature varies regularly around a mean temperature (e.g. day/night cycles); these generally correspond to non-operating phases (dormant mode, storage).
6. Each cycle starts and finishes at the same temperature; this point is particularly important when cycles are nested (cf. recommendation 10).
7. The description of nested temperature cycles is a frequent source of error; thermographs make it easy to identify them and to describe them correctly (see §2.5).
8. The ambient temperature is the temperature recorded for the longest time during the phase:
 - For “up-and-down” cycles, this is the plateau temperature
 - For “oscillating” cycles, this is the mean temperature (by approximation).
9. If the temperature cycle is not stable (as when the thermal time constant of the equipment is non-negligible relative to the cycle duration), it can be hard to estimate the ambient temperature. In this case, the Sedyakin principle can be used to estimate an equivalent ambient temperature; to simplify, the ambient temperature can generally be approximated by $T_{\text{reference}} + 0.7 \times \Delta T$, but this needs to be modulated if specific temperature management conditions apply.

2.11.4. Create a mission profile for each typical day/situation

Recommendations (ctd.):

10. As the creation of the mission profile for each typical day/situation is closely linked to thermographs, a subprofile should be created for each thermograph.

Watch points (ctd.):

11. A temperature cycle must be described once and only once (see §2.5); in other words, it must be attached to a single phase, even if it affects several.
12. Only by meticulous description of temperature cycles is it possible to correctly estimate the duration and number of phases and temperature cycles.
13. This meticulous description of temperature cycles – made comprehensible by the use of thermographs – accounts for the fact that during nested cycles $t_{\text{phase}} \neq \Theta_{\text{cy}} \times N_{\text{cy}}$ (an example is given in §3.2).
14. Thermographs for typical days/situations apply to a whole number of days; the number of temperature cycles must therefore also be a whole number.

Recommendations (ctd.):

15. A written record should be kept of the calculations for each mission profile.

2.11.5. Concatenate the different typical day/situation profiles

Watch points (ctd.):

16. After concatenation, the number of cycles must always be a whole number.
17. If there is only one typical day, including a storage phase, and the baseline mission duration profile is 1 year, there should be 365 “day/night” cycles.

2.11.6. Group/merge identical or similar phases

Watch points (ctd.):

18. Particular care is required to evaluate the duration of temperature cycles during the merge operation:

- If the temperature cycles of the phases to be merged have a single duration greater than 2 h (corresponding to the duration of the reference cycles of the Norris-Landzberg law modified for FIDES – cf. § 12.5.1), the duration of the new cycle is evaluated by the formula:

$$\theta_{\text{equiv}} = \left(\frac{\sum_{i=1}^n N_i \theta_i}{\sum_{i=1}^n N_i} \right) \text{ where } m \text{ is the number of cycles to be merged, } N_i \text{ the number of cycles in Phase } i, \text{ and } \theta_i \text{ the cycle duration attached to Phase } i.$$

- If the temperature cycles of the phases to be merged have a single duration less than 2 h, the duration of the new cycle is evaluated by the formula:

$$\theta_{\text{equiv}} = \left(\frac{1}{\sum_{i=1}^n N_i} \right)^3 \left(\sum_{i=1}^n N_i \theta_i^{\frac{1}{3}} \right)^3$$

where m is the number of cycles to be merged, N_i the number of cycles in Phase i , and θ_i the cycle duration attached to Phase i .

- In both cases, the number of the new cycle is the sum of the numbers of the merged cycles:

$$N_{\text{equiv}} = \sum_{i=1}^n N_i.$$

19. After merging, the number of cycles must always be a whole number.

Recommendations (ctd.):

20. If the number of phases is reasonable (less than 20) and the profile is understandable by all the stakeholders without any complementary information, it is advisable to store the profile corresponding to the concatenation step.

2.11.7. Don't forget the $\Pi_{\text{application}}$ factor

Recommendations (ctd.):

21. The $\Pi_{\text{application}}$ questionnaire must be filled in for each mission profile.

2.11.8. Simplify the final FIDES mission profile

Watch points (ctd.):

22. At this stage, the number of temperature cycles must always be a whole number, as no changes have been made to the number of typical days and/or situations. However, the mission profile may not correspond to what the customer expects; for example, the consideration of typical days/situations can lead to a mission profile that covers several years, although the customer wants, for whatever reason, to have an annual mission profile. In such cases, proportional time scaling can be used, applying the same coefficient to the phase duration and the number of cycles in order to reduce the multiannual profile to a single-year profile.

2.12. Standard mission profile

2.12.1. Principle

The standard mission profile is intended to be used “as is”, to provide a baseline reliability value for a COTS product (particularly in its datasheet), equipment or subassembly.

The standard mission profile resembles that of an electronic desktop device, but its purpose is not to describe a real use case. It is designed to be as close as possible to the reference conditions for each physical factor (it is not possible to reproduce the reference conditions for all physical factors at once). It describes a non-aggressive environment.

Note: If the standard life profile is used for a datasheet, it is recommended to specify on it that the profile must be adapted to the use case.

2.12.2. Description

This mission profile corresponds to a product used 10 hours a day for 365 days a year (i.e. 3 650 hours a year) under typical office or laboratory conditions of use. When not in use (14 hours a day, 5 110 hours a year) the product is switched off.

⇒ In a real mission profile, the actual duration and the number of times powered up must be taken into account.

When switched off, the average temperature of the product is 20°C and it is subjected to a moderate day/night temperature cycle representative of a non air-conditioned office ($\pm 5^\circ\text{C}$ around the average value of 20°C).

⇒ In a real mission profile, the appropriate climatic conditions should be considered. When the non-operating time predominates in the mission profile, it is important to refine the day/night temperature cycle levels.

In the operating phase, the product is subject to a temperature rise of 20°C due to its dissipated power. Its ambient temperature is therefore taken to be 40°C and a temperature cycle with an amplitude of 20°C is assumed. The number of cycles corresponds to the number of startups, which is 365 for this profile. Remark: The operating duration is subtracted from the duration of the day/night cycle.

⇒ In a real mission profile, the appropriate temperature rises should be considered.

The relative humidity is assumed to be 70% in ambient air at 20°C. When the product is in operation the RH drops to 22%, due to the rise in temperature.

⇒ In a real mission profile the appropriate climatic conditions should be considered. When the non-operating time predominates in the mission profile, it is important to refine the RH level in storage.

The vibration amplitude is considered to be zero during non-operation and moderate during operation.

⇒ In a real mission profile, the appropriate vibration levels should be considered. When the non-operating time predominates in the mission profile, it may be important to check if there is any residual vibration amplitude in this phase.

The pollution level is assumed to be low for this product. The system is not hermetically sealed and the product is therefore nonetheless subject to slight chemical stress.

⇒ In a real mission profile, the appropriate chemical stresses should be considered.

The exposure to accidental overstress is assumed to be very low for this baseline mission profile. Such a low level assumes, in particular, that the user is qualified and complies strictly with procedures.

⇒ The $\Pi_{\text{application}}$ factor plays a key role in reliability assessment. It is very important to evaluate it specifically for each mission profile and to break it down for each phase.

2.12.3. Table

Standard mission profile														
		Temperature and humidity			Thermal cycling				Mechanical	Chemical				Induced
Phase name	Calendar time (hours)	On/Off	Ambient temperature (°C)	Humidity level (%)	ΔT (°C)	Number of cycles (/year)	Cycle duration (hours)	Max. temp. during cycling (°C)	Random vibrations (Grms)	Saline pollution	Environmental pollution	Application pollution	Protection level	TT application
Off	5 110	Off	20	70	5	365	14	23	0.01	Low	Low	Low	Non-hermetic	1
On	3 650	On	40	22	20	365	10	40	0.5	Low	Low	Low	Non-hermetic	1.9

3. Examples of mission profiles

The level of detail with which the product's mission profile is described, as part of an operational system, determines the accuracy of the reliability assessment. This step of the predictive analysis should therefore be carried out with great care. The sample profiles presented here illustrate the profile building rules from the previous chapter. They are not meant to be used as is; rather, they should be seen as starting points to be adapted as required.

3.1. Mission profile of a helicopter navigation computer

There are a number of different methods for compiling a mission profile. This example of the mission profile of a navigation computer on board a VIP helicopter specifically illustrates the "typical day sequencing" method.

This example additionally describes in detail how to choose and database the phases of the mission profile based on the system mission sequence. This approach ensures that the process of building the mission profile is easily reproducible, particularly for determining the time- and climate-related aspects of the profile.

3.1.1. Quantification of time data

Sequencing of a year

Principle: Identify the different typical days experienced by the system and quantify their occurrence during the year.

The VIP helicopter experiences two different types of day:

- "Operating" days: 100 days a year
- "Non-operating" days: 265 days a year.

Remark: Climate is a parameter that can play a role in this characterization. To increase the accuracy of the mission profile, and thus of the prediction, typical days can be differentiated on the basis of variation in outside temperature. In this application case, assuming a temperate climate with significant variations in outside temperature and/or humidity throughout the year, it can be useful to increase the accuracy of predictions by dividing each of these typical days into 3 subtypes:

- An "operating – winter" day for 3 months, i.e. $100 \times 3/12 = 25$ days/year
- An "operating – spring/autumn" day for 6 months, i.e. $100 \times 6/12 = 50$ days/year
- An "operating – summer" day, 25 days/year
- A "non-operating – winter" day, $265 \times 3/12 = 66$ days/year
- A "non-operating – spring/autumn" day, 133 days/year
- A "non-operating – summer" day, 66 days/year.

This more detailed breakdown takes account of applied outside temperatures closer to reality. Its validity can be assessed retrospectively by comparing the difference in the final predictions between the detailed profile and the simplified profile.

Sequencing of days

Principle: Break each identified typical day down into a series of chronological steps in each of which the external stresses (temperature, vibration, humidity, pollution, type of application) can be taken as constant. This is done by identifying the events during the day that trigger changes in these external stresses.

The day starts with the helicopter in the “Parked” phase. On average, a VIP helicopter performs two flights (one outbound, one return) per day of operation. Each flight involves the following events, in sequence:

- Event 1: “Start navigation computer” (change in thermal stress)
- Event 2: “Start engine” (change in vibration stress)
- Event 3: “Takeoff” (change in vibration and thermal stress)
- Event 4: “Landing” (change in vibration and thermal stress)
- Event 5: “Stop engine” (change in vibration stress)
- Event 6: “Shut down computer” (change in thermal stress).

And return to the Parked phase.

There are therefore 7 steps, separated by 6 events:

Events		Steps	
1	Start navigation computer	1	Parked
2	Start engine	2	Pre-flight tests of electronic systems
3	Takeoff	3	Preparation for takeoff
4	Landing	4	Flight
5	Stop engine	5	Ground – Engine On
6	Shut down computer	6	Post-flight tests of electronic systems
		7	Parked

This chronology is followed twice per operating day (outward and return flights).

For “non-operating” days, there are no events to be considered other than the external temperature variation during the day. There is therefore only one phase: the Parked phase.

Quantification of step durations

Principle: Collect or predict the average duration of each identified step.

Non-operating days:

Step		Duration
1	Parked	1 440 minutes

Operating days:

Step		Duration
1	Parked	
2	Electronic systems tests 1	1 minute
3	Preparation for takeoff 1	11 minutes
4	Flight 1	60 minutes
5	Ground – Engine On 1	3 minutes
6	Systems tests and shutdown 1	1 minute
7	Parked	
8	Electronic systems tests 2	1 minute
9	Preparation for takeoff 2	11 minutes
10	Flight 2	60 minutes
11	Ground – Engine On 2	3 minutes
12	Systems tests and shutdown 2	1 minute
13	Parked	

Grouping steps into phases

Principle: Steps are grouped together in a three-stage process:

- Identification of use cycles.
Start by identifying, for each typical day, the steps in which the equipment returns to its initial status. The initial status generally corresponds to the system at rest.
- Identification of variations in external stress and causes of variation.
For each step, identify the variations in stress (thermal, vibration, humidity, chemical or type of application) relative to the initial status (which serves as a baseline) and the underlying causes.
- Identification of phases for grouping.
The phases to be grouped together are those situated inside the same cycle of use and which share the same causes of external stress.

Two steps can also be grouped together (to simplify the mission profile) if it can be anticipated that the impact on the final result of the prediction will not be significant. This may be the case where:

- The difference between the external stresses on the equipment in the two situations is only small
- The duration of one of the situations is sufficiently short to make the impact negligible once it has been time-weighted.

VIP helicopter:

1. Identification of use cycles.

The initial status of the equipment is the “Parked” step. In an operating day, there are three steps of this type that delimit use cycles: namely Steps 1, 7 and 13. There are consequently two use cycles in an operating day: a first use cycle from Phase 2 to Phase 6 (the first flight) and a second use cycle from Phase 8 to Phase 12 (the second flight).

2. Identification of variations in external stress and causes of variation.

The following table describes variations and causes for an operating day.

Steps		Stress variation relative to Parked phase	Causes of variation
1	Parked	None	None
2	Pre-flight tests of electronic systems 1	▪ Thermal ▪ Humidity	♦ Heat dissipation from equipment in the zone
3	Preparation for takeoff 1	▪ Thermal ▪ Humidity ▪ Vibration	♦ Heat dissipation from equipment in the zone ♦ Vibrations generated by the engine (on the ground)
4	Flight 1	▪ Thermal ▪ Humidity ▪ Vibration	♦ Heat dissipation from equipment in the zone ♦ Vibrations generated by the engine (in flight) ♦ Cooling due to increase in altitude
5	Ground – Engine On 1	▪ Thermal ▪ Humidity ▪ Vibration	♦ Heat dissipation from equipment in the zone ♦ Vibrations generated by the engine (on the ground)
6	Post-flight tests of electronic systems 1	▪ Thermal ▪ Humidity	♦ Heat dissipation from equipment in the zone
7	Parked	None	None
8	Pre-flight tests of electronic systems 2	▪ Thermal ▪ Humidity	♦ Heat dissipation from equipment in the zone
9	Preparation for takeoff 2	▪ Thermal ▪ Humidity ▪ Vibration	♦ Heat dissipation from equipment in the zone ♦ Vibrations generated by the engine (on the ground)
10	Flight 2	▪ Thermal ▪ Humidity ▪ Vibration	♦ Heat dissipation from equipment in the zone ♦ Vibrations generated by the engine (in flight) ♦ Cooling due to increase in altitude
11	Ground – Engine On 2	▪ Thermal ▪ Humidity ▪ Vibration	♦ Heat dissipation from equipment in the zone ♦ Vibrations generated by the engine (on the ground)
12	Post-flight tests of electronic systems 2	▪ Thermal ▪ Humidity	♦ Heat dissipation from equipment in the zone
13	Parked	None	None

“Humidity” stress varies in line with the “thermal” stress as heating air causes it to dry.

3. Identification of phases for grouping.
Steps 2 and 6, 8 and 12, 3 and 5, 9 and 11 belong, in pairs, to the same operating cycle and present identical causes of stress variation. They should therefore be grouped together. Likewise, the parked phases should be grouped together. Now we can constitute the FIDES mission profile, made up of 8 phases, as shown in the following tables.

Non-operating day:

Phase		Duration
1	Off-24h	1440 minutes

Operating day:

Phase		Duration
1-7-13	Parked, operating	The remainder
2-6	Electronic systems tests 1	1 + 1 = 2 minutes
3-5	Preparation for takeoff 1	11 + 3 = 14 minutes
4	Flight 1	60 minutes
8-12	Electronic systems tests 2	1 + 1 = 2 minutes
9-11	Preparation for takeoff 2	11 + 3 = 14 minutes
10	Flight 2	60 minutes

Additionally, as Steps 2-6 and 8-12 take place over very short durations (2 minutes), one can anticipate that their impact will be small, and consequently group them together, merging Step 2 into Step 3 and Step 8 into Step 9.

The phases of the simplified FIDES mission profile are now as shown below.

Non-operating day:

Phase		Duration
1	Off-24h	1440 minutes

Operating day:

Phase		Duration
1	Off	1440 - 152 = 1288 minutes
3	Ground-On 1	16 minutes
4	Flight 1	60 minutes
9	Ground-On 2	16 minutes
10	Flight 2	60 minutes

Calculation of phase durations

The final step is to calculate the time spent per year in each identified phase ($t_{\text{annual_phase}}$) by multiplying the daily phase durations by their annual occurrence.

Phase	Daily duration	N° of days a year	$t_{\text{annual_phase}}$
Off-24h	1440 minutes	265	6360 h
Off	1288 minutes	100	2 146 h
Ground-On 1	16 minutes	100	27 h
Flight 1	60 minutes	100	100 h
Ground-On 2	16 minutes	100	27 h
Flight 2	60 minutes	100	100 h

3.1.2. Quantification of temperature data

The component ambient temperatures are read off from the daily equipment temperature graphs. This calls for three types of input data:

- Climate data: The temperature and humidity level outside the system. These vary as a function of the climate in the geographic area of operation.
- Zone data: Data on the system zone holding the equipment; used to determine the temperature variations (relative to the outside temperature) that will affect the zone during the different phases of the mission profile.
- Equipment data: Data on the equipment, to determine its internal temperature rise (relative to the zone temperature) due to its heat dissipation and/or, where applicable, its hermeticity to humidity and chemical stresses.

When it comes to calculating T_{ambient} , there are two different cases:

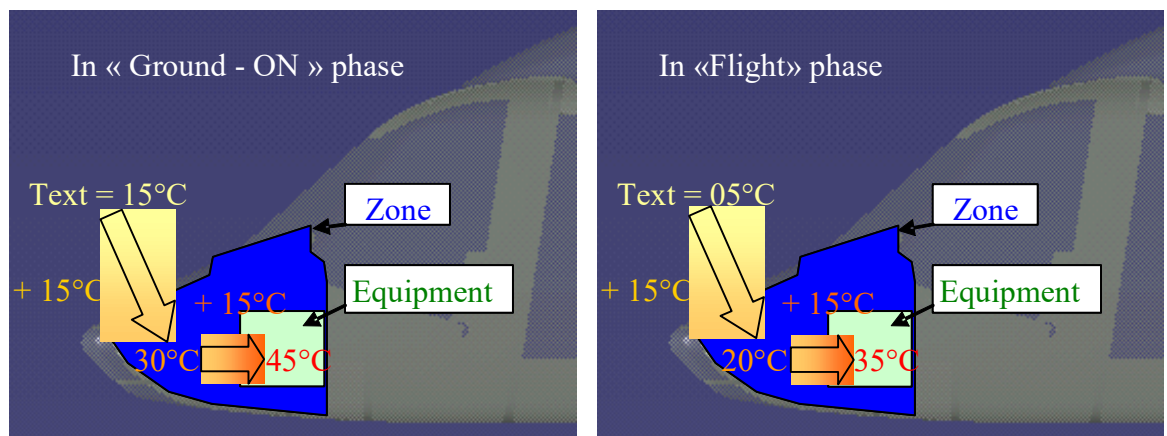
- Case 1: In the phase considered, the temperature varies from an initial temperature to a stable final temperature. The ambient temperature to be used is therefore the stable final temperature. However, if the phase is too short for the temperature to stabilize, the temperature to be used can be estimated as, for example, 70% of the final (non-stabilized) value reached during the phase.
- Case 2: In the phase considered, the temperature varies around a mean value. The ambient temperature to be used is therefore this mean temperature.

Remarks:

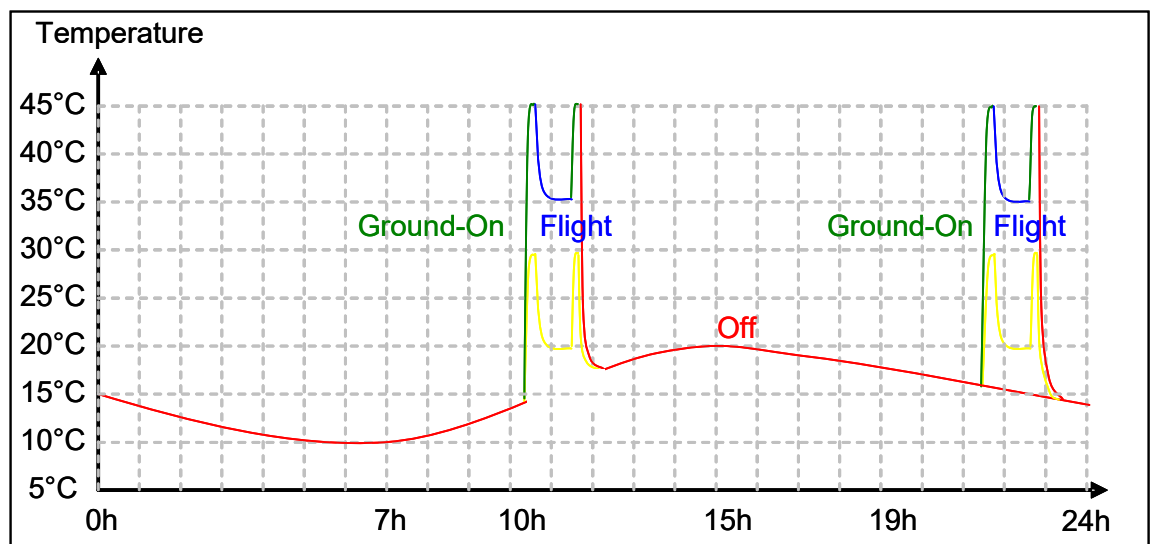
- The outside temperature varies as a function of altitude
- The zone temperature may be affected by kinetic heating.

VIP helicopter:

- Climate data: The mean outside temperature considered is 15°C, with a day/night variation of 10°C.
- Zone data: When the equipment in the zone containing the navigation computer is powered up, the total heat dissipation generates a local temperature increase of 15°C, which is reached in 5 minutes. In flight, climbing to an average of 1 500 m (5 000 feet) cools the outside temperature, and consequently the zone, by 10°C. There is no significant kinetic heating.
- Equipment data: Heat dissipation specifically from the computer generates a temperature rise in the equipment equal to 15°C in 3 minutes.



From this data, we can plot the following temperature graph:



The red curve represents the ambient temperature outside the system during the Parked phase.

The yellow curve represents the temperature in the system zone.

The other curves (green, blue) represent the internal temperature of the equipment during the various phases.

We can now quantify T_{ambient} for each phase of the mission profile:

- **Parked (red curve):** Over 24 hours, the temperature ranges from 10 to 20°C with an average of 15°C, but never stabilizes. In this case, $T_{\text{amb}} = 15^{\circ}\text{C}$.
- **Ground-On 1 (green curve):** The temperature starts at 15°C and increases by 30°C (zone +15°C and equipment +15°C) to approach 45°C. However, as the temperature does not have time to stabilize, the temperature considered for this phase is calculated as $15^{\circ}\text{C} + 0.70 \times 30^{\circ}\text{C} = 36^{\circ}\text{C}$. So $T_{\text{amb}} = 36^{\circ}\text{C}$.
- **Flight (blue curve):** the component temperature drops by 10°C due to increased altitude and stabilizes at 35°C. So $T_{\text{amb}} = 35^{\circ}\text{C}$.

3.1.3. Quantification of thermal cycling data

The thermal cycling data (ΔT , $T_{\text{max-cycling}}$, N_{cy} and θ_{cy}) is calculated in two steps:

- **Identification of the temperature cycles.**
A temperature cycle begins with a trigger event that generates a temperature variation and ends when the trigger event is no longer active and the element returns to its initial temperature. Consequently, each temperature cycle corresponds to a phase in the mission profile drawn up in §1.4.
- **The plotting of the temperature cycles on the temperature graphs.**
This enables the thermal cycling data to be quantified.

Remarks:

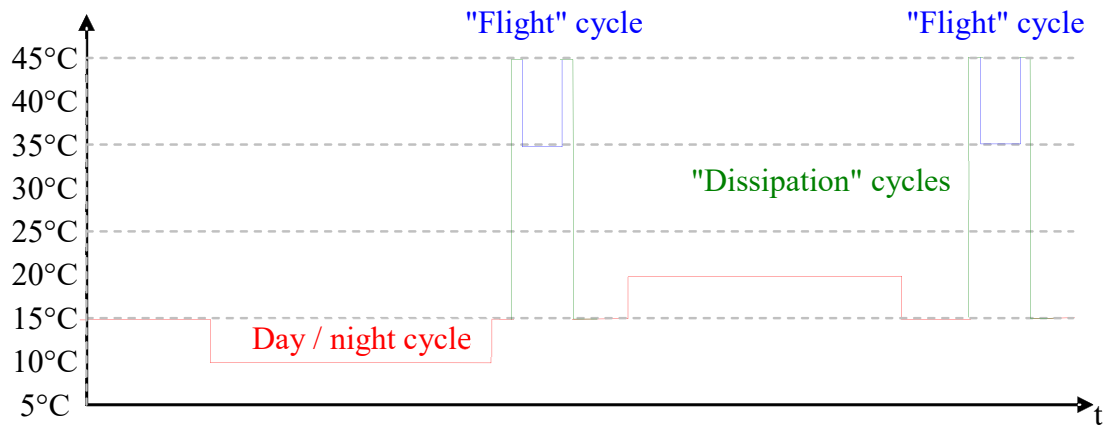
- The time taken by the system to return to its initial temperature is included in the cycle time θ_{cy} . The temperature cycle of one phase can therefore spill over into the next phase due to the thermal inertia of the system.
- It is possible for there to be no temperature cycle for a particular phase in the life profile, if the phase results from a non-thermal cause. If so, the thermal cycling data for this phase will consist of null values.

VIP helicopter:

1. **Identification of temperature cycles.**
The temperature cycles associated with each phase of the mission profile can be identified by taking the phase sequence defined in Table 4, and keeping only the causes of thermal variation.

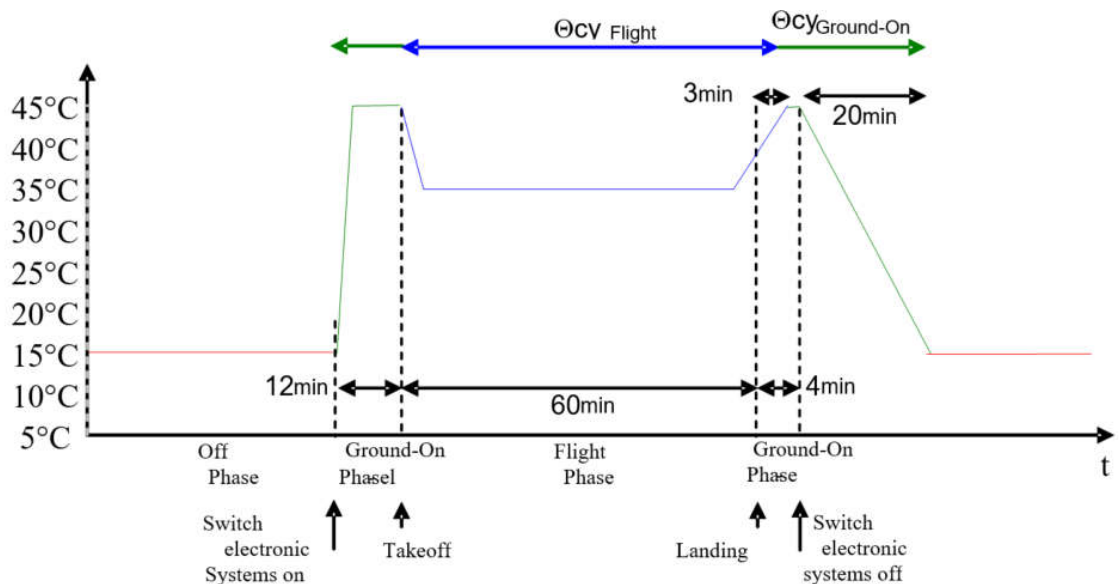
Phase	Trigger events generating thermal cycling	Causes of thermal cycling
Off-24h	Day/night variation	- Heating of outside temp. by the sun
Off	Day/night variation	- Heating of outside temp. by the sun
Ground-On 1	Startup/shutdown of electronic systems	- Heat dissipation from equipment
Flight 1	Takeoff/landing	- Heat dissipation from equipment - Variation in altitude
Ground-On 2	Startup/shutdown of electronic systems	- Heat dissipation from equipment
Flight 2	Takeoff/landing	- Heat dissipation from equipment - Cooling due to increase in altitude

2. Plotting of temperature cycles.
Taking the temperature spectrum of the “operating” day, we can now superimpose the temperature cycles over it.



Thermal inertia data:

- Once on the ground, the time to return to the initial temperature of 45°C is estimated at 3 minutes.
- Once the equipment has been switched off, the time to return to the initial temperature of 15°C is estimated at 20 minutes.



We can now quantify the duration of the temperature cycles: Θ_{cy} .

This figure shows that, due to the time the temperature takes to return to its initial value, the temperature cycle durations spill over into the next phase. Consequently:

- $\Theta_{cy}(\text{Flight}) = 60 + 3 = 63 \text{ min}$
- $\Theta_{cy}(\text{Ground-On}) = 12 + 4 + 20 - 3 = 33 \text{ min}$.

The “Flight” cycle, in blue, starts with the “Takeoff” event. This cycle includes part of the time spent in “Ground-On” getting back up to 45°C, as the cycle only finishes when there is a return to the initial temperature.

- $T_{\max_cycling} = 45^{\circ}\text{C}$
- $\Delta T = T_{\max} - T_{\min} = 10^{\circ}\text{C}$
- $N_{cy} = 100$ cycles per year
- $\Theta_{cy} = 60 + 3 \text{ minutes} = 63 \text{ minutes} = 1.05 \text{ h}$ (includes the time to return to 45°C during the Ground-On phase).

The “Dissipation” cycle starts with the “Start up electronic systems” event, is interrupted by the “Flight” cycle, then resumes until returning to its initial temperature of 5°C.

- $T_{\max_cycling} = 45^{\circ}\text{C}$
- $\Delta T = 30^{\circ}\text{C}$
- $N_{cy} = 100$ cycles per year
- $\Theta_{cy} = 12 + 4 + 20 - 3 = 33 \text{ min} = 0.55 \text{ h}$ (includes the time to return to 15°C during the Parked phase, but not the 3 minutes absorbed by the “Flight” cycle).

The “day / night” cycle, in red, covers the remainder of the time.

- $T_{\max_cycling} = 20^{\circ}\text{C}$
- $\Delta T = 10^{\circ}\text{C}$
- $N_{cy} = 100$ cycles per year
- $\Theta_{cy} = 24 - 2 \times 1.05 - 2 \times 0.55 = 20.8 \text{ h}$ (the total time for the 4 other cycles is subtracted from 24 hours).

3.1.4. Quantification of humidity data

To calculate the humidity level for the FIDES mission profile, we need to know:

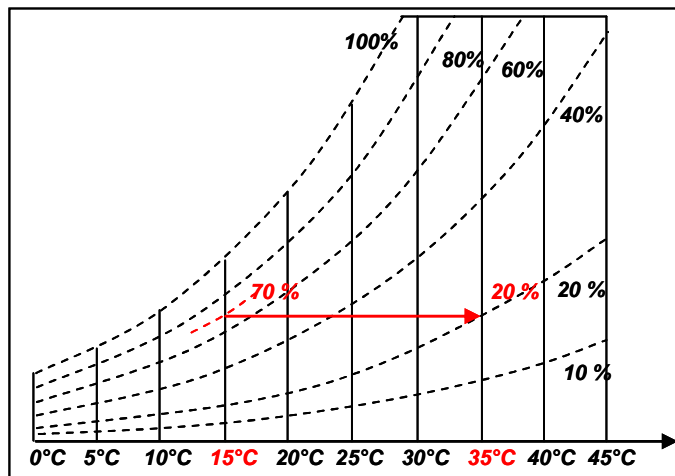
- The relative humidity outside the equipment
- The temperature outside the equipment
- The temperature rise inside the equipment.

VIP helicopter:

During the Parked phase, the mean relative humidity for a temperate climate is taken to be 70% at 15°C.

The ambient temperature due to heat generated by dissipation from the equipment is 36°C in Ground-On and 35 °C in Flight.

The humidity level as a result of drying can be read off the chart below as 20%.



3.1.5. Exposure to accidental overstress (application factor)

The application factor was established as follows:

Criterion	Parked phase	Ground-On phases	Flight phases
User type	Moderate (general compliance w/rules)	Moderate (general compliance w/rules)	Moderate (general compliance w/rules)
Level of user qualification	Moderate	Moderate	Favourable (pilot)
System mobility	Low	Low	Severe
Product handling	Favourable (product not handled)	Favourable (product not handled)	Favourable (product not handled)
Type of electrical network (system)	Favourable (not powered)	Severe (network disturbance)	Moderate (little network disturbance)
Exposure of product to human activity	Moderate (e.g. maintenance)	Low	Low
Exposure of product to machine disturbance	Low	Moderate (indirect exposure)	Moderate (indirect exposure)
Exposure of product to weather	Moderate (indirect exposure)	Moderate (indirect exposure)	Moderate (indirect exposure)
Value of $\Pi_{\text{application}}$	2.33	2.71	2.51

3.1.6. Simplification and finalization of the FIDES mission profile

If two lines in the mission profile have:

- The same physical stress values
- The same chemical pollution criteria
- The same values for $\Pi_{\text{application}}$

Then they have the same impact on reliability.

If so, the two lines can be merged into a single line that will have:

- The physical stress values common to both merged lines, or an average value weighted by duration (if the differences are small)

- For the cycle duration, the average value weighted by the number of cycles (again, if the differences are small, otherwise they should not be merged)
- A $t_{\text{annual phase}}$ value equal to the sum of the $t_{\text{annual phase}}$ values of the merged lines
- A N_{cy} value equal to the sum of the N_{cy} values of the merged lines.

VIP helicopter:

Considering a vibration amplitude of 0.5 Grms in Ground-On and 6 Grms in Flight, a $\Pi_{\text{application}}$ calculated for the conditions of a VIP helicopter, and the following chemical levels:

- Saline pollution: low
- Environmental pollution: moderate
- Application pollution: moderate
- Equipment: non-hermetic.

3.1.7. Table

Helicopter – Onboard navigation computer – VIP use														
Phase name	Calendar time (hours)	Temperature and humidity			Thermal cycling				Mechanical	Chemical				Induced
		On/Off	Ambient temperature (°C)	Humidity level (%)	ΔT (°C)	Number of cycles (/year)	Cycle duration (hours)	Max. temp. during cycling (°C)	Random vibrations (Grms)	Saline pollution	Environmental pollution	Application pollution	Protection level	TT application
Off -24h	6360	Off	15	70	10	265	24	20	0.01	Low	Moderate	Moderate	Non-hermetic	2.33
Off	2146	Off	15	70	10	100	20.6	20	0.01	Low	Moderate	Moderate	Non-hermetic	2.33
Ground-On 1	27	On	36	20	30	100	0.6	45	0.5	Low	Moderate	Moderate	Non-hermetic	2.71
Flight 1	100	On	35	20	10	100	1.1	45	6	Low	Moderate	Moderate	Non-hermetic	2.51
Ground-On2	27	On	36	20	30	100	0.6	45	0.5	Low	Moderate	Moderate	Non-hermetic	2.71
Flight 2	100	On	35	20	10	100	1.1	45	6	Low	Moderate	Moderate	Non-hermetic	2.51

The previous mission profile can be simplified without significantly changing the end result, by merging lines that describe similar stresses:

Helicopter – Computer in avionics bay – VIP use – Temperate climate (outside temperature 15°C)														
Phase name	Calendar time (hours)	Temperature and humidity			Thermal cycling				Mechanical	Chemical				Induced
		On/Off	Ambient temperature (°C)	Humidity level (%)	ΔT (°C)	Number of cycles (/year)	Cycle duration (hours)	Max. temp. during cycling (°C)	Random vibrations (Grms)	Saline pollution	Environmental pollution	Application pollution	Protection level	Π application
Off	8506	Off	15	70	10	365	23.1	20	0.01	Low	Moderate	Moderate	Non-hermetic	2.33
Ground-On	54	On	36	20	30	200	0.6	45	0.5	Low	Moderate	Moderate	Non-hermetic	2.71
Flight	200	On	35	20	10	200	1.1	45	6	Low	Moderate	Moderate	Non-hermetic	2.51

The principle used to create the previous life profile can be used to create life profiles for different equipment, for different uses of the same helicopter.

Influence of climate.

The following profile is applicable to the same product as above, but for a tropical climate. To emphasize the influence of the climate, the year is broken down into three seasons.

Helicopter – Computer in avionics bay – VIP use – Tropical climate														
Phase name	Calendar time (hours)	Temperature and humidity			Thermal cycling				Mechanical	Chemical				Induced
		On/Off	Ambient temperature (°C)	Humidity level (%)	ΔT (°C)	Number of cycles (/year)	Cycle duration (hours)	Max. temp. during cycling (°C)	Random vibrations (Grms)	Saline pollution	Environmental pollution	Application pollution	Protection level	TT application
Summer Off	2 127	off	30	90	12	91	23.1	36	0.01	Low	Low	Low	Non-hermetic	2.33
Summer Ground-On	13	on	51	35	30	50	0.6	60	0.5	Low	Low	Low	Non-hermetic	2.71
Summer Flight	50	on	50	35	10	50	1.1	60	6.0	Low	Low	Low	Non-hermetic	2.51
Spring/Autumn Off	4 253	off	25	90	12	183	23.1	31	0.01	Low	Low	Low	Non-hermetic	2.33
Spring/Autumn Ground-On	27	on	46	35	30	100	0.6	55	0.5	Low	Low	Low	Non-hermetic	2.71
Spring/Autumn Flight	100	on	45	35	10	100	1.1	55	6.0	Low	Low	Low	Non-hermetic	2.51
Winter Off	2 127	off	17	90	12	91	23.1	23	0.01	Low	Low	Low	Non-hermetic	2.33
Winter Ground-On	13	on	38	35	30	50	0.6	47	0.5	Low	Low	Low	Non-hermetic	2.71
Winter Flight	50	on	37	35	10	50	1.1	47	6.0	Low	Low	Low	Non-hermetic	2.51

Influence of location inside helicopter.

The following profile is applicable to the same VIP helicopter in a temperate climate, but concerns a product in the cockpit (rather than the avionics bay).

Helicopter – Screen in instrument panel – VIP use – Temperate climate (outside temperature 15°C)														
		Temperature and humidity			Thermal cycling*				Mechanical	Chemical				Induced
Phase name	Calendar time (hours)	On/Off	Ambient temperature (°C)	Humidity level (%)	ΔT (°C)	Number of cycles (/year)	Cycle duration (hours)	Max. temp. during cycling (°C)	Random vibrations (Grms)	Saline pollution	Environmental pollution	Application pollution	Protection level	TT application
Off	8 506	Off	15	70	10	365	23.2	20	0.01	Low	Moderate	Moderate	Non-hermetic	2.33
Ground-On	54	On	24	40	10	200	0.4	25	0.5	Low	Moderate	Moderate	Non-hermetic	3.21
Flight	200	On	25	40	-	-	-	-	3	Low	Moderate	Moderate	Non-hermetic	3.28

*: When a dash "-" is entered in the columns relating to the thermal cycling, it means that there is no cycling. For calculation needs, these dashes must be replaced by numerical values (0 for ΔT for example).

Influence of mission.

The following profile is applicable to the same product under the same climatic conditions as the initial example, but for a helicopter intended for a different mission (in this case Transport).

Helicopter – Computer in avionics bay – Transport use – Temperate climate (outside temperature 15°C)														
Phase name	Calendar time (hours)	Temperature and humidity			Thermal cycling				Mechanical	Chemical				Induced
		On/Off	Ambient temperature (°C)	Humidity level (%)	ΔT (°C)	Number of cycles (/year)	Cycle duration (hours)	Max. temp. during cycling (°C)	Random vibrations (Grms)	Saline pollution	Environmental pollution	Application pollution	Protection level	Π application
Off	8 035	Off	15	70	10	365	21.5	20	0.01	Low	Moderate	Moderate	Non-hermetic	2.33
Ground-On	125	On	36	20	30	600	0.5	45	0.5	Low	Moderate	Moderate	Non-hermetic	2.71
Flight	600	On	35	20	10	900	0.7	45	6	Low	Moderate	Moderate	Non-hermetic	2.51

Influence of mission.

The following profile is applicable to the same product under the same climatic conditions as the initial example, but for a helicopter intended for a different mission (in this case Offshore).

Helicopter – Computer in avionics bay – Offshore use – Temperate climate (outside temperature 15°C)														
Phase name	Calendar time (hours)	Temperature and humidity			Thermal cycling				Mechanical	Chemical				Induced
		On/Off	Ambient temperature (°C)	Humidity level (%)	ΔT (°C)	Number of cycles (/year)	Cycle duration (hours)	Max. temp. during cycling (°C)	Random vibrations (Grms)	Saline pollution	Environmental pollution	Application pollution	Protection level	TT application
Off	7 360	Off	15	70	10	365	19.7	20	0.01	Low	Moderate	Moderate	Non-hermetic	2.33
Ground-On	200	On	36	20	30	500	0.6	45	0.5	Low	Moderate	Moderate	Non-hermetic	2.71
Flight	1 200	On	35	20	10	1000	1.3	45	6	High	Low	Moderate	Non-hermetic	2.51

3.2. Mission profile of equipment in the avionics bay of a medium-haul civil aircraft

3.2.1. Description of the profile in number and duration of cycles

Definition of the profile

The “typical” mission profile of an equipment item (in the avionics bay) mounted in a medium-haul civil aircraft consists of the following phases:

- Operating phase on the ground when switching ON and OFF
- Operating phase on the ground during stopovers (equipment remaining in ON position)
- Taxiing phase (from the moment the aircraft leaves the boarding zone to the moment it prepares for takeoff)
- Flight phase during takeoff (climb) and landing (descent)
- Level flight phase (cruise)
- Non-operating phase on the ground: equipment in OFF position (daily shutdowns, and maintenance phases)
- It is assumed that a medium/long-haul civil aircraft is operated at a daily rhythm of 3 flights by day with 2 stopovers (and little or no change of time zone).

This type of aircraft is in service for 350 days a year; the remaining time may be spent on standby or in daily maintenance.

“Ground-Operating-ON/OFF” phase

The time spent preparing for the first flight of the day and shutting down the aircraft after the last flight of the day (the duration of the “Ground-Operating-ON/OFF” phase) is assumed to total 2 hours. This duration includes the first embarkation and final disembarkation of passengers, performed in parallel with service activities related to the operation of the aircraft.

The duration of the temperature cycle, however, is 13.6 hours. This is due to the fact that other phases, without temperature cycles – namely, 4 “Ground-Taxiing”, 4 “Flight-Climb/Descent” and 2 “Flight-Level” phases – are nested inside it.

With 1 daily cycle, the total duration of the phase is 700 hours per year (2 x 350).

“Ground-Operating-Stopover” phase

The average duration of this phase (i.e. the turnaround time) is assumed to be 2 hours (disembarking, cleaning, catering, re-embarking).

With 2 daily cycles (between 3 flights), the total number of cycles is 700 per year (2 x 350). The total duration of the phase is 1 400 hours per year (2 x 700).

“Ground-Taxiing” phase

The average taxiing duration is assumed to be 0.30 hours (18 minutes).

With a taxiing phase before and after each flight, the number of cycles is 6 per day, i.e. 2 100 per year (2 x 3 x 350). The total duration of the phase is 630 hours per year (2 100 x 0.30).

“Flight-Climb/Descent” phase

The duration of the “Flight-Climb/Descent” phase is assumed to be 1 hour/flight.

With 3 flights a day, the number of cycles is 1 050 per year (3 x 350). The total duration of the phase is 1 050 hours per year (1 050 x 1).

“Flight-Level” phase

The total duration of the average flight is assumed to be 4 hours, including 3 hours of “level flight”.

With 3 flights a day, the number of cycles is 1 050 per year (3×350). The total duration of the phase is 3 150 hours per year ($1\,050 \times 3$).

“Ground-Dormant” phase

This phase includes daily shutdowns and the 15 days a year of non-operation, for which no specific phase was created in this example.

The total duration of the phase is 1 830 hours per year, bringing the total duration of all the phases to 8 760 hours per year (i.e. 24×365). In this case, the cycle duration is 5.01 hours ($1\,830 / 365$).

3.2.2. Definition of the ON / OFF profile

For the most common equipment in the avionics bay, it is assumed that the equipment is OFF during the “Ground-Dormant” phase and ON during all other phases.

3.2.3. Definition of the temperature profile and thermal cycling

The baseline temperature cycle is the day/night cycle in the “Ground-Dormant” phase. The ambient temperature is taken to be 15°C with a thermal cycling differential (ΔT) of 10°C and a maximum cycling temperature of 20°C (the temperature consequently ranges between 10°C at night and 20°C during the day).

When the avionics bay is powered up, the fans begin to run. The transient phase at the beginning of startup (when the computers start to warm up before the ventilation has reached full efficiency) is ignored (besides, the computers can start up in a cold or warm ambient temperature before the temperature is regulated).

The internal temperature rise of the equipment (ambient temperature of the components relative to the ambient temperature outside the equipment) is assumed to be 15°C.

Taking into account the effect of ventilation, the ambient temperature for the “Ground-Operating-ON/OFF” phase is taken to be 40°C, which represents a cycling ΔT of 25°C relative to the ambient temperature in the “Ground-Dormant” phase (15°C). The maximum cycling temperature is equal to the ambient temperature (when ON).

The temperature is assumed to be constant for all the “Ground-Operating-ON/OFF”, “Taxiing”, “Flight-Climb/Descent” and “Flight-Level” phases. The chosen ambient temperature value is 40°C, which represents a cycling ΔT of 0°C relative to the ambient temperature in the “Ground-Operating-ON/OFF” phase. The maximum cycling temperature is equal to the ambient temperature.

For the “Ground-Operating-Stopover” phase, it is assumed that there is loss of efficacy in temperature regulation due to the aircraft doors being opened (passenger cabin and avionics bay). The ambient temperature considered (also the maximum cycling temperature) is 55°C, which represents a cycling ΔT of 15°C relative to the ambient temperature in the “Ground-Operating-ON/OFF”, “Taxiing”, “Flight-Climb/Descent” and “Flight-Level” phases.

3.2.4. Definition of the humidity profile

When the equipment is OFF (“Ground-Dormant” phase), the mean relative humidity is assumed to be in the order of 70%.

When the equipment is ON during the ground phases with the aircraft doors open (“Ground-Operating-ON/OFF” and “Ground-Operating-Stopover”), it is assumed that the internal temperature rise lowers the relative humidity to 30%.

When the equipment is ON during flight phases (“Flight-Climb/Descent” and “Flight-Level”) or ON during the ground phase with the aircraft doors closed (“Ground-Taxiing”), the relative humidity is assumed to drop to about 10% (very dry air due to the ventilation system).

3.2.5. Definition of the vibration profile

The vibration stress level is assumed to be zero in the “Ground-Dormant” phase.

The vibration stress level is considered to be very low during the “Ground-Operating-ON/OFF” and “Ground-Operating-Stopover” phases: 0.05 G_{RMS} .

The vibration stress during the “Taxiing” phase is assumed to be 5 G_{RMS} .

The vibration stress during the “Flight-Climb/Descent” and “Flight-Level” phases is assumed to be 0.6 G_{RMS} .

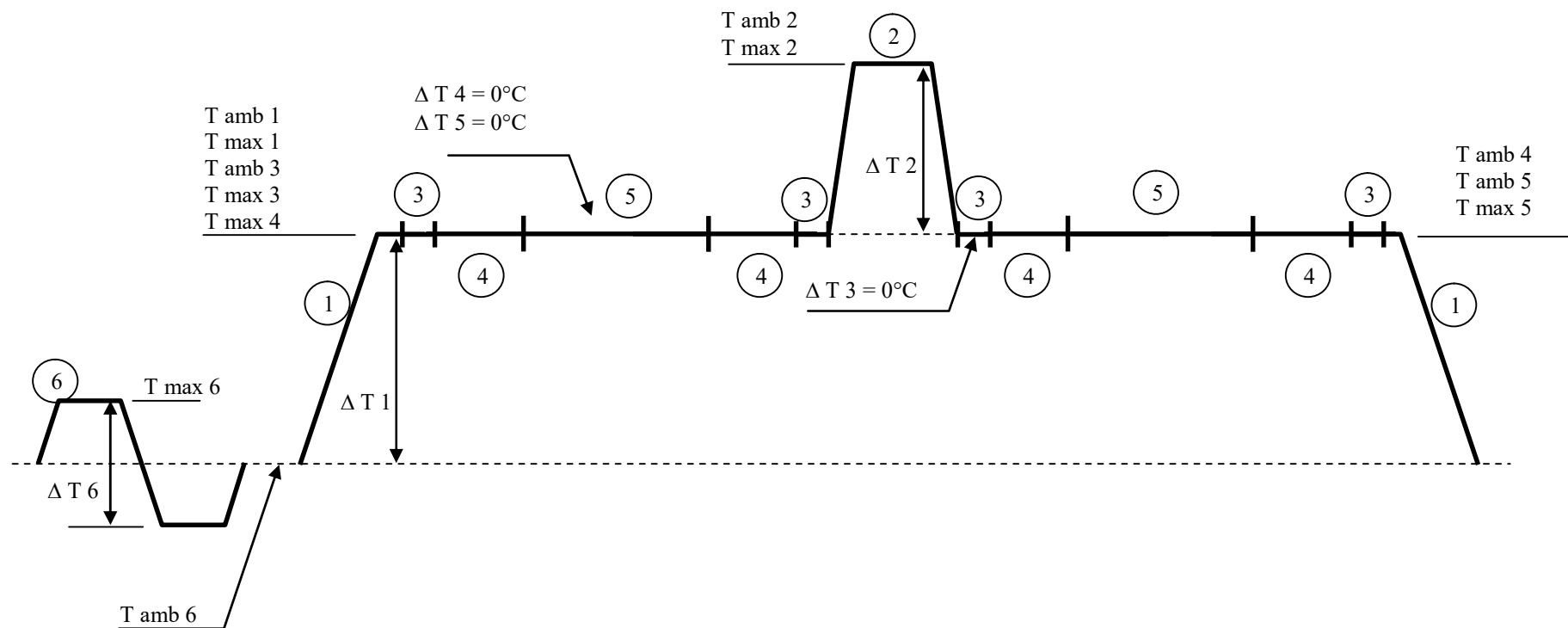
3.2.6. Definition of the chemical profile

The “environmental pollution” factor comes into play at the point where the aircraft is on the ground during the “Ground-Operating-ON/OFF”, “Ground-Operating-Stopover” and “Ground-Dormant” phases. In these cases, the equipment can be directly subjected to the external environment in an airport setting.

With regard to application pollution, only the “Ground-Dormant” phase might require action by persons close to equipment compartments.

3.2.7. Graph

The mission profile is illustrated by the following diagram:



3.2.8. Table

Civil aircraft, medium-haul, computer avionics bay														
Phase name	Calendar time (hours)	Temperature and humidity			Thermal cycling*				Mechanical	Chemical				Induced
		On/Off	Ambient temperature (°C)	Humidity level (%)	ΔT (°C)	Number of cycles (/year)	Cycle duration (hours)	Max. temp. during cycling (°C)	Random vibrations (Grms)	Saline pollution	Environmental pollution	Application pollution	Protection level	TT application
① Ground – Operating – ON/OFF	700	On	40	30	25	350	13.6	40	0.05	Low	Moderate	Moderate	Non-hermetic	4.8
② Ground – Operating Stopover	1400	On	55	30	15	700	2.00	55	0.05	Low	Moderate	Moderate	Non-hermetic	2.0
③ Ground – Taxiing	630	On	40	10	-	-	-	-	5	Low	Low	Moderate	Non-hermetic	1.2
④ Flight – Climb/Descent	1050	On	40	10	-	-	-	-	0.6	Low	Low	Moderate	Non-hermetic	1.1
⑤ Flight – Level	3150	On	40	10	-	-	-	-	0.6	Low	Low	Moderate	Non-hermetic	1.1
⑥ Ground – Dormant	1830	Off	15	70	10	365	5.01	20	0.01	Low	Moderate	Low	Non-hermetic	3.3

*: When a dash "-" is entered in the columns relating to the thermal cycling, it means that there is no cycling. For calculation needs, these dashes must be replaced by numerical values (0 for ΔT for example).

3.3. Mission profile of equipment in the avionics bay of a civil turboprop aircraft

3.3.1. Description of the profile in number and duration of cycles

Definition of the profile

The “typical” mission profile of an equipment item (in the avionics bay) fitted on a civil turboprop aircraft consists of the following phases:

- Operating phase on the ground when switching ON and OFF
- Operating phase on the ground during stopovers (equipment remaining in ON position)
- Taxiing phase (from the moment the aircraft leaves the boarding zone to the moment it prepares for takeoff)
- Flight phase: during takeoff (climb), level flight (cruise), and landing (descent)
- Non-operating phase on the ground: equipment in OFF position (daily shutdowns, and maintenance phases).

It is assumed that a civil turboprop aircraft is operated at a daily rhythm of 4 flights a day with 2 stopovers (short hops). The aircraft is switched OFF after each cycle of 2 flights with one intermediate stopover (and this cycle is repeated twice daily).

This type of aircraft is in service for 350 days a year; the remaining time may be spent on standby or in maintenance.

“Ground-Operating-ON/OFF” phase

This phase represents the time spent preparing for each flight (switching equipment ON: twice a day), and shutting down the aircraft (switching equipment OFF: twice a day). The duration of this “Ground-Operating-ON/OFF” phase is assumed to be 0.6 hours (36 minutes). This duration includes the passenger embarkation and disembarkation phases performed in parallel with service activities related to the operation of the aircraft.

With 2 daily cycles, the total number of cycles is 700 per year (2×350). The total duration of the phase is 420 hours per year (0.6×700).

“Ground-Operating-Stopover” phase

The average duration of this phase (the turnaround time or TAT) is assumed to be 0.5 hours, i.e. 30 minutes (disembarking, cleaning, catering, re-embarking).

With 2 daily cycles, the total number of cycles is 700 per year (2×350). The total duration of the phase is 350 hours per year (0.5×700).

“Ground-Taxiing” phase

The average taxiing duration is assumed to be 0.10 hours (6 minutes).

With a taxiing phase before and after each flight, the number of cycles is 4 per day, i.e. 2 800 per year ($2 \times 4 \times 350$). The total duration of the phase is 280 hours per year ($2\,800 \times 0.10$).

“Flight-Climb/Level/Descent” phase

The total duration of the average flight is assumed to be 1.2 hours, including approximately 0.2 hours (12 minutes) of “climb/descent”, and 1 hour of “level flight”.

No precise distinction can be made between the “Climb/Descent” and “Stable flight” phases for a typical civil turboprop aircraft profile, and considering the durations involved, it was decided to create a single phase called “Flight-Climb/Level/Descent”.

With 4 flights a day, the number of cycles is 1 400 per year (4×350). The total duration of the phase is 1 680 hours per year ($1\,400 \times 1.2$).

“Ground-Dormant” phase

This phase includes daily shutdowns and the 15 days a year of non-operation.

The total duration of the phase is 6 030 hours per year, bringing the total duration of all the phases to 8 760 hours per year (i.e. 24×365). In this case, the cycle duration is 16.52 hours ($6\,030 / 365$).

3.3.2. Definition of the ON / OFF profile

For the most common equipment in the avionics bay, it is assumed that the equipment is OFF during the “Ground-Dormant” phase and ON during all other phases.

3.3.3. Definition of the temperature profile and thermal cycling

The baseline temperature cycle is the day/night cycle in the “Ground-Dormant” phase. The ambient temperature is taken to be 15°C with a thermal cycling differential (ΔT) of 10°C and a maximum cycling temperature of 20°C (the temperature consequently ranges between 10°C at night and 20°C during the day).

When the avionics bay is powered up, the fans begin to run. The transient phase at the beginning of startup (when the computers start to warm up before the ventilation has reached full efficiency) is ignored (besides, the computers can start up in a cold or warm ambient temperature before the temperature is regulated).

The internal temperature rise of the equipment (the ambient temperature of the components relative to the ambient temperature outside the equipment) is assumed to be 15°C.

Taking into account the effect of ventilation, the ambient temperature for the “Ground-Operating-ON/OFF” phase is taken to be 40°C, which represents a cycling ΔT of 25°C relative to the ambient temperature in the “Ground-Dormant” phase (15°C). The maximum cycling temperature is equal to the ambient temperature (when ON).

The temperature is assumed to be constant for all the “Ground-Operating-ON/OFF”, “Taxiing” and “Flight-Climb/Level/Descent” phases. The chosen ambient temperature value is 40°C, which represents a cycling ΔT of 0°C relative to the ambient temperature in the “Ground-Operating-ON/OFF” phase. The maximum cycling temperature is equal to the ambient temperature.

For the “Ground-Operating-Stopover” phase, it is assumed that there is loss of efficacy in temperature regulation due to the aircraft doors being opened (passenger cabin and avionics bay). The ambient temperature considered (also the maximum cycling temperature) is 55°C, which represents a cycling ΔT of 15°C relative to the ambient temperature in the “Ground-Operating-ON/OFF”, “Taxiing” and “Flight-Climb/Level/Descent” phases.

3.3.4. Definition of the humidity profile

When the equipment is OFF (“Ground-Dormant” phase), the mean relative humidity is assumed to be in the order of 70%.

When the equipment is ON, whether on the ground (“Ground-Operating-ON/OFF”, “Ground-Operating-Stopover” and “Ground-Taxiing”) or in flight (“Flight-Climb/Level/Descent” phase), it is assumed that the internal temperature rise lowers the relative humidity to 30%.

3.3.5. Definition of the vibration profile

The vibration stress level is assumed to be zero in the “Ground-Dormant” phase.

The vibration stress level is considered to be very low during the “Ground-Operating-ON/OFF” and “Ground-Operating-Stopover” phases: $0.05 G_{RMS}$.

The vibration stress during the “Taxiing” phase is assumed to be $6 G_{RMS}$.

The vibration stress during the “Flight-Climb/Level/Descent” phase is assumed to be $1.2 G_{RMS}$.

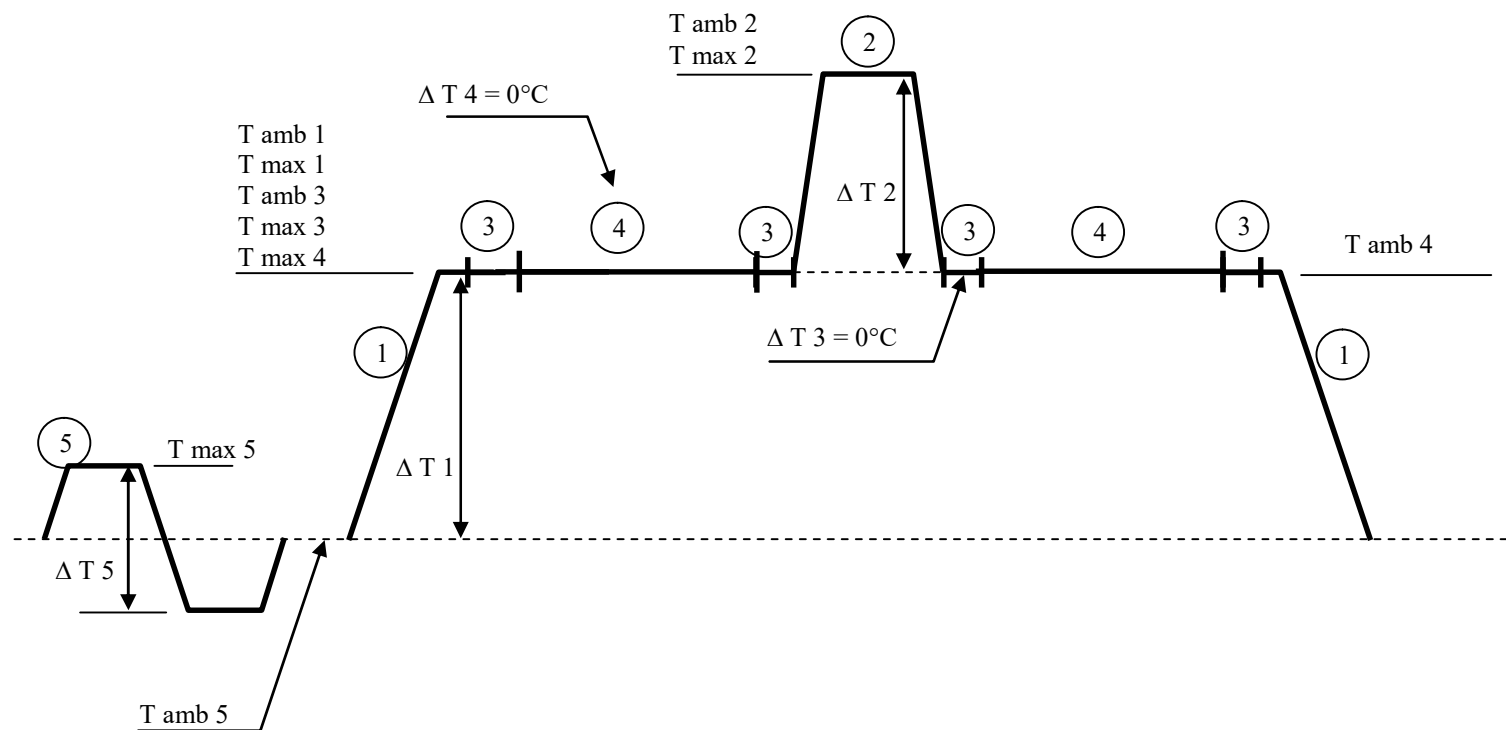
3.3.6. Definition of the chemical profile

The “environmental pollution” factor comes into play at the point where the aircraft is on the ground during the “Ground-Operating-ON/OFF”, “Ground-Operating-Stopover” and “Ground-Dormant” phases. In these cases, the equipment can be directly subjected to the external environment in an airport setting.

With regard to application pollution, only the “Ground-Dormant” phase might require action by people close to the equipment compartments.

3.3.7. Graph

The mission profile is illustrated by the following diagram:



3.3.8. Table

Civil aircraft, turboprop, avionics bay														
Phase name	Calendar time (hours)	Temperature and humidity			Thermal cycling*				Mechanical	Chemical				Induced
		On/Off	Ambient temperature (°C)	Humidity level (%)	ΔT (°C)	Number of cycles (/year)	Cycle duration (hours)	Max. temp. during cycling (°C)	Random vibrations (Grms)	Saline pollution	Environmental pollution	Application pollution	Protection level	TT application
① Ground – Operating – ON/OFF	420	On	40	30	25	700	0.60	40	0.05	Low	Moderate	Low	Non-hermetic	4.8
② Ground – Operating – Stopover	350	On	55	30	15	700	0.50	55	0.05	Low	Moderate	Low	Non-hermetic	2.0
③ Ground – Taxiing	280	On	40	30	-	-	-	-	6	Low	Moderate	Low	Non-hermetic	1.2
④ Flight – Climb / Level / Descent	1680	On	40	30	-	-	-	-	1.2	Low	Low	Low	Non-hermetic	1.1
⑥ Ground – Dormant	6030	Off	15	70	10	365	16.52	20	0.01	Low	Moderate	Moderate	Non-hermetic	3.3

*: When a dash "-" is entered in the columns relating to the thermal cycling, it means that there is no cycling. For calculation needs, these dashes must be replaced by numerical values (0 for ΔT for example).

3.4. Mission profile of industrial system equipment

3.4.1. Description of the profile in number and duration of cycles

Definition of the profile

The “typical” mission profile of industrial system equipment consists of the following phases:

A “Control-Monitoring” phase, which includes switching ON and OFF as well as control and monitoring tasks between actual operating phases

The system operating phase proper, in command mode

A non-operating phase: equipment in OFF position (daily shutdowns, maintenance phases).

It is assumed that industrial system equipment (e.g. a command/control module) is typically used for 17 working hours a day, in two consecutive shifts, with the “Command-Operation” phase being launched 4 times a day. The system is turned OFF at the end of each day.

This type of equipment is in service for 350 days a year; the remaining time may be spent in maintenance and/or repairs to the system.

“Control-Monitoring” phase ①

This phase represents the time spent preparing the system in the morning (switching equipment ON: once a day), and shutting it down at the end of each day (switching equipment OFF: once a day). It also includes the intermediate equipment control and monitoring phases (between “Command-Operation” cycles).

The duration of this “Control-Monitoring” phase is assumed to be 7 hours in total.

With 1 daily cycle, the total number of cycles is 350. The total duration of the phase is 2 450 hours per year (7 x 350).

“Command-Operation” phase ②

This phase represents the actual operating of the system in command mode. The average duration of this phase is assumed to be 2.5 hours.

With 4 daily cycles, the total number of cycles is 1 400 per year (4 x 350). The total duration of the phase is 3 500 hours per year (2.5 x 1 400).

“Dormant” phase ③

The number of cycles considered for this phase is 365 (day/night cycles over one year).

The total duration of the phase is 2 810 hours per year, bringing the total duration of all the phases to 8 760 hours per year (i.e. 24 x 365). In this case, the cycle duration is 7.7 hours (2 810 / 365).

3.4.2. Definition of the ON / OFF profile

For this type of industrial equipment, it is assumed that the equipment is OFF during the “Dormant” phase and ON during all other phases.

3.4.3. Definition of the temperature profile and thermal cycling

The baseline temperature cycle is the day/night cycle in the “Dormant” phase. The ambient temperature is taken to be 15°C with a thermal cycling differential (ΔT) of 10°C and a maximum cycling temperature of 20°C (the temperature consequently ranges between 10°C at night and 20°C during the day).

When the system is powered up, its temperature can be taken to be that of the workshop, i.e. around 15°C. The transient phase at the beginning of startup (while the system is warming up) is ignored.

The internal temperature rise of the equipment (the ambient temperature of the components relative to the ambient temperature outside the equipment) is assumed to be 15°C.

The ambient temperature for the “Control-Monitoring” phase is taken to be 30°C, which represents a cycling ΔT of 15°C relative to the ambient temperature of the “Dormant” phase (15°C). The maximum cycling temperature is equal to the ambient temperature (when ON).

For the “Command-Operation” phase, there is assumed to be a temperature rise due to the operation of the command. The ambient temperature considered (also the maximum cycling temperature) is 55°C, which represents a cycling ΔT of 25°C relative to the ambient temperature in the “Control-Monitoring” phase.

3.4.4. Definition of the humidity profile

When the equipment is OFF (“Dormant” phase), the mean relative humidity is assumed to be in the order of 80% (equipment located in a workshop).

When the equipment is ON (“Control-Monitoring” and “Command-Operation” phases), it is assumed that the internal temperature rise in the system lowers the relative humidity to 30%.

3.4.5. Definition of the vibration profile

The vibration stress level is assumed to be zero in the “Dormant” phase.

The vibration stress during the “Control-Monitoring” and “Command-Operation” phases is assumed to be 0.1 G_{RMS} .

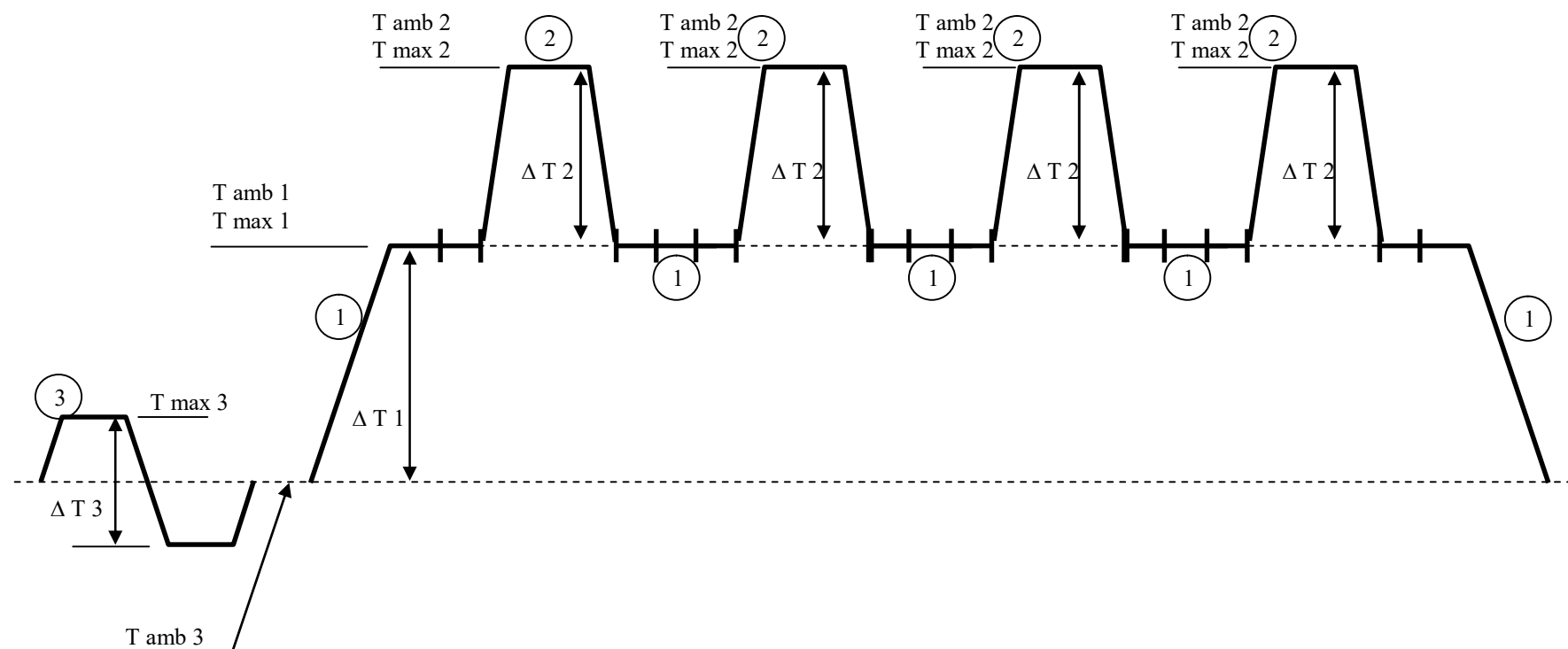
3.4.6. Definition of the chemical profile

As the equipment is situated in a production site workshop, the “environmental pollution” factor comes into play in all 3 phases: “Control-Monitoring”, “Command-Operation” and “Dormant”. The level is therefore set to “high”.

With regard to the application zone, all 3 phases can require action by people at equipment level. The zone value is therefore set to “moderate”.

3.4.7. Graph

The mission profile is illustrated by the following diagram:



3.4.8. Table

Specific industrial system														
Phase name	Calendar time (hours)	Temperature and humidity			Thermal cycling				Mechanical	Chemical				Induced
		On/Off	Ambient temperature (°C)	Humidity level (%)	ΔT (°C)	Number of cycles (/year)	Cycle duration (hours)	Max. temp. during cycling (°C)	Random vibrations (Grms)	Saline pollution	Environmental pollution	Application pollution	Protection level	TT application
① Control-Monitoring	2450	On	30	30	15	350	7.00	30	0.1	Low	High	Moderate	Non-hermetic	5.1
② Command-Operation	3500	On	55	30	25	1400	2.50	55	0.1	Low	High	Moderate	Non-hermetic	4.6
③ Dormant	2810	Off	15	80	10	365	7.70	20	0.01	Low	High	Moderate	Non-hermetic	2.6

3.5. Mission profile of a washing machine

3.5.1. Principle

This mission profile example is designed to show that the FIDES methodology is applicable even in cases where its definition appears complex due to the large number of phases to be defined (23, in this case). It was constructed by grouping together elementary tasks that presented identical physical stresses, based on the use of 150 wash cycles a year (roughly 3 a week) distributed as follows:

- 17 “fragile” cycles or programmes, with a duration of 58.5 minutes:
 - Cold wash (30°C)
 - Rinse
 - Short spin (includes a rinse).
- 100 “normal” cycles or programmes, with a duration of 104 minutes:
 - Normal wash (50°C)
 - Two rinses
 - Long spin (includes a rinse).
- 33 “non-delicate, very dirty” cycles or programmes, with a duration of 136.5 minutes:
 - Prewash
 - Hot wash (80°C)
 - Two rinses
 - Long high-speed spin (includes a rinse).

3.5.2. Values of the parameters

Temperatures, temperature cycles

The room in which the washing machine is located is assumed to be at 18°C on average, with a daily temperature cycle amplitude of 5°C.

The temperature values in this profile correspond to the temperature of the water in the drum, which is not necessarily representative for the electronic components in the washing machine. To create a realistic profile, one should take account of the temperature in the vicinity of the electronic components, which may be influenced by factors other than the water in the drum.

The temperature cycling parameters are deduced from the temperature change in the drum between the different steps of the wash:

- Heating, when the water is being heated
- Cooling, when filling with water at 18°C.

Relative humidity

Our theoretical washing machine is located in a laundry room where the humidity is normal during periods of inactivity (70%) and higher when washing is being done (85% at 18°C). Additionally, the humidity in the vicinity of the electronic is assumed to be influenced only by the air temperature.

Vibration

The random vibration amplitude was determined from the rotation speed of the drum in the phase considered.

Pollution

The pollution level was determined as follows:

- No saline pollution: low level
- Environmental pollution: moderate level corresponding to an urban zone
- Application pollution: moderate level corresponding to an inaccessible zone
- Protection level: non-hermetic (the machine's own water network is leak-proof).

Exposure to overstress (application factor)

The application factor was determined as follows:

Criterion	Inactive phase	Operating phases
User type	Moderate; general public user	Moderate; general public user
Level of user qualification	Favourable; the product does not require qualification	Favourable; the product does not require qualification
System mobility	Favourable; the product is stationary	Favourable; the product is stationary
Product handling	Low; the product is not handled when not in operation	Moderate; the product is handled moderately when used
Type of electrical network (system)	Favourable; the product is switched off in this phase	Moderate; the product is powered by a network with little disturbance
Exposure of product to human activity	Moderate; there is activity close to the product even when it is not being used	Moderate; there is activity close to the product when it is being used
Exposure of product to machine disturbance	Low; in a non-operating phase	Moderate; in an operating phase
Exposure of product to weather	Low; the product is in an indoor room	Low; the product is in an indoor room
Value of $\Pi_{\text{application}}$	1.9	2.7

3.5.3. Table

Washing machine														
Phase name	Calendar time (hours)	Temperature and humidity			Thermal cycling*				Mechanical	Chemical				Induced
		On/Off	Ambient temperature (°C)	Humidity level (%)	ΔT (°C)	Number of cycles (/year)	Cycle duration (hours)	Max. temp. during cycling (°C)	Random vibrations (Grms)	Saline pollution	Environmental pollution	Application pollution	Protection level	TT application
Inactive	8 502.52	Off	18	70	5	365	23.29	20.5	0.01	Low	Moderate	Moderate	Non-hermetic	1.9
Filling 18°C (drum at 18°C)	21.65	On	18	85	-	-	-	-	0.5	Low	Moderate	Moderate	Non-hermetic	2.7
Filling 18°C (drum at 30°C)	2.50	On	18	85	-	-	-	-	0.5	Low	Moderate	Moderate	Non-hermetic	2.7
Filling 18°C (drum at 50°C)	5.00	On	18	85	-	-	-	-	0.5	Low	Moderate	Moderate	Non-hermetic	2.7
Filling 18°C (drum at 80°C)	1.65	On	18	85	-	-	-	-	0.5	Low	Moderate	Moderate	Non-hermetic	2.7
Rinse	50.52	On	18	85	-	-	-	-	9.0	Low	Moderate	Moderate	Non-hermetic	2.7
Short spin, low-speed	0.99	On	18	85	-	-	-	-	14.0	Low	Moderate	Moderate	Non-hermetic	2.7
Long spin, low-speed	11.67	On	18	85	-	-	-	-	14.0	Low	Moderate	Moderate	Non-hermetic	2.7
Long spin, high-speed	3.85	On	18	85	-	-	-	-	16.0	Low	Moderate	Moderate	Non-hermetic	2.7
Waiting to open	2.50	On	18	85	-	-	-	-	0.5	Low	Moderate	Moderate	Non-hermetic	2.7
Emptying (drum at 18°C)	21.65	On	18	85	-	-	-	-	0.5	Low	Moderate	Moderate	Non-hermetic	2.7
Emptying (drum at 30°C)	2.50	On	30	85	-	-	-	-	0.5	Low	Moderate	Moderate	Non-hermetic	2.7
Emptying (drum at 50°C)	5.00	On	50	85	-	-	-	-	0.5	Low	Moderate	Moderate	Non-hermetic	2.7
Emptying (drum at 80°C)	1.65	On	80	85	-	-	-	-	0.5	Low	Moderate	Moderate	Non-hermetic	2.7
Short wash (drum at 30°C)	6.67	On	30	85	-	-	-	-	9.0	Low	Moderate	Moderate	Non-hermetic	2.7
Normal wash (drum at 50°C)	33.33	On	50	85	-	-	-	-	9.0	Low	Moderate	Moderate	Non-hermetic	2.7
Normal wash (drum at 80°C)	11.00	On	80	85	-	-	-	-	9.0	Low	Moderate	Moderate	Non-hermetic	2.7

Washing machine														
Phase name	Calendar time (hours)	Temperature and humidity			Thermal cycling*				Mechanical	Chemical				Induced
		On/Off	Ambient temperature (°C)	Humidity level (%)	ΔT (°C)	Number of cycles (/year)	Cycle duration (hours)	Max. temp. during cycling (°C)	Random vibrations (Grms)	Saline pollution	Environmental pollution	Application pollution	Protection level	TT application
Short rest, drum full (drum at 30°C)	6.67	On	30	85	-	-	-	-	0.5	Low	Moderate	Moderate	Non-hermetic	2.7
Normal rest, drum full (drum at 50°C)	33.33	On	50	85	-	-	-	-	0.5	Low	Moderate	Moderate	Non-hermetic	2.7
Normal rest, drum full (drum at 80°C)	11.00	On	80	85	-	-	-	-	0.5	Low	Moderate	Moderate	Non-hermetic	2.7
Heating to 30°C	2.50	On	30	85	12	50	0.05	30.0	0.5	Low	Moderate	Moderate	Non-hermetic	2.7
Heating to 50°C	13.33	On	50	85	32	100	0.13	50.0	0.5	Low	Moderate	Moderate	Non-hermetic	1.9
Heating to 80°C	8.53	On	80	85	62	33	0.26	80.0	0.5	Low	Moderate	Moderate	Non-hermetic	2.7

*: When a dash "-" is entered in the columns relating to the thermal cycling, it means that there is no cycling. For calculation needs, these dashes must be replaced by numerical values (0 for ΔT for example).

To make the mission profile easier to read, the number of cycles, cycle duration and maximum temperature parameters for each phase are completed even when there is no temperature cycle. Filling the drum with cold water causes cooling but does not constitute a cycle; this cooling terminates the cycle that began with heating.

3.6. Mission profile of multi-role combat aircraft external stores

3.6.1. Principle

For products used on combat aircraft, there is a convention of calculating an MTBF per flight hour in a daily mission profile that includes:

- One hour of flight per day
- Half an hour of aircraft maintenance per day
- The rest of the day shut down.

Using a conventional, and therefore arbitrary, mission profile inevitably impacts the realism of the prediction. But it can be practical for comparing different sets of predictive data. In this case, it is obviously impossible to compare predicted reliability directly with observed reliability.

The profile in the example below works within the limitations of the convention, while describing the aircraft's various missions (which may be shorter or longer than an hour). This profile was created using the "typical days" method described for the helicopter example.

3.6.2. Characteristics

Choice and duration of phases

There are several conventions for counting "flight hours" for combat aircraft. These conventions vary in the extent to which they include operating time on the ground. The one used here involves counting "flight hours" based on so-called "chock-to-chock" time, which includes on-ground operating time (other than maintenance) in "flight time".

Three types of mission are considered in this example:

- Patrol or escort mission, 1 flight hour (40% of missions):
 - Waiting on ground
 - Taxiing
 - Climb
 - Cruise (medium speed)
 - Descent
 - Taxiing
 - Waiting on ground.
- Low-altitude mission, 1.5 flight hours (30%):
 - Waiting on ground
 - Taxiing
 - Climb
 - Outbound cruise
 - Low-altitude mission (medium speed)
 - Return cruise
 - Descent
 - Taxiing
 - Waiting on ground.
- High-altitude mission, 0.75 flight hours (30%):
 - Waiting on ground
 - Taxiing
 - Climb
 - High-altitude mission (high speed)

- Descent
- Taxiing
- Waiting on ground.

Phase durations are chosen such that on average, the aircraft undergoes:

- One hour of flight per day
- Half an hour of maintenance per day
- The rest of the day shut down.

This mission profile may therefore be compatible with the usual convention. However, flights have been broken down into 6 phases in order to adequately describe the different types of environment. This breakdown shows that some flight phases are characterized by high thermal stresses, and others by almost no thermal stress.

Temperatures and temperature cycles

Multiple overlapping thermal phenomena are involved here:

- Day / night thermal cycling
- Heat dissipation from the product when powered up
- Temperature variation with altitude (depending on mission)
- Kinetic temperature rise (depending on speed, and therefore mission).

Humidity

The relative humidity is assumed to be 70% at 15°C on the ground. The only influence on humidity considered afterwards is the temperature.

Vibration

The vibration amplitude in flight and in taxiing is high. The vibration amplitude is assumed to be lower during the cruise phase.

Exposure to overstress – application factor

This factor was determined as follows:

Criterion	Ground-Off phase	Taxiing and flight phases	Maintenance phase
User type	Unfavourable Military use: mission constraints take precedence over hardware preservation	Unfavourable Military use: mission constraints take precedence over hardware preservation	Unfavourable Military use: mission constraints take precedence over hardware preservation
Level of user qualification	Moderate	Favourable (pilot)	Moderate
System mobility	Low	High	Low
Product handling	Favourable (product not handled)	Favourable (product not handled)	Moderate (product sometimes handled)
Type of electrical network (system)	Favourable (not powered)	Severe (network disturbance)	Severe (network disturbance)
Exposure of product to human activity	Moderate (e.g. stationing)	Low	Moderate (e.g. maintenance)
Exposure of product to machine disturbance	Low	Moderate (indirect exposure)	Low
Exposure of product to weather	Moderate (indirect exposure)	Moderate (indirect exposure)	Moderate (indirect exposure)
Value of $\Pi_{\text{application}}$	4.4	5.0	5.5

In the specific case of the “Flight – Waiting on ground, under power” phase, which precedes taxiing, we obtain an application factor of 4.8.

3.6.3. Table

Combat aircraft – Multi-role – External stores – Temperate climate														
Phase name	Calendar time (hours)	Temperature and humidity			Thermal cycling*				Mechanical	Chemical				Induced
		On/Off	Ambient temperature (°C)	Humidity level (%)	ΔT (°C)	Number of cycles (/year)	Cycle duration (hours)	Max. temp. during cycling (°C)	Random vibrations (Grms)	Saline pollution	Environmental pollution	Application pollution	Protection level	TT application
Ground – Non-operating	8 213	Off	15	70	10	365	22.50	20	0.01	Low	Moderate	Moderate	Non-hermetic	4.4
Flight – Waiting on ground under power	37	On	35	20	20	365	0.10	35	0.50	Low	Moderate	Moderate	Non-hermetic	4.8
Flight – Taxiing	37	On	35	20	-	-	-	-	8.00	Low	Moderate	Moderate	Non-hermetic	5.0
Flight – Climb and descent	73	On	35	20	-	-	-	-	8.00	Low	Low	Moderate	Non-hermetic	5.0
Flight – Cruise	131	On	- 2	90	37	219	0.80	35	4.00	Low	Low	Moderate	Non-hermetic	5.0
Flight – Low-altitude mission	37	On	50	10	52	73	0.50	50	8.00	Low	Low	Moderate	Non-hermetic	5.0
Flight – High-altitude mission	51	On	70	10	35	146	0.35	70	8.00	Low	Low	Moderate	Non-hermetic	5.0
Ground – Maintenance	183	On	35	20	20	365	0.50	35	0.50	Low	Moderate	Moderate	Non-hermetic	5.5

*: When a dash "-" is entered in the columns relating to the thermal cycling, it means that there is no cycling. For calculation needs, these dashes must be replaced by numerical values (0 for ΔT for example).

3.7. Other examples

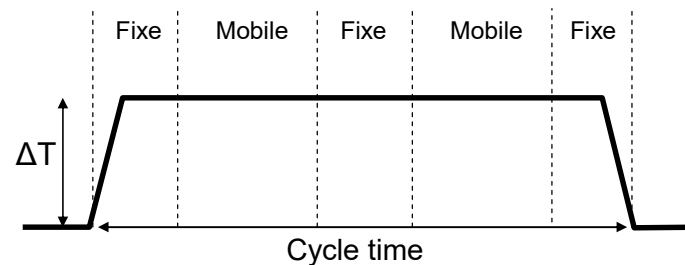
Tracked armoured vehicle														
Phase name	Calendar time (hours)	Temperature and humidity			Thermal cycling*				Mechanical	Chemical				Induced
		On/Off	Ambient temperature (°C)	Humidity level (%)	ΔT (°C)	Number of cycles (/year)	Cycle duration (hours)	Max. temp. during cycling (°C)	Random vibrations (Grms)	Saline pollution	Environmental pollution	Application pollution	Protection level	$\Pi_{\text{application}}$
Stationary powered	152	On	50	40	35	48	5.06	50	0.01	Low	Low	Low	Non-hermetic	6.2
Mobile powered	91	On	50	40	-	-	-	-	4	Low	Low	Low	Non-hermetic	6.7
Mobile non-powered (logistical transport)	100	Off	15	70	-	-	-	-	0.5	Low	Low	Low	Non-hermetic	5.2
Stationary non-powered	8417	Off	15	70	10	365	23.06	20	0.01	Low	Low	Low	Non-hermetic	7.5

*: When a dash "-" is entered in the columns relating to the thermal cycling, it means that there is no cycling. For calculation needs, these dashes must be replaced by numerical values (0 for ΔT for example).

Military portable radio														
Phase name	Calendar time (hours)	Temperature and humidity			Thermal cycling*				Mechanical	Chemical				Induced
		On/Off	Ambient temperature (°C)	Humidity level (%)	ΔT (°C)	Number of cycles (/year)	Cycle duration (hours)	Max. temp. during cycling (°C)	Random vibrations (Grms)	Saline pollution	Environmental pollution	Application pollution	Protection level	TT _{application}
Day without use – Protected storage	3960	Off	20	37	5	165	24	23	0.01	Low	Low	Moderate	Hermetic	4.1
In use – Stationary operating	600	On	30	20	15	200	6	30	0.5	Low	Low	Moderate	Hermetic	5.6
In use – Mobile operating	600	On	30	20	-	-	-	-	1.5	Low	Low	Moderate	Hermetic	7.7
In use – Stationary non-operating	3200	Off	15	50	10	200	18	20	0.01	Low	Low	Moderate	Hermetic	4.3
In use – Mobile non-operating	400	Off	15	50	-	-	-	-	1.5	Low	Low	Moderate	Hermetic	7.7

*: When a dash "-" is entered in the columns relating to the thermal cycling, it means that there is no cycling. For calculation needs, these dashes must be replaced by numerical values (0 for ΔT for example).

This mission profile describes an infantryman's radio set. It illustrates a case where a phase includes a temperature cycle that lasts longer than the phase itself. In this mission profile, the system is powered up while it is stationary (phase "In use – Stationary operating"). This phase therefore includes the powerup temperature cycle. Once under power, the system is transported, as described in another phase ("In use – Mobile operating"); this does not cause a new temperature cycle, but it does imply a more severe vibration amplitude. The duration of the powerup temperature cycle therefore corresponds to the stationary operating time plus the mobile operating time.



Desktop computer														
		Temperature and humidity			Thermal cycling				Mechanical	Chemical				Induced
Phase name	Calendar time (hours)	On/Off	Ambient temperature (°C)	Humidity level (%)	ΔT (°C)	Number of cycles (/year)	Cycle duration (hours)	Max. temp. during cycling (°C)	Random vibrations (Grms)	Saline pollution	Environmental pollution	Application pollution	Protection level	Π application
On	2860	On	50	10	30	220	13	50	0.1	Low	Moderate	Low	Non-hermetic	3.1
Off	2420	Off	20	50	5	220	11	23	0.01	Low	Moderate	Low	Non-hermetic	1.6
Day without use	3480	Off	20	50	5	145	24	23	0.01	Low	Moderate	Low	Non-hermetic	1.6

Mission profile for a desktop computer used 220 days a year. The other days, the computer is not powered up. On the days when it is used, the computer is powered up manually at 9:00 in the morning and shuts down automatically at 22:00 (10 pm). The internal temperature rise during operation is 30°C. The room in which the computer is located is air-conditioned during the day. The mean temperature is 20°C, with a daily temperature cycle of 5°C.

Automotive – Engine control device														
		Temperature and humidity			Thermal cycling				Mechanical	Chemical				Induced
Phase name	Calendar time (hours)	On/Off	Ambient temperature (°C)	Humidity level (%)	ΔT (°C)	Number of cycles (/year)	Cycle duration (hours)	Max. temp. during cycling (°C)	Random vibrations (Grms)	Saline pollution	Environmental pollution	Application pollution	Protection level	Π application
Non-operating / Full day	720	Off	14	70	10	30	24	19	0.01	Low	Moderate	Moderate	Hermetic	3.09
Non-operating / Night	7532	Off	14	70	10	335	22.5	19	0.01	Low	Moderate	Moderate	Hermetic	3.09
Start / Night	117	On	32	50	22	670	0.2	32	2	Low	Moderate	Moderate	Hermetic	4.80
Start / Day	58	On	32	60	18	1340	0.04	32	2	Low	Moderate	Moderate	Hermetic	4.80
Operating – full load	201	On	85	30	53	335	0.6	85	1	Low	Moderate	Moderate	Hermetic	4.80
Operating – motorway	131	On	60	30	28	30	4.4	60	2	Low	Moderate	Moderate	Hermetic	4.80

Mission profile for an engine control device in a car.

The car is used 335 days a year and is fully idle for 30 days a year. During days of use, there are two typical situations:

- The first (occurring twice a day) begins with a night-time start (temperature 10°C, rising to a reference temperature of 32°C) and continues with an operating phase on a motorway (during which the temperature stabilizes at 60°C) followed by an operating phase with a full load (during which the temperature stabilizes at 85°C).
- The second (occurring four times a day) begins with a daytime start (temperature 14°C, rising to a reference temperature of 32°C) and se continues with an operating phase on a motorway (during which the temperature stabilizes at 60°C) followed by an operating phase with a full load (during which the temperature stabilizes at 85°C).

Between each motorway operating phase and full-load operating phase, the temperature returns to its reference value.

Low-Earth orbit satellite – Payload equipment (observation camera)														
		Temperature and humidity			Thermal cycling*				Mechanical	Chemical				Induced
Phase name	Calendar time (hours)	On/Off	Ambient temperature (°C)	Humidity level (%)	ΔT (°C)	Number of cycles (/year)	Cycle duration (hours)	Max. temp. during cycling (°C)	Random vibrations (Grms)	Saline pollution	Environmental pollution	Application pollution	Protection level	Π application
Launch	2	Off	15	0	-	-	-	-	18	Low	Low	Low	Hermetic	1.95
Reaching orbit	4378	Off	15	0	10	2189	2	20	0.01	Low	Low	Low	Hermetic	1.23
Stabilization in orbit	2	Off	20	0	5	1	2	20	0.01	Low	Low	Low	Hermetic	1.13
Operating in orbit (OFF)	10950	Off	20	0	10	21900	0.5	25	0.01	Low	Low	Low	Hermetic	1.13
Operating in orbit (ON)	32850	On	25	0	5	65700	0.5	25	0.01	Low	Low	Low	Hermetic	1.27

*: When a dash "-" is entered in the columns relating to the thermal cycling, it means that there is no cycling. For calculation needs, these dashes must be replaced by numerical values (0 for ΔT for example).

This is a representative mission profile for an electronic device (e.g. a camera) in the payload of a low-orbit satellite circling the Earth at a distance of up to 2 000 km, typically an observation satellite.

The mission begins with a launch phase that lasts 2 hours and ends when the satellite reaches its injection orbit; the satellite is then switched off. The ambient temperature is 15°C without temperature variation (controlled temperature).

The mission continues with a transfer orbit phase lasting 6 months, during which the satellite will attain its target orbit (by electrical propulsion). During this phase, the payload equipment is switched off but is subject to the temperature cycles of the orbit in which it is located:

- The duration of an orbit (a complete circling of the Earth) is 120 minutes.
- The temperature variation is +/- 5°C around the reference temperature of 15°C.

Once the satellite is in its target orbit (which keeps the same characteristics: duration of 2 hours and temperature variation of +/- 5°C around the reference temperature of 15°C), the equipment is powered up whenever observation becomes possible. To build the above mission profile, 3 powerups are posited per orbit, at regular time intervals, followed by 30 minutes of image recording after each powerup.

Note: The “Stabilization in orbit” phase has no physical reality; it was added to enable transition between the “Reaching orbit” phase and the “Operating in orbit / Off” phase.

Geostationary orbit satellite – Platform equipment (computer)														
		Temperature and humidity			Thermal cycling*				Mechanical	Chemical				Induced
Phase name	Calendar time (hours)	On/Off	Ambient temperature (°C)	Humidity level (%)	ΔT (°C)	Number of cycles (/year)	Cycle duration (hours)	Max. temp. during cycling (°C)	Random vibrations (Grms)	Saline pollution	Environmental pollution	Application pollution	Protection level	Π application
Launch	2	Off	15	0	-	-	-	-	18	Low	Low	Low	Hermetic	1.95
Reaching orbit	48	On	25	0	10	1	48	25	0.01	Low	Low	Low	Hermetic	1.23
Stabilization in orbit	32850	On	25	0	5	5475	24	27.5	0.01	Low	Low	Low	Hermetic	1.27

*: When a dash "-" is entered in the columns relating to the thermal cycling, it means that there is no cycling. For calculation needs, these dashes must be replaced by numerical values (0 for ΔT for example).

This is a representative mission profile for a device in a geostationary satellite orbiting the Earth at about 36 000 km; the device considered is an item of platform equipment (such as a computer).

The mission begins with a launch phase that lasts 2 hours and ends when satellite reaches its injection orbit; the satellite is then switched off. The ambient temperature is 15°C without temperature variation (controlled temperature).

The mission continues with a transfer orbit phase lasting 48 hours, during which the satellite will attain its target orbit (by electrical propulsion). During this phase, the equipment is under power and is subject to the temperature cycles of the orbit:

- The duration of an orbit (a complete circling of the Earth) is 24 hours.
- The temperature variation during each orbit is +/- 2.5°C around the reference temperature.

Finally, the satellite reaches its target orbit, where it will stay for 15 years. It remains under power and subject to the temperature cycles of its orbit:

- The duration of an orbit (a complete circling of the Earth) is 24 hours.
- The temperature variation during each orbit is +/- 2.5°C around the reference temperature.

III

Reliability – Failure Rate Prediction Guide

Calculation Sheets

Electronic components

Induced factors

Factors contributing to overstress

$$\Pi_{induced-i} = (\Pi_{placement} \times \Pi_{application-i} \times \Pi_{ruggedizing})^{0.511 \times \ln(e^{C_{sensitivity}})}$$

The index i denotes the phase considered.

Contributions to the $C_{sensitivity}$ factor

The $C_{sensitivity}$ factor is given in the datasheet for each type of item.

The relative sensitivities to EOS (Electrical OverStress), TOS (Thermal OverStress) and MOS (Mechanical OverStress) are given for information only, to indicate the relative sensitivity of product families to the different types of overstress.

$C_{sensitivity}$ is calculated by the following formula:

$$C_{sensitivity} = \alpha \times EOS + \beta \times MOS + \gamma \times TOS$$

Where:

- $\alpha=72.5\%$
- $\beta=22.5\%$
- $\gamma=5\%$.

Note: α , β and γ are based on field return from a dedicated study on the behalf of the French MoD.

Contributions to the $\Pi_{Placement}$ factor

$\Pi_{Placement}$ must be assigned according to the electronic function served by the component, and not according to the nature or technology of the component itself.

The values of the $\Pi_{Placement}$ parameter therefore differ depending on the type of function (digital, low-level analog or power analog) and on the position (in the interface or not).

	$\Pi_{Placement}$
Digital non-interface function	1.0
Digital interface function	1.6
Low-level analog non-interface function	1.3
Low-level analog interface function	2.0
Power analog non-interface function	1.6
Power analog interface function	2.5

The main entry-points for assigning $\Pi_{Placement}$ are generally the electronic diagram and the identification of its functional blocks, which is done for each technical function/zone: all the components in the same functional block inherit the same $\Pi_{Placement}$ value.

Of course, the identification of the functional blocks on a circuit board can be more or less detailed depending on the level of precision required: from a single functional block for a very high-level preliminary study (one circuit board = one functional block) to several functional blocks and sub-blocks for a detailed study.

To determine $\Pi_{\text{Placement}}$ one must answer two questions:

1. Interface or non-interface?

An interface is the junction that provides interconnection between two systems. The concept of interface must be considered from the electrical point of view; it depends very much on the architecture in which the product is situated. An item must be treated as an “interface” if it is more exposed to accidental electrical hazards because of its position in the system. The items that act as a link between internal equipment and external systems are part of interface functions.

On a circuit board, the interface components are often (but not necessarily) protections (transils, transorbs), filters (capacitors, inductors, resistors), or insulation components (optocouplers). From an electrical point of view, interface components are often similar to connectors.

2. Digital, low-level analog or power analog?

The digital functions are usually easy to identify.

The threshold between low-level analog and power analog corresponds approximately to a current of 1 A. But other elements can influence the choice of $\Pi_{\text{Placement}}$, such as the voltage and, especially, the type of function:

- In this guide, the low-level analog functions mainly involve discrete input/outputs, measurement signals, and analog logics.
- The power functions mainly involve power supply and power transmission components.

Contributions to the $\Pi_{\text{application}}$ factor

The $\Pi_{\text{application}}$ parameter is evaluated by scoring a series of criteria. Each criterion can have three levels, corresponding to a favourable, moderate or unfavourable situation. Each criterion has a particular impact on the over stresses (W_{os}):

$\Pi_{\text{application}}$: Table 1

Criterion	Description	Levels	Examples and comments	Weight W_{os}
User-related risk in the phase considered	Represents users' aptitude for compliance with procedures when faced with operational constraints	0: Favourable 1: Moderate 2: Unfavourable	0: The user complies scrupulously with the rules of use of the product, particularly for quality and safety reasons 1: The user generally complies with the rules of use of the product, but cost takes precedence 2: The user may not comply with the rules of use of the product, due to a difficult context of use where the only thing that counts is to succeed in the mission or to achieve an objective	20
Level of user qualification in the phase considered	Represents the level of expertise of the user or technician with regard to the operational context	0: Favourable 1: Moderate 2: Unfavourable	0: Highly qualified 1: Qualified 2: Underqualified or inexperienced In some phases, the user to be considered is the person who does the maintenance or servicing	10
System mobility	Represents hazards relating to potential movement of the system	0: Light 1: Moderate 2: Severe	0: Few hazards (fixed or stable environment) 1: Moderate hazards 2: Severe hazards, large variability (automotive)	4
Product handling	Represents the risk of incorrect handling, shocks, falls, etc.	0: Light 1: Moderate 2: Severe	0: Not handled 1: Handling without displacement or disassembly 2: Handling with displacement or disassembly The severe level should be adopted if there is a possibility of maintenance being carried out on the product during the phase considered	15
Type of electrical network (system)	Represents the level of electrical interference expected on power supplies, signals and electrical lines: powerups, power supply switching, connection/disconnection	0: Light 1: Moderate 2: Severe	0: Network not disturbed (dedicated and regulated power supply) 1: Little network disturbance 2: Network subject to disturbance (onboard network) The type of network is a system-level parameter but it can be broken down to product level	4
Exposure of product to human activity	Represents exposure to hazards relating to human activity: shocks, use for other purposes, etc.	0: Light 1: Moderate 2: Severe	0: Uninhabitable zone 1: Possible activity in the product zone 2: Normal activity in the product zone The product may be exposed to human activity even if it is not itself handled during normal use	8
Exposure of product to machine disturbance	Represents hazards relating to the operation of machines, engines and actuators: shocks, overheating, electrical disturbance, pollutants, etc.	0: Light 1: Moderate 2: Severe	0: Null (telephone) 1: Indirect exposure (product in hold) 2: High or direct exposure (product in engine area)	3
Exposure of product to weather	Represents exposure to rain, hail, frost, sandstorms, lightning, dust, etc.	0: Light 1: Moderate 2: Severe	0: Null (home use) 1: Indirect exposure (luggage hold, station concourse) 2: Outdoor (car engine)	2

Contributions to the $\Pi_{\text{application}}$ factor (ctd.)

Responses are required for each criterion (user type, system mobility, etc.) in order to define a light, moderate or severe level:

- It is important to determine the $\Pi_{\text{application}}$ value for each phase of use. Exposure to overstress is very context-dependent. For example, it is useful to represent increased exposure during maintenance phases (where applicable).
- Some criteria apply at the level of the product (the entity, usually an equipment item, of which the reliability is being assessed) and others at the level of the system (i.e. the assembly into which the product is integrated, such as an aircraft or a car). It is important to bear this distinction in mind when evaluating the criteria.

Each level (light, moderate or severe) is weighted specifically as defined in the following table:

$\Pi_{\text{application}}$: **Table 2**

Level	Weighting of scores (W_{scores})
0: Favourable or light	1
1: Moderate	3.2
2: Unfavourable or severe	10

On the basis of these tables and the answers to the criteria, $\Pi_{\text{application}}$ is obtained by the formula:

$$\Pi_{\text{application}} = \frac{1}{66} \times \sum_{k=\text{Criteria}} W_{\text{scores}_k} \times W_{os_k}$$

Where:

W_{scores_k} is the weightings for the scores assigned to each criterion ($\Pi_{\text{application}}$: Table 2).

W_{os_k} is the weight of each criterion ($\Pi_{\text{application}}$: Table 1).

Contributions to the $\Pi_{\text{Ruggedizing}}$ factor

The $\Pi_{\text{Ruggedizing}}$ factor is determined by the answers to the following questionnaire.

The answers and the justifications provided by the auditee are used to determine a **level of compliance** with the recommendation (levels L1 to L4):

- L1 = Recommendation not applied → Definite risks with regard to reliability
- L2 = Recommendation only partially applied → Potential risks with regard to reliability
- L3 = Recommendation generally applied → Few risks with regard to reliability
- L4 = Recommendation applied in full and proceduralized → Reliability is under control.

Sheet	Recommendation	Weight
169	Write complete procedures for all product implementation and maintenance operations	7
157	Provide training and maintain skills in product implementation and maintenance	7
158	Ensure compliance with product-specific procedures and function-specific rules by means of an appropriate monitoring system	7
168	Review maintenance operations performed by end users and address their recommendations	4
156	Ensure that environmental specifications are complete. Criteria for checking the completeness of specifications: analysis, tests, field data, compliance with standards	4
164	Demonstrate compliance with environmental specifications	4
165	Implement a product improvement process (e.g. accelerated stress tests) to limit the product's sensitivity to environmental stresses (disturbance, severe environments, overstress)	7
167	Conduct a process analysis of implementation and maintenance operations	4
170	Comply with a standard on power supplies (such as EN2282, which defines possible disturbances and variations). The standard must be complied with both for electricity generation and for electricity consumption	4
166	Conduct an analysis of failure cases that could lead to failure propagation	4
163	Include the production, storage and maintenance environments in the product's environmental specifications	4
160	Analyse and manage the risk of the product under test being damaged due to test or maintenance equipment failures. Criteria: risks analysed in the design of the test equipment and the unit under test; implementation of suitable prevention measures	4
162	Identify reasonably foreseeable abnormal uses and take appropriate preventive measures	4
161	Identify reasonably foreseeable weather-related hazards and take appropriate preventive measures	4
159	Design dependable electrical protection devices: - Identify the electrical protection devices - Make sure they are testable and maintainable - Integrate the use cases for these devices into the definition of the maintenance policy	4
171	Comply with a standard on conducted and radiated electromagnetic interference: this applies both to the product and to the system into which it is integrated	3

Detailed recommendation sheets on the evaluation by audit of $\Pi_{\text{Ruggedizing}}$ are provided with all the recommendation sheets for the reliability process control and audit guide.

Contributions to the $\Pi_{Ruggedizing}$ factor (ctd.)

The score for each level is determined as follows:

Level	Score
L1	0
L2	1
L3	2
L4	3

Each recommendation is weighted by a specific **Recom_Weight**.

The $\Pi_{Ruggedizing}$ factor is calculated as follows:

$$\Pi_{Ruggedizing} = e^{0.7 \times (1 - Recom_Grade)}$$

Where:

$$Recom_Grade = \frac{1}{225} \times \sum_i^{Recommendations} (Recom_Weight_i \times Satisfaction_Score_i)$$

Where:

- Recom_Weight is the weight associated with a recommendation
- Compliance_Score is the score obtained for this recommendation (0, 1, 2 or 3).

Remarks:

- The Recom_Grade factor ranges from 0 (worst case: no recommendation applied) to 1 (best case)
- Factor “225” corresponds to the score obtained if the highest score is given to each recommendation. If one or more recommendations are deemed to be inapplicable and not relevant to a given project, then this total can be updated in the same way as for the process factor calculation.

If $\Pi_{Ruggedizing}$ is not evaluated, a default value of 1.7 is proposed. Using the default value may reduce the accuracy of the final results.

Part Manufacturing factor

General model associated with the part manufacturing factor Π_{PM} :

$$\Pi_{PM} = e^{1.39 \times (1 - \text{Part_Grade}) - 0.69}$$

Where, for active components (integrated circuits, discrete semiconductors, LEDs, optocouplers):

$$\text{Part_Grade} = \left\lceil \frac{(QA_{\text{manufacturer}} + QA_{\text{component}} + RA_{\text{component}}) \times \varepsilon}{36} \right\rceil$$

And for all other items:

$$\text{Part_Grade} = \left\lceil \frac{(QA_{\text{manufacturer}} + QA_{\text{component}}) \times \varepsilon}{24} \right\rceil$$

The general formula for calculating Π_{PM} can be particularized, and is then specified case-by-case.

Model associated with the $QA_{\text{manufacturer}}$ factor

This factor is common to all items.

Manufacturer quality assurance level	Position relative to state of the art	$QA_{\text{manufacturer}}$
Certified according to standard: IATF/TS 16949	Higher	3
Certified according to one of the following standards: TL9000, ISO/TS 22163, EN9100, AS9100, JISQ9100, AQAP 2110, AQAP 2120, AQAP 2130, ISO/TS 22163, IEC TS 62239, ESA/ECSS QML/QPL, MIL-PRF-38535 QML, MIL-PRF-19500 and any standards in accordance with ISO 9001 version 2015	Equivalent	2
Certified ISO 9001 version 2015	Lower	1
No information	Much lower	0

Model associated with the $QA_{\text{component}}$ factor

The parameter $QA_{\text{component}}$ is defined for each item family. It takes account mainly of the qualification methodology without considering the severity of the tests defined in the listed standards. Test severities for active components are taken into account by the $RA_{\text{component}}$ factor.

Component quality assurance level	Position relative to state of the art	$QA_{\text{component}}$
Level criteria are defined for each item family	Higher	3
	Equivalent	2
	Lower	1
	Much lower	0

Model associated with the $RA_{component}$ factor

The $RA_{manufacturer}$ factor is defined for integrated circuits and discrete semiconductors. It is quantified on the basis of the results and severity of the tests performed by the manufacturer

	Risk $RA_{component}$
Very reliable Level A	3
Very reliable Level B	2
Reliable	1
Not reliable	0

Model associated with the experience factor ϵ

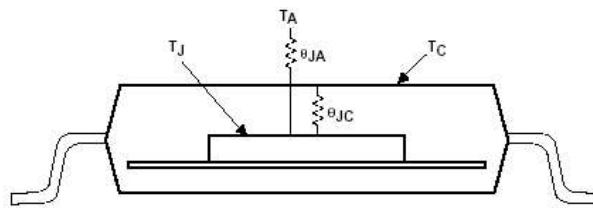
The ϵ factor is intended to represent the experience that the buyer of the component has of its supplier, and is therefore specific to each manufacturer. Its role as a multiplication factor in the model reflects the importance of knowledge about suppliers in component reliability. This factor is common to all items, but in some case specific indications are proposed for its determination.

Manufacturer-related risk	Value of the ϵ factor
Known manufacturer – Mature processes for the item considered	4
Known manufacturer – Processes not analysed or not mature for the item considered	3
Unknown manufacturer (for example, never audited or audited more than 6 years ago) or low volume production	2
Previous disqualification or feedback issue	1

Thermal resistance of components

For active components, the thermal stress model uses the component junction temperature. This requires calculating the rise in the junction temperature relative to the ambient temperature. This evaluation is usually based on the power dissipated by the component and the thermal resistance between the junction and the ambient temperature.

The thermal resistance data published by component suppliers should be used where possible. By default, a method is proposed here for evaluating thermal resistance in active components.



Integrated circuits

$$R_{JA_0m/s} = C_{type} \cdot Np^{-0.58} \cdot K$$

$$R_{JA_2m/s} = \frac{R_{JA_0m/s}}{1.5}$$

R_{JA_V} Junction-to-ambient thermal resistance, as a function of airflow speed

V 0 m/s or 2 m/s = Airflow speed, depending on environmental convection (0 m/s = natural convection)

C_{type} Constant dependent on type of package

Np # of pins/package; for QFN packages, the same formula applies with Np = package area in mm².

K Constant depending on the thermal conductivity value in the board plane ($k_x=k_y$)

Note:

- Low conductivity: $k_x < 15 \frac{W}{m.K}$
- High conductivity: $k_x \geq 15 \frac{W}{m.K}$

Type of package	C_{type}	Range		
QFN (area in mm ²)	300	3	< Np <	148
CerDIP / CDIP	320	8	< Np <	48
Power QFP (HQFP, RQFP, etc.)	340	160	< Np <	304
PDIP	360	8	< Np <	68
PPGA	380	28	< Np <	447
PLCC	390	20	< Np <	84
SOIC	400	8	< Np <	32
SOJ	400	24	< Np <	44
CPGA	410	68	< Np <	655
SOP	410	8	< Np <	32
Power BGA-1.27mm (SBGA, TBGA, etc.)	450	256	< Np <	956
J-CLCC	470	28	< Np <	84
CBGA	780	255	< Np <	1156
Cerpack	480	20	< Np <	56
TQFP, VQFP, LQFP	480	32	< Np <	208
PBGA-1.27mm	530	119	< Np <	729
PBGA-1mm	670	20	< Np <	1950
PBGA-0.8mm	850	60	< Np <	480
PBGA-0.5mm	800	60	< Np <	480
PBGA-0.4mm	1000	60	< Np <	480
Flip-Chip BGA 1mm	520	484	< Np <	1932
Power BGA-1mm (SBGA, TBGA, etc.)	550	256	< Np <	1508
SSOP	560	16	< Np <	64
CQFP	560	64	< Np <	256
PQFP	570	44	< Np <	304
TSSOP	650	8	< Np <	64
TSOP	750	5	< Np <	56
LGA	260	14	< Np <	190
WLP	500	4	< Np <	196

Thermal conductivity of board	K
Low conductivity	1.15
High Conductivity	0.94

For BGA packages, given the diversity of possible forms, it may be preferable to refer to the manufacturer data.

Discrete semiconductors

R_{JA} Junction-to-ambient thermal resistance (for natural convection only, airflow = 0 m/s) in °C/W

R_{JC} Junction-to-case thermal resistance in °C/W

N_p Number of pins in package

k_x Thermal conductivity in the board plane ($k_x = k_y$) in W/m.K

Low conductivity: $k_x < 15 \frac{W}{m.K}$

High conductivity: $k_x \geq 15 \frac{W}{m.K}$

Type of package	Also known as	N_p	R_{JA} Low conductivity	R_{JA} High conductivity	R_{JC}
DO15	DO-204AC	2	60	~42	~5
DO27	DO-201AA	2	41	~30	~1
DO35	DO-204AH	2	378	~241	~134
DO41	DO-204AL	2	73	~50	45
DO92		3	195	~126	150
DO220 *		3	65	~45	4
DPAK *	TO-252AA, SC63, SOT428	4	97	71	4
D2PAK *	TO-263, SC83A, SMD-220	4	58	~40	1
IPACK *	TO-251AA	3	96	50	3
I2PAK		3	63	~44	1
ISOTOP *	SOT227, TO-244, Half-Pak	4	35	~26	1
F126		2	40	~29	~1
SIL	SIL, ZIP	(See manufacturer specification)			
SIP	SIL, ZIP	(See manufacturer specification)			
SOD6	DO-214AA, SMB-J	2	88	~59	27
SOD15	DO-214AB, SMC-J	2	67	~46	2
SOD80	Mini-MELF, DO213AA	2	568	~361	~172
SOD87	DO-214AC, SMA-J	2	110	~73	~41
SOD110		2	315	~202	~119
SOD123		2	337	~216	130
SOD323	SC76	2	428	~273	~146
SOD523	SC79	2	93	~62	~31
SOT23	TO-236AB	3	443	360	130
SOT23	SC74A, SOT25	5	285	136	106
SOT23	SC74, SOT26, SOT457	6	212	133	110
SOT82 *	TO225	3	100	~67	8
SOT89	SC62, TO-243AA	4	142	125	100
SOT90B		6	500	~318	~160
SOT143	TO-253AA, SC61B	4	473	250	~155
SOT223	SC73, TO261AA	4	84	57	21
SOT323	SC70	3	516	~328	~164
SOT343	SC82	4	215	~139	~88
SOT346	SC59, TO-236AA	3	500	~318	~160
SOT353	SC70-5, SC88A	5	358	229	144
SOT363	SC70-6, SC88	6	553	~351	~164
TO18	TO-71, TO-72, SOT31, SOT18	3	475	~302	150
TO39	SOT5	3	219	~142	58
TO92	SOT54, SC43, TO226AA	3	180	~117	66
TO126	SOT32, TO-225AA	3	95	~64	3
TO218 *	ISOWATT218	3	40	~29	1

Type of package	Also known as	Np	R _{JA} Low conductivity	R _{JA} High conductivity	R _{JC}
TO220 *	TO220-5, ISOWATT220, TO220XX	3	58	~40	4
TO247 *	Max247, Super247, SOT429	3	47	~34	1

Note:

1. Data precede by “~” are orders of magnitude derived from regression analyses based on averages for each type of package.
There are no standardized tests for measuring the thermal resistances of discrete packages; the thermal performance data for these components depends on the manufacturer alone. The *data in italics* is provided here for information only; for the packages concerned, it is highly advisable to refer to the manufacturer data listed in the specifications.
2. *: For power packages (TO218, DPAK, ISOTOP, etc.), the junction-to-ambient thermal resistance “R_{JA}” should be applied only if the package is mounted directly on the board; otherwise, if (for example) the package is screwed onto a metallic structure or equipped with a heat sink, the recommendation is to apply the junction-to-case thermal resistance “R_{JC}”.
3. If the ΔT of the component is very high ($\Delta T = R_{JA} \times P_{Dissipated} > 150^\circ\text{C}$), it is preferable to look up the thermal measurement conditions in the specification and apply the thermal resistance value “R_{JA}” supplied by the manufacturer, if it is lower than the one obtained by the FIDES method; if not, the thermal resistance “R_{JC}” should be applied (as it takes account of the better metallization under the component).

Light-emitting diodes (LEDs)

For LEDs, given the diversity of size across the different types of package, it is preferable to refer to the manufacturer data. Value ranges are given here for information.

R_{JA} Junction-to-ambient thermal resistance in °C/W

R_{JC} Junction-to-case thermal resistance in °C/W

Current DC direct maximal	SMD or through-hole	Type of package	Number of pins	R_{JA} °C/W	R_{JC} °C/W
$I_F < 150\text{mA}$	Through-hole	T1-x	2 to 4	300-500	160-290
		High flux	4	200	90-155
	SMD	Chip	2	450-800	260-450
		PLCC	Mini 2	460-580	230-330
			2	360-400	180-280
			3	270-290	130-190
			4	270-560	130-180
			6	210-500	130-160
		Round	2	420-530	250
		LGA	2	380-630	180-360
		Other	Any	-	
$150\text{mA} \leq I_F < 300\text{mA}$	SMD			90-140	15-25
$300\text{mA} \leq I_F < 1\text{A}$				39-65	8-23
$I_F \geq 1\text{A}$				30-50	3-17

Integrated circuits

General model associated with the family

$$\lambda = \lambda_{physical} \times \Pi_{PM} \times \Pi_{Process}$$

Where:

$$\lambda_{Physical} = \sum_i^{Phases} \left(\frac{t_{phase}}{T_{Total}} \right)_i \times \left(\begin{array}{l} \lambda_{0TH} \times \Pi_{Thermal} \\ + \lambda_{0TCyCase} \times \Pi_{TCyCase} \\ + \lambda_{0TCySolderJoints} \times \Pi_{TCySolderJoints} \\ + \lambda_{0RH} \times \Pi_{RH} \\ + \lambda_{0Mech} \times \Pi_{Mech} \end{array} \right)_i \times (\Pi_{Induced})_i$$

C_{sensitivity} factor

	Relative sensitivity (score out of 10)			C _{sensitivity}
	EOS	MOS	TOS	
Integrated circuit	10	2	1	7.75

Π_{PM}: Model associated with the QA_{component} factor

Component quality assurance level	Position relative to state of the art	QA _{component}
Qualification according to one of the following standards: AEC Q100, MIL-PRF-38535 class V/Y, MIL-PRF-38510 class S, ESCC 90xx, NASDA-QTS-xxxx classe I, NPSL NASA level 1	Higher	3
Qualification according to one of the following standards: MIL-PRF-38535 class Q, MIL-PRF-38535 class M, MIL-PRF-38535 class N, MIL-PRF-38510 class B, NASDA-QTSxxxx class II, NPSL NASA levels 2 and 3	Equivalent	2
Qualification program internal to the manufacturer and unidentified manufacturing sites	Lower	1
No information	Much lower	0

II_{PM}: Model associated with the RA_{component} factor

Accelerated ageing test	High Temperature Operating Life (HTOL)	Pre-conditioning before TCy, THB or HAST	Thermal cycling (TCy)	Pressure Cooker Test (PCT)	Highly Accelerated Stressed Tests (HAST)	Temperature Humidity Biased (THB)	
Reference standard	EIA JESD-22-A108 A or equivalent	EIA JESD-22-A113A or equivalent	EIA JESD-22-A104 or equivalent	EIA JESD-22-A102 or equivalent	EIA JESD-22-A110 or equivalent	EIA JESD-22-A101 or equivalent	
	Test results						RA _{component} risk
Very reliable Level A	1000 h, 125°C, Vmax; 231/0 ⁽¹⁾ 1500/0*	Done	1000 cycles - 55°C /+150°C or 500 cycles - 65°C/+150°C 231/0 or 1000 cycles -55°C/125°C 385/0	168 h at 121°C / 100%RH 231/0	168 h at 130°C / 85%RH 231/0	168 h at 130°C / 85%RH 231/0	3
Very reliable Level B	1000 h, 125°C, Vmax; 154/0 ⁽¹⁾ 900/0*	Done	1000 cycles - 55°C /+125°C, 154/0	96 h at 121°C / 100%RH; 154/0	96 h at 130°C / 85%RH; 154/0	96 h at 130°C / 85%RH; 154/0	2
Reliable	1000 h, 125°C, Vmax; 77/0 ⁽¹⁾ 231/0*	Done	500 cycles -55°C /+125°C 154/0	96 h at 121°C / 100%RH; 77/0	96 h at 130°C 85%RH; 77/0	1000 h at 85°C/85%RH, 154/0	1
Not reliable	Dimensions below reliable level	Not done	Dimensions below reliable level				0

Each cell in the table contains a description of the test conditions, with the expected result in the form XXX/Y, where XXX is the number of parts under test and Y the number of defects (in practice Y=0)

(1): Applicable to an item or a front-end process for a given package

*: Applicable to all the front-end processes for a given package.

Where different types of tests result in different reliability levels, the lowest level must be selected.

Basic failure rates associated with packages

The basic failure rates for the different physical stresses are obtained by the equation:

$$\lambda_{0_Stress} = e^{-a} \times Np^b$$

Where:

- a and b are constants that depend on the type of package and the number of pins, as listed in the table below.
- Np is the number of pins in the package.

Usual name	Description	Np	λ_{0RH}		λ_{0TCy_Case}		λ_{0TCy_Solder} Joints		$\lambda_{0Mechanical}$		Note 4
			a	b	a	b	a	b	a	b	
PDIP	Plastic Dual In line Package	8 à 68	6,27	0,69	10,23	0,95	8,29	0,92	12,90	0,92	
CERDIP, CDIP	Ceramic Dual-In-Line Package	8 à 20	-	-	12,68	2,27	8,29	0,92	11,51	0,92	
		>20 à 48					7,96	0,92	11,18	0,92	
PQFP	Plastic Quad Flatpack, L lead	44 à 240	10,94	1,57	13,72	1,62	8,29	0,92	12,21	0,92	
		>240 à 304					7,96	0,92	11,87	0,92	
TQFP, LQFP	Plastic Shrink (thickness) Quad Flatpack, L lead Plastic Thin Quad Flatpack, L lead	32 à 120	6,62	0,52	13,05	1,30	8,29	0,92	12,90	0,92	
		>120 à 208					7,20	0,92	11,80	0,92	
Power QFP	Plastic Quad Flatpack with heat sink, L lead	160 à 240	14,17	2,41	11,80	1,36	8,29	0,92	12,21	0,92	
		>240 à 304					7,96	0,92	11,87	0,92	
CERPACK	Ceramic Package	20 à 56	-	-	8,14	1,01	8,29	0,92	11,51	0,92	
CQFP	Ceramic Quad Flat Pack	20 à 132	-	-	8,14	1,01	8,29	0,92	11,51	0,92	
		>132 à 256					6,68	0,92	9,90	0,92	
PLCC	Plastic Leaded Chip Carrier, J-Lead	20 à 52	10,50	1,92	19,45	3,29	8,29	0,92	12,43	0,92	
		>52 à 84					7,20	0,92	11,29	0,92	
J-LCC	J-Lead Ceramic Leaded, Chip Carrier	4 à 32	-	-	8,07	0,93	8,29	0,92	11,51	0,92	
		44					7,95	0,92	11,17	0,92	
		52					7,19	0,92	10,41	0,92	
		68					6,21	0,92	9,43	0,92	
		84					5,58	0,92	8,80	0,92	
CLCC	Ceramic Leadless Chip Carrier	4 à 10	-	-	8,07	0,93	7,19	0,92	10,41	0,92	
		20					5,89	0,92	9,11	0,92	
		32					5,58	0,92	8,80	0,92	
		44 à 52					5,07	0,92	8,29	0,92	
		68 à 84					4,36	0,92	7,58	0,92	

Usual name	Description	Np	λ_{0RH}		λ_{0TCy_Case}		λ_{0TCy_Solder} Joints		$\lambda_{0Mechanical}$		Note 4
			a	b	a	b	a	b	a	b	
SOJ	Plastic Small Outlines, J-Lead	24 à 44	4,33	0,83	8,76	1,49	8,29	0,92	12,90	0,92	
SO	Plastic Small Outlines, L lead	8 à 14	11,45	1,95	16,80	2,94	8,29	0,92	12,90	0,92	
		16 à 18					8,29	0,92	12,90	0,92	
		20 à 28					7,96	0,92	12,56	0,92	
		32					7,96	0,92	12,56	0,92	
TSOP	Thin Small Outlines, leads on small edges, L lead Thin Small Outlines, leads on long edges, L lead	5 à 16	6,87	1,10	9,60	0,83	8,29	0,92	12,90	0,92	
		>16 à 32					7,20	0,92	11,80	0,92	
		>32 à 44					6,68	0,92	11,29	0,92	
		>44 à 56					6,17	0,92	10,77	0,92	
SSOP, QSOP	Plastic Shrink (pitch) Small Outlines, L lead	16 à 64	17,70	3,35	20,88	3,38	7,96	0,92	12,56	0,92	
TSSOP, MSOP	Thin Shrink Small Outlines, L lead Mini Small Outline Package	8 à 28	11,25	1,57	14,93	1,87	8,29	0,92	12,90	0,92	
		>28 à 48					7,96	0,92	12,56	0,92	
		56					7,20	0,92	11,80	0,92	
		64					6,17	0,92	10,77	0,92	
QFN, DFN ≥ 0.5 mm	Quad Flat No lead (package without lead)	8 à 24	8,84	0,77	12,03	0,94	6,68	0,92	9,90	0,92	
		28 à 56					6,17	0,92	9,38	0,92	
		64 à 80					5,95	0,92	9,64	0,92	
QFN 0.4 mm	Quad Flat No lead (package without lead)	8 à 40	6,22	0,78	9,65	0,91	6,17	0,92	9,38	0,92	
		44 à 80					5,95	0,92	9,17	0,92	
LGA Plastic	Plastic Land Grid Array	6 à 20	5,3	0,84	9,02	1,02	6,68	0,92	9,90	0,92	
		>20					6,17	0,92	9,38	0,92	
PBGA <0.5 mm	Plastic Ball Grid Array with solder ball pitch <0.5 mm	<290	4,25	0,14	5,61	0,34	5,64	0,92	9,33	0,92	X
PBGA 0.5 mm	Plastic Ball Grid Array with solder ball pitch =0.5 mm	<100	4,25	0,14	8,78	0,53	5,95	0,92	9,64	0,92	X
		100 à 400					5,64	0,92	9,33	0,92	X
PBGA 0.65 mm	Plastic Ball Grid Array with solder ball pitch =0.65 mm	<400	5,59	0,40	14,15	1,57	6,17	0,92	9,85	0,92	X
		400 à 625					5,95	0,92	9,64	0,92	X
PBGA BT 0.8 mm	Plastic Ball Grid Array with solder ball pitch =0.8 mm	48 à 96	4,99	0,34	7,31	0,22	6,68	0,92	10,37	0,92	X
		100 à 384					6,17	0,92	9,85	0,92	X
PBGA flex 0.8 mm	Plastic Ball Grid Array with solder ball pitch =0.8 mm et 0.75 mm	48 à 288	9,70	1,50	8,03	0,29	5,95	0,92	9,64	0,92	X
PBGA BOC 0.8 mm (memories)	Plastic Ball Grid Array with solder ball pitch =0.8 mm	48 à 96	5,59	0,40	14,15	1,57	5,95	0,92	9,64	0,92	X
PBGA BT 1.00 mm		64 à 484	4,77	0,40	10,46	0,76	6,68	0,92	10,37	0,92	

Usual name	Description	Np	λ_{0RH}		λ_{0TCy_Case}		$\lambda_{0TCy_Solder\ Joints}$		$\lambda_{0Mechanical}$		Note 4
			a	b	a	b	a	b	a	b	
	Plastic Ball Grid Array with solder ball pitch =1.0 mm	>484 à 1156					6,17	0,92	9,85	0,92	
PBGA 1.27 mm	Plastic Ball Grid Array, with solder ball pitch =1.27 mm	119 à 352	6,85	0,80	10,29	0,86	7,20	0,92	10,87	0,92	
		>352 à 432					6,68	0,92	10,37	0,92	
		>432 à 729					6,17	0,92	9,85	0,92	
Power BGA	Tape BGA, PBGA with heat sink, die top down pitch =1.27 mm Super BGA, PBGA with heat sink, die top down pitch =1.27 mm	256 à 352	5,59	0,61	11,01	0,79	7,20	0,92	10,87	0,92	
		>352 à 956					6,68	0,92	10,37	0,92	
BGA WLP <0.5 mm	BGA with solder ball pitch <0.5 mm, Wafer Level Packaging	4 à 169	6,96	0,78	6,60	0,07	5,64	0,92	9,33	0,92	X
BGA WLP 0.5 mm	BGA with solder ball pitch =0.5 mm, Wafer Level Packaging	4 à 36	6,96	0,78	10,31	1,02	5,95	0,92	9,64	0,92	X
		>36					5,64	0,92	9,33	0,92	X
FC-PBGA 0.8 mm	Flip Chip PBGA with solder ball pitch =0.8 mm	<300	8,39	0,79	9,9	0,53	6,68	0,92	10,37	0,92	X
		300 à 1089					6,17	0,92	9,85	0,92	X
FC-PBGA 1 mm	Flip Chip PBGA with solder ball pitch =1 mm	<672	8,39	0,79	9,82	0,52	7,20	0,92	10,87	0,92	
		672 à 1704					6,68	0,92	10,37	0,92	
CBGA	Ceramic Ball Grid Array	255 à 560	4,41	0,63	10,60	1,12	5,12	0,92	8,80	0,92	
		>560 à 1156					4,36	0,92	7,35	0,92	
DBGAs	Dimpled BGA	255 à 1156	4,41	0,63	10,60	1,12	6,68	0,92	10,37	0,92	
CI CGA	Ceramic Land GA + interposer, Ceramic column GA	255 à 1156	4,41	0,63	10,60	1,12	5,95	0,92	9,64	0,92	
CPGA	Ceramic Pin Grid Array	68 à 250	-	-	8,63	1,02	7,96	0,92	11,62	0,92	
		>250 à 655					6,68	0,92	10,37	0,92	

Note 1: For hermetic packages, the failure rate due to atmospheric humidity is null ($\lambda_{0RH} = 0$).

Note 2: The basic failure rates for solder joints are estimated on the basis of assumptions about the type of printed circuit (the selected type is FR4), the CTE differential between the PCB and the component, the material used for the pins, the curvature of the CQFP pins, and the type of substrate used for CBGA, Flex BGA and PBGA arrays. These parameters influence reliability but cannot usually be addressed in a predictive reliability study.

Note 3: Some discrete semiconductor packages are also used for integrated circuits, in particular the “SMD, small signal, L^Qlead, plastic”, “SMD, medium power, small heatsink, L^Qlead, plastic”, “Through-hole, power, plastic” and “SMD, power, large heatsink, L^Qlead, plastic” types. For the failure rates of these packages, refer to the active discrete components datasheet.

Note 4: For eligible packages, the $\lambda_{0TCy_SolderJoints}$ and $\lambda_{0Mechanical}$ values are divided by 3 when the underfill process is used (this injects a resin between the package and the board to improve the reliability of the solder joints on ball or bump grid arrays, such as BGAs or WLPs).

Basic failure rates associated with the chip

Technology families	λ_{0TH}
FPGA, CPLD, FPGA Antifuse, PAL	0.076
Analog and mixed circuit (MOS, Bipolar, BiCMOS)	0.086
Microprocessor, Microcontroller, DSP	0.075
Flash, EEPROM, EPROM	0.060
SRAM	0.053
DRAM	0.047
Digital circuit (MOS, Bipolar, BiCMOS)	0.021

Note 1: Mixed = analog and digital.

Note 2: For ASICs, refer to the ASIC model.

Mission profile data

T_{phase}	Time associated with each phase (hours)
$RH_{ambient}$	Humidity level associated with a phase (%)
$T_{ambient-board}$	Mean temperature of the board during a phase (°C)
$\Delta T_{cycling}$	Amplitude of temperature variation associated with a cycling phase (°C)
$T_{max-cycling}$	Maximum temperature on the board during a cycling phase (°C)
N_{cy}	Number of cycles associated with each cycling phase (cycles)
θ_{cy}	Cycle duration (hours)
G_{RMS}	Vibration amplitude associated with each random vibration phase (Grms)

Application data

$T_{J_component}$	Component junction temperature during an operating phase (°C)
$T_{J_component} = T_{ambient} + R_{JA} \cdot P_{dissipated}$	
$P_{dissipated}$	Power dissipated by the component during the phase (W)

Factors contributing to physical stresses

$\Pi_{Thermal}$	$11604 \times 0.7 \times \left[\frac{1}{293} - \frac{1}{(T_{j-component} + 273)} \right]$ In an operating phase: e In a non-operating phase: $\Pi_{Thermal} = 0$
$\Pi_{TCyCase}$	$\left(\frac{12 \times N_{cy}}{T_{phase}} \right) \times \left(\frac{\Delta T_{cycling}}{20} \right)^4 \times e^{1414 \times \left[\frac{1}{313} - \frac{1}{T_{max-cycling} + 273} \right]}$
Π_{TCy} <i>Solder Joints</i>	$\left(\frac{12 \times N_{cy}}{T_{phase}} \right) \times \left(\frac{\min(\theta_{cy} \cdot 2)}{2} \right)^{\frac{1}{3}} \times \left(\frac{\Delta T_{cycling}}{20} \right)^{1.9} \times e^{1414 \times \left[\frac{1}{313} - \frac{1}{T_{max-cycling} + 273} \right]}$
Π_{Mech}	$\left(\frac{G_{RMS}}{0.5} \right)^{1.5}$
Π_{RH}	$\left(\frac{RH_{ambient}}{70} \right)^{4.4} \times e^{11604 \times 0.9 \times \left[\frac{1}{293} - \frac{1}{(T_{ambient-board} + 273)} \right]}$ In an operating phase: $\Pi_{RH} = 0$

Factoring in the lifetime of DSM components

In the specific case of deep-submicron (DSM) planar components, when both of the following conditions are met:

$$\begin{cases} 100^{\circ}C < T_{junction} \leq 125^{\circ}C \\ V = \frac{V_{utilization}}{V_{nominal}} \leq 1.1 \end{cases}$$

A supplementary model is proposed to cater for a change of dominant failure mechanism (ageing) when it is no longer negligible (cf. Annex “Recommendations on the use of DSM components”).

Application-Specific Integrated Circuits (ASICs)

General model associated with the family

$$\lambda_{ASIC} = \lambda_{Physical} \times \Pi_{ASIC_Process} \times \Pi_{PM} \times \Pi_{Process}$$

ASICs are modelled in the same way as other integrated circuits, but with certain specificities described in this datasheet.

The $\lambda_{Physical}$ and $\Pi_{Process}$ factors are as defined for the Integrated Circuits family. The λ_{0TH} values for the different types of ASIC are detailed below.

Model associated with the $\Pi_{ASIC_Process}$ factor

The $\Pi_{ASIC_Process}$ factor represents:

- Compliance with a formal development methodology (such as DO254, COCISPER, etc.) and the project manager's level of control over subcontractors, for projects that involve multiple contributors (silicon manufacturer, assembler, test house, etc.)
- Effective production control and appropriate monitoring during the ASIC manufacturing process.

$$\Pi_{ASIC_Process} = e^{1.39 \times (1 - ASIC_Grade)}$$

If $\Pi_{ASIC_Process}$ is not evaluated, a default value of 2.5 is proposed. Using the default value may reduce the accuracy of the final results.

The ASIC_Grade factor is calculated from a questionnaire about the ASIC development process.

$$ASIC_Grade = \frac{\sum Scores}{100}$$

Where "Scores" is defined by the values of the corresponding scores in the table below.

N°	Factors influencing the ASIC_Grade	Value (if true)	Value (if false)
Development and design: recommendations on ASIC design and project management			
1	Application of a formal methodology during the design phase (DO254, COCISPER, etc.)	10	0
2	Existence of a subcontractor management plan (project participants)	10	0
3	Subcontractor management plan covering all project participants (control over the entire life cycle)	10	0
4	Selection of subcontractors with experience in the target technology, functionality and complexity level	10	0
5	Selection of subcontractors experienced in handling the complexity of the industrial organization of the project	5	0
6	Formal control (with reviews) by the design centre over subcontractors (silicon manufacturer, assembler, and “test house”)	5	0
7	Previous history of industrial organization	5	0
8	No subcontractors with unfavourable feedback	5	0
Manufacture: Recommendations on manufacturing the monitoring of the ASIC manufacturing process			
9	Implementation of a mature non-ageing technology	15	0
10	Implementation of wafer annealing	10	0
11	Functional test of the ASIC at 3 temperatures	15	0

The range of variation of the $\Pi_{ASIC_Process}$ factor is from 1 (for the best process) to 4 (for the worst process).

Model associated with the Π_{PM} factor

$$\Pi_{PM} = e^{1.39 \times (1 - Part_Grade) - 0.69}$$

$$Part_{Grade} = \left\lceil \frac{(QA_{manufacturer} + QA_{component} + RA_{component}) \times \varepsilon}{36} \right\rceil$$

The $QA_{manufacturer}$, $QA_{component}$ and $RA_{component}$ factors are as defined for the Integrated Circuits family.

Model associated with the experience factor ε

As with integrated circuits, the ε factor is intended to represent the buyer's experience of the supplier. In the case of ASICs, particular attention must be paid to manufacturers who propose aging technologies, with little support, or technologies that are too recent. Such case should be assimilated to a “small production series” (with $\varepsilon=2$).

Manufacturer-related risk	ε
Known manufacturer – Mature processes for the product considered	4
Known manufacturer – Processes not analysed or not mature for the product considered	3
Unknown manufacturer (for example, never audited or audited more than 6 years ago) or manufacturing of small series or of an ageing or immature ASIC technology	2
Previous disqualification or major feedback issue	1

Basic failure rates associated with the chip

Type of technology	Type of ASIC function	λ_{0TH}
Silicon MOS	Digital ASIC, simple function	0.021
	Digital ASIC, complex function (with IP and/or μP core, memory blocks)	0.075
	Analog, mixed ASIC	0.123
Silicon bipolar, BICMOS	Digital ASIC	0.021
	Mixed, analog ASIC	0.123

Discrete semiconductors: diodes, low-power transistors and bipolar power transistors

General model associated with the family

$$\lambda = \lambda_{physical} \times \Pi_{PM} \times \Pi_{process}$$

Where:

$$\lambda_{Physical} = \sum_i^{Phases} \left(\frac{t_{phase}}{T_{Total}} \right)_i \times \left(\begin{array}{l} \lambda_{0TH} \times \Pi_{Thermal} \\ + \lambda_{0TCyCase} \times \Pi_{TCyCase} \\ + \lambda_{0TCySolderJoints} \times \Pi_{TCySolderJoints} \\ + \lambda_{0RH} \times \Pi_{RH} \\ + \lambda_{0Mech} \times \Pi_{Mech} \end{array} \right)_i \times (\Pi_{Induced})_i$$

C_{sensitivity} factor

	Relative sensitivity (score out of 10)			C _{sensitivity}
	EOS	MOS	TOS	
Discrete semiconductor circuit	8	2	1	6.30

Π_{PM}: Model associated with the QA_{component} factor

Component quality assurance level	Position relative to state of the art	QA _{component}
Qualification according to one of the following standards: AEC Q101, AEC Q102, MIL-PRF-19500 JANS, ESCC 5000, ESCC 5010 level B, NASDA-QTS-xxxx class I, JAXA-QTS Class I (NASDA-QTS-2030)	Higher	3
Qualification according to one of the following standards: MIL-PRF-19500 JANTX or JANTXV, ESCC 5010 level C, NASDA-QTS-xxxx class II, JAXA-QTS Class II	Equivalent	2
Qualification according to one of the following standards: MIL-PRF-19500 JAN or qualification program internal to the manufacturer and unidentified manufacturing sites	Lower	1
No information	Much lower	0

Π_{PM} : Model associated with the $RA_{component}$ factor

Accelerated ageing test	High Temperature Reverse Bias (HTRB)	High Temperature Gate Bias (HTGB)	Intermittent Operating Life ⁽²⁾ or Power and Temperature cycle ⁽²⁾	Pre-conditioning before TCy, THB or HAST	Thermal cycling (TCy)	Pressure Cooker Test (PCT)	High Humidity High Temperature Reverse Bias (H ³ TRB)	
Reference standard	EIA JESD-22-A108 A or equivalent	EIA JESD-22-A108 A or equivalent	MIL-STD-750 Method 1037 EIA JESD22 A-105	EIA JESD-22-A113A or equivalent	EIA JESD-22-A104 or equivalent	EIA JESD-22-A102 or equivalent	EIA JESD-22-A101 or equivalent	
	Test results							$RA_{component}$ risk
Very reliable Level A	1000 h, 125°C, 80% to 100% of the rated voltage, 231/0 ⁽¹⁾ 1500/0*	1000 h, 150°C, 80% to 100% of the rated voltage 231/0 ⁽¹⁾ 1500/0*	Ta=25°C. Component biased to obtain $\Delta T_j \geq 100^\circ\text{C}$ (without reaching absolute max. ratings) 231/0 ⁽¹⁾ 1500/0*	Done	1000 cycles -55°C /+150°C or 500 cycles - 65°C/+150°C 231/0 or 1000 cycles -55°C/125°C 385/0	2000 h at 85°C/85% RH 154/0	168 h at 130°C / 85%RH 231/0	3
Very reliable Level B	1000 h, 125°C, 80% to 100% of the rated voltage, 154/0 ⁽¹⁾ 900/0*	1000 h, 150°C, 80% to 100% of the rated voltage 154/0 ⁽¹⁾ 900/0*	Ta=25°C. Component biased to obtain $\Delta T_j \geq 100^\circ\text{C}$ (without reaching absolute max. ratings) 154/0 ⁽¹⁾ 900/0*	Done	1000 cycles -55°C /+125°C; 154/0	96 h at 121°C / 100%RH; 154/0	2000 h at 85°C/85%RH 154/0	2
Reliable	1000 h, 150°C, 80% to 100% of the rated voltage, 77/0 ⁽¹⁾ 231/0*	1000 h, 150°C, 80 to 100% of the rated voltage; 77/0 ⁽¹⁾ 231/0*	Ta=25°C. Component biased to obtain $\Delta T_j \geq 100^\circ\text{C}$ (without reaching absolute max. ratings); 77/0 ⁽¹⁾ 231/0*	Done	500 cycles -55°C /+125°C, 154/0	96 h at 121°C / 100%RH; 77/0	1000 h at 85°C/85%RH, 154/0	1
Not reliable	Dimensions below reliable level			Not done	Dimensions below reliable level			0

Each cell in the table contains a description of the test conditions, with the expected result in the form XXX/Y, where XXX is the number of parts under test and Y the number of defects (in practice Y=0)

(1): Applicable to an item or a Front End process for a given package.

(2): Test conditions as defined in AEC-Q101.

*: Applicable to all the Front End processes for a given package.

Where different types of tests result in different reliability levels, the lowest level must be selected.

Basic failure rates associated with packages

Package	Also known as	Designation	λ_{0RH}	λ_{0TCy_Case}	$\lambda_{0TCy_Solder\ Joints}$	$\lambda_{0Mechanical}$
CB417		Through-hole, small signal, plastic	0.005713	0.000298	0.001491	0.000030
CB429						
DO13	DO202AA					
DO15	DO204AC					
DO27	DO201AA					
DO35	DO204AH					
DO41	DO204AL					
DO92						
F126						
SIL, SIP	SIL, SIP, ZIP					
TO92	SOT54, SC43, TO226AA					
TO126	SOT32, TO225AA					
TO202						
SOT23-3	TO236AB	SMD, small signal, L ⁺ lead, plastic	0.002467	0.000048	0.000242	0.000005
SOT23-5	SC74A, SOT25					
SOT23-6	SC74, SOT26, SOT457					
SOT143	TO253AA, SC61B					
SOT323	SC70					
SOT346	SC59, TO236AA					
SOT353	SC70-5, SC88A					
SOT363	SC70-6, SC88					
SOD123						
SOD323	SC76					
SOD523	SC79					
SOT223	SC73, TO261AA	SMD, medium power, small heatsink, L ⁺ lead, plastic	0.003009	0.000747	0.003734	0.000075
SOT243						
SOT343	SC82					
SOT89	SC62, TO243AA					
SOT194						
TO218	ISOWATT218	Through-hole, power, plastic	0.039622	0.001553	0.007764	0.000155
TO220	TO220-5, ISOWATT220, TO220XX					
TO247	Max247, Super247, SOT429					
ISOWATT						
DO220						
IPACK	TO251AA					
SOT82	TO225					
SOD6	DO214AA, SMB-J	SMD, small signal, C ⁺ lead, plastic	0.010994	0.000590	0.002949	0.000059
SOD15	DO214AB, SMC-J					
DDPAK	TO252AA, SC63, SOT428	SMD, power, large heatsink, L ⁺ lead, plastic	0.027131	0.001296	0.006479	0.000130
D2PAK	TO263, SC83A, SMD220					
D3PAK	TO268					
ISOTOP	SOT227, TO244, Half-Pak	SMD, high-power, screw, plastic	0.487813	0.013371	0.066853	0.001337
SOD80	Mini-MELF, DO213AA	SMD, hermetically sealed glass	-	0.004788	0.023938	0.000479
SOD87	MELF, DO213AB					

Package	Also known as	Designation	λ_{0RH}	λ_{0TCy_Case}	$\lambda_{0TCy_Solder\ Joints}$	$\lambda_{0Mechanical}$
TO18	TO71, TO72, SOT31, SOT18	Through-hole, metal	-	0.004247	0.021234	0.000425
TO39	SOT5					
TO52						
SOT553		SMD, medium power, small heatsink, flat lead, plastic	0.033236	0.003246	0.016230	0.000325
SOT563						
SOT563F						
SOT663						
SOT666						
SOT66						
SOT665						
SOT490						

Notes:

- For hermetic packages, the failure rate due to atmospheric humidity is null.
- Some Integrated Circuit packages are also used for discrete semiconductors, in particular the “Thin Shrink Small Outlines, L^Qlead, plastic (TSSOP)”, “Thin Small Outlines, leads on long edge, L^Qlead, plastic (TSOP)” and “Plastic Small Outlines, L^Qlead, plastic (SO)” types. For the failure rates of these packages, refer to the integrated circuit components datasheet.

Basic failure rates associated with the chip

Low-power diodes	λ_{0TH}	Power diodes	λ_{0TH}
Signal diodes up to 1A (PIN, Schottky, signal, varactor)	0.0044	Thyristors, triacs over 3A	0.1976
Rectifier diodes 1A to 3A	0.0100	Rectifier diodes > 3A	0.1574
Zener regulation diodes up to 1.5W	0.0080	Zener regulation diodes over 1.5W	0.0954
Protection diodes up to 3kW (peak 10ms / 100ms) (TVS)	0.0210	Protection diodes over 3kW (peak 10ms / 100ms) (TVS)	1.4980

Low-power transistors	λ_{0TH}	Power transistors	λ_{0TH}
Silicon, bipolar < 5W	0.0138	Silicon, bipolar > 5W	0.0478
Silicon, MOS < 5W	0.0145		
Silicon, JFET < 5W	0.0143		

When N elements (diodes, transistors) are installed in a single package, the value of λ_{0TH} must be multiplied by $\sqrt{N}$.

Note: Silicon, MOS (>5W) and IGBT power components are covered by a dedicated model (see page 138).

Mission profile data

T_{phase}	Time associated with each phase (hours)
RH_{ambient}	Humidity level associated with a phase (%)
$T_{\text{ambient-board}}$	Mean temperature of the board during a phase (°C)
$\Delta T_{\text{cycling}}$	Amplitude of temperature variation associated with a cycling phase (°C)
$T_{\text{max-cycling}}$	Maximum temperature on the board during a cycling phase (°C)
N_{cy}	Number of cycles associated with each cycling phase (cycles)
θ_{cy}	Cycle duration (hours)
G_{RMS}	Vibration amplitude associated with each random vibration phase (Grms)

Application data

$T_{\text{J-component}}$	Component junction temperature during an operating phase (°C)
$T_{\text{J-component}} = T_{\text{ambient}} + R_{\text{JA}} \cdot P_{\text{dissipated}}$	
$P_{\text{dissipated}}$	Power dissipated by the component during the phase (W)
V_{applied}	Inverse voltage applied during the phase (for signal diodes only) (V)

Technical data

V_{rated}	Rated inverse voltage (V) (for signal diodes only)
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Factors contributing to physical stresses

Π_{Thermal}	$11604 \times 0.7 \times \left[\frac{1}{293} - \frac{1}{273 + T_{\text{J-component}}} \right]$ In an operating phase: $\Pi_{\text{El}} \times e$ For signal diodes up to 1A (PIN, Schottky, signal, varactor): $\Pi_{\text{El}} = \left(\frac{V_{\text{applied}}}{V_{\text{rated}}} \right)^{2.4} \text{ if } \frac{V_{\text{applied}}}{V_{\text{rated}}} > 0.3$ $\Pi_{\text{El}} = 0.056 \text{ if } \frac{V_{\text{applied}}}{V_{\text{rated}}} \leq 0.3$ For other types of item: $\Pi_{\text{El}} = 1$ In a non-operating phase: $\Pi_{\text{Thermal}} = 0$
Π_{TCyCase}	$\left(\frac{12 \times N_{\text{cy}}}{T_{\text{phase}}} \right) \times \left(\frac{\Delta T_{\text{cycling}}}{20} \right)^4 \times e^{1414 \times \left[\frac{1}{313} - \frac{1}{T_{\text{max-cycling}} + 273} \right]}$
Π_{TCy} <i>Solder Joints</i>	$\left(\frac{12 \times N_{\text{cy}}}{T_{\text{phase}}} \right) \times \left(\frac{\min(\theta_{\text{cy}}, 2)}{2} \right)^{\frac{1}{3}} \times \left(\frac{\Delta T_{\text{cycling}}}{20} \right)^{1.9} \times e^{1414 \times \left[\frac{1}{313} - \frac{1}{T_{\text{max-cyclage}} + 273} \right]}$
Π_{Mech}	$\left(\frac{G_{\text{RMS}}}{0.5} \right)^{1.5}$
Π_{RH}	$\left(\frac{RH_{\text{ambient}}}{70} \right)^{4.4} \times e^{11604 \times 0.9 \times \left[\frac{1}{293} - \frac{1}{T_{\text{ambient-board}} + 273} \right]}$ In an operating phase: $\Pi_{\text{RH}} = 0$

Discrete power semiconductors: Silicon, MOS > 5W and IGBT

General model associated with the family

$$\lambda = \lambda_{physical} \times \Pi_{PW} \times \Pi_{PM} \times \Pi_{process}$$

Where:

$$\lambda_{Physical} = \sum_i^{Phases} \left(\frac{t_{phase}}{T_{Total}} \right)_i \times \left(\begin{array}{c} \lambda_{0TH} \times \Pi_{Thermal} \\ + \lambda_{0TCyCase} \times \Pi_{TCyCase} \\ + \lambda_{0TCySolderJoints} \times \Pi_{TCySolderJoints} \\ + \lambda_{0RH} \times \Pi_{RH} \\ + \lambda_{0Mech} \times \Pi_{Mech} \end{array} \right)_i \times (\Pi_{Induced})_i$$

C_{sensitivity} factor

The C_{sensitivity} factor is as defined for the “Discrete semiconductors: diodes, low-power transistors and bipolar power transistors” family.

Model associated with the QA_{component} factor

The QA_{component} factor is as defined for the “Discrete semiconductors: diodes, low-power transistors and bipolar power transistors” family.

Model associated with the RA_{component} factor

The RA_{component} factor is as defined for the “Discrete semiconductors: diodes, low-power transistors and bipolar power transistors” family.

Model associated with the Π_{PW} factor

The formula for calculating Π_{PW} is:

$$\Pi_{PW} = \exp(\delta \cdot (1 - \text{Process_Grade}) - \alpha)$$

Where:

$$\begin{aligned} \delta &= 3.401 \\ \alpha &= 0 \end{aligned}$$

The *Process_Grade* parameter is estimated from an audit of how the factors influencing the design rules are managed, as described below.

Each item is assigned a weight and an answer: TRUE (score = 1) or FALSE (score = 0). The value of *Process_Grade* is then:

$$\text{Process_Grade} = \frac{\sum_{j=1}^{10} \text{Weight_answer}_j}{82}$$

Process_Grade equals 1 if all the design rules in the audit are applied fully, and 0 if none are applied. The range of variation of the Π_{PW} factor is [1;30]. If Π_{PW} is not evaluated, a default value of 10 is proposed.

The Π_{PW} audit questionnaire is as follows:

Family	Item	Questionnaire	Weight
Electrical derating	1.1	Is a derating of at least 20% applied to the V_{ds} (MOSFET) or V_{ce} (IGBT) voltage?	6
	1.2	Is a derating of at least 20% applied to the V_{gs} (MOSFET) or V_{ge} (IGBT) voltage?	6
	1.3	Is a derating of at least 20% applied to the current max I_d (MOSFET) or I_c (IGBT)?	6
Thermal derating	2.1	Is a derating of at least 25°C applied to T_{jmax} , or a derating of at least 20% to max power?	12
	2.2	Has allowance been made for variability in thermal contact resistance (thermal contact quality, solder voids, screw torque, spring push force, electrical insulation thickness, etc.)	12
Transient overstress	3.1	Have the maximum switching speeds (dV/dt , dI/dt) associated with parasitic wiring elements been considered?	10
	3.2	Has consideration been given to protection from short-circuits/overstresses (power/current/voltage) (with due attention to transient modes on resonant structures / soft switching)? Note: Indicate whether justified by analysis or by testing.	10
	3.3	Has consideration been given to the effects of the component's intrinsic diode recovery, including in the transient modes (start, off, overvoltage, overstress, short-circuit, etc.)?	10
Accidental overstress	4.1	Is a gate resistor used to avoid high frequency oscillation on switching, particularly in the case of parallel use?	5
	4.2	Is voltage protection used, in case of excessive bus voltage with high-current switching?	5

Basic failure rates associated with the packages

The basic failure rate associated with the packages is as defined for the “Discrete semiconductors: diodes, low-power transistors and bipolar power transistors” family.

Basic failure rates associated with the chip

Power transistors	λ_{0TH}
Silicon, MOS > 5W	0.56
IGBT	0.56

When N elements (diodes, transistor) are installed in a single package, the value of λ_{0TH} must be multiplied by $\sqrt{N}$.

Mission profile data

T_{phase}	Time associated with each phase (hours)
$RH_{ambient}$	Humidity level associated with a phase (%)
$T_{ambient-board}$	Mean temperature of the board during a phase (°C)
$\Delta T_{cycling}$	Amplitude of temperature variation associated with a cycling phase (°C)
$T_{max-cycling}$	Maximum temperature on the board during a cycling phase (°C)
N_{cy}	Number of cycles associated with each cycling phase (cycles)
θ_{cy}	Cycle duration (hours)
G_{RMS}	Vibration amplitude associated with each random vibration phase (Grms)

Application data

$T_{J_component}$	Component junction temperature during an operating phase (°C). The maximum value of this temperature will be 175°C.
$T_{J_component} = T_{ambient} + R_{JA} \cdot P_{dissipated}$	
$P_{dissipated}$	Power dissipated by the component during the phase (W)

Technical data

T_{Ref}	60°C
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Factors contributing to physical stresses

$\Pi_{Thermal}$	In an operating phase: $e^{11604 \times 0.7 \times \left[\frac{1}{T_{Ref}+273} - \frac{1}{T_{j-component}+273} \right]}$ In a non-operating phase: $\Pi_{Thermal} = 0$
$\Pi_{TCyCase}$	$\left(\frac{12 \times N_{cy}}{T_{phase}} \right) \times \left(\frac{\Delta T_{cycling}}{20} \right)^4 \times e^{1414 \times \left[\frac{1}{313} - \frac{1}{T_{max-cycling}+273} \right]}$
Π_{TCy} <i>Solder Joints</i>	$\left(\frac{12 \times N_{cy}}{T_{phase}} \right) \times \left(\frac{\min(\theta_{cy}, 2)}{2} \right)^{\frac{1}{3}} \times \left(\frac{\Delta T_{cycling}}{20} \right)^{1.9} \times e^{1414 \times \left[\frac{1}{313} - \frac{1}{T_{max-cycling}+273} \right]}$
Π_{Mech}	$\left(\frac{G_{RMS}}{0.5} \right)^{1.5}$
Π_{RH}	$\left(\frac{RH_{ambient}}{70} \right)^{4.4} \times e^{11604 \times 0.9 \times \left[\frac{1}{293} - \frac{1}{273+T_{ambient-board}} \right]}$ In an operating phase: $\Pi_{RH} = 0$

Light-emitting diodes (LEDs)

General model associated with the family

ⓘ **Caution: Limited lifetime**

$$\lambda = \lambda_{physical} \times \Pi_{PM} \times \Pi_{Process}$$

Where:

$$\lambda_{Physical} = \sum_i^{\text{Phases}} \left(\frac{t_{\text{phase}}}{T_{\text{Total}}} \right)_i \times \left(\begin{array}{c} \lambda_{0TH} \times \Pi_{Thermal} \\ + \lambda_{0TCyCase} \times \Pi_{TCyCase} \\ + \lambda_{0TCySolderJoints} \times \Pi_{TCySolderJoints} \\ + \lambda_{0RH} \times \Pi_{RH} \\ + \lambda_{0Mech} \times \Pi_{Mech} \end{array} \right)_i \times (\Pi_{Induced})_i$$

C_{sensitivity} factor

	Relative sensitivity (score out of 10)			C_{sensitivity}
	EOS	MOS	TOS	
Light-emitting diode (LED)	7	2	3	5.68

Model associated with the Part Manufacturing factor Π_{PM}

The Π_{PM} factor is the same as for discrete semiconductors.

Basic failure rates associated with packages

Max. direct current I_F	SMD or through-hole	Type of package	Number of pins	λ_{0RH}	$\lambda_{0TCyCase}$	λ_{0Tcy} Solder Joints	λ_0 Mechanical
$I_F < 150mA$	Through-hole	T1-x	2 to 4	0.0034	0.0104	0.0520	0.0052
		High flux	4				
	SMD	Chip	2				
		PLCC	Mini 2				
			2				
			3				
			4				
			6				
		Round	2			0.1560	0.0624
		LGA	Plastic			0.2080	0.0832
			Ceramic			0.3640	0.1820
		Other	Plastic			0.1560	0.0624
			Ceramic			0.3640	0.1820
$I_F \geq 150mA$	SMD	Plastic	Any	0.0031	0.0042	0.0420	0.0064
		Ceramic				0.1470	0.0735

Basic failure rates associated with the chip

Light-emitting diode (LED)	λ_{0TH}
COLOUR	0.01
WHITE	0.05

When N diodes (colour or white) are installed in a single package, λ_{0TH} must be multiplied by $\sqrt{N}$. Some white diodes are made up of three colour diodes; for these white diodes, do not calculate the individual λ_{0TH} from the λ_{0TH} of 3 colour diodes; instead, use the λ_{0TH} value provided in the table.

Mission profile data

t_{phase}	Time associated with each phase (hours)
RH_{ambient}	Humidity level associated with a phase (%)
$T_{\text{ambient-board}}$	Mean temperature of the board during a phase (°C)
$\Delta T_{\text{cycling}}$	Amplitude of temperature variation associated with a cycling phase (°C)
$T_{\text{max-cycling}}$	Maximum temperature on the board during a cycling phase (°C)
N_{cy}	Number of cycles associated with each cycling phase (cycles)
θ_{cy}	Cycle duration (hours)
G_{RMS}	Vibration amplitude associated with each random vibration phase (Grms)

Application data

$T_{\text{J-component}}$	Component junction temperature during an operating phase (°C)
$T_{\text{J-component}} = T_{\text{ambient}} + R_{\text{JA}} \cdot P_{\text{dissipated}}$	
$P_{\text{dissipated}}$	Power dissipated by the component during the phase (W)
V_{applied}	Inverse voltage applied during the phase (for signal diodes only) (V)

Factors contributing to physical stresses

Π_{Thermal}	$^{11604 \times 0.4 \times \left[\frac{1}{293} - \frac{1}{(273 + T_{\text{J-component}})} \right]}$ In an operating phase: e In a non-operating phase: $\Pi_{\text{Thermal}} = 0$
Π_{TCyCase}	$\left(\frac{12 \times N_{\text{cy}}}{T_{\text{phase}}} \right) \times \left(\frac{\Delta T_{\text{cycling}}}{20} \right)^4 \times e^{1414 \times \left[\frac{1}{313} - \frac{1}{T_{\text{max-cycling}} + 273} \right]}$
$\Pi_{\text{TCy Solder Joints}}$	$\left(\frac{12 \times N_{\text{cy}}}{T_{\text{phase}}} \right) \times \left(\frac{\min(\theta_{\text{cy}}, 2)}{2} \right)^{\frac{1}{3}} \times \left(\frac{\Delta T_{\text{cycling}}}{20} \right)^{1.9} \times e^{1414 \times \left[\frac{1}{313} - \frac{1}{T_{\text{max-cycling}} + 273} \right]}$
Π_{Mech}	$\left(\frac{G_{\text{RMS}}}{0.5} \right)^{1.5}$
Π_{RH}	$\left(\frac{RH_{\text{ambient}}}{70} \right)^{4.4} \times e^{11604 \times 0.9 \times \left[\frac{1}{293} - \frac{1}{273 + T_{\text{ambient-board}}} \right]}$ In an operating phase: $\Pi_{\text{RH}} = 0$

Optocouplers

General model associated with the family

$$\lambda = \lambda_{physical} \times \Pi_{PM} \times \Pi_{Process}$$

Where:

$$\lambda_{Physical} = \sum_i^{Phases} \left(\frac{t_{phase}}{T_{Total}} \right)_i \times \left(\begin{array}{c} \lambda_{0TH} \times \Pi_{Thermal} \\ + \lambda_{0TCyCase} \times \Pi_{TCyCase} \\ + (\lambda_{0TCySolderJoints} + \lambda_{0TCyChip}) \times \Pi_{TCySolderJoints} \\ + \lambda_{0RH} \times \Pi_{RH} \\ + (\lambda_{0MechCase} + \lambda_{0MechChip}) \times \Pi_{Mech} \end{array} \right)_i \times (\Pi_{Induced})_i$$

C_{sensitivity} factor

	Relative sensitivity (score out of 10)			C _{sensitivity}
	EOS	MOS	TOS	
Optocoupler	7	2	2	5.63

Model associated with the Part Manufacturing factor Π_{PM}

The Π_{PM} factor is the same as for discrete semiconductors.

Basic failure rates associated with the chip

Component description	Activation energy (eV)	λ_{0_Th}	$\lambda_{0_TCY_chip}$	$\lambda_{0_MECH_chip}$
Photodiode optocoupler	0.4	0.05	0.01	0.005
Phototransistor optocoupler	0.4	0.11	0.021	0.011

When N optocouplers are installed in a single package, λ_{0_TH} , λ_{0TCY_chip} , and $\lambda_{0_MECH_chip}$ must be multiplied by $\sqrt{N}$.

The values of $\lambda_{0_Tcy_Packages}$, $\lambda_{0_Tcy_Solder\ joints}$, $\lambda_{0_Mech_package}$ and λ_{0_RH} can be found in the basic failure rate tables for integrated circuit or discrete semiconductor packages.

Mission profile data

t_{phase}	Time associated with each phase (hours)
$RH_{ambient}$	Humidity level associated with a phase (%)
$T_{ambient-board}$	Mean temperature of the board during a phase (°C)
$\Delta T_{cycling}$	Amplitude of temperature variation associated with a cycling phase (°C)
$T_{max-cycling}$	Maximum temperature on the board during a cycling phase (°C)
N_{cy}	Number of cycles associated with each cycling phase (cycles)
θ_{cy}	Cycle duration (hours)

Application data

$T_{J_component}$	Component junction temperature during an operating phase (°C)
$T_{J_component} = T_{ambient} + R_{JA} \cdot P_{dissipated}$	
$P_{dissipated}$	Power dissipated by the component during the phase (W)

Factors contributing to physical stresses:

$\Pi_{Thermal}$	$e^{11604 \times 0.4 \times \left[\frac{1}{293} - \frac{1}{(T_{j-component} + 273)} \right]}$ In an operating phase: $\Pi_{Thermal} = 0$
$\Pi_{TCyCase}$	$\left(\frac{12 \times N_{cy}}{T_{phase}} \right) \times \left(\frac{\Delta T_{cycling}}{20} \right)^4 \times e^{1414 \times \left[\frac{1}{313} - \frac{1}{T_{max-cycling} + 273} \right]}$
Π_{TCy} <i>Solder Joints</i>	$\left(\frac{12 \times N_{cy}}{T_{phase}} \right) \times \left(\frac{\min(\theta_{cy}, 2)}{2} \right)^{\frac{1}{3}} \times \left(\frac{\Delta T_{cycling}}{20} \right)^{1.9} \times e^{1414 \times \left[\frac{1}{313} - \frac{1}{T_{max-cycling} + 273} \right]}$
Π_{Mech}	$\left(\frac{G_{RMS}}{0.5} \right)^{1.5}$
Π_{RH}	$\left(\frac{RH_{ambient}}{70} \right)^{4.4} \times e^{11604 \times 0.9 \times \left[\frac{1}{293} - \frac{1}{(273 + T_{ambient-board})} \right]}$ In an operating phase: $\Pi_{RH} = 0$

Resistors

General model associated with the family

$$\lambda = \lambda_{physical} \times \Pi_{PM} \times \Pi_{process}$$

Where:

$$\lambda_{Physical} = \lambda_{0_Resistor} \sum_i^{Phases} \left(\frac{t_{phase}}{T_{Total}} \right)_i \times (\Pi_{Thermo-electrical} + \Pi_{TCy} + \Pi_{Mechanical} + \Pi_{RH})_i \times (\Pi_{Induced})_i$$

C_{sensitivity} factor

	Relative sensitivity (score out of 10)			C _{sensitivity}
	EOS	MOS	EOS	
“MiniMELF” fixed resistor, low-power, thin film, current use (RC), high-stability (RS)	4	2	4	3.55
Fixed resistor, high-power (≥1W), thick film	2	4	1	2.40
Fixed resistor, low-power, wirewound, precision	2	1	3	1.83
Fixed resistor, high-power (≥1W), wirewound	2	4	1	2.40
Potentiometer, non-wirewound	1	5	2	1.95
Fixed resistor, SMD, thick film	4	3	5	3.83
Resistor network, SMD	3	5	3	3.45
Fixed resistor, thin film, precision, high-stability	5	5	4	4.95

QA_{component} factor

Component quality assurance level	Position relative to state of the art	QA _{component}
Qualification according to one of the following standards: AEC Q200, MIL-PRF-xxxx level T, MIL-PRF-xxxx level S, MIL-PRF-xxxx level R, ESCC 400x, NASDA-QTS-xxxx class I (JAXA-QTS-2050D)	Higher	3
Qualification according to one of the following standards: MIL-PRF-xxx level P, NASDA-QTS-xxxx class II with identification of manufacturing sites for these standards, qualification according to approved CECC standards.	Equivalent	2
Qualification according to MIL-PRF-xxxx level M, or qualification program internal to the manufacturer and unidentified manufacturing sites	Lower	1
No information	Much lower	0

Note: This factor is determined in this way for all passive components. The MIL-PRF or NASDA-QTS standard appropriate to the item family concerned must be considered.

Basic failure rates associated with the component

Component description			$\lambda_{0-\text{Resistor}}$	A (°C)	$\gamma_{\text{TH-EL}}$	γ_{TCy}	γ_{Mech}	γ_{RH}
“MiniMELF” fixed resistor, low-power, thin film, current use (RC), high-stability (RS)			0.1	85	0.04	0.89	0.01	0.06
Fixed resistor, high-power ($\geq 1\text{W}$), thick film			0.4	130	0.04	0.89	0.01	0.06
Fixed resistor, low-power, wirewound, precision			0.3	30	0.02	0.96	0.01	0.01
Fixed resistor, high-power ($\geq 1\text{W}$), wirewound			0.4	130	0.01	0.97	0.01	0.01
Potentiometer, non-wirewound			0.3	65	0.42	0.35	0.22	0.01
Fixed resistor, SMD, thick film			0.01	70	0.01	0.97	0.01	0.01
Resistor network, SMD			$0.01 \times \sqrt{N_R}$	70	0.01	0.97	0.01	0.01
Fixed resistor, thin film, precision, high-stability	SMD	$<10\text{k}\Omega$	0.18	85	0.14	0.53	0.07	0.26
		$10\text{k}\Omega < \dots < 100\text{k}\Omega$	0.21	85	0.10	0.54	0.06	0.30
		$>100\text{k}\Omega$	0.25	85	0.07	0.55	0.05	0.33
	Through-hole	$<10\text{k}\Omega$	0.14	85	0.18	0.43	0.08	0.31
		$10\text{k}\Omega < \dots < 100\text{k}\Omega$	0.18	85	0.12	0.44	0.07	0.37
		$>100\text{k}\Omega$	0.21	85	0.08	0.45	0.06	0.41

For resistor networks, N_R is the number of resistors in the network.

Mission profile data

t_{phase}	Time associated with each phase (hours)
$RH_{ambient}$	Humidity level associated with a phase (%)
$T_{ambient-board}$	Mean temperature of the board during a phase (°C)
$\Delta T_{cycling}$	Amplitude of temperature variation associated with a cycling phase (°C)
$T_{max-cycling}$	Maximum temperature on the board during a cycling phase (°C)
N_{cy}	Number of cycles associated with each cycling phase (cycles)
θ_{cy}	Cycle duration (hours)
G_{RMS}	Vibration amplitude associated with each random vibration phase (Grms)

Application data

$P_{dissipated}$	Power dissipated by the component during the phase (W)
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Technical data

P_{rated}	Maximum power supported by the component, as specified by the supplier (W)
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Factors contributing to physical stresses

$\Pi_{Thermo-electrical}$	In an operating phase: $\gamma_{TH-EL} \times e^{11604 \times 0.15 \times \left[\frac{1}{293} - \frac{1}{T_{ambient-board} + 273 + A \times \frac{P_{applied}}{P_{rated}}} \right]}$ In a non-operating phase: $\Pi_{Thermo-electrical} = 0$
Π_{TCy}	$\gamma_{TCy} \times \left(\frac{12 \times N_{cy}}{T_{phase}} \right) \times \left(\frac{\min(\theta_{cy}, 2)}{2} \right)^{\frac{1}{3}} \times \left(\frac{\Delta T_{cycling}}{20} \right)^{1.9} \times e^{1414 \times \left[\frac{1}{313} - \frac{1}{T_{max-cycling} + 273} \right]}$
$\Pi_{Mechanical}$	$\gamma_{Mech} \times \left(\frac{G_{RMS}}{0.5} \right)^{1.5}$
Π_{RH}	$\left(\frac{RH_{ambient}}{70} \right)^{4.4} \times e^{11604 \times 0.9 \times \left[\frac{1}{293} - \frac{1}{273 + T_{ambient-board}} \right]}$ In an operating phase: $\Pi_{RH} = 0$

Fuses

Preliminary remarks

Predicting the reliability of fuses poses a particular problem. Unlike other components, there is very little correlation between the replacement of a fuse and its failure. Generally, when a fuse opens and has to be replaced, that means that it has performed its function correctly. A fuse that did not open when it should have will not necessarily be diagnosed as having failed. Care must therefore be taken when using predictive reliability with fuses.

General model associated with the family

$$\lambda = \lambda_{physical} \times \Pi_{PM} \times \Pi_{Process}$$

Where

$$\lambda_{Physical} = \lambda_{0_Fuse} \sum_i^{Phases} \left(\frac{t_{phase}}{T_{Total}} \right)_i \times \left(\frac{\Pi_{Thermo-electrical} + \Pi_{TCy}}{\Pi_{Mechanical} + \Pi_{RH} + \Pi_{Chi}} \right)_i \times (\Pi_{induced})_i$$

C_{sensitivity} factor

	Relative sensitivity (score out of 10)			C _{sensitivity}
	EOS	MOS	TOS	
Fuse	6	6	4	5.90

QA_{component} factor

Component quality assurance level	Position relative to state of the art	QA _{component}
Qualification according to one of the following standards: ESCC 4008, MIL-PRF-23419/08/12 or equivalent	Higher	3
Qualification according to one of the following standards: IEC 60127 or equivalent	Equivalent	2
Qualification program internal to the manufacturer and unidentified manufacturing sites	Lower	1
No information	Much lower	0

Basic failure rate associated with the component

$$\lambda_{0_Fuse} = 0.5$$

The λ_{0_Fuse} value applies to all types of fuse: glass tube fuses, ceramic tube fuses, plug-in fuses, through-hole fuses, SMD fuses, and chip fuses (with FR4 or ceramic substrate).

The model does not include a fuse holder (which can be modelled as a component support connector with 2 contacts).

Mission profile data

t_{phase}	Time associated with each phase (hours)
$RH_{ambient}$	Humidity level associated with a phase (%)
$T_{ambient-board}$	Mean temperature of the board during a phase (°C)
$\Delta T_{cycling}$	Amplitude of temperature variation associated with a cycling phase (°C)
$T_{max-cycling}$	Maximum temperature on the board during a cycling phase (°C)
N_{cy}	Number of cycles associated with each cycling phase (cycles)
θ_{cy}	Cycle duration (hours)
G_{RMS}	Vibration amplitude associated with each random vibration phase (Grms)

Pollution level (see tables)

Saline pollution level	Π_{sal}
Low	1
High	2

Product protection level	Π_{prot}
Hermetic	0
Non-hermetic	1

Application pollution level	Π_{zone}
Low	1
Moderate	2
High	4

Environmental pollution level	Π_{envir}
Low	1
Moderate	1.5
High	2

Application data

$I_{applied}$	Current in the fuse during the phase (A)
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Technology-related Information

I_{rated}	Rated current supported by the fuse without opening (A)
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Factors contributing to physical stresses

$\Pi_{Thermo-electrical}$	In an operating phase: $0.13 \times \left(\frac{1}{0.8} \times \frac{e I_{applied}}{I_{rated}} \right)^{1.5} \times e^{11604 \times 0.15 \times \left[\frac{1}{293} - \frac{1}{273 + T_{ambient-board}} \right]}$ In a non-operating phase: $\Pi_{Thermo-electrical} = 0$
Π_{TCy}	$0.51 \times \left(\frac{12 \times N_{cy}}{T_{phase}} \right) \times \left(\frac{\min(\theta_{cy}, 2)}{2} \right)^{\frac{1}{3}} \times \left(\frac{\Delta T_{cycling}}{20} \right)^{1.9} \times e^{1414 \times \left[\frac{1}{313} - \frac{1}{T_{max-cycling} + 273} \right]}$
$\Pi_{Mechanical}$	$0.06 \times \left(\frac{G_{RMS}}{0.5} \right)^{1.5}$
Π_{RH}	$0.24 \times \left(\frac{RH_{ambient}}{70} \right)^{4.4} \times e^{11604 \times 0.9 \times \left[\frac{1}{293} - \frac{1}{273 + T_{ambient-board}} \right]}$ In an operating phase: $\Pi_{RH} = 0$
Π_{CHI}	$0.06 \times \Pi_{Sal} \times \Pi_{Envir} \times \Pi_{Zone} \times \Pi_{Prot}$

Ceramic capacitors

General model associated with the family

$$\lambda = \lambda_{physical} \times \Pi_{PM} \times \Pi_{Process}$$

Where:

$$\lambda_{Physical} = \lambda_{0_Capacitor} \times \sum_i^{Phases} \left(\frac{t_{phase}}{T_{Total}} \right)_i \times (\Pi_{Thermo-electrical} + \Pi_{TCy} + \Pi_{Mechanical})_i \times (\Pi_{Induced})_i$$

C_{sensitivity} factor

	Relative sensitivity (score out of 10)			C _{sensitivity}
	EOS	MOS	TOS	
Ceramic capacitor with defined temperature coefficient (Type I)	7	5	2	6.30
Ceramic capacitor with non-defined temperature coefficient (Type II X5R)	7	6	1	6.48
Ceramic capacitor with non-defined temperature coefficient (Type II X7R)	7	6	1	6.48
Ceramic capacitor with polymer terminations and a non-defined temperature coefficient (Type II X5R)	7	4	2	6.08
Ceramic capacitor with polymer terminations and a non-defined temperature coefficient (Type II X7R)	7	4	1	6.03

Model associated with the QA_{component} factor

Component quality assurance level	Position relative to state of the art	QA _{component}
Qualification according to one of the following standards: MIL-PRF-xxxx level T, MIL-PRF-xxxx level S, MIL-PRF-xxxx level R, ESCC 300x, NASDA-QTS-xxxx class I (JAXA-QTS-2040E)	Higher	3
Qualification according to one of the following standards: AEC Q200, MIL-PRF-xxx level P, NASDA-QTS-xxxx class II with identification of manufacturing sites for these standards, qualification according to approved CECC standards.	Equivalent	2
Qualification according to one of the following MIL-PRF-xxxx level M, or qualification program internal to the manufacturer and unidentified manufacturing sites	Lower	1
No information	Much lower	0

Basic failure rates associated with the component

Component description	$\lambda_{0_Capacitor}$	Activation energy (eV)	$S_{reference}$	γ_{TH-EL}	γ_{TCy}	γ_{Mech}
Ceramic capacitor Type I, Category 1	0.03	0.1	0.3	0.70	0.28	0.02
Ceramic capacitor Type I, Category 2	0.05	0.1	0.3	0.70	0.28	0.02
Ceramic capacitor Type I, Category 3	0.40	0.1	0.3	0.69	0.26	0.05
Ceramic capacitor Type II, Category 1	0.08	0.1	0.3	0.70	0.28	0.02
Ceramic capacitor Type II, Category 2	0.15	0.1	0.3	0.70	0.28	0.02
Ceramic capacitor Type II, Category 3	1.20	0.1	0.3	0.44	0.51	0.05
Ceramic capacitor with polymer terminations Type II, Category 1	0.08	0.1	0.3	0.70	0.28	0.02
Ceramic capacitor with polymer terminations Type II, Category 2 or 3	0.15	0.1	0.3	0.70	0.28	0.02

Note: Type I capacitors are capacitors with a defined temperature coefficient; Type II capacitors are capacitors with a non-defined temperature coefficient.

Note: $S_{reference}$ represents the reference level of electrical stress.

CV product

	Type I	Type II
Category 1	CV product lower than $5 \cdot 10^{-8}$ V.F.	CV product lower than $5 \cdot 10^{-6}$
Category 2	CV product strictly higher than $5 \cdot 10^{-8}$ V.F. and lower than $1 \cdot 10^{-6}$ V.F. or CV product strictly higher than $1 \cdot 10^{-6}$ V.F. for a non-top-end capacitor	CV product strictly higher than $5 \cdot 10^{-6}$ V.F. and lower than $1 \cdot 10^{-4}$ V.F. or CV product strictly higher than $1 \cdot 10^{-4}$ for a non-top-end capacitor
Category 3	Top-end capacitor with a CV product strictly higher than $1 \cdot 10^{-6}$ V.F.	Top-end capacitor with a CV product strictly higher than $1 \cdot 10^{-4}$ V.F.

Note: A capacitor is described here as “top-end” if the manufacturer does not offer a component with a higher capacity in the same range (package, type, dielectric, voltage).

Mission profile data

t_{phase}	Time associated with each operating phase (hours)
$T_{ambient-board}$	Mean temperature of the board during a phase (°C)
$\Delta T_{cycling}$	Amplitude of temperature variation associated with a cycling phase (°C)
$T_{max-cycling}$	Maximum temperature on the board during a cycling phase (°C)
N_{cy}	Number of cycles associated with each cycling phase (cycles)
θ_{cy}	Cycle duration (hours)
G_{RMS}	Stress associated with each random vibration phase (Grms)

Application data

$V_{applied}$	Voltage applied to the component during the phase (V)
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Technology-related Information

V_{rated}	Maximum voltage applicable to the component, as specified by the supplier (V)
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Factors contributing to physical stresses

$\Pi_{Thermo-electrical}$	In an operating phase: $\gamma_{TH-EL} \times \left(\frac{1}{S_{reference}} \times \frac{V_{applied}}{V_{rated}} \right)^3 \times e^{11604 \times E_a \times \left[\frac{1}{293} - \frac{1}{T_{ambient-board} + 273} \right]}$ In a non-operating phase: $\Pi_{Thermo-electrical} = 0$
Π_{TCy}	$\gamma_{TCy} \times \left(\frac{12 \times N_{cy}}{T_{phase}} \right) \times \left(\frac{\min(\theta_{cy}, 2)}{2} \right)^{\frac{1}{3}} \times \left(\frac{\Delta T_{cycling}}{20} \right)^{1.9} \times e^{1414 \times \left[\frac{1}{313} - \frac{1}{T_{max-cycling} + 273} \right]}$
$\Pi_{Mechanical}$	$\gamma_{Mech} \times \left(\frac{G_{RMS}}{0.5} \right)^{1.5}$

Aluminium capacitors

General model associated with the family

ⓘ **Caution: Limited lifetime**

$$\lambda = \lambda_{physical} \times \Pi_{PM} \times \Pi_{process}$$

Where:

$$\lambda_{Physical} = \lambda_{0_Capacitor} \times \sum_i^{Phases} \left(\frac{t_{phase}}{T_{Total}} \right)_i \times (\Pi_{Thermo-electrical} + \Pi_{TCy} + \Pi_{Mechanical})_i \times (\Pi_{Induced})_i$$

C_{sensitivity} factor

	Relative sensitivity (score out of 10)			C _{sensitivity}
	EOS	MOS	TOS	
Wet aluminium electrolytic capacitor	7	7	1	6.70
Solid aluminium electrolytic capacitor	7	7	1	6.70

Model associated with the QA_{component} factor

Component quality assurance level	Position relative to state of the art	QA _{component}
Qualification according to one of the following standards: AEC Q200, MIL-PRF-xxxx level T, MIL-PRF-xxxx level B, ESCC 300x, NASDA-QTS-xxxx class I (JAXA-QTS-2040E)	Higher	3
Qualification according to one of the following standards: MIL-PRF-xxxx level C, NASDA-QTS-xxxx class II with identification of manufacturing sites for these standards. Qualification according to approved CECC standards.	Equivalent	2
Qualification according to one of the following: MIL-PRF-xxxx level D, or qualification program internal to the manufacturer and unidentified manufacturing sites	Lower	1
No information	Much lower	0

Basic failure rates associated with the component

Component description	$\lambda_{0_Capacitor}$	Activation energy Ea (eV)	S _{reference}	γ_{TH-EL}	γ_{TCy}	γ_{Mech}
Wet aluminium electrolytic capacitor	0.21	0.40	0.5	0.85	0.14	0.01
Solid aluminium electrolytic capacitor	0.4	0.40	0.55	0.85	0.14	0.01

Note: S_{reference} represents the reference level of electrical stress.

Mission profile data

t_{phase}	Time associated with each operating phase (hours)
$T_{ambient-board}$	Mean temperature of the board during a phase (°C)
$\Delta T_{cycling}$	Amplitude of temperature variation associated with a cycling phase (°C)
$T_{max-cycling}$	Maximum temperature on the board during a cycling phase (°C)
N_{cy}	Number of cycles associated with each cycling phase (cycles)
θ_{cy}	Cycle duration (hours)
G_{RMS}	Stress associated with each random vibration phase (Grms)

Application data

$V_{applied}$	Voltage applied to the component during the phase (V)
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Technology-related Information

V_{rated}	Maximum voltage applicable to the component, as specified by the supplier (V)
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Factors contributing to physical stresses

$\Pi_{Thermo-electrical}$	In an operating phase: $\gamma_{TH-EL} \times \left(\frac{1}{S_{reference}} \times \frac{V_{applied}}{V_{rated}} \right)^3 \times e^{11604 \times E_a \times \left[\frac{1}{293} - \frac{1}{T_{ambient-board} + 273} \right]}$ In a non-operating phase: $\Pi_{Thermo-electrical} = 0$
Π_{TCy}	$\gamma_{TCy} \times \left(\frac{12 \times N_{cy}}{T_{phase}} \right) \times \left(\frac{\min(\theta_{cy}, 2)}{2} \right)^{\frac{1}{3}} \times \left(\frac{\Delta T_{cycling}}{20} \right)^{1.9} \times e^{1414 \times \left[\frac{1}{313} - \frac{1}{T_{max-cycling} + 273} \right]}$
$\Pi_{Mechanical}$	$\gamma_{Mech} \times \left(\frac{G_{RMS}}{0.5} \right)^{1.5}$

Tantalum capacitors

General model associated with the family

$$\lambda = \lambda_{physical} \times \Pi_{PM} \times \Pi_{process}$$

Where:

$$\lambda_{Physical} = \lambda_{0_Capacitor} \times \sum_i^{Phases} \left(\frac{t_{phase}}{T_{Total}} \right)_i \times (\Pi_{Thermo-electrical} + \Pi_{TCy} + \Pi_{Mechanical})_i \times (\Pi_{Induced})_i$$

C_{sensitivity} factor

	Relative sensitivity (score out of 10)			C _{sensitivity}
	EOS	MOS	TOS	
Tantalum capacitor (solid or gel electrolyte)	8	7	1	7.43

Model associated with the QA_{component} factor

Component quality assurance level	Position relative to state of the art	QA _{component}
Qualification according to one of the following standards: AEC Q200, MIL-PRF-xxxx level T, MIL-PRF-xxxx level B, ESCC 300x, NASDA-QTS-xxxx class I (JAXA-QTS-2040E)	Higher	3
Qualification according to one of the following standards: MIL-PRF-xxxx level C, NASDA-QTS-xxxx class II with identification of manufacturing sites for these standards. Qualification according to approved CECC standards.	Equivalent	2
Qualification according to one of the following: MIL-PRF-xxxx level D, or qualification program internal to the manufacturer and unidentified manufacturing sites	Lower	1
No information	Much lower	0

Basic failure rates associated with the component

Tantalum capacitor with gel electrolyte

Component description	$\lambda_{0_Capacitor}$	Activation energy Ea (eV)	S _{reference}	γ_{TH-EL}	γ_{TCy}	γ_{Mech}
Wet tantalum capacitor, silver case, elastomer-sealed	0.77	0.15	0.6	0.87	0.01	0.12
Wet tantalum capacitor, silver case, glass-sealed	0.33	0.15	0.6	0.81	0.01	0.18
Wet tantalum capacitor, tantalum case, glass-sealed	0.05	0.15	0.6	0.88	0.04	0.08

Tantalum capacitor with solid electrolyte

Component description	$\lambda_{0_Capacitor}$	Activation energy Ea (eV)	S _{reference}	γ_{TH-EL}	γ_{TCy}	γ_{Mech}
Solid tantalum capacitor, bead packaging	1.09	0.15	0.4	0.86	0.12	0.02
Solid tantalum capacitor, SMD packaging	0.54	0.15	0.4	0.84	0.14	0.02
Solid tantalum capacitor, axial metal packaging	0.25	0.15	0.4	0.94	0.04	0.02

Note: S_{reference} represents the reference level of electrical stress.

Remarks:

- For wet tantalum capacitors, the default configuration is silver case, glass-sealed.
- For solid tantalum capacitors, the default configuration is the SMD packaging.

Mission profile data

t_{phase}	Time associated with each operating phase (hours)
$T_{ambient-board}$	Mean temperature of the board during a phase (°C)
$\Delta T_{cycling}$	Amplitude of temperature variation associated with a cycling phase (°C)
$T_{max-cycling}$	Maximum temperature on the board during a cycling phase (°C)
N_{cy}	Number of cycles associated with each cycling phase (cycles)
θ_{cy}	Cycle duration (hours)
G_{RMS}	Stress associated with each random vibration phase (Grms)

Application data

$V_{applied}$	Voltage applied to the component during the phase (V)
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Technical data

V_{rated}	Maximum voltage applicable to the component, as specified by the supplier (V)
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Factors contributing to physical stresses

$\Pi_{Thermo-electrical}$	In an operating phase: $\gamma_{TH-EL} \times \left(\frac{1}{S_{reference}} \times \frac{V_{applied}}{V_{rated}} \right)^3 \times e^{11604 \times Ea \times \left[\frac{1}{293} - \frac{1}{(T_{ambient-board} + 273)} \right]}$ In a non-operating phase: $\Pi_{Thermo-electrical} = 0$
Π_{TCy}	$\gamma_{TCy} \times \left(\frac{12 \times N_{cy}}{T_{phase}} \right) \times \left(\frac{\min(\theta_{cy}, 2)}{2} \right)^{\frac{1}{3}} \times \left(\frac{\Delta T_{cycling}}{20} \right)^{1.9} \times e^{1414 \times \left[\frac{1}{313} - \frac{1}{(T_{max-cycling} + 273)} \right]}$
$\Pi_{Mechanical}$	$\gamma_{Mech} \times \left(\frac{G_{RMS}}{0.5} \right)^{1.5}$

Plastic film capacitors

General model associated with the family

$$\lambda = \lambda_{\text{Physical}} \times \Pi_{\text{Film}} \times \Pi_{\text{PM}} \times \Pi_{\text{Process}}$$

Where:

$$\lambda_{\text{Physical}} = \lambda_{0_Capacitor} \times \sum_i^{\text{Phases}} \left(\frac{t_{\text{phase}}}{T_{\text{Total}}} \right)_i \times \left(\frac{\Pi_{\text{Thermo-electrical}} + \Pi_{\text{TCy}}}{\Pi_{\text{Mechanical}} + \Pi_{\text{RH}}} \right)_i \times (\Pi_{\text{Induced}})_i$$

C_{sensitivity} factor

	Relative sensitivity (Score out of 10)			C _{sensitivity}
	EOS	MOS	TOS	
Plastic film capacitor	7	6	1	6.05

Model associated with the “ Π_{Film} ” factor

$$\Pi_{\text{Film}} = e^{1.39 \times \text{FILM_Grade}}$$

If Π_{Film} is not evaluated, a default value of 2.5 is proposed. Using the default value may reduce the accuracy of the final results.

The Π_{Film} factor is calculated from a questionnaire about the environmental conditions around the choice of a plastic film capacitor.

$$\text{FILM_Grade} = \frac{\sum \text{Values in table below}}{100}$$

N°	Factors influencing FILM_Grade	Value
1	Plastic film capacitor used in AC voltages	50
	Plastic film capacitor used in DC voltages	0
2	Plastic film capacitor used in humidity environments above 90%RH without being specifically selected for such an environment	30
	Plastic film capacitor used in humidity environments above 90%RH and selected specifically for such an environment	15
	Plastic film capacitor used in humidity environments between 70%RH and 90%RH without being specifically selected for such an environment	15
	Plastic film capacitor used in humidity environments between 70%RH and 90%RH and selected specifically for such an environment	0
	Plastic film capacitor used in humidity environments below 70%RH	0
3	Plastic film capacitor not developed specifically for this application, not validated by the manufacturer and not validated by waiver	20
	Plastic film capacitor not developed specifically for this application but validated by the manufacturer or validated by waiver	10
	Plastic film capacitor developed specifically for this application	0

The range of variation of the factor is from 1 (for the choice of the most appropriate capacitor) to 4 (for the choice of the least appropriate capacitor).

Model associated with the $QA_{\text{component}}$ factor

Component quality assurance level	Position relative to state of the art	$QA_{\text{component}}$
Qualification according to one of the following standards: AEC Q200, MIL-PRF-xxxx level T, MIL-PRF-xxxx level S, MIL-PRF-xxxx level R, ESCC 400x, NASDA-QTS-xxxx class I (JAXA-QTS-2050D)	Higher	3
Qualification according to one of the following standards: MIL-PRF-xxx level P, NASDA-QTS-xxxx class II with identification of manufacturing sites for these standards, qualification according to approved CECC standards.	Equivalent	2
Qualification according to MIL-PRF-xxxx level M, or qualification program internal to the manufacturer and unidentified manufacturing sites	Lower	1
No information	Much lower	0

Basic failure rates associated with the component

Component description	$\lambda_{0_Capacitor}$	Activation energy (eV)	$S_{\text{reference}}$	Y_{TH_EL}	Y_{TCy}	Y_{Mech}	Y_{RH}
Polypropylene (PP) film capacitor	0.02	0.65	0.3	0.18	0.14	0.02	0.66
Polyethylene terephthalate (PET) film capacitor	0.06	0.48	0.3	0.18	0.14	0.02	0.66
Polyethylene naphthalate (PEN) film capacitor	0.03	0.55	0.3	0.18	0.14	0.02	0.66
Polyphenylene sulfide (PPS) film capacitor	0.02	0.55	0.3	0.18	0.14	0.02	0.66
Teflon (PTFE) film capacitor	0.03	0.55	0.3	0.18	0.14	0.02	0.66

Note: $S_{\text{reference}}$ represents the reference level of electrical stress.

Mission profile data

t_{phase}	Time associated with each operating phase (hours)
$T_{\text{ambient-board}}$	Mean temperature of the board during a phase (°C)
RH_{ambient}	Humidity level associated with a phase (%)
$\Delta T_{\text{cycling}}$	Amplitude of temperature variation associated with a cycling phase (°C)
$T_{\text{max-cycling}}$	Maximum temperature on the board during a cycling phase (°C)
N_{cy}	Number of cycles associated with each cycling phase (cycles)
θ_{cy}	Cycle duration (hours)
G_{RMS}	Stress associated with each random vibration phase (G_{rms})

Application data

V_{applied}	Voltage applied to the component during the phase (V)
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Technical data

V_{rated}	Maximum voltage applied to the component, as specified by the supplier (V)
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Factors contributing to physical stresses

$\Pi_{Thermo\text{-}\acute{e}lectrique}$	In an operating phase: $\gamma_{TH-EL} \times \left(\frac{1}{S_{reference}} \times \frac{V_{applied}}{V_{rated}} \right)^6 \times e^{11604 \times Ea \times \left[\frac{1}{293} - \frac{1}{(T_{ambient-board} + 273)} \right]}$ In a non-operating phase: $\Pi_{Thermo-electrical} = 0$
Π_{TCy}	$\gamma_{TCy} \times \left(\frac{12 \times N_{cy}}{t_{phase}} \right) \times \left(\frac{\min(\theta_{cy}, 2)}{2} \right)^{\frac{1}{3}} \times \left(\frac{\Delta T_{cycling}}{20} \right)^{1.9} e^{11604 \times 0.12 \times \left[\frac{1}{313} - \frac{1}{(T_{max-cycling} + 273)} \right]}$
$\Pi_{M\acute{e}canique}$	$\gamma_{Mech} \times \left(\frac{G_{RMS}}{0.5} \right)^{1.5}$
Π_{RH}	$\gamma_{RH} \times \left(\frac{RH_{ambient}}{70} \right)^{4.4} \times e^{11604 \times 0.9 \times \left[\frac{1}{293} - \frac{1}{(T_{ambient-board} + 273)} \right]}$

Magnetic components: inductors and transformers

General model associated with the family

$$\lambda = \lambda_{physical} \times \Pi_{PM} \times \Pi_{Process}$$

Where:

$$\lambda_{Physical} = \lambda_{0_Magnetic} \times \sum_i^{Phases} \left(\frac{t_{phase}}{T_{Total}} \right)_i \times (\Pi_{Thermo-electrical} + \Pi_{TCy} + \Pi_{Mechanical})_i \times (\Pi_{Induced})_i$$

C_{sensitivity} factor

	Relative sensitivity (score out of 10)			C _{sensitivity}
	EOS	MOS	TOS	
Low-current (or low-level) wirewound inductor	5	4	4	4.73
High-current (or power) wirewound inductor	7	6	3	6.58
Multilayer inductor	4	6	1	4.30
Low-power (or low-level) transformer	6	5	3	5.63
High-power transformer	6	7	4	6.13

Model associated with the QA_{component} factor

Component quality assurance level	Position relative to state of the art	QA _{component}
Qualification according to one of the following standards: AEC Q200, MIL-STD-981 class S, MIL-PRF-xxx level T, ESCC 320x, NASDA-QTS-xxxx class I	Higher	3
Qualification according to one of the following standards: MIL-STD-981 class B, MIL-PRF-xxx level M, NASDA-QTS- xxxx class II with identification of manufacturing sites for these standards, qualification according to approved CECC standards.	Equivalent	2
Qualification according to one of the following MIL-PRF-xxxx level C, or qualification program internal to the manufacturer and unidentified manufacturing sites	Lower	1
No information	Much lower	0

Basic failure rates associated with the component

Component description	$\lambda_{0\text{-Magnetic}}$	Activation energy Ea (eV)	$\gamma_{\text{TH-EL}}$	γ_{TCy}	γ_{Mech}	ΔT (°C)
Low-current (or low-level) wirewound inductor	0.025	0.15	0.01	0.73	0.26	10
High-current (or power) wirewound inductor	0.05	0.15	0.09	0.79	0.12	30
Multilayer inductor	0.05	0.15	0.71	0.28	0.01	10
Low-power (or low-level) transformer	0.125	0.15	0.01	0.73	0.26	10
High-power transformer	0.25	0.15	0.15	0.69	0.16	30

Mission profile data

t_{phase}	Time associated with each phase (hours)
$T_{\text{ambient-board}}$	Mean temperature of the board during a phase (°C)
$\Delta T_{\text{cycling}}$	Amplitude of temperature variation associated with a cycling phase (°C)
$T_{\text{max-cycling}}$	Maximum temperature on the board during a cycling phase (°C)
N_{cy}	Number of cycles associated with each cycling phase (cycles)
θ_{cy}	Cycle duration (hours)
G_{RMS}	Stress associated with each random vibration phase (Grms)

Application data

ΔT	Temperature rise of component relative to ambient (°C). The previous table lists typical ΔT values that can be used in the absence of a better estimate.
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Factors contributing to physical stresses

$\Pi_{\text{Thermo-electrical}}$	In an operating phase: $\gamma_{\text{TH-EL}} \times e^{11604 \times Ea \times \left[\frac{1}{293} - \frac{1}{(T_{\text{ambient-board}} + \Delta T + 273)} \right]}$ In a non-operating phase: $\Pi_{\text{Thermo-electrical}} = 0$
Π_{TCy}	$\gamma_{\text{TCy}} \times \left(\frac{12 \times N_{\text{cy}}}{T_{\text{phase}}} \right) \times \left(\frac{\min(\theta_{\text{cy}}, 2)}{2} \right)^{\frac{1}{3}} \times \left(\frac{\Delta T_{\text{cycling}}}{20} \right)^{1.9} \times e^{1414 \times \left[\frac{1}{313} - \frac{1}{T_{\text{max-cycling}} + 273} \right]}$
$\Pi_{\text{Mechanical}}$	$\gamma_{\text{Mech}} \times \left(\frac{G_{\text{RMS}}}{0.5} \right)^{1.5}$

Piezoelectric components: quartz crystal resonators and oscillators

General model associated with the family

$$\lambda = \lambda_{physical} \times \Pi_{PM} \times \Pi_{Process}$$

Where:

$$\lambda_{Physical} = \lambda_{0_Piezoelectrical} \times \sum_i^{Phases} \left(\frac{t_{phase}}{T_{Total}} \right)_i \times (\Pi_{Thermo-electrical} + \Pi_{TCy} + \Pi_{Mechanical} + \Pi_{RH})_i \times (\Pi_{Induced})_i$$

	Relative sensitivity (score out of 10)			
	EOS	MOS	TOS	C _{sensitivity}
Quartz crystal resonator	2	10	5	3.95
Quartz crystal oscillator	7	9	3	7.25

Model associated with the QA_{component} factor

Component quality assurance level	Position relative to state of the art	QA _{component}
Qualification according to one of the following standards: AEC Q200 (Resonator) MIL-PRF-38534 class K (oscillator), MIL-PRF-55310 class S (Oscillator), ESCC 3503 (oscillator)/3501 (resonator) or equivalent	Higher	3
Qualification according to one of the following standards: MIL-PRF-38534 class H, MIL-PRF-55310 class B	Equivalent	2
Qualification according to one of the following MIL-PRF-xxxx level M	Lower	1
No information	Much lower	0

Basic failure rates associated with the component

Component description	$\lambda_{0_Piezoelectrical}$	γ_{TH-EL}	γ_{TCy}	γ_{Mech}	γ_{RH}
Quartz crystal resonator (HCxx package, through-hole)	0.82	0.16	0.46	0.27	0.11
Quartz crystal resonator (surface-mounted)	0.79	0.16	0.59	0.15	0.1
Quartz crystal oscillator (XO package through-hole)	1.6	0.32	0.42	0.14	0.12
Quartz crystal oscillator (XO, MCSO package, surface-mounted)	1.63	0.31	0.53	0.07	0.09

Mission profile data

t_{phase}	Time associated with each phase (hours)
RH_{ambient}	Humidity level associated with a phase (%)
$T_{\text{ambient-board}}$	Mean temperature of the board during a phase (°C)
$\Delta T_{\text{cycling}}$	Amplitude of temperature variation associated with a cycling phase (°C)
$T_{\text{max-cycling}}$	Maximum temperature on the board during a cycling phase (°C)
N_{cy}	Number of cycles associated with each cycling phase (cycles)
θ_{cy}	Cycle duration (hours)
G_{RMS}	Vibration amplitude associated with each random vibration phase (Grms)

Application data

I_{output}	Current supplied by the component when operating in the application (A)
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Technical data

$I_{\text{max-output}}$	Maximum current that the component can supply in operation (A)
$T_{\text{max-ambient-manuf}}$	Maximum temperature specified by the manufacturer (°C)

Factors contributing to physical stresses

$\Pi_{\text{Thermo-electrical}}$	In an operating phase: $\gamma_{TH-EL} \times \Pi_{\text{rating_TH}} \times \Pi_{\text{rating_EL}}$ <table> <tr> <td>Conditions of use</td><td>Value of $\Pi_{\text{rating_TH}}$</td></tr> <tr> <td>$T_{\text{ambient-board}} < (T_{\text{max-ambient-manuf}} - 40^{\circ}\text{C})$</td><td>1</td></tr> <tr> <td>$T_{\text{ambient-board}} \geq (T_{\text{max-ambient-manuf}} - 40^{\circ}\text{C})$</td><td>5</td></tr> </table> <table> <tr> <td>Conditions of use</td><td>Value of $\Pi_{\text{rating_EL}}$</td></tr> <tr> <td>Quartz crystal resonator:</td><td>1</td></tr> <tr> <td>Oscillator : $I_{\text{output}} < 0.8 \times I_{\text{max-output}}$</td><td>1</td></tr> <tr> <td>Oscillator : $I_{\text{output}} \geq 0.8 \times I_{\text{max-output}}$</td><td>5</td></tr> </table> In a non-operating phase: $\Pi_{\text{Thermo-electrical}} = 0$	Conditions of use	Value of $\Pi_{\text{rating_TH}}$	$T_{\text{ambient-board}} < (T_{\text{max-ambient-manuf}} - 40^{\circ}\text{C})$	1	$T_{\text{ambient-board}} \geq (T_{\text{max-ambient-manuf}} - 40^{\circ}\text{C})$	5	Conditions of use	Value of $\Pi_{\text{rating_EL}}$	Quartz crystal resonator:	1	Oscillator : $I_{\text{output}} < 0.8 \times I_{\text{max-output}}$	1	Oscillator : $I_{\text{output}} \geq 0.8 \times I_{\text{max-output}}$	5
Conditions of use	Value of $\Pi_{\text{rating_TH}}$														
$T_{\text{ambient-board}} < (T_{\text{max-ambient-manuf}} - 40^{\circ}\text{C})$	1														
$T_{\text{ambient-board}} \geq (T_{\text{max-ambient-manuf}} - 40^{\circ}\text{C})$	5														
Conditions of use	Value of $\Pi_{\text{rating_EL}}$														
Quartz crystal resonator:	1														
Oscillator : $I_{\text{output}} < 0.8 \times I_{\text{max-output}}$	1														
Oscillator : $I_{\text{output}} \geq 0.8 \times I_{\text{max-output}}$	5														
Π_{TCy}	$\gamma_{TCy} \times \left(\frac{12 \times N_{cy}}{T_{\text{phase}}} \right) \times \left(\frac{\min(\theta_{cy}, 2)}{2} \right)^{\frac{1}{3}} \times \left(\frac{\Delta T_{\text{cycling}}}{20} \right)^{1.9} \times e^{1414 \times \left[\frac{1}{313} - \frac{1}{T_{\text{max-cycling}} + 273} \right]}$														
$\Pi_{\text{Mechanical}}$	$\gamma_{\text{Mech}} \times \left(\frac{G_{\text{RMS}}}{0.5} \right)^{1.5}$														
Π_{RH}	$\gamma_{RH} \times \left(\frac{RH_{\text{ambient}}}{70} \right)^{4.4} \times e^{11604 \times 0.9 \times \left[\frac{1}{293} - \frac{1}{273 + T_{\text{ambient-board}}} \right]}$ In an operating phase: $\Pi_{RH} = 0$														

Monostable electromechanical relays

General model associated with the family

ⓘ Caution: Limited lifetime

$$\lambda = \lambda_{physical} \times \Pi_{PM} \times \Pi_{process}$$

Where:

$$\lambda_{Physical} = \lambda_{0 \text{ Relay}} \times \sum_i^{\text{Phases}} \left(\frac{t_{\text{phase}}}{T_{\text{Total}}} \right)_i \times (\Pi_{\text{Thermal}} + \Pi_{\text{Electrical}} + \Pi_{\text{TCy}} + \Pi_{\text{Mechanical}} + \Pi_{\text{RH}})_i \times (\Pi_{\text{Induced}})_i$$

C_{sensitivity} factor

	Relative sensitivity (score out of 10)			C _{sensitivity}
	EOS	MOS	TOS	
Electromechanical relay	7	10	2	7.43

QA_{component} factor

Component quality assurance level	Position relative to state of the art	QA _{component}
Qualification according to one of the following standards: ESCC 360x, NASDA-QTS-39016A or specific manufacturer specifications based on ESCC	Higher	3
Qualification according to one of the following standards: MIL-PRF-39016 (or 83536 or 6106) level R, MIL-PRF-39016 (or 83536 or 6106) level P, NASDA-QTS-6106A	Equivalent	2
Qualification according to one of the following approved EIA, IEC, SAE, BS	Lower	1
No information	Much lower	0

Basic failure rate associated with the component

$$\lambda_{0_Relay} = 1.1$$

Mission profile data

T_{amb}	Temperature of the item associated with an operating phase (°C)
t_{phase}	Time associated with each phase (hours)
$\Delta T_{cycling}$	Amplitude of temperature variation associated with a cycling phase (°C)
$T_{max-cycling}$	Maximum temperature on the board during a cycling phase (°C)
N_{cy}	Number of cycles associated with each cycling phase (cycles)
θ_{cy}	Cycle duration
G_{RMS}	Vibration amplitude associated with each random vibration phase (Grms)
$RH_{ambient}$	Humidity level associated with a phase (%)

Pollution level during the phase (see tables)

Saline pollution level	π_{sal}
Low	1
High	2

Product protection level	π_{prot}
Hermetic	0
Non-hermetic	1

Application pollution level	π_{zone}
Low	1
Moderate	2
High	4

Environmental pollution level	π_{envir}
Low	1
Moderate	1.5
High	2

Application data

$V_{contact}$	Voltage at contact terminals (V)
$I_{contact}$	Current passing through contact (A)
U_{coil}	Relay control voltage (V)
ΔT_{relay}	Temperature rise in the relay (°C). By default, for an always-on relay, use $\Delta T_{relay} = 45^{\circ}\text{C}$.
Number of operations per hour during the phase (see table)	
Load type (see table).	

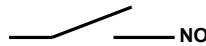
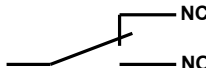
Technical data

V_{rated}	Rated voltage specified at contact terminals (V)
I_{rated}	Rated current specified at contact terminals (A)
U_{rated}	Rated control voltage specified for the relay (V)
N_{ST}	Number of Single Throw (ST) poles, in which only the normally-open (NO) contact is used
N_{DT}	Number of Double Throw (DT) poles, in which the normally-open (NO) and normally-closed (NC) contacts are used
Finish type of relay contact (see table)	
Relay breaking capacity (see table)	
Relay hermeticity type (see table).	

Technological attributes

$$\pi_{\text{pole}} = 1.25 \times N_{\text{DT}} + 0.5 \times N_{\text{ST}} + 0.5$$

Number of poles and type of contact	π_{pole}
SPST	1
DPST	1.5
3PST	2
4PST	2.5
SPDT	1.75
DPDT	3
3PDT	4.25
4PDT	5.5
6PDT	8

Typical diagram of an ST pole with NO contact only	
Typical diagram of a DT pole with NO and NC contacts	

Load type	$\pi_{\text{load type}}$	S_v	S_l
Resistive	0.3	1	$\frac{I_{\text{contact}}}{I_{\text{rated}}}$
Inductive	8	1	$\frac{I_{\text{contact}}}{I_{\text{rated}}}$
Incandescent lamp	4	$\frac{V_{\text{contact}}}{V_{\text{rated}}}$	$\frac{I_{\text{contact}}}{I_{\text{rated}}}$
Capacitive	6	$\frac{V_{\text{contact}}}{V_{\text{rated}}}$	1

$\frac{V_{\text{contact}}}{V_{\text{rated}}}$	m_1
≤ 1	3
> 1	8.8

$\frac{I_{\text{contact}}}{I_{\text{rated}}}$	m_2
≤ 1	3
> 1	5.9

Contact type	$\pi_{\text{contact ME}}$	$\pi_{\text{contact RH}}$
Gold-plated	1.5	1
Silver	1	2

Breaking capacity	$\pi_{\text{breaking TH}}$	$\pi_{\text{breaking EL}}$	$\pi_{\text{breaking ME}}$
Breaking capacity $< 2\text{A}$	1.8	1.5	3
Breaking capacity $> 2\text{A}$	1.2	1.2	1

Number of operations per hour	$\pi_{\text{operations}}$
≤ 1	1
> 1	$\sqrt{(\text{Number of operations per hour})}$

Relay protection level	$\Pi_{\text{prot CHEM}}$	$\Pi_{\text{prot TCY}}$
Hermetic	0.01	1
Sealed	0.6	3
Not sealed	1	3

Factors contributing to physical stresses

Π_{Thermal}	$0.29 \times (1 + \Pi_{\text{chemical}}) \times \Pi_{\text{operations}} \times \Pi_{\text{contact TH}} \times \Pi_{\text{breaking TH}}$ $\times e^{11604 \times 0.25 \times \left[\frac{1}{313} - \frac{1}{273 + T'} \right]}$ If $T_{\text{amb}} + \Delta T_{\text{relay}} \leq 125^{\circ}\text{C}$ then $\Pi_{\text{contact TH}} = 1$ If $T_{\text{amb}} + \Delta T_{\text{relay}} > 125^{\circ}\text{C}$ then $\Pi_{\text{contact TH}} = \Pi_{\text{contact type ME}} \times \Pi_{\text{pole}}$ In an operating phase: If $T_{\text{amb}} + \Delta T_{\text{relay}} \leq 0^{\circ}\text{C}$ then $T' = 40 - 85/55 \times (T_{\text{amb}} + \Delta T_{\text{relay}})$ If $0 < T_{\text{amb}} + \Delta T_{\text{relay}} \leq 40^{\circ}\text{C}$ then $T' = 40^{\circ}\text{C}$ If $T_{\text{amb}} + \Delta T_{\text{relay}} > 40^{\circ}\text{C}$ then $T' = T_{\text{amb}} + \Delta T_{\text{relay}}$ In a non-operating phase: $\Pi_{\text{Thermal}} = 0$
$\Pi_{\text{Electrical}}$	In an operating phase: $0.55 \times \Pi_{\text{pole}} \times \Pi_{\text{breaking EL}} \times \Pi_{\text{load type}} \times \Pi_{\text{operations}} \times S_V^{m_1} \times S_I^{m_2} \times \left(\frac{U_{\text{rated}}}{U_{\text{coil}}} \right)$ In a non-operating phase: $\Pi_{\text{Electrical}} = 0$
Π_{TCY}	$0.02 \times \Pi_{\text{prot TCY}} \times \left(\frac{12 \times N_{\text{cy}}}{T_{\text{phase}}} \right) \times \left(\frac{\min(\theta_{\text{cy}}, 2)}{2} \right)^{\frac{1}{3}} \times \left(\frac{\Delta T_{\text{cycling}}}{20} \right)^{1.9} \times e^{1414 \times \left[\frac{1}{313} - \frac{1}{T_{\text{max-cycling}} + 273} \right]}$
$\Pi_{\text{Mechanical}}$	$0.05 \times (1 + \Pi_{\text{chemical}}) \times \Pi_{\text{pole}} \times \Pi_{\text{contact ME}} \times \Pi_{\text{operations}} \times \Pi_{\text{breaking ME}} \times \left(\frac{G_{\text{RMS}}}{0.5} \right)^{1.5}$
Π_{RH}	$0.09 \times \Pi_{\text{pole}} \times \Pi_{\text{chemical}} \times \Pi_{\text{contact RH}} \times \left(\frac{RH_{\text{ambient}}}{70} \right)^{4.4} \times e^{11604 \times 0.9 \times \left[\frac{1}{293} - \frac{1}{T_{\text{ambient}} + 273} \right]}$ $\Pi_{\text{chemical}} = \Pi_{\text{sal}} \times \Pi_{\text{zone}} \times \Pi_{\text{envir}} \times \Pi_{\text{prot}} \times \Pi_{\text{prot CHEM}}$

Switches

General model associated with the family:

ⓘ Caution: Limited lifetime

$$\lambda = \lambda_{physical} \times \Pi_{PM} \times \Pi_{process}$$

Where:

$$\lambda_{physical} = \lambda_{0 \text{ Switch}} \times \sum_i^{\text{Phases}} \left(\frac{t_{\text{phase}}}{T_{\text{Total}}} \right)_i \times \left(\frac{\Pi_{\text{Thermal}} + \Pi_{\text{Electrical}}}{+\Pi_{\text{TCy}} + \Pi_{\text{Mechanical}} + \Pi_{\text{RH}}} \right)_i \times (\Pi_{\text{Induced}})_i$$

C_{sensitivity} factor

	Relative sensitivity (score out of 10)			C _{sensitivity}
	EOS	MOS	TOS	
Switches	7	10	1	7.38

QA_{component} factor

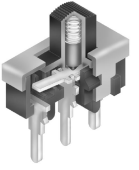
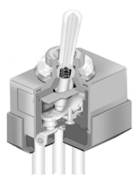
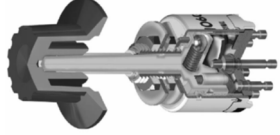
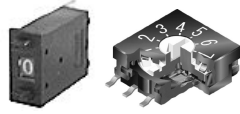
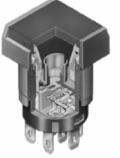
Component quality assurance level	Position relative to state of the art	QA _{component}
Qualification according to one of the following standards: ESCC 370x, MIL-PRF-8805	Much higher	3
Qualification according to one of the following standards: MIL-PRF-24236, MIL-C-xxxx	Higher	2
Qualification according to one of the following approved EIA, IEC, SAE, BS	Equivalent	1
No information	Lower	0

Basic failure rates associated with the component

Limit switch

Type	C _{TH}	C _{TCy}	C _{ME}	C _{RH}	C _{EL}	λ_{0_switch}
Limit switch, microcontacts	1	1	1	1	1	0.85

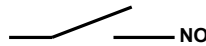
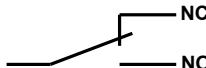
Manually operated switch

Type	Illustration	C _{TH}	C _{TCy}	C _{ME}	C _{RH}	C _{EL}	λ_{0_switch}
Toggle		1.11	0.56	1.11	0.56	0.56	0.85
Slide		1.11	0.56	1.11	0.56	0.56	0.85
Lever		1.11	0.56	1.11	0.56	0.56	0.85
DIP		1.11	0.56	1.11	0.56	0.56	0.85
Rotary		1.78	1.19	1.78	1.19	1.19	0.85
Encoder		1.78	1.19	1.78	1.19	1.19	0.85
Momentary push button (monostable)		1	1	1	1	1	0.85
Permanent push button (bistable)		1.11	0.56	1.11	0.56	0.56	0.85

Technological attributes

$$\pi_{\text{pole}} = 1.25 \times N_{\text{DT}} + 0.5 \times N_{\text{ST}} + 0.5$$

Number of poles and type of contact	π_{pole}
SPST	1
DPST	1.5
3PST	2
4PST	2.5
SPDT	1.75
DPDT	3
3PDT	4.25
4PDT	5.5
6PDT	8

Typical diagram of an ST pole with NO contact only	
Typical diagram of a DT pole with NO and NC contacts	

Remark: For rotary and encoder switches, all poles should be counted as DT (Double Throw) regardless of the number of contacts per pole.

Load type	$\pi_{\text{load type}}$	S_v	S_I
Resistive	0.3	1	$\frac{I_{\text{contact}}}{I_{\text{rated}}}$
Inductive	8	1	$\frac{I_{\text{contact}}}{I_{\text{rated}}}$
Incandescent lamp	4	$\frac{V_{\text{contact}}}{V_{\text{rated}}}$	$\frac{I_{\text{contact}}}{I_{\text{rated}}}$
Capacitive	6	$\frac{V_{\text{contact}}}{V_{\text{rated}}}$	1

$\frac{V_{\text{contact}}}{V_{\text{rated}}}$	m_1	$\frac{I_{\text{contact}}}{I_{\text{rated}}}$	m_2
≤ 1	3	≤ 1	3
> 1	8.8	> 1	5.9

Contact type	$\pi_{\text{contact ME}}$	$\pi_{\text{contact RH}}$
Gold-plated	1.5	1
Silver	1	2

Breaking capacity	$\pi_{\text{breaking EL}}$	$\pi_{\text{breaking ME}}$
Breaking capacity $< 2\text{A}$	1.5	3
Breaking capacity $> 2\text{A}$	1.2	1

Number of operations per hour	$\pi_{\text{operations}}$
≤ 1	1
> 1	$\sqrt{\text{Number of operations per hour}}$

Protection level switch	$\pi_{\text{prot_CHEM}}$	$\pi_{\text{prot_TCY}}$
Hermetic	0.01	1
Sealed	0.6	3
Not sealed	1	3

Mission profile data

T_{amb}	Temperature of the item associated with an operating phase (°C)
t_{phase}	Time associated with each phase (hours)
$\Delta T_{\text{cycling}}$	Amplitude of temperature variation associated with a cycling phase (°C)
$T_{\text{max-cycling}}$	Maximum temperature on the board during a cycling phase (°C)
N_{cy}	Number of cycles associated with each cycling phase (cycles)
θ_{cy}	Cycle duration
G_{RMS}	Vibration amplitude associated with each random vibration phase (Grms)
RH_{ambient}	Humidity level associated with a phase (%)

Pollution level during the phase (see tables)

Saline pollution level	π_{sal}
Low	1
High	2

Product protection level	π_{prot}
Hermetic	0
Non-hermetic	1

Application pollution level	π_{zone}
Low	1
Moderate	2
High	4

Environmental pollution level	π_{envir}
Low	1
Moderate	1.5
High	2

Application data

V_{contact}	Voltage at the contact terminals
I_{contact}	Current passing through contact
	Number of operations per hour during the phase (see table)
	Load type (see table).

Technical data

V_{rated}	Rated voltage specified at contact terminals
I_{rated}	Rated current specified at contact terminals
N_{ST}	Number of Single Throw (ST) poles, in which only the normally-open (NO) contact is used
N_{DT}	Number of Double Throw (DT) poles, in which the normally-open (NO) and normally-closed (NC) contacts are used)
	Finish type of the switch contact (see table)
	Switch breaking capacity (see table)
	Switch hermeticity type (see table).

Factors contributing to physical stresses

Π_{Thermal}	$0.21 \times C_{TH} \times (1 + \Pi_{\text{chemical}}) \times \Pi_{\text{operations}} \times \Pi_{\text{contact TH}} \times e^{11604 \times 0.25 \times \left[\frac{1}{313} - \frac{1}{273 + T'} \right]}$ If $T_{\text{amb}} \leq 125^{\circ}\text{C}$ then $\Pi_{\text{contact TH}} = 1$ If $T_{\text{amb}} > 125^{\circ}\text{C}$ then $\Pi_{\text{contact TH}} = \Pi_{\text{contact type ME}} \times \Pi_{\text{pole}}$ In an operating phase: If $T_{\text{amb}} \leq 0^{\circ}\text{C}$ then $T' = 40 - 85/55 \times T_{\text{amb}}$ If $0 < T_{\text{amb}} \leq 40^{\circ}\text{C}$ then $T' = 40^{\circ}\text{C}$ If $T_{\text{amb}} > 40^{\circ}\text{C}$ then $T' = T_{\text{amb}}$ In a non-operating phase: $\Pi_{\text{Thermal}} = 0$
$\Pi_{\text{Electrical}}$	In an operating phase: $0.59 \times C_{EL} \times \Pi_{\text{pole}} \times \Pi_{\text{breakage EL}} \times \Pi_{\text{load type}} \times \Pi_{\text{operations}} \times S_V^{m_1} \times S_I^{m_2}$ In a non-operating phase: $\Pi_{\text{Electrical}} = 0$
Π_{TCY}	$0.02 \times C_{TCY} \times \Pi_{\text{pole}} \times \Pi_{\text{prot TCY}} \times \left(\frac{12 \times N_{cy}}{T_{\text{phase}}} \right) \times \left(\frac{\min(\theta_{cy}, 2)}{2} \right)^{\frac{1}{3}} \times \left(\frac{\Delta T_{\text{cycling}}}{20} \right)^{1.9}$ $\times e^{1414 \times \left[\frac{1}{313} - \frac{1}{T_{\text{max-cycling}} + 273} \right]}$
$\Pi_{\text{Mechanical}}$	$0.06 \times C_{MECH} \times (1 + \Pi_{\text{chemical}}) \times \Pi_{\text{pole}} \times \Pi_{\text{contact ME}} \times \Pi_{\text{operations}} \times \Pi_{\text{breakage ME}}$ $\times \left(\frac{G_{RMS}}{0.5} \right)^{1.5}$
Π_{RH}	$0.12 \times C_{RH} \times \Pi_{\text{pole}} \times \Pi_{\text{chemical}} \times \Pi_{\text{contact RH}} \times \left(\frac{RH_{\text{ambient}}}{70} \right)^{4.4} \times e^{11604 \times 0.9 \times \left[\frac{1}{293} - \frac{1}{T_{\text{amb}} + 273} \right]}$ $\Pi_{\text{chemical}} = \Pi_{\text{sal}} \times \Pi_{\text{zone}} \times \Pi_{\text{envir}} \times \Pi_{\text{prot}} \times \Pi_{\text{ProtCHEM}}$

Printed circuit boards (PCBs)

General model associated with the family

$$\lambda = \lambda_{physical} \times \Pi_{PM} \times \Pi_{Process}$$

Where:

$$\lambda_{physical} = \lambda_{0\text{ PCB}} \times \sum_i^{\text{Phases}} \left(\frac{t_{\text{phase}}}{T_{\text{Total}}} \right)_i \times (\Pi_{TCy} + \Pi_{\text{Mechanical}} + \Pi_{RH} + \Pi_{\text{Chemical}})_i \times (\Pi_{\text{Induced}})_i$$

C_{sensitivity} factor

	Relative sensitivity (score out of 10)			C _{sensitivity}
	EOS	MOS	TOS	
Printed circuit board (PCB)	4	10	8	5.55

Π_{Placement} factor

For PCBs, the placement factor is fixed at Π_{Placement} = 1

QA_{component} factor

Component quality assurance level	Position relative to state of the art	QA _{component}
Qualification according to one of the following standards: MIL-PRF-31032, MIL-PRF-55110, MIL-PRF-50884, ESCC-Q-ST-70-10, JAXA-QTS-2140	Higher	3
Qualification according to following standard: IPC-9701 with identification of manufacturing sites for these standards	Equivalent	2
Qualification according to one of the following standard: BS CECC 23000, IEC 61189-6 or qualification program internal to the manufacturer and unidentified manufacturing sites	Lower	1
No information	Much lower	0

Basic failure rates associated with the item

$$\lambda_{0\text{ PCB}} = 5 \cdot 10^{-4} \times (N_{\text{layers}})^{\frac{1}{2}} \times \left(\frac{N_{\text{mount}}}{2} \right) \times \Pi_{\text{Class}} \times \Pi_{\text{Techno-PCB}}$$

Technical data

N_{layers} Number of layers on the printed circuit

N_{mount} Number of mounting points (surface-mounted + through-hole)

Printed circuit technology	Value of $\Pi_{\text{Techno-PCB}}$
Through-holes	0.25
Blind holes	0.5
Micro-via technology	1
Pad on via technology	2.5

Minimum conductor width (μm) / Minimum spacing between conductors or pads (μm)	Value of Π_{Class}
800 / 800	1
500 / 500	1
310 / 310	2
210 / 210	3
150 / 150	4
125 / 125	5
100 / 100	6

For a multilayer PCB, measure the layer with the highest density. Within any one layer, measure the area with the highest density.

Mission profile data

t_{phase}	Time associated with each phase (hours)
RH_{ambient}	Humidity level associated with a phase (%)
$T_{\text{ambient-board}}$	Mean temperature of the board during a phase (°C)
$\Delta T_{\text{cycling}}$	Amplitude of temperature variation associated with a cycling phase (°C)
$T_{\text{max-cycling}}$	Maximum temperature on the board during a cycling phase (°C)
N_{cy}	Number of cycles associated with each cycling phase (cycles)
θ_{cy}	Cycle duration (hours)
G_{RMS}	Stress associated with each random vibration phase (Grms)

Pollution level (see tables)

Saline pollution level	Π_{sal}
Low	1
High	2

Product protection level	Π_{prot}
Hermetic	0
Non-hermetic	1

Application pollution level	Π_{zone}
Low	1
Moderate	2
High	4

Environmental pollution level	Π_{envir}
Low	1
Moderate	1.5
High	2

Ambient temperature range	Π_{TV}
$T_{\text{ambient-board}} < 110^{\circ}\text{C}$	1
$T_{\text{ambient-board}} > 110^{\circ}\text{C}$	$e^{0.2 \times (T_{\text{ambient-board}} - 110)}$

Factors contributing to physical stresses

Π_{TCy}	$0.6 \times \Pi_{\text{TV}} \times \left(\frac{12 \times N_{\text{cy}}}{T_{\text{phase}}} \right) \times \left(\frac{\min(\theta_{\text{cy}}, 2)}{2} \right)^{\frac{1}{3}} \times \left(\frac{\Delta T_{\text{cycling}}}{20} \right)^{1.9} \times e^{1414 \times \left[\frac{1}{313} - \frac{1}{T_{\text{max-cycling}} + 273} \right]}$
$\Pi_{\text{Mechanical}}$	$0.2 \times \Pi_{\text{TV}} \times \left(\frac{G_{\text{RMS}}}{0.5} \right)^{1.5}$
Π_{RH}	$0.18 \times \Pi_{\text{TV}} \times \left(\frac{RH_{\text{ambient}}}{70} \right)^{4.4} \times e^{11604 \times 0.8 \times \left[\frac{1}{293} - \frac{1}{T_{\text{ambient-board}} + 273} \right]}$
Π_{Chemical}	$0.02 \times \Pi_{\text{TV}} \times \Pi_{\text{sal}} \times \Pi_{\text{envir}} \times \Pi_{\text{zone}} \times \Pi_{\text{prot}}$

Connectors

General model associated with the family

$$\lambda = \lambda_{physical} \times \Pi_{PM} \times \Pi_{Process}$$

Where:

$$\lambda_{physical} = \lambda_{0_Connector} \times \sum_i^{Phases} \left(\frac{t_{phase}}{T_{Total}} \right)_i \times \left(\frac{\Pi_{Thermal} + \Pi_{TCy}}{\Pi_{Mechanical} + \Pi_{RH} + \Pi_{Chemical}} \right)_i \times (\Pi_{Induced})_i$$

Contributions to the $C_{sensitivity}$ factor

	Relative sensitivity (score out of 10)			$C_{sensitivity}$
	EOS	MOS	TOS	
Connectors	1	10	3	3.13

$\Pi_{Placement}$ factor

For connectors, the placement factor is fixed at $\Pi_{Placement} = 1$

$QA_{component}$ factor

Component quality assurance level	Position relative to state of the art	$QA_{component}$
Qualification according to one of the following standards: ESCC 340x level B, NASDA-QTS-xxxx class 1, MIL-DTL-xxxxx, JAXA-QTS-2060E, GSFC	Much higher	3
Qualification according to one of the following standards: Telcordia GR1217-CORE, ARINC 600 and 80x (not space adapted), AECMA, SAE (39029)	Higher	2
Qualification according to one of the following standards: EIA, IEC, SAE, BS	Equivalent	1
No information	Lower	0

Basic failure rates associated with the item

$$\lambda_{0_Connector} = \lambda_{Type} \times \Pi_{mount} \times \Pi_{contact} \times \Pi_{cycle}$$

The model presented here applies to a half-pair of connectors.

Technical data

Connector type	λ_{type}
Round and rectangular connectors	0.05
Coaxial connectors	0.07
Connectors for printed (and assimilated) circuits	0.1
Component supports	0.1

Mount type	Π_{mount}
Insertion (press fit)	1
Soldered (through-hole)	6
Soldered (SMD)	10
Wrapping (braid)	3
Wrapping (wire)	2

Number of contacts

$$\Pi_{Contact} = (N_{Contact})^{0.5}$$

Where $N_{contact}$ is the number of connector contacts.

Connection frequency

$$\Pi_{Cycles} = 0.2 \times (N_{Cycles})^{0.25}$$

Where N_{cycles} is the number of cycles (a cycle comprises one connection and one disconnection). If $N_{cycles} < 1$ per year, use $\Pi_{cycles} = 0.2$.

Insert temperature rise

The insert temperature rise is usually taken from a thermal analysis, and is therefore supplied by the Design Office.

By default, it can be calculated by the formula:

$$\Delta T_{insert} = a \times I_{contact}^{1.85}$$

Where $I_{contact}$ is the average current across a pin (in amperes).

The value of a is given in the table below for the most common gauge values:

AWG	32	30	28	24	22	20	18	16	12
a	3.256	2.856	2.286	1.345	0.989	0.64	0.429	0.274	0.1

The gauge (AWG) depends on the cross-section of the connector pin. The formula for obtaining the cross-section of the pin is recalled in the tables below:

Rectangular connector pin	
Pin length (mm)	L
Pin width (mm)	l
Cross-section (mm ²)	$s = L \times l$

Cylindrical connector pin	
Diameter (mm)	D
Cross-section (mm ²)	$s = \pi \times \frac{D^2}{4}$

The gauge is obtained by the formula:

$$AWG = 36 - \left(\ln \left(\frac{s}{0.0127} \right) \times 19.5 / \ln(92) \right)$$

The following table (available on the internet), for round pins, is provided for assistance:

American Wire Gauge (AWG)	Diameter (inches)	Diameter (mm)	Cross Sectional Area (mm ²)
0000	0.46	11.68	107.16
000	0.4096	10.40	84.97
00	0.3648	9.27	67.40
0	0.3249	8.25	53.46
1	0.2893	7.35	42.39
2	0.2576	6.54	33.61
3	0.2294	5.83	26.65
4	0.2043	5.19	21.14
5	0.1819	4.62	16.76
6	0.162	4.11	13.29
7	0.1443	3.67	10.55
8	0.1285	3.26	8.36
9	0.1144	2.91	6.63
10	0.1019	2.59	5.26
11	0.0907	2.30	4.17
12	0.0808	2.05	3.31
13	0.072	1.83	2.63
14	0.0641	1.63	2.08
15	0.0571	1.45	1.65
16	0.0508	1.29	1.31
17	0.0453	1.15	1.04
18	0.0403	1.02	0.82
19	0.0359	0.91	0.65
20	0.032	0.81	0.52
21	0.0285	0.72	0.41
22	0.0254	0.65	0.33
23	0.0226	0.57	0.26
24	0.0201	0.51	0.20
25	0.0179	0.45	0.16
26	0.0159	0.40	0.13

Mission profile data

t_{phase}	Time associated with each phase (hours)
$RH_{ambient}$	Humidity level associated with a phase (%)
Note: The RH value for connectors at equipment interfaces may differ from that of other items.	
$T_{ambient-board}$	Mean temperature of the board during a phase (°C)
$\Delta T_{cycling}$	Amplitude of temperature variation associated with a cycling phase (°C)
$T_{max-cycling}$	Maximum temperature during a cycling phase (°C)
N_{cy}	Number of cycles associated with each cycling phase (cycles)
θ_{cy}	Cycle duration (hours)
G_{RMS}	Vibration amplitude associated with each random vibration phase (Grms)

Pollution level (see tables):

Saline pollution level	Π_{sal}
Low	1
High	2

Product protection level	Π_{prot}
Hermetic	0
Non-hermetic	1

Application pollution level	Π_{zone}
Low	1
Moderate	2
High	4

Environmental pollution level	Π_{envir}
Low	1
Moderate	1.5
High	2

Application data

ΔT Insert temperature rise

Factors contributing to physical stresses

$\Pi_{Thermal}$	In an operating phase: $0.58 \times e^{11604 \times 0.1 \times \left[\frac{1}{293 - (T_{ambient-board} + \Delta T + 273)} \right]}$ In a non-operating phase: $\Pi_{thermal} = 0$
Π_{Tcy}	$0.04 \times \left(\frac{12 \times N_{cy}}{T_{phase}} \right) \times \left(\frac{\min(\theta_{cy}, 2)}{2} \right)^{\frac{1}{3}} \times \left(\frac{\Delta T_{cycling}}{20} \right)^{1.9} \times e^{1414 \times \left[\frac{1}{313 - T_{max-cycling} + 273} \right]}$
$\Pi_{Mechanical}$	$0.05 \times \left(\frac{G_{RMS}}{0.5} \right)^{1.5}$
Π_{RH}	$0.13 \times \left(\frac{RH_{ambient}}{70} \right)^{4.4} \times e^{11604 \times 0.8 \times \left[\frac{1}{293 - (T_{ambient-board} + 273)} \right]}$
$\Pi_{Chemical}$	$0.20 \times \Pi_{Sal} \times \Pi_{Envir} \times \Pi_{Zone} \times \Pi_{Prot}$

Hybrids and multi-chip modules

**Chapter not modified in the 2020
version (October 2021)**

General model

Hybrids and Multi Chip Modules (MCMs) are miniaturized assemblies of components on different types of substrate, with different types of encapsulation (moulding, packaging). The failure rate of a hybrid is calculated in a similar way to that of a circuit board. It is based on an inventory of microcomponents inside the hybrid or MCM and takes account of connections, wiring, encapsulation, and different technological or process control attributes. Active components (integrated circuits, transistors, diodes) integrated into hybrids and MCMs may take the form of bare chips or of micro-packages.

General model associated with the family

$$\lambda_{H\&M} = \sum_{\mu\text{component}} \left(\lambda_{\mu\text{component}} \times \Pi_{PM_{\mu\text{component}}} \right) \times \Pi_{H\&M_process} \times \Pi_{process} \\ + \left(\lambda_{wiring} + \lambda_{Case+Substrate} + \lambda_{External_connections} \right) \times \Pi_{H\&M_process} \times \Pi_{process}$$

Where, for each basic element (microcomponent, wiring, case/substrate, external connections):

$$\lambda_{element} = \sum_i^{Phases} \left[\left(\frac{t_{phase}}{T_{Total}} \right)_i \times \left(\sum_{stresses} (\lambda_{0\ stress} \times \Pi_{stress}) \right)_i \times (\Pi_{induced})_i \right]$$

The $\Pi_{Process}$ factor is the factor calculated for the product into which the hybrid or MCM is integrated.

The $\Pi_{PM_{\mu\text{component}}}$ factor is calculated by the method established to determine the Π_{PM} of the corresponding components (integrated circuits, discrete semiconductors, resistors, capacitors, inductors).

Induced factor

Factors contributing to overstress

$$\Pi_{\text{induced-}i} = \left(\Pi_{\text{placement}} \times \Pi_{\text{application-}i} \times \Pi_{\text{ruggedizing}} \right)^{0.511 \times \ln(C_{\text{sensitivity}})}$$

The index i denotes the phase considered.

Contributions to the $C_{\text{sensitivity}}$ factor

A single $C_{\text{sensitivity}}$ factor is defined for the whole of the hybrid or MCM, based on the type of substrate and encapsulation.

	Relative sensitivity (score out of 10)			$C_{\text{sensitivity}}$
	EOS	TOS	MOS	
Metal case, ceramic case, ceramic substrate	6	1	5	5.5
Glass/epoxy substrate with moulding	6	1	2	4.1
Glass/epoxy substrate without moulding	6	1	4	4.8

The relative sensitivities to EOS, TOS, MOS (Electrical Over-Stress, Thermal Over-Stress, Mechanical Over-Stress) are given for information only, to show the relative sensitivity of item families to different types of overstress. They are not used in the calculations.

Contributions to the $\Pi_{\text{Placement}}$ factor

This contribution is determined in the same way as for the components.
The placement to be considered is placement of the hybrid in the product.

	$\Pi_{\text{Placement}}$
Digital non-interface function	1
Digital interface function	1.3
Low-level analog non-interface function	1.2
Low-level analog interface function	1.5
Power analog non-interface function	1.3
Power analog interface function	1.8

Contributions to the $\Pi_{\text{application}}$ factor

This contribution is determined in the same way as for the components.

Contributions to the $\Pi_{\text{ruggedizing}}$ factor

This contribution is determined in the same way as for the components.

H&M process factor

The $\Pi_{\text{Process_H\&M}}$ factor takes account of how control over the design and manufacturing of hybrids or MCMs influences reliability.

$$\Pi_{\text{H\&M_process}} = e^{1.39 \times (1 - \text{H\&M_process_grade})}$$

$$\text{Where } H \& M _ \text{process_grade} = \frac{\sum \text{values_from_table_below}}{140}$$

	Value if true	Value if false
Application of a formal method ¹ to take account of hybrid or MCM manufacturing capabilities during the design phase	15	0
Application of a formal method ¹ to take account of hybrid or MCM board mounting capabilities during the design phase	15	0
Previous experience in the development of hybrids and MCMs, with favourable feedback	10	0
Internal interconnection technology(ies) already implemented in a previous development	3	0
Case and substrate technology(ies) already implemented in a previous development	3	0
External interconnection technology(ies) already implemented in a previous development	4	0
Production line qualified for space applications (ESA/CNES) – (Class K according to the MIL-PRF 38534F classification, or Class I according to JESD93), or equivalent	30 See note 2	0
Production line qualified for military applications – (Class H according to the MIL-PRF 38534F classification or Class II according to JESD93), or equivalent	20 See note 2	0
Class G, E or D according to the MIL-PRF 38534F classification, or class below II according to JESD93, or manufacturer in-house qualification programme	10 See note 2	0
Application of the three reliability improvement methods: 1. Burn-in, 2. Batch-by-batch DPA, 3. Three-temperature operating test	30	0
Application of two out of three from: 1. Burn-in, 2. Batch-by-batch DPA, 3. Three-temperature operating test	20	0
Application of one out of three from: 1. Burn-in, 2. Batch-by-batch DPA, 3. Three-temperature operating test	10	0

Notes:

1. “Formal method” may refer to the use of a guide or specification that takes account of technical manufacturing capabilities.
2. These three cases are mutually exclusive.

If $\Pi_{\text{process_H\&M}}$ is not evaluated, a default value of 2.5 is proposed. Using the default value may reduce the accuracy of the final results.

Microcomponents

Failure rate of bare chips (integrated circuits, transistors, diodes, etc.)

$$\lambda_{\text{chip}} = \left(\lambda_{0\text{TH}} \times \Pi_{\text{Thermal}} \right) + \left(\lambda_{0\text{chip_TCy}} \times C_{\text{moulding}} \times C_{\text{chip_surface}} \times \Pi_{\text{TCy_case}} \right)$$

Thermal

For the values of $\lambda_{0\text{TH}}$, refer to the basic failure rate associated with the chip for integrated circuits or discrete semiconductors.

Chip thermal cycling

This is the acceleration factor for thermal cycling $\Pi_{\text{TCy_case}}$ (thermal amplitude to the power of 4.0) applicable to chips for this type of stress.

$$\lambda_{0\text{chip_TCy}} = 0.011$$

	C_{moulding}
Non-moulded hermetic circuit	1
Moulded circuit, silicon-embedded	1.4
Moulded circuit, polyurethane-embedded	1.6
Moulded circuit, epoxy-embedded	2

$$C_{\text{chip_surface}} = (1 + S^d)$$

S: individual surface area of each chip in mm²

	d
Silicon digital integrated circuits (MOS, Bipolar and BiCMOS)	0.35
Silicon analog integrated circuits (MOS, Bipolar and BiCMOS)	0.2
Discrete circuits	0.1

If the surface area of the chips is unknown, use the following default values for each chip:

	Chip_area (mm²)
Logical	75
Analog	4
Discrete low-power	0.8
Discrete power	3

Failure rate of micro-packaged components (integrated circuits, transistors, diodes)

$$\lambda_{\text{micro-package}} = \left(\begin{array}{l} \lambda_{0\text{ TH}} \times \Pi_{\text{Thermal}} + \lambda_{0\text{ TCy Case}} \times \Pi_{\text{TCy Case}} \\ + C_{\text{moulding}} \times \lambda_{0\text{ TCy SolderJoints}} \times \Pi_{\text{TCy SolderJoints}} \\ + C_{\text{hermeticity}} \times \lambda_{0\text{ RH}} \times \Pi_{\text{RH}} \end{array} \right)$$

Thermal

For the values of $\lambda_{0\text{ TH}}$, refer to the basic failure rate associated with the chips for integrated circuits or discrete semiconductors.

Thermal cycling

The basic failure rates for solder joints are calculated as a function of the type of substrate.

For integrated circuit (IC) packages the basic failure rates are obtained by the equation:

$$\lambda_{0\text{ Stress}} = e^{-a} \times Np^b$$

Where:

- a and b are constants that depend on the type of package and the number of pins, as listed in the table below.
- Np is the number of pins in the package.

Package (usual name)	Description	Np	$\lambda_{0\text{ TCy Case}}$		$\lambda_{0\text{ TCy SolderJoints}}$			
					Glass/epoxy substrate		Ceramic substrate	
			a	b	a	b	a	b
PQFP	Plastic Quad FlatPack, L□lead	44 to 240	12.41	1.46	10.80	1.46	9.41	1.46
		>240 to 304			10.11	1.46	8.61	1.46
SQFP TQFP, VQFP, LQFP	Plastic Shrink (thickness) Quad FlatPack, L□lead Plastic Thin Quad FlatPack, L□lead	32 to 120	8.57	0.73	6.96	0.73	5.57	0.73
		>120 to 208			5.57	0.73	4.65	0.73
Power QFP (RQFP, HQFP, PowerQuad, EdQuad, etc.)	Plastic Quad FlatPack with heat sink, L□lead	160 to 240	15.11	1.96	13.50	1.96	12.11	1.96
		>240 to 304			12.81	1.96	11.31	1.96
CERPACK		20 to 56	12.41	1.46	10.80	1.46	10.8	1.46
CQFP, Cerquad	Ceramic Quad Flat Pack	64 to 132	12.41	1.46	10.80	1.46	10.8	1.46
		>132 to 256			9.19	1.46	10.8	1.46
PLCC	Plastic Leaded Chip Carrier J□lead	20 to 52	18.52	3.15	16.91	3.15	16.91	3.15
		>52 to 84			15.52	3.15	15.52	3.15
J-CLCC	J□lead Ceramic Leaded Chip Carrier	4	8.07	0.93	6.46	0.93	6.46	0.93
		20			6.46	0.93	6.46	0.93
		32			6.46	0.93	6.46	0.93
		44			5.77	0.93	6.46	0.93
		52			5.36	0.93	6.46	0.93
		68			4.85	0.93	6.46	0.93

Package (usual name)	Description	Np	λ_{0TCy_Case}		$\lambda_{0TCy_SolderJoints}$			
					Glass/epoxy substrate		Ceramic substrate	
			a	b	a	b	a	b
CLCC	Ceramic Leadless Chip Carrier	4	8.07	0.93	5.07	0.93	6.46	0.93
		20			4.51	0.93	6.46	0.93
		32			4.38	0.93	6.46	0.93
		44			4.26	0.93	6.46	0.93
		52			4.26	0.93	6.46	0.93
		68			4.16	0.93	5.77	0.93
		84			4.16	0.93	5.77	0.93
SOJ	Plastic Small Outlines, J $\square$ lead	24 to 44	8.36	1.39	6.75	1.39	4.96	1.39
SO, SOP, GROUND, SOIC, SOW	Plastic Small Outlines, L $\square$ lead	8 to 14	13.36	2.18	11.75	2.18	9.67	2.18
		16 to 18			11.06	2.18	9.45	2.18
		20 to 28			10.36	2.18	9.16	2.18
		32			10.14	2.18	9.45	2.18
TSOP I TSOP II	Thin Small Outlines, leads on short edge, L $\square$ lead Thin Small Outlines, leads on long edge, L $\square$ lead	5 and 6	9.05	0.76	7.44	0.76	7.44	0.76
		>6 to 16			7.44	0.76	6.34	0.76
		>16 to 32			6.05	0.76	5.13	0.76
		>32 to 44			5.83	0.76	5.14	0.76
		>44 to 56			5.36	0.76	5.14	0.76
SSOP, VSOP, QSOP	Plastic Shrink (pitch) Small Outlines, L $\square$ lead	16 to 64	16.28	2.60	14.67	2.60	13.28	2.60
TSSOP, MSOP, μSO, μMAX, TVSOP	Thin Shrink Small Outlines, L $\square$ lead	8 to 28	15.56	2.66	13.95	2.66	12.0	2.66
		>28 to 48			13.21	2.66	11.71	2.66
		56			12.56	2.66	11.64	2.66
		64			12.16	2.66	11.65	2.66
PBGA CSP BT 0.8 and 0.75 mm	Plastic Ball Grid Array with solder ball pitch = 0.8 mm and 0.75 mm	48 to 384	12.13	1.49	9.13	1.49	10.52	1.49
PBGA flex 0.8 mm	Plastic Ball Grid Array with solder ball pitch = 0.8 mm and 0.75 mm	48 to 288	12.13	1.49	8.57	1.49	9.82	1.49
PBGA BT 1.00 mm	Plastic Ball Grid Array with solder ball pitch = 1.0 mm	64 to 1156	10.89	1.00	7.67	1.00	7.67	1.00
PBGA 1.27mm	Plastic Ball Grid Array, with solder ball pitch = 1.27 mm	119 to 352	10.36	0.93	7.36	0.93	7.36	0.93
		>352 to 432			7.14	0.93	7.14	0.93
		>432 to 729			6.67	0.93	6.67	0.93
Power BGA (TBGA SBGA, etc.)	Tape BGA; PBGA with heat sink, die top-down, Pitch=1.27 mm Super BGA; PBGA with heat sink, die top-down, Pitch=1.27 mm	256 to 352	15.73	1.68	12.73	1.68	12.51	1.68
		>352 to 956			12.33	1.68	12.18	1.68
CBGA	Ceramic Ball Grid Array	255 to 1156	15.37	1.87	11.56	1.87	13.76	1.87
DBGA	Dimpled BGA	255 to 1156	15.37	1.87	12.15	1.87	13.76	1.87
CGA IC	Ceramic Land GA + interposer, Ceramic column GA	255 to 1156	15.37	1.87	11.81	1.87	13.76	1.87

The basic failure rates for discrete semiconductor packages are listed below:

Package	Also known as	Designation	$\lambda_{0Tcy_Package}$	$\lambda_{0Tcy_SolderJoints}$	
				FR4	Ceramic
SOT23-3	TO236AB	SMD, small signal, L ⁺ lead, plastic	0.00057	0.00285	0.00285
SOT23-5	SC74A, SOT25				
SOT23-6	SC74, SOT26, SOT457				
SOT143	TO253AA, SC61B				
SOT323	SC70				
SOT346	SC59, TO236AA				
SOT353	SC70-5, SC88A				
SOT363	SC70-6, SC88				
SOD123					
SOD323	SC76				
SOD523	SC79	SMD, medium power, small heatsink, L ⁺ lead, plastic	0.00091	0.00455	0.0091
SOT223	SC73, TO261AA				
SOT243					
SOT343	SC82				
SOT89	SC62, TO243AA				
SOT194					
SOD6	DO214AA, SMB-J	SMD, small signal, C ⁺ lead, plastic	0.00091	0.00455	0.0091
SOD15	DO214AB, SMC-J				
DKAK	TO252AA, SC63, SOT428	SMD, power, large heatsink, L ⁺ lead, plastic	0.00413	0.02065	0.0413
D2PAK	TO263, SC83A, SMD220				
D3PAK	TO268				
ISOTOP	SOT227, TO244, Half-Pak	SMD, high-power, screw, plastic	0.03333	0.16665	0.199
SOD80	Mini-MELF, DO213AA	SMD, hermetically sealed glass	0.00781	0.03905	0.00781
SOD87	MELF, DO213AB				

Remarks:

- Some discrete semiconductor packages are also used for integrated circuits.
- Some integrated circuit packages are also used for discrete semiconductors.
- The packages used in hybrids and MCMs are generally micro-packages. The tables nonetheless list all packages, regardless of size, that might conceivably be used in a hybrid or MCM.

Humidity

For the values of λ_{0RH} , refer to the basic failure rates associated with the packages described in the integrated circuits or discrete semiconductors chapters.

	C_{hermeticity}
Hermetic circuit (with or without internal coating)	0.05
Sealed-cavity circuit (non-hermetic)	0.5
Moulded circuit	1

Failure rate of internal microcomponents: passive devices (resistors, capacitors, inductors)

Factors common to all the passive microcomponents

The influence of thermal cycling depends on how the component is mounted on the board:

Mount type	Model of Π_{TCy}
By gluing (conductive glue)	Π_{TCy} case
By soldering (solder)	Π_{TCy} solder joints

Moulding factor

	$C_{moulding}$
Hermetic circuit non-moulded	1
Moulded circuit, silicon-embedded	1.4
Moulded circuit, polyurethane-embedded	1.6
Moulded circuit, epoxy-embedded	2

Resistor chips (SMD)

$$\lambda_{resistor} = \lambda_{0_resistor} \times C_{moulding} \times \Pi_{TCy}$$

Where:

$$\lambda_{0_resistor} = 0.01$$

Deposited resistors

$$\lambda_{deposited_resistors} = \lambda_{0_deposited_resistors} \times C_{moulding} \times C_{tolerance} \times \Pi_{TCy}$$

Where:

$$\lambda_{0_deposited_resistances} = (0.01 \times R_e + 0.04 \times R_m)$$

R_e : Number of thick-film deposited resistors.

R_m : Number of thin-film deposited resistors.

Tolerance	$C_{tolerance}$
Tolerance above 5%	1
Tolerance from 1 to 5%	1.5 (default value)
Tolerance less than 1%	2

Resistor networks

$$\lambda_{\text{network_resistor}} = \lambda_{0_network_resistor} \times C_{\text{moulding}} \times \Pi_{\text{TCy}}$$

Where:

$$\lambda_{0_network_resistor} = 0.059$$

Capacitors

$$\lambda_{\text{capacitor}} = \lambda_{0_capacitor} \times \lambda \left(C_{\text{moulding}} \times \gamma_{\text{TCy}} \times \Pi_{\text{TCy}} + \gamma_{\text{TH_EL}} \times \Pi_{\text{Voltage}} \times \Pi_{\text{TH_EL}} \right)$$

The applicable $\Pi_{\text{TH_EL}}$ factor should be selected for each capacitor on the basis of the activation energy (Ea).

Where:

Component description	$\lambda_{0_Capacitor}$	Activation energy (eV)	S _{reference}	$\gamma_{\text{TH-EL}}$	γ_{TCy}
Ceramic capacitor Type I, Category 1	0.03	0.1	0.3	0.70	0.28
Ceramic capacitor Type I, Category 2	0.05	0.1	0.3	0.70	0.28
Ceramic capacitor Type I, Category 3	0.40	0.1	0.3	0.69	0.26
Ceramic capacitor Type II, Category 1	0.08	0.1	0.3	0.70	0.28
Ceramic capacitor Type II, Category 2	0.15	0.1	0.3	0.70	0.28
Ceramic capacitor Type II, Category 3	1.20	0.1	0.3	0.44	0.51
Ceramic capacitor with polymer terminations Type II, Category 1	0.08	0.1	0.3	0.70	0.28
Ceramic capacitor with polymer terminations Type II, Category 2 or 3	0.15	0.1	0.3	0.70	0.28
Solid tantalum capacitor (SMD packaging)	0.54	0.15	0.4	0.85	0.15
Deposited capacitor	0.1	0.1	0.3	0.71	0.29

For the capacitor category selection criteria, refer to the ceramic capacitor datasheet.

$$\Pi_{\text{Voltage}} = \left(\frac{1}{S_{\text{reference}}} \times \frac{V_{\text{applied}}}{V_{\text{rated}}} \right)_i^3$$

The index i designates the phase. The Π_{Voltage} factor needs to be calculated for each phase.

V_{applied} : voltage applied to the component during the phase (V)

V_{rated} : maximum voltage applicable to the component, as specified by the supplier (V).

Multilayer inductors

$$\lambda_{\text{inductor}} = \lambda_{0_inductor} \times \left(C_{\text{moulding}} \times \gamma_{\text{TCy}} \times \Pi_{\text{TCy}} + \gamma_{\text{TH_EL}} \times \Pi_{\text{TH_EL}} \right)$$

Where:

	Ea (eV)	λ_0	$\gamma_{\text{TH_EL}}$	γ_{TCy}
Inductor	0.15	0.05	0.71	0.29

Other components

If other types of component are present in the hybrid or MCM, the model for these components will be used, with the following adaptations:

Thermal cycling:

- If the model does not make a distinction for thermal cycling of cases:
The C_{moulding} factor will be used to weight the Π_{TCy} factor, as with the other microcomponents. The acceleration law for thermal cycling will be chosen on the basis of the mount type (gluing or soldering). See the “Model of Π_{TCy} ” table.
- If the model distinguishes between thermal cycling of cases and thermal cycling of soldered joints, the previous rule will be applied only to the $\Pi_{\text{TCy_solder joint}}$ factor.

Humidity:

The $C_{\text{hermeticity}}$ factor will be used to weight the Π_{RH} factor, as in the case of micro-packages, to take account of the protection against humidity that may be provided by the hermeticity of the hybrid or MCM.

Wiring, case, substrate, external connections

Failure rate of internal wiring by wires (bonding), bumps (flip-chip) or strips

$$\lambda_{\text{wiring}} = \left(\lambda_{0_wiring} \times \left(c_{\text{moulding}} \times \gamma_{\text{TCy}} \times \Pi_{\text{TCy}} + c_{\text{particle}} \times \gamma_{\text{ME}} \times \Pi_{\text{ME}} \right) + \lambda_{0_Chip_RH} \times C_{\text{hermeticity}} \times \Pi_{\text{RH}} \right)$$

Where:

$$\lambda_{0_wiring} = 1.04 \cdot 10^{-4} \text{Nb}_{\text{wires}}^{0.93}$$

Nb_{wires}: Total number of wires, bumps or strips inside the hybrid or MCM.

If the Nb_{wires} factor is unknown, it can be estimated by default from the number of input/outputs (Nb_{I/O}) of the hybrid or MCM:

- For hybrids: $\text{Nb}_{\text{wires}} = \text{Max}(6.15 \times \text{Nb}_{\text{I/O}} - 24.55 ; \text{Nb}_{\text{I/O}})$
- For MCMs: $\text{Nb}_{\text{wires}} = 2.9 \times \text{Nb}_{\text{I/O}}$

$$\gamma_{\text{TCy}} = 0.65$$

$$\gamma_{\text{ME}} = 0.35$$

Type of wiring	Model of Π_{TCy}
Wires and strips (aluminium and gold)	Π_{TCy} package
Bumps	Π_{TCy} solder joints

	C_{moulding}
Hermetic circuit non-moulded	1
Moulded circuit, silicon-embedded	1.4
Moulded circuit, polyurethane-embedded	1.6
Moulded circuit, epoxy-embedded	2

	C_{particle}
Moulded circuit or chip interconnection by flip-chip, with underfill	0
Non-moulded hermetic circuit with particle trap and gold wire*	0.5
Non-moulded hermetic circuit with particle trap and aluminium wire* or chip interconnection by flip-chip without underfill	0.3
Non-moulded hermetic circuit without particle trap and gold wire*	1.5
Non-moulded hermetic circuit without particle trap and aluminium wire*	1

(*): wire or strip

Humidity

$$\lambda_{0_chip_RH} = 7.01 \cdot 10^{-7} \times Nb_{wires}^{2.41}$$

Where

Nb_{wires} : Total number of wires, bumps or strips inside the hybrid or MCM.

If the Nb_{wires} factor is unknown, it can be estimated by default from the number of input/outputs ($Nb_{I/O}$) of the hybrid or MCM:

- For hybrids: $Nb_{wires} = \text{Max}(6.15 \times Nb_{I/O} - 24.55 ; Nb_{I/O})$
- For MCMs: $Nb_{wires} = 2.9 \times Nb_{I/O}$

$Nb_{I/O}$: Number of pins in the hybrid or MCM.

Remark: Where a case has two distinct parts (for example, a hermetic face and an unprotected face), each face must be calculated separately.

$C_{hermeticity}$ factor

	$C_{hermeticity}$
Hermetic circuit (with or without internal coating)	0.05
Sealed cavity circuit (non-hermetic)	0.5
Moulded circuit	1

Failure rate of the case and substrate

$$\lambda_{CS} = \lambda_{0_CS} \times \left(\gamma_{TCy} \times \Pi_{TCy_case} + C_{ME} \times \gamma_{ME} \times \Pi_{ME} + \gamma_{RH} \times \Pi_{RH} + \gamma_{chemical} \times \Pi_{chemical} \right)$$

Where:

$$\lambda_{0_CS} = \lambda_{0_substrate} \times \left(N_{layers} \right)^{\frac{1}{2}} \times \left(\frac{N_{tracks}^b}{2} \right) \times \Pi_{Class} \times \Pi_{Techno-substrate}$$

N_{tracks} = number of tracks

b see table on next page

Default value: $N_{tracks} = (Nb_{wires} / 2)$

For the default estimate of Nb_{wires} , see the proposed calculation based on the number of input/outputs.

N_{layers} = number of layers

Substrate type	Technology	Value of $\Pi_{Techno_substrate}$
Ceramic		0.25
Glass/epoxy	Through-holes	0.25
	Blind holes	0.5
	Micro-via technology	1
	Pad on via technology	2.5

Minimum conductor width (μm) / Minimum spacing between conductors or pads (μm)	Value of Π_{Class}
800 / 800	1
500 / 500	1
310 / 310	2
210 / 210	3
150 / 150	4
125 / 125	5
100 / 100	6

	$\lambda_{0_substrate}$	b	γ_{TCy}	γ_{ME}	γ_{RH}	$\gamma_{Chemical}$	C_{ME}
Hermetic ceramic case (cofired MCM package, etc.)	$2.08 \cdot 10^{-4}$	0.93	0.32	0.66	0.01	0.01	$1 + 0.1\sqrt{S_{case}}$
Alumina substrate with hermetic metal case	$2.08 \cdot 10^{-4}$	0.93	0.32	0.66	0.01	0.01	$1 + 0.1\sqrt{S_{case}}$
Alumina substrate with moulding	$2.08 \cdot 10^{-4}$	0.93	0.6	0.35	0.04	0.01	$1 + 0.1\sqrt{S_{case}}$
Alumina substrate without case or moulding	$2.08 \cdot 10^{-4}$	0.93	0.3	0.58	0.1	0.02	$1 + 0.1\sqrt{S_{case}}$
Organic substrate (glass/epoxy) with hermetic metal case	$5 \cdot 10^{-4}$	1	0.48	0.5	0.01	0.01	$1 + 0.1\sqrt{S_{case}}$
Organic substrate (glass/epoxy) with moulding	$5 \cdot 10^{-4}$	1	0.72	0.18	0.09	0.01	1
Glass/epoxy substrate without case or moulding	$5 \cdot 10^{-4}$	1	0.6	0.2	0.18	0.02	1

S_{case} = surface area of the case in cm²

If a hybrid or MCM has more than one type of substrate, the “ $\lambda_{0_substrate}$ ” and “b” parameters of a “glass/epoxy” substrate must be selected for the whole of the hybrid or MCM.

If a package contains parts in different configurations (e.g. an MCM with a hermetic face and an unprotected face), the weighted average failure rate must be used for each part (each face or each cavity), where the S_{case} parameter corresponds to the surface of the part considered.

The model then becomes:

$$\lambda_{CS} = \sum_{Cavities} \frac{\lambda_{0_CS} \times S_{cavity}}{S_{total}} \times \left(\gamma_{TCy} \times \Pi_{TCy_case} + C_{ME} \times \gamma_{ME} \times \Pi_{ME} + \gamma_{RH} \times \Pi_{RH} + \gamma_{chemical} \times \Pi_{chemical} \right)$$

and

$$C_{ME} = 1 + 0.1\sqrt{S_{cavity}}$$

Where

S_{cavity} = surface area of each cavity (part, or face) in cm²

S_{total} = sum of surface areas of each cavity in cm²

Failure rate associated with the external connections

$$\lambda_{\text{connections}} = \lambda_{0_{\text{TCy}}} \prod_{\text{TCy_solder_joints}} + \lambda_{0_{\text{ME}}} \prod_{\text{ME}}$$

Thermal cycling

$$\lambda_{0_{\text{TCy}}} = 20.5 \cdot 10^3 \times (K \times (D \times \Delta\text{CTE})^2)^{1.1}$$

K: connection rigidity parameter

Mounting type	K
Soldered surface-mount component with flat pins With copper or copper alloy pins	$\min \left(5000 \cdot \left(\frac{S_{\text{pin}}}{L_{\text{pin}}} - 0.01 \right), 200 \right)$ if negative, use K=0 (low-rigidity connection)
Soldered surface-mount component with flat pins With iron-nickel alloy pins (alloy 42, Kovar, etc.)	$\min \left(6150 \cdot \left(\frac{S_{\text{pin}}}{L_{\text{pin}}} - 0.01 \right), 200 \right)$ if negative, use K=0 (low-rigidity connection)
Soldered through-hole component with pins With copper or copper alloy pins	$\min \left(30 \cdot \left(\frac{S_{\text{pin}}}{0.196} \right)^{1.1}, 100 \right)$
Soldered through-hole component with pins With iron-nickel alloy pins (alloy 42, Kovar, etc.)	$\min \left(37 \cdot \left(\frac{S_{\text{pin}}}{0.196} \right)^{1.1}, 100 \right)$
Surface-mount soldered component without pins	500
Component without pins, assembled by contact	5

S_{pin} Pin cross-section in mm²

L_{pin} Pin length in mm

D Space between the most distant connections in the module (mm)

ΔCTE Difference between the CTE of the package and that of the PCB

Where, by default:

Nature of package	PCB	
	Printed circuit (FR4, polyimide) without Cu/In/Cu*	Printed circuit (FR4, polyimide) with Cu/In/Cu*
Metal (Kovar)	$\Delta\text{CTE} = 9 \cdot 10^{-6} / ^\circ\text{C}$	$\Delta\text{CTE} = 7 \cdot 10^{-6} / ^\circ\text{C}$
Ceramic (alumina or cofired)	$\Delta\text{CTE} = 8 \cdot 10^{-6} / ^\circ\text{C}$	$\Delta\text{CTE} = 6 \cdot 10^{-6} / ^\circ\text{C}$
Moulded (organic substrate or ceramic)	$\Delta\text{CTE} = 2 \cdot 10^{-6} / ^\circ\text{C}$	$\Delta\text{CTE} = 4 \cdot 10^{-6} / ^\circ\text{C}$

(*) The Cu/In/Cu (copper/Invar/copper) layers are deposited on external layers of the printed circuit, to reduce the CTE at the surface.

Mechanical

$$\lambda_{0_{ME}} = \frac{1}{a} \times \left(7 \cdot 10^{-4} \times \frac{M^{1.6}}{S_{pin} \cdot Nb_{pins}^{0.5}} + 1 \cdot 10^{-6} \times \frac{D^2}{S_{pin}} \right)$$

Where:

Mode of attachment	Example	a
None (allows for a degree of movement between the component and the printed circuit)	Component simply mounted on the board	1
Flexible (allows for limited movement of the component on its support)	Gluing	2
Rigid (allows for little or no movement between the component and the printed circuit)	Clamping Screwing	4

S_{pin} : Pin cross-section in mm²

Nb_{pins} Number of pins of the hybrid or MCM

D Space between the most distant connections in the module (mm)

M Mass of the package (grams)

In the absence of information on the mass of a hybrid in a metal package, use:

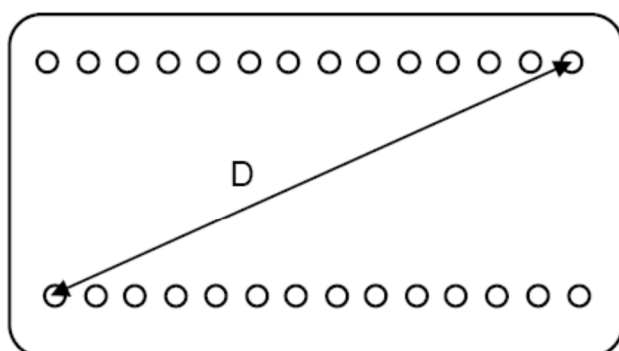
$M \text{ (gr)} = 0.003 \times \text{Volume of the hybrid (mm}^3\text{)}$

In the absence of information on the mass of an MCM in a ceramic package, use:

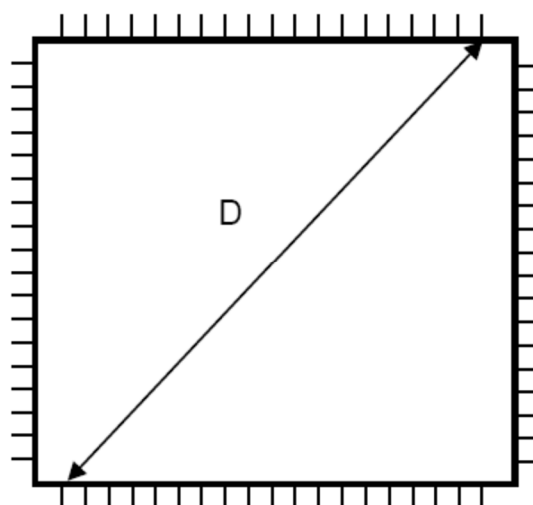
$M \text{ (gr)} = 0.004 \times \text{Volume of substrate (mm}^3\text{)}$

Remark: The distance D between the furthest connections of the module is often close to the diagonal of the package, as can be seen in the diagrams below.

Example of a platform package:



Example of a QFP package:



Physical stresses

Mission profile data

t_{annual}	Time associated with each phase over one year (hours)
$T_{\text{ambient_board}}$	Mean temperature of the board during the phase (°C)
RH_{ambient}	Humidity level associated with a phase (%)
$\Delta T_{\text{cycling}}$	Amplitude of temperature variation associated with a cycling phase (°C)
$T_{\text{max-cycling}}$	Maximum temperature on the board during a cycling phase (°C)
$N_{\text{cy-annual}}$	Number of cycles associated with each cycling phase over one year (cycles)
θ_{cy}	Cycle duration (hours)
Grms	Vibration amplitude associated with each random vibration phase (Grms)

Pollution level (see tables)

Saline pollution level	Π_{sal}
Low	1
High	2

Protection level system	Π_{prot}
Hermetic	0
Non-hermetic	1

Application pollution level	Π_{zone}
Low	1
Moderate	2
High	4

Environmental pollution level	Π_{envir}
Low	1
Moderate	1.5
High	2

Application data

$T_{\text{H\&M}}$	Mean temperature of the hybrid or MCM during the phase (°C)
$T_{\text{J_component}}$	Component junction temperature during the phase (°C)
$P_{\text{dissipated}}$	Power dissipated during the phase by the component, hybrid, MCM or microcomponent, as appropriate (W)

Calculation of the component $T_{\text{H\&M}}$ and T_{J} values for hybrids and MCMs:

$$T_{\text{J_component}} = T_{\text{H\&M}} + R_{\text{JC}} \cdot P_{\text{dissipated from component}}$$

$$T_{\text{H\&M}} = T_{\text{ambient}} + R_{\text{CA}} \cdot P_{\text{dissipated from H\&M}}$$

Where:

R_{JC} is the thermal resistance between the junction and the substrate of the hybrid or MCM.

R_{CA} is the thermal resistance between the hybrid or MCM and the ambient.

Factors contributing to physical stresses

$\Pi_{Thermal}$ Active components	In an operating phase: $\Pi_{El} \times e^{11604 \times 0,7 \times \left[\frac{1}{293} - \frac{1}{(T_{j-composant} + 273)} \right]}$ For signal diodes up to 1A (PIN, Schottky, signal, varactor): $\Pi_{El} = \left(\frac{V_{applied}}{V_{rated}} \right)^{2.4} \text{ if } \frac{V_{applied}}{V_{rated}} > 0.3$ $\Pi_{El} = 0.056 \text{ if } \frac{V_{applied}}{V_{rated}} \leq 0.3$ For other types of item: $\Pi_{El} = 1$ In a non-operating phase: $\Pi_{Thermal} = 0$
$\Pi_{Thermo-electrical}$ $Ea = 0.1 \text{ eV}$ Capacitors other than solid tantalum	$11604 \times 0.1 \times \left[\frac{1}{293} - \frac{1}{(T_{H\&M} + 273)} \right]$ In an operating phase: e In a non-operating phase: $\Pi_{Thermo-electrical} = 0$
$\Pi_{Thermo-electrical}$ $Ea = 0.15 \text{ eV}$ Solid tantalum capacitors, inductors	$11604 \times 0.15 \times \left[\frac{1}{293} - \frac{1}{(T_{H\&M} + 273)} \right]$ In an operating phase: e In a non-operating phase: $\Pi_{Thermo-electrical} = 0$
$\Pi_{TCy_solder_joints}$	$\left(\frac{12 \times N_{cy}}{T_{phase}} \right) \times \left(\frac{\min(\theta_{cy}, 2)}{2} \right)^{\frac{1}{3}} \times \left(\frac{\Delta T_{cycling}}{20} \right)^{1.9} \times e^{1414 \times \left[\frac{1}{313} - \frac{1}{T_{max-cycling} + 273} \right]}$
Π_{TCy_Case}	$\left(\frac{12 \times N_{cy}}{T_{phase}} \right) \times \left(\frac{\Delta T_{cycling}}{20} \right)^4 \times e^{1414 \times \left[\frac{1}{313} - \frac{1}{T_{max-cycling} + 273} \right]}$
$\Pi_{Mechanical}$	$\left(\frac{G_{RMS}}{0.5} \right)^{1.5}$
Π_{RH}	$\left(\frac{RH_{ambient}}{70} \right)^{4.4} \times e^{11604 \times 0.9 \times \left[\frac{1}{293} - \frac{1}{(T_{ambient} + 273)} \right]}$ In an operating phase: $\Pi_{RH} = 0$
$\Pi_{Chemical}$	$\Pi_{Sal} \times \Pi_{Envir} \times \Pi_{zone} \times \Pi_{Prot}$

Microwave (MW) and radiofrequency (RF) components

RF and MW process factor

$\Pi_{\text{RFMW process}}$ factor

The $\Pi_{\text{RFMW process}}$ factor takes account of how control over the life cycle of a radiofrequency (RF) or microwave (MW) product influences reliability.

This factor differs depending on the RF/MW model used.

The $\Pi_{\text{RFMW process}}$ factor is a value between 0.5 and 4.

When the data is unknown, the default value is $\Pi_{\text{RFMW process}} = 2.2$.

This factor is complementary to the Π_{process} factor, which remains applicable to the microwave and radiofrequency products.

The $\Pi_{\text{RFMW process}}$ factor takes the form:

$$\Pi_{\text{ProcessRFMW}} = e^{1.504 \cdot (1 - \text{RFMW_grade})} - 0.5$$

Where $\text{RFMW_grade} = \text{Sum (Values in table below if True, else 0)} / (\text{Maximum applicable score})$.

N°	Criterion	GaN	GaAs Si ICs SiGe Si & SiGe DSs	Passive RFMW
General				
1.1	The manufacturer has proven experience in the development of a radiofrequency (or microwave) board or function, with positive feedback on the reliability of the function.	8	8	8
1.2	Presence of protection against antenna disconnection around RF and MW power transistors and amplifiers.	7	8	N/A
1.3	Presence of protection for EMC compatibility (out-of-band management) around RF and MW power transistors and amplifiers. For example: circulators, filters, etc.	6	6	N/A
1.4	Presence of protection against load mismatch around RF and MW power transistors and amplifiers. For example: circulators, isolators, filters, etc.	7	8	N/A
Power supply management				
2.1	Power sequencing management has been taken into account for powering up and for the shutdown phase.	10	7	N/A
2.2	In operational mode and in the case of a pulsed power supply, the timings have been verified.	8	6	N/A

N°	Criterion	GaN	GaAs Si ICs SiGe Si & SiGe DSs	Passive RFMW
Burn-In				
3.1	The component manufacturer performs a burn-in. If not, the user integrates a burn-in phase.	7	N/A	N/A
3.2	If burn-in is recommended by the manufacturer, it will be complied with and implemented by the component user.	7	7	N/A
3.3	Possible changes in the component during the first hours of operation have been taken into account and verified.	5	N/A	N/A
Junction temperature				
4.1	The Tj averaged across the surface of the chip, determined from the component Rth and from the power dissipation of the application, complies with the manufacturer's ROR.	7	7	N/A
4.2	Particular care has been taken in estimating the component Tj (or in-plane temperature for passive devices) by comparing the value of Tj obtained from the manufacturer Rth with that obtained using an electrical (or any other relevant) method.	8	6	N/A
4.3	A safety margin has been taken into account relative to the manufacturer's ROR. The derating values are typically 15% for power components and 10% for low-power components.	6	6	N/A
4.4	The risk of thermal runaway has been taken into account.	9	N/A	N/A
Compression, DC/RF performance and robustness				
5.1	Compliance with all of the ROR has been verified in the context of the application.	10	10	10
5.2	Where applicable, the application design complies, throughout the TOS and temperature range, with a radio-frequency grid current lower than the maximum admissible grid current, ensuring the reliability of the component: 'I _g forward'=f(P _{in})<I _{gmax} and 'I _{gs} reverse' as per the manufacturer datasheet (if indicated).	10	10	N/A
5.3	The application design precludes any temporary excursion of the parameters beyond the ROR throughout the TOS and operational temperature range.	7	6	6

N°	Criterion	GaN	GaAs Si ICs SiGe Si & SiGe DSs	Passive RFMW
5.4	In operational mode, the compression does not exceed 6dB relative to the maximum gain and/or is compatible with the technology used (as assessed by RF step stress tests), with the smallest value being selected (max admissible level of compression in the ROR).	8	N/A	N/A
5.5	Tests have been conducted under conditions of maximum application mismatch (with an additional margin). Either there is no drift, or any drifts have been analysed by the user and factored into the design of the function.	9	N/A	N/A
Environment & qualification tests				
6.1	The manufacturer's qualification tests on the component cover the mission profile of the product as defined in the guide; alternatively, they will be conducted by the user (packaged component or bare chip).	6	6	6
6.2	The component's sensitivity to ESD is known and precautionary measures are taken throughout the supply chain, from reception to the final test before shipping.	8	8	6
6.3	The component mounting method is known and is compatible with the component (package, MSL, temperature, etc.).	8	8	N/A
Reliability				
7.1	Reliability tests under RF polarization conditions representative of the application are planned by the user. Any drift will be factored into the design of the function.	9	9	N/A
7.2	The component lifetime, in its different operating scenarios, is compatible with the requirement and includes a margin.	9	9	9

GaN RF/MW components

General model associated with the family

$$\lambda = \lambda_{Physical} \times \Pi_{PM} \times \Pi_{Process} \times \Pi_{RFHFProcess}$$

Where:

$$\lambda_{Physical} = \sum_{i=1}^{\text{Number of phases}} \left(\frac{t_{\text{phase}}}{T_{\text{Total}}} \right)_i \times \left(\begin{array}{c} \lambda_{0TH} \times \Pi_{Thermal} \\ + \lambda_{0TCyCase} \times \Pi_{TCyCase} \\ + \lambda_{0TCySolderJoints} \times \Pi_{TCySolderJoints} \\ + \lambda_{0RH} \times \Pi_{RH} \\ + \lambda_{0Mech} \times \Pi_{Mech} \end{array} \right)_i \times (\Pi_{Induced})_i$$

C_{sensitivity} factor

	Relative sensitivity (score out of 10)			C _{sensitivity}
	EOS	MOS	TOS	
GaN, MMIC components	8	3	4	6.9
GaN components, discrete semiconductors (incl. Power Bar)	8	3	4	6.9

Part manufacturing factor Π_{PM}

For MMICs, the Π_{PM} factor is determined in the same way as for non-RF/MW integrated circuits. For discrete semiconductors (including Power Bar), the Π_{PM} factor is determined in the same way as for non-RF/MW discrete semiconductors.

For the choice of the ε factor, it is recommended that a factor of 4 be applied for large-volume applications (telecommunications, general public) and a factor of 2 for low-volume applications (defence, industrial, aerospace).

Basic failure rates associated with the packages

Refer to the failure rates provided for non-RF/MW integrated circuit packages.

Basic failure rates associated with the chip

Basic material	Type	λ_{0TH}
GaN	GaN MMICs and discrete semiconductors (incl. Power Bar)	0.3033

Mission profile data

t_{phase}	Time associated with each phase (hours)
RH_{ambient}	Humidity level associated with a phase (%)
$T_{\text{ambient-board}}$	Mean temperature of the board during a phase (°C)
$\Delta T_{\text{cycling}}$	Amplitude of temperature variation associated with a cycling phase (°C)
$T_{\text{max-cycling}}$	Maximum temperature on the board during a cycling phase (°C)
N_{cy}	Number of cycles associated with each cycling phase (cycles)
θ_{cy}	Cycle duration (hours)
G_{RMS}	Vibration amplitude associated with each random vibration phase (Grms)

Application data

$T_{\text{j-component}}$ Component junction temperature during an operating phase (°C)

$T_{\text{i-component}} = T_{\text{ambient}} + \Delta T$

$\Delta T = P_{\text{dissipated}} \times R_{\text{JA}}$

ΔT Component temperature rise

R_{JA} Junction-to-ambient thermal resistance

$P_{\text{dissipated}}$ Power dissipated by the component during the phase (W)

In the case of pulse operation, it is important to factor in the duty cycle when calculating Π_{Thermal} . For short pulses (shorter than the thermal time constant of the chip), ΔT must be determined using the thermal impedance Z instead of R_{JA} .

η Duty cycle during the phase

Technical data

T_{Ref} 150°C

Factors contributing to physical stresses

Π_{Thermal}	In an operating phase: If $T_{\text{j-component}} > T_{\text{Ref}}$: $\eta \times e^{11604 \times 1.5 \times \left[\frac{1}{T_{\text{Ref}} + 273} - \frac{1}{(T_{\text{j-component}} + 273)} \right]}$ If $T_{\text{j-component}} \leq T_{\text{Ref}}$: $\eta \times e^{11604 \times 1.5}$ In a non-operating phase: $\Pi_{\text{Thermal}} = 0$ In a non-pulse operating phase, $\eta = 1$
Π_{TCyCase}	$\left(\frac{12 \times N_{\text{cy}}}{t_{\text{total}}} \right) \times \left(\frac{\Delta T_{\text{cycling}}}{20} \right)^4 \times e^{1414 \times \left[\frac{1}{313} - \frac{1}{(T_{\text{max-cycling}} + 273)} \right]}$
Π_{TCy} <i>Solder Joints</i>	$\left(\frac{12 \times N_{\text{cy}}}{t_{\text{total}}} \right) \times \left(\frac{\min(\theta_{\text{cy}}, 2)}{2} \right)^{\frac{1}{3}} \times \left(\frac{\Delta T_{\text{cycling}}}{20} \right)^{1.9} \times e^{1414 \times \left[\frac{1}{313} - \frac{1}{(T_{\text{max-cycling}} + 273)} \right]}$
Π_{Mech}	$\left(\frac{G_{\text{RMS}}}{0.5} \right)^{1.5}$
Π_{RH}	$\left(\frac{RH_{\text{ambient}}}{70} \right)^{4.4} \times e^{11604 \times 0.9 \times \left[\frac{1}{293} - \frac{1}{(T_{\text{ambient-board}} + 273)} \right]}$ In an operating phase: $\Pi_{\text{RH}} = 0$

GaAs RF/MW components

General model associated with the family

$$\lambda = \lambda_{Physical} \times \Pi_{PM} \times \Pi_{Process} \times \Pi_{RFHFprocess}$$

Where:

$$\lambda_{Physical} = \sum_{i=1}^{\text{Number of phases}} \left(\frac{t_{\text{phase}}}{T_{\text{Total}}} \right)_i \times \left(\begin{array}{c} \lambda_{0TH} \times \Pi_{Thermal} \\ + \lambda_{0TCyCase} \times \Pi_{TCyCase} \\ + \lambda_{0TCySolderJoints} \times \Pi_{TCySolderJoints} \\ + \lambda_{0RH} \times \Pi_{RH} \\ + \lambda_{0Mech} \times \Pi_{Mech} \end{array} \right)_i \times (\Pi_{Induced})_i$$

C_{sensitivity} factor

	Relative sensitivity (score out of 10)			C _{sensitivity}
	EOS	MOS	TOS	
GaAs components, MMICs (incl. LNAs)	9	3	5	7.4
GaAs components, discrete semiconductors (incl. transistors)	9	3	5	7.4

Part manufacturing factor Π_{PM}

For MMICs (including LNAs), the Π_{PM} factor is determined in the same way as for non-RF/MW discrete semiconductors.

For discrete semiconductors (including transistors), the Π_{PM} factor is determined in the same way as for non-RF/MW discrete semiconductors.

For the choice of the ε factor, it is recommended that a factor of 4 be applied for large-volume applications (telecommunications, general public) and a factor of 2 for low-volume applications (defence, industrial, aerospace).

Basic failure rates associated with the packages

Refer to the failure rates provided for non-RF/MW integrated circuit packages.

Basic failure rate associated with the chip

Basic material	Type	λ_{0TH}
GaAs	MMICs (including LNAs) and discrete semiconductors (including transistors)	0.3756

Mission profile data

t_{phase}	Time associated with each phase (hours)
RH_{ambient}	Humidity level associated with a phase (%)
$T_{\text{ambient-board}}$	Mean temperature of the board during a phase (°C)
$\Delta T_{\text{cycling}}$	Amplitude of temperature variation associated with a cycling phase (°C)
$T_{\text{max-cycling}}$	Maximum temperature on the board during a cycling phase (°C)
N_{cy}	Number of cycles associated with each cycling phase (cycles)
θ_{cy}	Cycle duration (hours)
G_{RMS}	Vibration amplitude associated with each random vibration phase (Grms)

Application data

$T_{\text{j-component}}$ Component junction temperature during an operating phase (°C)

$$T_{\text{j-component}} = T_{\text{ambient}} + \Delta T$$

$$\Delta T = P_{\text{dissipated}} \times R_{\text{JA}}$$

ΔT Component temperature rise

R_{JA} Junction-to-ambient thermal resistance

$P_{\text{dissipated}}$ Power dissipated by the component during the phase (W)

In the case of pulse operation, it is important to factor in the duty cycle when calculating Π_{Thermal} . For short pulses (shorter than the thermal time constant of the chip), ΔT must be determined using the thermal impedance Z instead of R_{JA} .

η Duty cycle during the phase

Technical data

T_{Ref} 125°C

Factors contributing to physical stresses

Π_{Thermal}	In an operating phase: If $T_{\text{j-component}} > T_{\text{Ref}}$: $\eta \times e^{11604 \times 1.3 \times \left[\frac{1}{T_{\text{Ref}} + 273} - \frac{1}{(T_{\text{j-component}} + 273)} \right]}$ If $T_{\text{j-component}} \leq T_{\text{Ref}}$: $\eta \times e^{11604 \times 1.5}$ In a non-operating phase: $\Pi_{\text{Thermal}} = 0$ In a non-pulse operating phase, $\eta = 1$
Π_{TCyCase}	$\left(\frac{12 \times N_{\text{cy}}}{t_{\text{total}}} \right) \times \left(\frac{\Delta T_{\text{cycling}}}{20} \right)^4 \times e^{1414 \times \left[\frac{1}{313} - \frac{1}{(T_{\text{max-cycling}} + 273)} \right]}$
$\Pi_{\text{TCy Solder Joints}}$	$\left(\frac{12 \times N_{\text{cy}}}{t_{\text{total}}} \right) \times \left(\frac{\min(\theta_{\text{cy}}, 2)}{2} \right)^{\frac{1}{3}} \times \left(\frac{\Delta T_{\text{cycling}}}{20} \right)^{1.9} \times e^{1414 \times \left[\frac{1}{313} - \frac{1}{(T_{\text{max-cycling}} + 273)} \right]}$
Π_{Mech}	$\left(\frac{G_{\text{RMS}}}{0.5} \right)^{1.5}$
Π_{RH}	$\left(\frac{RH_{\text{ambient}}}{70} \right)^{4.4} \times e^{11604 \times 0.9 \times \left[\frac{1}{293} - \frac{1}{(T_{\text{ambient-board}} + 273)} \right]}$ In an operating phase: $\Pi_{\text{RH}} = 0$

Si and SiGe RF/MW integrated circuits

General model associated with the family

$$\lambda = \lambda_{physical} \times \Pi_{PM} \times \Pi_{Process} \times \Pi_{RF/MWprocess}$$

Where:

$$\lambda_{Physical} = \sum_{i=1}^{\text{Number of phases}} \left(\frac{t_{\text{phase}}}{T_{\text{Total}}} \right)_i \times \left(\begin{array}{c} \lambda_{0TH} \times \Pi_{Thermal} \\ + \lambda_{0TCyCase} \times \Pi_{TCyCase} \\ + \lambda_{0TCySolderJoints} \times \Pi_{TCySolderJoints} \\ + \lambda_{0RH} \times \Pi_{RH} \\ + \lambda_{0Mech} \times \Pi_{Mech} \end{array} \right)_i \times (\Pi_{Induced})_i$$

C_{sensitivity} factor

	Relative sensitivity (score out of 10)			C _{sensitivity}
	EOS	MOS	TOS	
Si and SiGe integrated circuits	9	3	5	7.4

Part manufacturing factor Π_{PM}

The factor Π_{PM} is determined in the same way as for non-RF/MW integrated circuits.

For the choice of the ε factor, it is recommended that a factor of 4 be applied for large-volume applications (telecommunications, general public) and a factor of 2 for low-volume applications (defence, industrial, aerospace).

Basic failure rates associated with the packages

Refer to the failure rates provided for non-RF/MW integrated circuit packages.

Basic failure rates associated with the chip

Basic material	Type	λ_{0TH}
Si	Analog circuit (power amplifier) RF and MW (MOS)	0.5300
Si, SiGe	Analog and mixed circuit (MOS, Bipolar, BiCMOS, MESFET, PHEMT), RF and MW	0.1900
Si, SiGe	Digital circuit (MOS, Bipolar, BiCMOS), RF and MW	0.0400

Notes:

- Mixed = analog and digital.
- The distinction between “power” and “low-level” for amplifiers is indicated in the table below:

Family (chip with or without package)	Frequency	P1dB (dBm)
Power amplifiers & power transistors	≤ 20 GHz	≥ 30
Power amplifiers & power transistors	> 20 GHz	≥ 20

Mission profile data

t_{phase}	Time associated with each phase (hours)
RH_{ambient}	Humidity level associated with a phase (%)
$T_{\text{ambient-board}}$	Mean temperature of the board during a phase (°C)
$\Delta T_{\text{cycling}}$	Amplitude of temperature variation associated with a cycling phase (°C)
$T_{\text{max-cycling}}$	Maximum temperature on the board during a cycling phase (°C)
N_{cy}	Number of cycles associated with each cycling phase (cycles)
θ_{cy}	Cycle duration (hours)
G_{RMS}	Vibration amplitude associated with each random vibration phase (G_{RMS})

Application data

$T_{\text{j-component}}$ Component junction temperature during an operating phase (°C)

$T_{\text{j-component}} = T_{\text{ambient}} + \Delta T$

$\Delta T = P_{\text{dissipated}} \times R_{\text{JA}}$

ΔT Component temperature rise

R_{JA} Junction-to-ambient thermal resistance

$P_{\text{dissipated}}$ Power dissipated by the component during the phase (W)

In the case of pulse operation, it is important to factor in the duty cycle when calculating Π_{Thermal} . For short pulses (shorter than the thermal time constant of the chip), ΔT must be determined using the thermal impedance Z instead of R_{JA} .

η Duty cycle during the phase

Factors contributing to physical stresses

Π_{Thermal}	In an operating phase: $e^{11604 \times 0.7 \times \left(\frac{1}{293} - \frac{1}{273 + T_{\text{j-component}}} \right)}$ In a pulse operating phase: $\eta \times e^{11604 \times 0.7 \times \left(\frac{1}{293} - \frac{1}{273 + T_{\text{j-component}}} \right)}$ In a non-operating phase: $\Pi_{\text{Thermal}} = 0$
Π_{TCyCase}	$\left(\frac{12 \times N_{\text{cy}}}{t_{\text{total}}} \right) \times \left(\frac{\Delta T_{\text{cycling}}}{20} \right)^4 \times e^{1414 \times \left[\frac{1}{313} - \frac{1}{(T_{\text{max-cycling}} + 273)} \right]}$
Π_{TCy} <i>Solder Joints</i>	$\left(\frac{12 \times N_{\text{cy}}}{t_{\text{total}}} \right) \times \left(\frac{\min(\theta_{\text{cy}}, 2)}{2} \right)^{\frac{1}{3}} \times \left(\frac{\Delta T_{\text{cycling}}}{20} \right)^{1.9} \times e^{1414 \times \left[\frac{1}{313} - \frac{1}{(T_{\text{max-cycling}} + 273)} \right]}$
Π_{Mech}	$\left(\frac{G_{\text{RMS}}}{0.5} \right)^{1.5}$
Π_{RH}	$\left(\frac{RH_{\text{ambient}}}{70} \right)^{4.4} \times e^{11604 \times 0.9 \times \left[\frac{1}{293} - \frac{1}{(T_{\text{ambient-board}} + 273)} \right]}$ In an operating phase: $\Pi_{\text{RH}} = 0$

Si and SiGe RF/MW discrete semiconductors

General model associated with the family

$$\lambda = \lambda_{Physical} \times \Pi_{PM} \times \Pi_{Process} \times \Pi_{RF/MWprocess}$$

Where:

$$\lambda_{Physical} = \sum_{i=1}^{\text{Number of phases}} \left(\frac{t_{\text{phase}}}{T_{\text{Total}}} \right)_i \times \left(\begin{array}{c} \lambda_{0TH} \times \Pi_{Thermal} \\ + \lambda_{0TCyCase} \times \Pi_{TCyCase} \\ + \lambda_{0TCySolderJoints} \times \Pi_{TCySolderJoints} \\ + \lambda_{0RH} \times \Pi_{RH} \\ + \lambda_{0Mech} \times \Pi_{Mech} \end{array} \right)_i \times (\Pi_{Induced})_i$$

C_{sensitivity} factor

	Relative sensitivity (score out of 10)			C _{sensitivity}
	EOS	MOS	TOS	
Discrete semiconductor circuit, Si and SiGe, RF and MW	10	2	1	6.3
Discrete semiconductor circuit, GaAs, RF and MW	10	4	5	7.4

Part manufacturing factor Π_{PM}

The Π_{PM} factor is determined in the same way as for non-RF/MW discrete semiconductors.

For the choice of the ε factor, it is recommended that a factor of 4 be applied for large-volume applications (telecommunications, general public) and a factor of 2 for low-volume applications (defence, industrial, aerospace).

Basic failure rates associated with the packages

Refer to the failure rates provided for non-RF/MW discrete semiconductor packages.

Basic failure rates associated with the chip

Low-power diodes	λ_{0TH}
PIN, Schottky, Tunnel, varactor diodes (RF/MW)	0.0120
Low-power transistors	
Silicon, bipolar < 5W, SiGe, bipolar < 1W	0.0138
Power transistors	
Silicon, bipolar > 5W	0.0478
Silicon, MOS > 5W	0.0202

Mission profile data

t_{phase}	Time associated with each phase (hours)
N_{cy}	Number of cycles associated with each cycling phase over one year (cycles)
θ_{cy}	Cycle duration (hours)
G_{RMS}	Vibration amplitude associated with each random vibration phase (G_{RMS})

Application data

$T_{j_component}$ Component junction temperature during an operating phase (°C)

$$T_{j_component} = T_{\text{ambient}} + \Delta T$$

$$\Delta T = P_{\text{dissipated}} \times R_{JA}$$

ΔT Component temperature rise

R_{JA} Junction-to-ambient thermal resistance

$P_{\text{dissipated}}$ Power dissipated by the component during the phase (W)

V_{applied} Inverse voltage applied during the phase (for signal diodes only) (V)

In the case of pulse operation, it is important to factor in the duty cycle when calculating Π_{Thermal} . For short pulses (shorter than the thermal time constant of the chip), ΔT must be determined using the thermal impedance Z instead of R_{JA} .

η Duty cycle during the phase

Technical data

V_{rated} Rated inverse voltage (V) (for signal diodes only)

Factors contributing to physical stresses

$\Pi_{Thermal}$	In an operating phase: $\pi_{El} \cdot e^{11604 \times 0.7 \times \left(\frac{1}{293} - \frac{1}{273 + T_j - \text{component}} \right)}$ In a pulse operating phase: $\eta \cdot \pi_{El} \cdot e^{11604 \times 0.7 \times \left(\frac{1}{293} - \frac{1}{273 + T_j - \text{component}} \right)}$ For signal diodes up to 1A (PIN, Schottky, signal, varactor): $\pi_{El} = \left(\frac{eV_{\text{applied}}}{V_{\text{rated}}} \right)^{2.4} \text{ if } \frac{eV_{\text{applied}}}{V_{\text{rated}}} > 0.3$ $\pi_{El} = 0.056 \text{ if } \frac{eV_{\text{applied}}}{V_{\text{rated}}} \leq 0.3$ For other types of item: $\pi_{El} = 1$ In a non-operating phase: $\Pi_{Thermal} = 0$
$\Pi_{TCyCase}$	$\left(\frac{12 \times N_{cy}}{t_{\text{total}}} \right) \times \left(\frac{\Delta T_{\text{cycling}}}{20} \right)^4 \times e^{1414 \times \left[\frac{1}{313} - \frac{1}{T_{\text{max-cycling}} + 273} \right]}$
Π_{TCy} <i>Solder Joints</i>	$\left(\frac{12 \times N_{cy}}{t_{\text{total}}} \right) \times \left(\frac{\min(\theta_{cy}, 2)}{2} \right)^{\frac{1}{3}} \times \left(\frac{\Delta T_{\text{cycling}}}{20} \right)^{1.9} \times e^{1414 \times \left[\frac{1}{313} - \frac{1}{T_{\text{max-cycling}} + 273} \right]}$
Π_{Mech}	$\left(\frac{G_{RMS}}{0.5} \right)^{1.5}$
Π_{RH}	$\left(\frac{RH_{\text{ambient}}}{70} \right)^{4.4} \times e^{11604 \times 0.9 \times \left[\frac{1}{293} - \frac{1}{T_{\text{ambient-board}} + 273} \right]}$ In an operating phase: $\Pi_{RH} = 0$

RF/MW passive components

General model associated with the family

$$\lambda = \lambda_{Physical} \times \Pi_{PM} \times \Pi_{Process} \times \Pi_{RF/MWprocess}$$

Where:

For RF and MW specific functions:

$$\lambda_{Physical} = \lambda_{0_Passive\ RF/MW} \times \sum_i^{\text{Phases}} \left(\frac{t_{\text{phase}}}{T_{\text{Total}}} \right)_i \times (\Pi_{\text{Thermo-electrical}} + \Pi_{\text{TCy}} + \Pi_{\text{Mechanical}} + \Pi_{\text{RH}})_i \times (\Pi_{\text{Induced}})_i$$

For surface-mounted RF and MW resistors, apply the $\lambda_{Physical}$ calculation for resistor chips described in the datasheet for (non RF/MW) resistors.

For surface-mounted RF and MW ceramic capacitors, apply the $\lambda_{Physical}$ calculation for ceramic capacitors with a defined temperature coefficient (Type I) described in the datasheet for (non RF/MW) ceramic capacitors.

For RF and MW inductors, apply the $\lambda_{Physical}$ calculation for multilayer inductors described in the datasheet for (non RF/MW) magnetic components.

C_{sensitivity} factor

	Relative sensitivity (score out of 10)			C _{sensitivity}
	EOS	MOS	TOS	
Passive components for microwave applications, fixed: attenuator, load (50 Ω), filter, power divider (combiner, splitter)	2	4	1	2.4
Passive components for microwave applications, variable: variable attenuator, tuneable filter	2	4	1	2.4
Passive components for microwave applications, with ferrites: circulator, isolator, phase shifter	2	4	1	2.4
Passive components for microwave applications: surface wave filters	6	7	5	6.2

Part manufacturing factor Π_{PM}

The Π_{PM} factor is determined in the same way as for non-RF/MW discrete semiconductors.

For the choice of the ε factor, it is recommended that a factor of 4 be applied for large-volume applications (telecommunications, general public) and a factor of 2 for low-volume applications (defence, industrial, aerospace).

Basic failure rates associated with the component

Component description	$\lambda_{0_passive}$ RF/MW	γ_{TH-EL}	γ_{TCy}	γ_{Mech}	γ_{RH}
Passive components for microwave applications, fixed: attenuator, load (50 Ω), filter, power divider (combiner, splitter)	0.5	0.01	0.67	0.30	0.02
Passive components for microwave applications, variable: variable attenuator, tuneable filter	1	0.01	0.67	0.30	0.02
Passive components for microwave applications, with ferrites: circulator, isolator, phase shifter	1	0.01	0.69	0.30	0
Surface wave filters	3.75	0.01	0.67	0.30	0.02

Mission profile data

t_{phase}	Time associated with each phase (hours)
RH_{ambient}	Humidity level associated with a phase (%)
$T_{\text{ambient-board}}$	Mean temperature of the board during a phase (°C)
$\Delta T_{\text{cycling}}$	Amplitude of temperature variation associated with a cycling phase (°C)
$T_{\text{max-cycling}}$	Maximum temperature on the board during a cycling phase (°C)
N_{cy}	Number of cycles associated with each cycling phase (cycles)
θ_{cy}	Cycle duration (hours)
G_{RMS}	Vibration amplitude associated with each random vibration phase (G_{RMS})

Application data

$$\Delta T = P_{\text{dissipated}} \times R_{\text{CA}}$$

ΔT Component temperature rise

R_{CA} Component-ambient thermal resistance

$P_{\text{dissipated}}$ Power dissipated by the component during the phase (W)

In the case of pulse operation, it is important to factor in the duty cycle when calculating $\Pi_{\text{Thermo-electrical}}$.

For short pulses (shorter than the thermal time constant of the chip), ΔT must be determined using the thermal impedance Z instead of R_{CA} .

η Duty cycle during the phase

Factors contributing to physical stresses

$\Pi_{\text{Thermo-electrical}}$	In an operating phase: $\gamma_{TH-EL} \times e^{11604 \times 0.15 \times \left[\frac{1}{293} - \frac{1}{(T_{\text{ambient-board}} + \Delta T + 273)} \right]}$ In a pulse operating phase: $\eta \times \gamma_{TH-EL} \times e^{11604 \times 0.15 \times \left[\frac{1}{293} - \frac{1}{(T_{\text{ambient-board}} + \Delta T + 273)} \right]}$ In a non-operating phase: $\Pi_{\text{Thermo-electrical}} = 0$
Π_{TCy}	$\gamma_{TCy} \times \left(\frac{12 \times N_{cy}}{t_{\text{total}}} \right) \times \left(\frac{\min(\theta_{cy}, 2)}{2} \right)^{\frac{1}{3}} \times \left(\frac{\Delta T_{\text{cycling}}}{20} \right)^{1.9} \times e^{1414 \times \left[\frac{1}{313} - \frac{1}{(T_{\text{max-cycling}} + 273)} \right]}$
$\Pi_{\text{Mechanical}}$	$\gamma_{Meca} \times \left(\frac{G_{\text{RMS}}}{0.5} \right)^{1.5}$
Π_{RH}	$\gamma_{RH} \times \left(\frac{RH_{\text{ambient}}}{70} \right)^{4.4} \times e^{11604 \times 0.9 \times \left[\frac{1}{293} - \frac{1}{(T_{\text{ambient-board}} + 273)} \right]}$ In an operating phase: $\Pi_{RH} = 0$

COTS boards

**Chapter not updated in the
2020 version (October 2021)**

General

The COTS board model is intended for commercially available boards that perform standard electronic functions. This model is particularly useful for:

- Estimating the reliability of COTS boards for which the manufacturer has not provided any reliability information
- Estimating the reliability of COTS boards in environments other than the one for which the manufacturer has specified a reliability value
- Estimating the reliability of a set of COTS boards from different sources by reference to a common framework; when COTS board manufacturers provide reliability data, they do not necessarily specify the source, or the conditions under which the data is applicable.

One of the practical objectives of this model is to be applicable using information directly available on the COTS board. This is usually limited to the board's technical datasheet. However, the level of detail in COTS board technical datasheets is often low, which places certain limitations on the COTS board model:

- The proposed breakdown into electronic functions provides a means of describing boards that perform standard functions; it is not suitable for the description of specific boards that are not COTS boards
- The realism of the prediction should be assessed against the low level of information used.

It is recommended that COTS board manufacturers who would like to publish reliability information about their boards should use the component method rather than this board method.

Induced factor

Factors contributing to overstress

$$\Pi_{\text{induced}_i} = (\Pi_{\text{placement}} \times \Pi_{\text{application}_i} \times \Pi_{\text{ruggedizing}})^{0.511 \times \ln(C_{\text{sensitivity}})}$$

The index i denotes the phase considered.

Contributions to the $\Pi_{\text{placement}}$ and $C_{\text{sensitivity}}$ factors

Functions common to all boards	$\Pi_{\text{placement}}$	$C_{\text{sensitivity}}$
Common functions	1.3	6.1

Central digital functions	$\Pi_{\text{placement}}$	$C_{\text{sensitivity}}$
CPU function	1.0	6.1
Flash memory boot function (NOR)		
Flash memory storage function (NAND)		
DRAM memory function (DDR-SDRAM, SGRAM)		
L2, L3 cache or SRAM function		
SCSI controller function		
Chipset function (Northbridge, Southbridge)		

Peripheral digital functions	$\Pi_{\text{placement}}$	$C_{\text{sensitivity}}$
Ethernet control function (LAN)	1.3	6.1
Graphic control function (VGA)		
Fieldbus control function (ADC, ARINC, 1553)		
Wireless control function (Bluetooth, Wi-Fi)		
Analog-to-digital or digital-to-analog conversion function		

Input/outputs	$\Pi_{\text{placement}}$	$C_{\text{sensitivity}}$
Parallel bus digital line	1.6	6.1
Analog or discrete line	2.0	
Serial peripheral line (RS232, RS485, RS422, USB, mouse, keyboard, Ethernet)		
Serial bus line (ADC, ARINC, 1553)		
Input/output isolation by optocoupling		
Input/output isolation or switching by electromechanical relay		

Contribution to the $\Pi_{\text{application}}$ factor

This contribution is determined in the same way as for the components.

Contribution to the $\Pi_{\text{ruggedizing}}$ factor

This contribution is determined in the same way as for the components.

Part manufacturing factor

Model associated with the Π_{PM} factor

$$\Pi_{PM} = e^{1.39 \times (1 - \text{Part_Grade}) - 0.69}$$

Where:

$$\text{Part_Grade} = \left[\frac{(QA_{\text{manufacturer}} + QA_{\text{item}}) \times \varepsilon}{24} \right]$$

$QA_{\text{manufacturer}}$ factor

This factor is determined in the same way as for the components.

QA_{item} factor

Part quality assurance level	Position relative to state of the art	QA_{item}
Severe environment and accelerated stress tests performed	Higher	3
Known qualification / ESS procedure internal to the manufacturer	Equivalent	1
No information	Lower	0

Experience factor ε :

This factor is determined in the same way as for the components.

Embedded electronic functions

General model associated with the family

$$\lambda = \lambda_{\text{Physique}} \times \Pi_{\text{PM}} \times \Pi_{\text{Process}}$$

Where:

$$\lambda_{\text{Physical}} = \sum_i^{\text{Phases}} \left(\left(\frac{t_{\text{phase}}}{T_{\text{Total}}} \right) \times \lambda_{\text{Board}} \right)_i$$

$$\lambda_{\text{Board}} = \lambda_{\text{common functions}} + \sum_j^{\text{Functions}} \lambda_{\text{function } j}$$

$$\lambda_{\text{common functions}} = \gamma_{0 \text{ common functions}} \times \sum_k^{\text{Stress}} \left(\gamma_{\text{common functions}} \times \Pi_{\text{Acceleration}} \right)_k \times \left(\Pi_{\text{induced}} \right)_{\text{common functions}}$$

$$\lambda_{\text{function}} = \lambda_{0 \text{ function}} \times \sum_k^{\text{Stress}} \left(\gamma_{\text{function}} \times \Pi_{\text{Acceleration}} \right)_k \times \left(\Pi_{\text{induced}} \right)_{\text{function}}$$

Note: As with the other models, $\left(\Pi_{\text{induced}} \right)_{\text{common functions}}$ and $\left(\Pi_{\text{induced}} \right)_{\text{function}}$ need to be calculated for each phase.

Determination of distribution factors by stress

Function type	γ_{TH}	γ_{TCy} Solder joints	$\gamma_{\text{TCy Case}}$	γ_{Mech}	γ_{RH}	γ_{Chi}
Functions common to all boards	0.54	0.24	0.02	0.05	0.08	0.07
Central digital functions	0.38	0.50	0.04	0.03	0.05	0.00
Peripheral digital functions	0.38	0.50	0.04	0.03	0.05	0.00
Input/output functions						
• Input/output line (parallel bus, serial bus, discrete or analog line)	0.49	0.40	0.03	0.02	0.05	0.01
• Input/output isolation by optocoupling						
• Input/output isolation or switching by electromechanical relay	0.60	0.23	0.02	0.05	0.10	0.01

Basic failure rates

Functions common to all boards	λ_0 function
Common functions	155 x FF

Central digital functions	λ_0 function
CPU function	11
Flash memory boot function (NOR)	17 x D_flash_boot
Flash memory storage function (NAND)	19 x D_flash_stock
DRAM memory function (DDR-SDRAM, SGRAM)	23 x D_DRAM
L2, L3 cache or SRAM function	11 x D_SRAM
SCSI controller function	6
Chipset function (Northbridge, Southbridge)	16

Peripheral digital functions	λ_0 function
Ethernet control function (LAN)	12
Graphic control function (VGA)	24
Fieldbus control function (ADC, ARINC, 1553)	12
Wireless control function (Bluetooth, Wi-Fi)	11
Analog-to-digital or digital-to-analog conversion function	10

Input/output lines	λ_0 function
Parallel bus digital line	1.0 x M _{parallel}
Analog line or discrete digital line	1.2 x M _{analog}
Serial peripheral line (RS232, RS485, RS422, USB, mouse, keyboard, Ethernet)	2 x M _{serial}
Serial bus line (ADC, ARINC, 1553)	3 x M _{serial}
Input/output isolation by optocoupling	1 x quantity
Input/output isolation or switching by electromechanical relay	3 x quantity

Determination of the Form Factor (FF)

Board format	Length (mm)	Width (mm)	FF
lpack	99	45	0.12
PC104	96	90	0.23
PMC	149	74	0.30
EPIC	165	115	0.51
3U	160	100	0.43
mini ITX	170	170	0.78
6U	233	160	1.00
Flex-ATX	228	190	1.16
micro ATX	244	244	1.60
ATX	304	244	2.00

$$FF = \frac{\text{Width} \times \text{Length}}{37280}$$

Where Width and Length are the main dimensions of the board in millimetres. If no data is available on the size of the board, FF is taken as equal to 1.

Determination of memory density (D) factors

Flash memory boot NOR	
Size (MB)	D_flash_boot
≤ 4	1.00
8	1.41
16	2.00
32	2.83
64	4.00
128	5.66

$D_flash_boot = 1$ if size ≤ 4Mo

$$D_flash_boot = \left(\frac{\text{Size(MB)}}{4} \right)^{\frac{1}{2}} \text{ if size} > 4\text{Mo}$$

Flash memory storage NAND	
Size (MB)	D_flash_stock
≤ 512	1
1024	1.41
2048	2
4096	2.82

$D_flash_stock = 1$ if size ≤ 512Mo

$$D_flash_stock = \left(\frac{\text{Size(MB)}}{512} \right)^{\frac{1}{2}} \text{ if size} > 512\text{Mo}$$

DRAM memory	
Size (MB)	D_DRAM
≤ 256	1
512	1.41
1024	2
2048	2.82
4096	4

$D_DRAM = 1$ if size ≤ 256Mo

$$D_DRAM = \left(\frac{\text{Size(MB)}}{256} \right)^{\frac{1}{2}} \text{ if size} > 256\text{Mo}$$

SRAM memory	
Size (KB)	D_SRAM
≤ 512	1
1024	1.41
2048	2
4096	2.82

$D_SRAM = 1$ if size ≤ 512Ko

$$D_SRAM = \left(\frac{\text{Size(KB)}}{512} \right)^{\frac{1}{2}} \text{ if size} > 512\text{Ko}$$

If no technological data is available for the listed function, D is taken as equal to 1.

Determination of the input/output line multiplicity (M) factor

For parallel bus input/outputs: $M_{\text{parallel}} = \sum_{\text{Parallel bus}} (\text{Number of lines in bus})^{1/2}$

If the number of parallel input/outputs is not available, M_{parallel} is taken as equal to 4.

For analog input/outputs: $M_{\text{analog}} = (\text{Number of lines})^{1/2}$

If the number of parallel input/outputs is not available, M_{analog} is taken as equal to 3.

For serial bus input/outputs: $M_{\text{serial}} = \text{Number of serial bus interfaces}$

If the number of lines is not available, M_{serial} is taken as equal to 2.

The number of serial bus interfaces must take into consideration all the protocols present.

For example, if the board has 3 ADC buses and one ARINC bus, the relevant M_{serial} value will be 4.

Guide for the enumeration of functions

The analysis used to identify electronic functions of the COTS board being studied must be based on the specifications given in manufacturer datasheets. The recommended process is as follows:

1	Identify the dimensions of the board under study: calculate the form factor FF
2	Identify the central digital functions: these will necessarily be associated with the presence on the board of components that control part of the architecture
3	Identify the technological generation of the various memory resources on the board under study: calculate the memory density factors D
4	Identify the peripheral digital functions: these will necessarily be associated with the presence on the board of components that specifically manage a communication or signal processing protocol
5	Identify the number of buses and of parallel bus input/output lines: calculate the M_{parallel} factor Example: for a board that manages 4 parallel buses with 16 lines each, $M_{\text{parallel}} = 4 \times (16)^{1/2} = 16$
6	Identify the number of analog input/output lines: calculate the M_{analog} factor Example: for a board that manages 16 analog inputs, $M_{\text{analog}} = (16)^{1/2} = 4$
7	Identify the number of serial bus input/output lines: calculate the M_{serial} factor Example: a board that manages 3 accesses to an ADC bus and one access to an ARINC bus will have an ADC controller and an ARINC controller (and therefore 2 fieldbus control functions), 3 ADC serial bus lines and 1 ARINC serial bus line (M_{serial} factor = 4)
8	Identify input/output lines that have isolation barriers (optocoupling or electromechanical) and add the failure rates to the input/output lines concerned Example: for a board with an analog input that can be switched by 8 relays to 8 external sources, we will have $M_{\text{analog}} = 1$, plus 8 electromechanical relay-based barrier isolation functions

Mission profile data

t_{annual}	Time associated with each phase over one year (hours)
RH_{ambient}	Humidity level associated with a phase (%)
$T_{\text{ambient-board}}$	Mean temperature of the board during a phase (°C)
$\Delta T_{\text{cycling}}$	Amplitude of temperature variation associated with a cycling phase (°C)
$T_{\text{max-cycling}}$	Maximum temperature on the board during a cycling phase (°C)
N_{cy}	Number of cycles associated with each cycling phase (cycles)
θ_{cy}	Cycle duration (hours)
G_{RMS}	Stress associated with each random vibration phase (G_{RMS})

Pollution level (see tables)

Saline pollution level	Π_{sal}
Low	1
High	2

Product protection level	Π_{prot}
Hermetic	0
Non-hermetic	1

Application pollution level	Π_{zone}
Low	1
Moderate	2
High	4

Environmental pollution level	Π_{envir}
Low	1
Moderate	1.5
High	2

Factors contributing to physical stresses

$\Pi_{Thermal}$	In an operating phase: $e^{11604 \times 0.45 \times \left[\frac{1}{293} - \frac{1}{(T_{ambient-board} + 273)} \right]}$ In a non-operating phase: $\Pi_{thermal} = 0$
$\Pi_{Tcy_solder\ joints}$	$\left(\frac{12 \times N_{cy}}{t_{total}} \right) \times \left(\frac{\Delta T_{cycling}}{20} \right)^4 \times e^{1414 \times \left[\frac{1}{313} - \frac{1}{(T_{max-cycling} + 273)} \right]}$
$\Pi_{Tcy_packages}$	$\left(\frac{12 \times N_{cy}}{t_{total}} \right) \times \left(\frac{\min(\theta_{cy}, 2)}{2} \right)^{\frac{1}{3}} \times \left(\frac{\Delta T_{cycling}}{20} \right)^{1.9} \times e^{1414 \times \left[\frac{1}{313} - \frac{1}{(T_{max-cycling} + 273)} \right]}$
$\Pi_{Mechanical}$	$\left(\frac{G_{RMS}}{0.5} \right)^{1.5}$
Π_{RH} - Common functions - Input/output lines (parallel bus, analog or discrete, serial peripheral, serial bus) - Input/output isolation or switching by electromechanical relay	In a non-operating phase: $\left(\frac{RH_{ambient}}{70} \right)^{4.4} \times e^{11604 \times 0.9 \times \left[\frac{1}{293} - \frac{1}{(T_{ambient-board} + 273)} \right]}$ In an operating phase: $0.6 \times \left(\frac{RH_{ambient}}{70} \right)^{4.4} \times e^{11604 \times 0.9 \times \left[\frac{1}{293} - \frac{1}{(T_{ambient-board} + 273)} \right]}$
Π_{RH} Other functions	$\left(\frac{RH_{ambient}}{70} \right)^{4.4} \times e^{11604 \times 0.9 \times \left[\frac{1}{293} - \frac{1}{(T_{ambient-board} + 273)} \right]}$ In an operating phase: $\Pi_{RH} = 0$
$\Pi_{Chemical}$	$\Pi_{Sal} \times \Pi_{Indus} \times \Pi_{Zone} \times \Pi_{Prot}$

Miscellaneous subassemblies

**Chapter not modified in
the 2020 version**

General

Warning on the theoretical domain of applicability

Due to the way in which some of the subassembly models presented in this chapter are constructed, their behaviour in extreme environments (whether severe or, like storage, entirely non-aggressive) is not necessarily as representative as is expected for the component models.

These models should consequently be used with caution for extreme environments (whether severe or non-aggressive).

The subassemblies concerned are:

- LCD screens
- Hard drives
- CRT monitors
- Lithium and nickel batteries
- Fans
- Keyboards.

Operating life

Modelling

A product's operating life (often expressed in terms of the mean time to failure or MTTF), linked to ageing phenomena, is reflected by an increase in the failure rate, which can be modelled using a Weibull equation.

$$\lambda_{\text{Ageing}}(t) = \beta \cdot \frac{t^{\beta-1}}{\eta^{\beta}}$$

Where:

t Time
 β Shape factor
 η Scale factor.

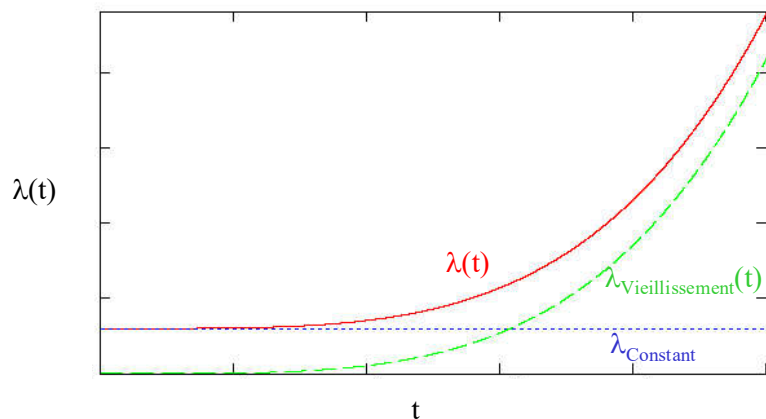
The lifetime is generally expressed using the parameter “L10”, which corresponds to the time by which 10% of wear-out failures occur.

$$\lambda_{\text{Ageing}}(t) = \beta \cdot \text{Ln}\left(\frac{1}{0.9}\right) \cdot \frac{t^{\beta-1}}{L10^{\beta}}$$

These failures – $\lambda_{\text{Ageing}}(t)$ – are added to the “random” failures modelled using an exponential equation with a constant failure rate.

$$\lambda(t) = \lambda_{\text{Constant}} + \lambda_{\text{Ageing}}(t)$$

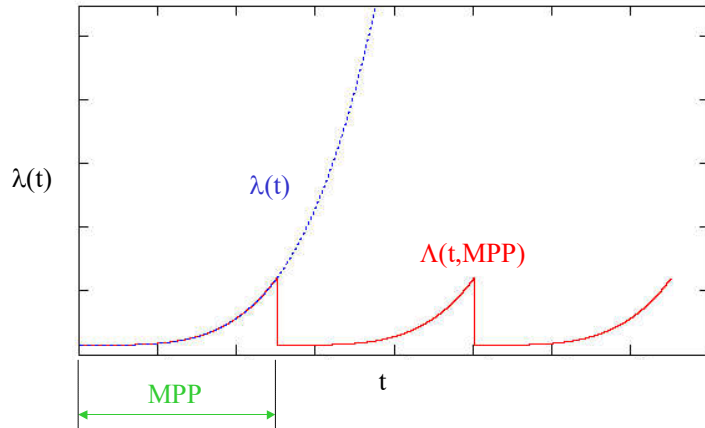
As illustrated by the following graph:



Preventive maintenance

To avoid cascading failures due to ageing, it is often useful to set up periodic preventive maintenance with a periodicity of “PrPM”.

The failure rate is then represented as follows:



When determining PrPM, one must consider cases where the subassembly has a longer lifetime than the system as a whole (for example the hard drive in a desktop computer). In this case, PrPM will be augmented by the lifetime of the complete system.

The PrPM should be seen as the periodicity with which the subassembly is replaced in the system. The subassembly itself is considered here to be a non-repairable item.

Mean failure rate

Strictly speaking, the failure rate should be modelled as a function of time. However, in light of the requirements, it is much more practical to use a mean failure rate.

$$\lambda = \lambda_{\text{Constant}} + \text{Ln}\left(\frac{1}{0.9}\right) \cdot \frac{\text{Pr PM}^{\beta-1}}{L10^{\beta}}$$

When the maintenance policy does not include preventive maintenance, the mean failure rate is calculated over the average period between two corrective maintenance (CM) operations due to wear-out failures.

$$\text{CM} = L10 \cdot \left(\frac{-\text{Ln}(0.5)}{\text{Ln}\left(\frac{1}{0.9}\right)} \right)^{\frac{1}{\beta}}$$

Some item models (subassemblies) therefore draw on the parameter λ_{Ageing} , which is constructed using the principles set out here.

Weibull shape factor β

The shape factor β is used to model the type of ageing. A value of β is proposed by default for every model with an λ_{Ageing} parameter. When the supplier of the subassembly provides a value for β , this is to be preferred to the default value.

Induced factor and part manufacturing factor

Factors contributing to overstress: Π_{Induced}

The Π_{Induced} factor is calculated in the same way as for the components.

The method for determining the $\Pi_{\text{placement}}$ parameter is defined in each item datasheet.

Model associated with the part manufacturing factor Π_{PM}

For all subassemblies, the Π_{PM} factor is calculated in the same way as for COTS boards.

LCD (TFT, STN) screens

General model associated with the family

ⓘ Caution: Limited lifetime

$$\lambda = \lambda_{\text{Cst}} + \lambda_{\text{Ageing}}$$

Where:

$$\lambda_{\text{Cst}} = \lambda_{\text{Physical}} \times \Pi_{\text{PM}} \times \Pi_{\text{Process}}$$

and:

$$\lambda_{\text{Physical}} = \sum_i^{\text{Phases}} \left(\frac{t_{\text{phase}}}{T_{\text{Total}}} \right)_i \times (\lambda_{\text{Screen_Thermal}} \times \Pi_{\text{Thermal}} + \lambda_{\text{Screen_Mechanical}} \times \Pi_{\text{Mechanical}})_i \times (\Pi_{\text{Induced}})_i$$

and:

$$\lambda_{\text{Ageing}} = 0.105 \cdot 10^9 \times \frac{\min(\text{PrPM}, \text{CM})^{\beta-1}}{\left(\frac{\text{OL}}{T_{\text{use}}} \right)^{\beta}}$$

where:

- OL: operating life (L10).
LCD screens have a limited lifetime, due mainly to their back lighting lamps.
In the absence of manufacturer data, use OL = 40 000 hours of operation.
- PrPM: mean calendar time between periodic preventive maintenance operations.
If the maintenance policy does not include preventive maintenance, assume PrPM = CM.
- CM: mean calendar time between corrective maintenance operations due solely to wear-out

failures:

$$\text{CM} = \frac{\text{OL}}{T_{\text{use}}} \times 6.579^{\frac{1}{\beta}}$$

- T_{use} : rate of use during the mission profile (sum of operating times divided by total time).
- β : Weibull shape factor; by default $\beta = 3$.

Contributions to the $C_{\text{sensitivity}}$ factor

LCD screens		EOS	MOS	TOS	$C_{\text{sensitivity}}$
	TFT	7	2	1	2.40
	STN	3	2	1	1.80

Contributions to the $\Pi_{\text{placement}}$ factor

LCD screens		$\Pi_{\text{placement}}$
	Portable	1.6
	Fixed	1.0

Failure rates associated with the subassembly

Description of subassembly	$\lambda_{\text{Screen_Mechanical}}$	$\lambda_{\text{Screen_Thermal}}$	Ea (eV) activation energy
TFT LCD screens	$\Pi_c \times (126 \times D^{1.11})$	$\Pi_c \times \left(193 \times e^{\frac{P}{115}} \right)$	0.6
STN LCD screens	$\Pi_c \times (11 \times D^{2.48})$	$\Pi_c \times \left(96.5 \times e^{\frac{P}{10.8}} \right)$	0.5

Technical data

D Screen size, diagonal (inches). $6'' < D_{\text{TFT}} < 70''$ and $6'' < D_{\text{STN}} < 17''$

P Power (Watts). $P_{\text{TFT}} < 300\text{W}$ and $P_{\text{STN}} < 40\text{W}$

Remark: If P is unknown, use $P(D) = 2.4 \times e^{0.18 \times D}$, for $6'' < D < 20''$.

Determination of the class factor Π_c

Simplified classification in accordance with ISO13406-2:

Class	Maximum number of pixels, sub-pixels or dead clusters per million pixels					Class factor Π_c
	On pixels Type 1	Off pixels Type 2	Sub-pixels Type 3	Clusters Type 1 or 2	Clusters Type 3	
I	0	0	0	0	0	2.48
II	2	2	5	0	2	1.00
III	5	15	50	0	5	0.46
IV	50	150	500	5	50	0.28

This table shows the number of defects beyond which a screen is considered defective. For example, in Class II, a screen with only 2 bad pixels per million is not considered defective.

Mission profile data:

t_{annual}	Time associated with each phase over one year (hours)
T_{ambient}	Mean ambient temperature associated with a phase (°C)
G_{RMS}	Stress associated with each random vibration phase (G_{RMS})

Factors contributing to physical stresses:

Π_{Thermal}	In an operating phase: $e^{11604 \times E_a \times \left[\frac{1}{293} - \frac{1}{(T_{\text{ambient}} + 273)} \right]}$ In a non-operating phase: $\Pi_{\text{thermal}} = 0$
$\Pi_{\text{Mechanical}}$	$\left(\frac{G_{\text{RMS}}}{0.5} \right)^{1.5}$

Hard drives (EIDE, SCSI)

General model associated with the family:

ⓘ Caution: Limited lifetime

$$\lambda = \lambda_{\text{Cst}} + \lambda_{\text{Ageing}}$$

Where:

$$\lambda_{\text{Cst}} = \lambda_{\text{Physical}} \times \Pi_{\text{PM}} \times \Pi_{\text{Process}}$$

and:

$$\lambda_{\text{Physical}} = \sum_i^{\text{Phases}} \left(\frac{t_{\text{annual}}}{T_{\text{Total}}} \right)_i \times \left(\lambda_{\text{Hard-drive_Thermal}} \times \Pi_{\text{Thermal}} + \lambda_{\text{Hard-drive_Mechanical}} \times \Pi_{\text{Mechanical}} \right)_i \times (\Pi_{\text{Induced}})_i$$

and:

$$\lambda_{\text{Ageing}} = 0.105 \cdot 10^9 \times \frac{\min(\text{PrPM}, \text{MC})^{\beta-1}}{\left(\frac{\text{DDV}}{T_{\text{use}}} \right)^{\beta}}$$

Where:

- OL: operating life (L10).
Hard drives have a limited lifetime, due mainly to wear on mechanical moving parts.
In the absence of manufacturer data, use OL = 50 000 hours of operation.
- PrPM: mean calendar time between periodic preventive maintenance operations.
If the maintenance policy does not include preventive maintenance, assume PrPM = CM.
- CM: mean calendar time between corrective maintenance operations due solely to wear-out

failures:

$$\text{CM} = \frac{\text{OL}}{T_{\text{use}}} \times 6.579^{\frac{1}{\beta}}$$

- T_{use} : rate of use during the mission profile (sum of operating times divided by total time).
- β : Weibull shape factor; by default $\beta = 4.5$.

Contributions to the $C_{\text{sensitivity}}$ factor

			<i>EOS</i>	<i>MOS</i>	<i>TOS</i>	$C_{\text{sensitivity}}$
Hard drives	Normal use	Protection against shocks/vibrations	2	6	2	4.00
		Qualification for shocks/vibrations	2	8	2	5.00
		No specific protection or qualification	2	10	2	6.00
	Intensive use, ventilated	Protection against shocks/vibrations	2	6	5	5.05
		Qualification for shocks/vibrations	2	8	5	6.05
		No specific protection or qualification	2	10	5	7.05
	Intensive use, non-ventilated	Protection against shocks/vibrations	2	6	8	6.10
		Qualification for shocks/vibrations	2	8	8	7.10
		No specific protection or qualification	2	10	8	8.10

Contributions to the $\Pi_{\text{placement}}$ factor

		$\Pi_{\text{placement}}$
Hard drives	Portable or rack	2.5
	Fixed	1.8

Failure rate associated with the subassembly

Description of subassembly	$\lambda_{\text{Hard-drive_Mechanical}}$	$\lambda_{\text{Hard-drive_Thermal}}$
IDE hard drive family (IDE, EIDE, E-IDE, ATA, SATA, S-ATA, Ultra ATA, DMA, Ultra DMA, etc.)	$[425 - 208 \times \ln(Ft)]$	$\left[5,2 + \left(\frac{Ta}{9,6} \right)^{4,97} \right]$
SCSI hard drive family (SCSI-1, 2, 3, Ultra Wide SCSI 1, 2, 3, 4, SAS, etc.)	$[205 - 100 \times \ln(Ft)]$	$\left[2,5 + \left(\frac{Ta}{11,1} \right)^{4,97} \right]$

Description of technological factors

Ft Format of hard drive (inches). $1'' < Ft < 5.25''$

Ta Mean access time (ms). $Ta < 20\text{ms}$

Pc Platter count

Remark: if Pc is unknown, use: $Pc = \text{Whole_part} \left(\frac{1 + Nh}{2} \right)$ where Nh: Number of heads.

Application data: Calculation of the load factor Π_L

$$\Pi_L(P_c, D_c) = \frac{P_c \times D_c + 3}{4}$$

Where:

P_c Platter count

D_c Duty cycle, defined by:

$$D_c = \frac{\left(\sum_a \text{Access_time} + \sum_b \text{Read_time} + \sum_c \text{Write_time} \right)}{\text{Time_in_use}}$$

However, in the absence of data on the duty cycle, the following default values, based on type of use, can be adopted:

Type of use	Dc
Office or home application, excluding high-speed data transfers	10%
Net server (applications, data)	50%
High speed data transfers (server or client)	100%

Mission profile data

t_{annual} Time associated with each phase over one year (hours)
 T_{ambient} Mean ambient temperature associated with a phase (°C)
 G_{RMS} Stress associated with each random vibration phase (G_{RMS})

Factors contributing to physical stresses:

Π_{Thermal}	In an operating phase: $\Pi_L \times e^{11604 \times 0.785 \times \left[\frac{1}{293} - \frac{1}{(T_{\text{ambient}} + 273)} \right]}$ In a non-operating phase: $\Pi_{\text{thermal}} = 0$
$\Pi_{\text{Mechanical}}$	$\Pi_L \times \left(\frac{G_{\text{RMS}}}{0.5} \right)^{1.5}$

CRT monitors

General model associated with the family

① Caution: Limited lifetime

$$\lambda = \lambda_{\text{Cst}} + \lambda_{\text{Ageing}}$$

Where:

$$\lambda_{\text{Cst}} = \lambda_{\text{Physical}} \times \Pi_{\text{PM}} \times \Pi_{\text{Process}}$$

and:

$$\lambda_{\text{Physical}} = \sum_i^{\text{Phases}} \left(\frac{t_{\text{phase}}}{T_{\text{Total}}} \right)_i \times \left(\begin{array}{c} \lambda_{\text{Moni_Thermal}} \times \Pi_{\text{Thermal}} \\ + \lambda_{\text{Moni_TCy}} \times \Pi_{\text{TCy}} \\ + \lambda_{\text{Moni_Mechanical}} \times \Pi_{\text{Mechanical}} \\ + \lambda_{\text{Moni_RH}} \times \Pi_{\text{RH}} \end{array} \right)_i \times (\Pi_{\text{Induced}})_i$$

and:

$$\lambda_{\text{Ageing}} = 0.105 \cdot 10^9 \times \frac{\min(\text{PrPM}, \text{CM})^{\diamond-1}}{\left(\frac{\text{OL}}{T_{\text{use}}} \right)^{\diamond}}$$

where:

- OL, operating life (L10).
CRT monitors have a limited lifetime, due mainly to loss of precision in electron bombardment and, in particular, to deterioration of the phosphor layer.
In the absence of manufacturer data, use OL = 20 000 hours of operation.
- PrPM: mean calendar time between periodic preventive maintenance operations;
If the maintenance policy does not include preventive maintenance, assume PrPM = CM.
- CM: mean calendar time between corrective maintenance operations due solely to wear-out

failures:

$$\text{CM} = \frac{\text{OL}}{T_{\text{use}}} \times 6.579^{\frac{1}{\beta}}$$

- T_{use} , rate of use during the mission profile (sum of operating times divided by total time).
- β : Weibull shape factor; by default $\beta = 2.5$.

Contributions to the $C_{\text{sensitivity}}$ factor

	EOS	MOS	TOS	$C_{\text{sensitivity}}$
CRT monitors	2	5	1	3.15

Contribution to the $\Pi_{\text{placement}}$ factor

	$\Pi_{\text{placement}}$
CRT monitors	1.4

Failure rate associated with the subassembly

$\lambda_{\text{Moni_Mechanical}}$	$\lambda_{\text{Moni_TCy}}$	$\lambda_{\text{Moni_Thermal}}$	$\lambda_{\text{Moni_RH}}$
$\left(262 + e^{\frac{Wt-14.4}{3.97}} \right)$	$\left(524 + e^{\frac{Wt-11.7}{3.97}} \right)$	$\left[13.6 + \left(\frac{P}{40.7} \right)^{2.5} + \frac{215}{\sqrt{Fh}} \right]$	$\left(128 + e^{\frac{D-16.9}{1.18}} \right)$

Description of technological factors

Wt	Weight of monitor without casing (kg): $Wt < 40\text{kg}$
D	Screen size, diagonal (inches): $D < 25''$
Fh	Maximum horizontal scanning frequency (kHz): $30\text{kHz} < Fh < 150\text{ kHz}$
P	Maximum operating power (Watts): $P < 200\text{W}$

Remark: If P is unknown, use: $P(D) = 0.78 \times D^{1.72}$

Mission profile data

t_{phase}	Time associated with each phase over one year (hours)
T_{ambient}	Mean ambient temperature associated with a phase (°C)
RH_{ambient}	Humidity level associated with a phase (%)
G_{RMS}	Stress associated with each random vibration phase (G_{RMS})

Application data

Π_{Prot}	Protection level of the subassembly
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Factors contributing to physical stresses

Π_{Thermal}	In an operating phase: $e^{11604 \times 0.35 \times \left[\frac{1}{293} - \frac{1}{T_{\text{ambient}} + 273} \right]}$ In a non-operating phase: $\Pi_{\text{thermal}} = 0$						
Π_{TCy}	$\left(\frac{12 \times N_{\text{cy}}}{t_{\text{total}}} \right) \times \left(\frac{\min(\theta_{\text{cy}}, 2)}{2} \right)^{\frac{1}{3}} \times \left(\frac{\Delta T_{\text{cycling}}}{20} \right)^{1.9} \times e^{1414 \times \left[\frac{1}{313} - \frac{1}{T_{\text{max-cycling}} + 273} \right]}$						
$\Pi_{\text{Mechanical}}$	$\left(\frac{G_{\text{RMS}}}{0.5} \right)^{1.5}$						
Π_{RH}	$\Pi_{\text{Prot}} \times \left(\frac{RH_{\text{ambient}}}{70} \right)^{4.4} \times e^{11604 \times 0.8 \times \left[\frac{1}{293} - \frac{1}{T_{\text{ambient}} + 273} \right]}$ <table> <tr> <td>Protection level of the subassembly:</td><td>Value of Π_{Prot}</td></tr> <tr> <td>Hermetic</td><td>0</td></tr> <tr> <td>Non-hermetic</td><td>1</td></tr> </table>	Protection level of the subassembly:	Value of Π_{Prot}	Hermetic	0	Non-hermetic	1
Protection level of the subassembly:	Value of Π_{Prot}						
Hermetic	0						
Non-hermetic	1						

AC/DC and DC/DC voltage converters

General model associated with the family

$$\lambda = \lambda_{\text{Physical}} \times \Pi_{\text{PM}} \times \Pi_{\text{Process}}$$

Where:

$$\lambda_{\text{Physical}} = \sum_i^{\text{Phases}} \left(\frac{t_{\text{phase}}}{T_{\text{Total}}} \right)_i \times \left(\lambda_{0\text{TH-TCy}} \times (\gamma_{\text{TH}} \times \Pi_{\text{TH}} + \gamma_{\text{TCy}} \times \Pi_{\text{TCy}}) + \lambda_{0\text{M-RH}} \times (\gamma_{\text{M}} \times \Pi_{\text{M}} + \gamma_{\text{RH}} \times \Pi_{\text{RH}}) \right)_i \times (\Pi_{\text{Induced}})_i$$

Contributions to the $C_{\text{sensitivity}}$ factor

	EOS	MOS	TOS	$C_{\text{sensitivity}}$
AC/DC and DC/DC converters	8.4	3.4	1	5.90

Contributions to the $\Pi_{\text{placement}}$ factor

	$\Pi_{\text{placement}}$
AC/DC and DC/DC converters	1.6

Basic failure rates associated with the subassembly

	λ_0 AC/DC	λ_0 DC/DC	γ_{TH}	γ_{TCy}	γ_{M}	γ_{RH}
$\lambda_{0\text{TH-TCy}}$	$\sqrt{1150 + 86 \cdot \sqrt{P} \cdot \ln(P)}$	$(3.4 + 0.27 \cdot \sqrt{P})^2$	0.359	0.523		
$\lambda_{0\text{M-RH}}$	$\frac{1}{6.48 \cdot 10^{-3} + 0.296 \cdot \frac{\ln(\text{Vol})}{\text{Vol}}}$	$6.4 \cdot \text{Vol}^{0.38} - 0.79$			0.090	0.028

Description of technological factors

	AC/DC	DC/DC
P: Output power (W)	5 W < P < 7 000 W	0.5 W < P < 1 000 W
Vol: Volume in cm ³	25 cm ³ < Vol < 10 000 cm ³	0.5 cm ³ < Vol < 3 500 cm ³
If Vol is unknown, use:	$\text{Vol}(\text{Wt}) = 1.4 \times \text{Wt}$	$\text{Vol}(\text{Wt}) = \left(\frac{\text{Wt}}{4.2} \right)^{1.3}$
Where Wt: Weight in g	20 g < Wt < 7 000 g	10 g < Wt < 2 200 g

Mission profile data

t_{annual}	Time associated with each phase over one year (hours)
RH_{ambient}	Humidity level associated with a phase (%)
T_{ambient}	Mean temperature of the subassembly during a phase (°C)
$\Delta T_{\text{cycling}}$	Amplitude of temperature variation associated with a cycling phase (°C)
$T_{\text{max-cycling}}$	Maximum temperature of the subassembly during a cycling phase (°C)
N_{cy}	Number of cycles associated with each cycling phase (cycles)
θ_{cy}	Cycle duration (hours)
G_{RMS}	Vibration amplitude associated with each random vibration phase (G_{RMS})

Technical data

Type of case Moulded or other case.

Factors contributing to physical stresses

Π_{Thermal}	In an operating phase: $e^{11604 \times 0.44 \times \left[\frac{1}{293} - \frac{1}{(T_{\text{ambient}} + \Delta T + 273)} \right]}$ Where: moulded cases, $\Delta T = 15^\circ\text{C}$ other packages, $\Delta T = 10^\circ\text{C}$ In a non-operating phase: $\Pi_{\text{Thermal}} = 0$
Π_{TCy}	$\left(\frac{12 \times N_{\text{cy}}}{t_{\text{total}}} \right) \times \left(\frac{\min(\theta_{\text{cy}}, 2)}{2} \right)^{\frac{1}{3}} \times \left(\frac{\Delta T_{\text{cycling}}}{20} \right)^{2.5} \times e^{1414 \times \left[\frac{1}{313} - \frac{1}{(T_{\text{max-cycling}} + 273)} \right]}$
$\Pi_{\text{Mechanical}}$	$\left(\frac{G_{\text{RMS}}}{0.5} \right)^{1.5}$
Π_{RH}	In a non-operating phase: $\left(\frac{RH_{\text{ambient}}}{70} \right)^{4.4} \times e^{11604 \times 0.6 \times \left[\frac{1}{293} - \frac{1}{(T_{\text{ambient}} + 273)} \right]}$ In an operating phase: $\Pi_{\text{RH}} = 0$

Lithium and nickel batteries

General model associated with the family

ⓘ **Caution: Limited lifetime**

$$\lambda = \lambda_{\text{Cst}} + \lambda_{\text{Ageing}}$$

Where:

$$\lambda_{\text{Cst}} = \lambda_{\text{Physical}} \times \Pi_{\text{PM}} \times \Pi_{\text{Process}}$$

and:

$$\lambda_{\text{Physical}} = \lambda_{0_Battery} \times N_{\text{cells}} \times \sum_i^{\text{Phases}} \left(\frac{t_{\text{phase}}}{T_{\text{Total}}} \right)_i \times (\Pi_{\text{Thermal}} + \Pi_{\text{TCy}} + \Pi_{\text{Mechanical}})_i \times (\Pi_{\text{Induced}})_i$$

where:

- N_{cells} is the number of cells in the battery. If N_{cells} is unknown, use $N_{\text{cells}} = 1$.

and:

$$\lambda_{\text{Ageing}} = 0.105 \cdot 10^9 \times \frac{\min(\text{PrPM}, \text{CM})^{\beta-1}}{\min(\text{OL}_1, \text{OL}_2)^{\beta}}$$

where:

- OL_1, OL_2 : calendar life durations (L10).
Batteries have a limited lifetime (OL_1) due to the number of charge/discharge cycles. The life (OL_2) of lithium batteries is also limited in time, independently of the number of charge/discharge cycles. The failure criterion used for estimating these life durations is generally a battery capacity below a specified threshold (usually between 60% and 80% of initial capacity).

In the absence of manufacturer data, use the following:

		Specific energy density (Wh/Kg)	NbCD: # charge/discharge cycles	OL ₂ in calendar hours
Lithium	Li-ion Polymer	-	300	20 000
	Li-ion (1)	≤ 100	300	20 000
		100 < ≤ 200	500	
		> 200	1 000	
	Li-metal Phosphate	-	2 000	35 000
	Nano Titanium	-	10 000	150 000
Nickel	NiMH Metal Hydride (2)	≤ 50	300	NA
		> 50	1 000	
	Ni-Zn Zinc	-	1 000	
	Ni-Cd Cadmium	-	2 000	

(1) and (2): if the specific energy density is unknown, use NbCD = 500

$$OL_1 = \frac{NbCD}{NbCD_{Annual}} \times 8760 \text{ (8760, number of hours per year for an annual mission profile)}$$

Where:

- $NbCD_{Annual}$: Number of charge/discharge cycles per year.
- PrPM: mean calendar time between periodic preventive maintenance operations.
If the maintenance policy does not include preventive maintenance, assume $PrPM = CM$.
- CM: mean calendar time between corrective maintenance operations due solely to wear-out failures:
$$CM = \min(OL_1, OL_2) \times 6.579^{\frac{1}{\beta}}$$
- β : Weibull shape factor; by default $\beta = 5.0$.

Contributions to the $C_{sensitivity}$ factor

	EOS	MOS	TOS	$C_{sensitivity}$
Lithium and nickel batteries	7	7	1	6.40

Contribution to the $\Pi_{placement}$ factor

	$\Pi_{placement}$
Lithium and nickel batteries	1.3

Basic failure rates associated with the subassembly

Description of subassembly	λ_0 - Battery	Activation energy (eV)	γ_{TH}	γ_{TCy}	γ_{Mech}
Nickel: NiMH Metal Hydride, Ni-Zn Zinc, Ni-Cd Cadmium	0.21	0.40	0.85	0.14	0.01
Lithium: Li-ion, Nano Titanium					
Lithium: Li-ion Polymer	0.29				
Lithium: Li-metal Phosphate	0.40				

Mission profile data

t_{phase}	Time associated with each operating phase (hours)
$T_{\text{ambient-board}}$	Mean temperature of the board during a phase (°C)
$\Delta T_{\text{cycling}}$	Amplitude of temperature variation associated with a cycling phase (°C)
$T_{\text{max-cycling}}$	Maximum temperature on the board during a cycling phase (°C)
N_{cy}	Number of cycles associated with each cycling phase (cycles)
θ_{cy}	Cycle duration (hours)
G_{RMS}	Stress associated with each random vibration phase (G_{RMS})

Factors contributing to physical stresses

Π_{Thermal}	In an operating phase: $\gamma_{\text{TH}} \times e^{11604 \times E_a \times \left[\frac{1}{293} - \frac{1}{(T_{\text{ambient-board}} + 273)} \right]}$ In a non-operating phase: $\Pi_{\text{Thermal}} = 0$
Π_{TCy}	$\gamma_{\text{TCy}} \times \left(\frac{12 \times N_{\text{cy}}}{t_{\text{total}}} \right) \times \left(\frac{\min(\theta_{\text{cy}}, 2)}{2} \right)^{\frac{1}{3}} \times \left(\frac{\Delta T_{\text{cycling}}}{20} \right)^{1.9} \times e^{1414 \times \left[\frac{1}{313} - \frac{1}{(T_{\text{max-cycling}} + 273)} \right]}$
$\Pi_{\text{Mechanical}}$	$\gamma_{\text{Mech}} \times \left(\frac{G_{\text{RMS}}}{0.5} \right)^{1.5}$

Fans

General model associated with the family

ⓘ Caution: Limited lifetime

$$\lambda = \lambda_{\text{Cst}} + \lambda_{\text{Ageing}}$$

Where:

$$\lambda_{\text{Cst}} = \lambda_{\text{Physical}} \times \Pi_{\text{PM}} \times \Pi_{\text{Process}}$$

and:

$$\lambda_{\text{Physical}} = \lambda_{0_Fan} \times \sum_i^{\text{Phases}} \left(\frac{t_{\text{phase}}}{T_{\text{Total}}} \right)_i \times (\Pi_{\text{Thermo-electrical}} + \Pi_{\text{TCy}} + \Pi_{\text{Mechanical}} + \Pi_{\text{RH}})_i \times (\Pi_{\text{Induced}})_i$$

and:

$$\lambda_{\text{Ageing}} = 0.105 \cdot 10^9 \times \frac{\min(\text{PrPM}, \text{CM})^{\beta-1}}{(\text{OL}_{\text{calendar}})^{\beta}}$$

where:

- $\text{OL}_{\text{calendar}}$, lifetime in calendar hours.
Fans have a limited lifetime (OL), due mainly to wear on bearings. As fan technologies can vary considerably, it is preferable to take the manufacturer's data, which is generally available in the form of an operating life (L10). This operating life is assumed to depend on the operating temperature. The lifetime in calendar hours is deduced from the lifetime in each of the different life phases by means of the following formula:

$$\text{OL}_{\text{calendar}} = \left(\sum_i^{\text{Phases}} \left(\frac{t_{\text{annual}}}{8760} \right)_i \times \left(\frac{1}{\text{OL}_i} \right) \right)^{-1}$$

Where:

$$\left(\frac{1}{\text{OL}_i} \right) = 0 \text{ for non-operating phases.}$$

For operating phases, $\left(\frac{1}{\text{OL}} \right)$ is calculated from the operating life (L10) provided by the manufacturer, where available. In the absence of manufacturer data, use:

$$\text{OL} = 79200 \times \Pi_{\text{Type}} \times \left[3.53 - 0.744 \times \ln(N) \right] \times e^{-Ea \cdot 11604 \cdot \left(\frac{1}{313} - \frac{1}{T+273} \right)} \times \left(\frac{\text{RS}}{3000} \right)^{-m}$$

Where:

N, Noise in dBA

RS, Rotation speed in revs/minute (rpm)

T, Temperature around the bearing in the phase considered (°C)

If T is unknown, use $T = 30^\circ\text{C}$

And $T = 1.1 \times T_{\text{amb}} + 12.5$

Where T_{amb} is the ambient temperature in the phase considered (°C)

10 dBA < N < 90 dBA

1500 rpm < RS < 5000 rpm

30°C < T < 90°C

for $-40^\circ\text{C} \leq T_{\text{amb}} \leq 16^\circ\text{C}$

for $16^\circ\text{C} < T_{\text{amb}} \leq 70^\circ\text{C}$

And:

Type of bearing	Π_{type}		Ea	m
	AC	DC		
Sealed sleeve bearing	0.50	0.46	0.46	1.60
Single ball bearing (ball/sleeve) Hydrodynamic (hypro)	0.70	0.64	0.40	1.23
Ball bearing (dual)	1.0	0.92	0.28	0.93
Ceramic bearing	1.4	1.3	0.21	0.57

- PrPM: mean calendar time between periodic preventive maintenance operations.
If the maintenance policy does not include preventive maintenance, assume PrPM = CM.
- CM: mean calendar time between corrective maintenance operations due solely to wear-out failures:

$$CM = OL_{calendar} \times 6.579^{\frac{1}{\beta}}$$

- β : Weibull shape factor; by default $\beta = 2.2$.

C_{sensitivity} factor

	Relative sensitivity (score out of 10)			C _{sensitivity}
	EOS	MOS	TOS	
Fan	3.7	9.1	2.4	5.5

Contributions to the $\Pi_{placement}$ factor

	$\Pi_{placement}$
Fan	1.6

Basic failure rates associated with the subassembly

Component description	λ_{0-Fan}	γ_{Th}	γ_{TCy}	γ_{Mech}	γ_{Rh}
Fan	0.17	0.51	0.31	0.08	0.11

Mission profile data

t_{phase}	Time associated with each phase over one year (hours)
T_{ambient}	Mean ambient temperature during a phase (°C)
$\Delta T_{\text{cycling}}$	Amplitude of temperature variation associated with a cycling phase (°C)
$T_{\text{max-cycling}}$	Maximum temperature on the board during a cycling phase (°C)
N_{cy}	Number of cycles associated with each cycling phase over one year (cycles)
θ_{cy}	Cycle duration (hours)
G_{RMS}	Stress associated with each random vibration phase (Grms)
RH_{ambient}	Humidity level associated with a phase (%)

Factors contributing to physical stresses

Π_{Thermal}	In an operating phase: $\gamma_{\text{Th}} \times e^{11604 \times 0.15 \times \left[\frac{1}{293} - \frac{1}{(T+273)} \right]}$ In a non-operating phase: $\Pi_{\text{Thermal}} = 0$ T: temperature in the vicinity of the bearing during a phase (°C) If T is unknown, use: For $-40^{\circ}\text{C} \leq T_{\text{ambient}} \leq 16$ $T = 30^{\circ}\text{C}$ For $16^{\circ}\text{C} < T_{\text{ambient}} \leq 70^{\circ}\text{C}$ $T = 1.1 \times T_{\text{ambient}} + 12.5$
Π_{Tcy}	$\gamma_{\text{Tcy}} \times \left(\frac{12 \times N_{\text{cy}}}{t_{\text{total}}} \right) \times \left(\frac{\min(\theta_{\text{cy}}, 2)}{2} \right)^{\frac{1}{3}} \times \left(\frac{\Delta T_{\text{cycling}}}{20} \right)^{1.9} \times e^{1414 \times \left[\frac{1}{313} - \frac{1}{(T_{\text{max-cycling}}+273)} \right]}$
$\Pi_{\text{Mechanical}}$	$\gamma_{\text{M}} \times \left(\frac{G_{\text{RMS}}}{0.5} \right)^{1.5}$
Π_{RH}	$\gamma_{\text{Rh}} \times \left(\frac{RH_{\text{ambient}}}{70} \right)^{4.4} \times e^{11604 \times 0.8 \times \left[\frac{1}{293} - \frac{1}{(T_{\text{ambient}}+273)} \right]}$

Keyboards

General

ⓘ Caution: Limited lifetime

This model describes short-stroke and long-stroke membrane keyboards.

The keyboard model includes the mechanical part of the keyboard (the keys) and the keyboard circuit board that provides the electrical interface. The two elements are separate in the model.

The keyboard model does not take account of the different options that may be included in some keyboards (card readers, wireless functionality, pointing device, etc.).

General model associated with the family

$$\lambda = \lambda_{Cst} + \lambda_{Ageing}$$

Where:

$$\lambda_{Cst} = \lambda_{Physical} \times \Pi_{PM} \times \Pi_{Process}$$

and:

$$\lambda_{Physical} = \lambda_{Keyboard} + \lambda_{Board}$$

$$\lambda_{Keyboard} = \lambda_{0_Keyboard} \times \sum_i^{Phases} \left(\frac{t_{phase}}{T_{Total}} \right)_i \times \left(\frac{\Pi_{Th_Keyboard} + \Pi_{TCy_Keyboard}}{\Pi_{Rh_Keyboard} + \Pi_{Mechanical_Keyboard}} \right)_i \times (\Pi_{Induced})_i$$

$$\lambda_{Board} = \lambda_{0_Board} \times \sum_i^{Phases} \left(\frac{t_{phase}}{T_{Total}} \right)_i \times \left(\frac{\Pi_{Th_Board} + \Pi_{TCy_Board} + \Pi_{Rh_Board}}{\Pi_{Mechanical_Board} + \Pi_{Chemical_Board}} \right)_i \times (\Pi_{Induced})_i$$

and:

$$\lambda_{Ageing} = 0.105 \cdot 10^9 \times \frac{\min(PrPM, CM)^{\beta-1}}{OL^{\beta}}$$

where:

- OL, calendar lifetime (L10).
Membrane keyboards have a limited lifetime. It is always preferable to use the manufacturer data, which is generally available. In the absence of information, however, use:

$$OL = \frac{N_{Strikes}}{Frq}$$

Where:

- $N_{Strikes}$ Number of keyboard strikes before failure.
 - Frq Frequency of keyboard strikes (number of strikes/calendar hour).
- If $N_{Strikes}$ is unknown, use:

$$N_{Strikes} = \left(0.69 + 0.34 \times Weight^{2.5} \right) \times \left(N_{Keys} \right)^{1-\frac{1}{\beta}} \times 10^6$$

Where N_{Keys} is the number of keys on the keyboard.

- Weight is the weight of the keyboard (kg) 0.1 Kg < Wt < 2 Kg.
- β : Weibull shape factor; by default $\beta = 2.4$.

- PrPM: mean calendar time between periodic preventive maintenance operations. If the maintenance policy does not include preventive maintenance, assume PrPM = CM.
- CM: mean calendar time between corrective maintenance operations due solely to wear-out failures:

$$CM = OL \times 6.579^{\frac{1}{\beta}}$$

Example: A desktop computer keyboard for which the number of cycles before failure is $N_{\text{Strikes}}=10^6$ strikes has an average keystroke frequency in the order of 3000 keystrokes per day, i.e. $Frq=3000/24=125$ keystrokes per calendar hour. Its operating life is therefore: $OL=10^6/125=80\,000$ calendar hours.

The same keyboard, used at a rate of 180 strikes/minute for 2 hours a day, corresponding to a frequency $Frq=900$ strikes per calendar hour, would have an operating life of: $OL=10^6/900=11\,000$ calendar hours.

Contributions to the $C_{\text{sensitivity}}$ factor

	$C_{\text{sensitivity}}$
Membrane keyboard	7.2
Keyboard circuit board	4.7

Contribution to the $\Pi_{\text{placement}}$ factor

	$\Pi_{\text{placement}}$
Keyboard and keyboard circuit board	1.6

Basic failure rates associated with the keyboard subassembly

Component description		λ_0 -Keyboard	γ_{Th} -Keyboard	γ_{Tcy} -Keyboard	γ_{RH} -Keyboard	γ_{Mech} -Keyboard
Membrane keyboard	Up to 20 keys	1.0	0.27	0.13	0.32	0.28
	From 20 to 70 keys	2.0	0.38	0.10	0.19	0.33
	From 70 to 95 keys	3.0	0.45	0.10	0.12	0.33
	From 95 to 120 keys	3.6	0.45	0.10	0.12	0.33
	More than 120 keys	4.5	0.45	0.10	0.12	0.33

Basic failure rates associated with the keyboard circuit board subassembly

Component description	Phases	λ_0 -Board	γ_{Th} -Board	γ_{Tcy_B} -Board	$\gamma_{\text{Tcy}_{JB}}$ -Board	γ_{RH} -Board	γ_{Mech} -Board	γ_{Chem} -Board
Keyboard circuit board	ON	2.9	0.21	0.24	0.23	0.03	0.07	0.22
	OFF	2.9	0	0.24	0.23	0.24	0.07	0.22

Mission profile data

t_{phase}	Time associated with each phase (hours)
T_{ambient}	Mean ambient temperature during a phase (°C)
$\Delta T_{\text{cycling}}$	Amplitude of temperature variation associated with a cycling phase (°C)
$T_{\text{max-cycling}}$	Maximum temperature on the board during a cycling phase (°C)
N_{cy}	Number of cycles associated with each cycling phase (cycles)
θ_{cy}	Cycle duration (hours)
G_{RMS}	Stress associated with each random vibration phase (G_{RMS})
RH_{ambient}	Humidity level associated with a phase (%)

Pollution level (see tables):

Saline pollution level	Π_{sal}
Low	1
High	2

Product protection level	Π_{prot}
Hermetic	0
Non-hermetic	1

Application pollution level	Π_{zone}
Low	1
Moderate	2
High	4

Environmental pollution level	Π_{envir}
Low	1
Moderate	1.5
High	2

Factors contributing to physical stresses

$\Pi_{Th \text{ Keyboard}}$	$\gamma_{Th\text{-Keyboard}} \times e^{11604 \times 0.25 \times \left[\frac{1}{293} - \frac{1}{(T_{\text{ambient}} + 273)} \right]}$
$\Pi_{Tcy \text{ Keyboard}}$	$\gamma_{Tcy\text{-Keyboard}} \times \left(\frac{12 \times N_{cy}}{T_{\text{phase}}} \right) \times \left(\frac{\min(\theta_{cy}, 2)}{2} \right)^{\frac{1}{3}} \times \left(\frac{\Delta T_{\text{cycling}}}{20} \right)^{1.9} \times e^{1414 \times \left[\frac{1}{313} - \frac{1}{T_{\text{max-cycling}} + 273} \right]}$
$\Pi_{Mechanical \text{ Keyboard}}$	$\gamma_{Mech\text{-Keyboard}} \times \left(\frac{G_{\text{RMS}}}{0.5} \right)^{1.5}$
$\Pi_{RH \text{ Keyboard}}$	$\gamma_{RH\text{-Keyboard}} \times \left(\frac{RH_{\text{ambient}}}{70} \right)^{4.4} \times e^{11604 \times 0.86 \times \left[\frac{1}{293} - \frac{1}{(T_{\text{ambient}} + 273)} \right]}$
$\Pi_{Th \text{ Board}}$	$\gamma_{Th\text{-Board}} \times e^{11604 \times 0.27 \times \left[\frac{1}{293} - \frac{1}{(T_{\text{ambient}} + 273)} \right]}$ $\gamma_{Th\text{-Board}}$ differs between operating and non-operating mode
$\Pi_{Tcy \text{ Board}}$	$\Pi_{Tcy} = \Pi_{Tcy_B} + \Pi_{Tcy_JB}$ $\Pi_{Tcy_B} = \gamma_{Tcy_B\text{-carte}} \times \left(\frac{12 \times N_{cy}}{T_{\text{phase}}} \right) \times \left(\frac{\min(\theta_{cy}, 2)}{2} \right)^{\frac{1}{3}} \times \left(\frac{\Delta T_{\text{cyclage}}}{20} \right)^{1.27} \times e^{1414 \times \left[\frac{1}{313} - \frac{1}{T_{\text{max-cyclage}} + 273} \right]}$ $\Pi_{Tcy_JB} = \gamma_{Tcy_JB\text{-carte}} \times \left(\frac{12 \times N_{cy}}{T_{\text{phase}}} \right) \times \left(\frac{\min(\theta_{cy}, 2)}{2} \right)^{\frac{1}{3}} \times \left(\frac{\Delta T_{\text{cyclage}}}{20} \right)^{1.9} \times e^{1414 \times \left[\frac{1}{313} - \frac{1}{T_{\text{max-cyclage}} + 273} \right]}$
$\Pi_{Mechanical \text{ Board}}$	$\gamma_{Mech\text{-Board}} \times \left(\frac{G_{\text{RMS}}}{0.5} \right)^{1.5}$
$\Pi_{RH \text{ Board}}$	$\gamma_{RH\text{-Board}} \times \left(\frac{RH_{\text{ambient}}}{70} \right)^{4.4} \times e^{11604 \times Ea \times \left[\frac{1}{293} - \frac{1}{(T_{\text{ambiente}} + 273)} \right]}$ $\gamma_{RH\text{-Board}}$ differs between operating and non-operating mode In an operating phase: $Ea = 0.80$ In a non-operating phase: $Ea = 0.84$
$\Pi_{Chemical \text{ Board}}$	$\gamma_{Chem\text{-Board}} \times \Pi_{\text{Sal}} \times \Pi_{\text{Envir}} \times \Pi_{\text{Zone}} \times \Pi_{\text{Prot}}$

Parts-count reliability prediction

General principles

The “parts count” method of calculation is a simplified approach to reliability assessment, established by taking averages for the various parameters listed in the component datasheets. This simplified method does not therefore necessarily produce pessimistic or cautious results. The failure rate values obtained for a set of components will be that much closer to a detailed calculation where there is a large quantity, and above all diversity, of components. Conversely, larger variances may emerge in the case of assemblies made up of a small number of different component types. The stresses applied to the components are almost always determined by default, on the basis of commonly observed levels that comply with standard practice.

The parts count prediction method allows for rapid reliability assessments so that efforts to study reliability and build reliability models can then be focused on the most sensitive areas. This method is therefore especially useful for the reliability studies on very large systems for which it is not necessary to describe millions of components in detail.

Note: This method should be distinguished from the COTS board method, which remains the most suitable method for evaluating COTS boards.

Mission profile and physical stresses

The mission profile should be described in the same way as when applying the complete detailed method. The only simplification made concerns information about chemical stresses, which are set at default values and do not need to be input.

Process audit, Π_{process}

Π_{process} should be considered in the same way as for the complete application of the FIDES methodology. Using the default value may reduce the accuracy of the final results.

Induced factor, Π_{induced}

Π_{induced} should be considered in the same way as for the complete application of the FIDES methodology, except for $\Pi_{\text{placement}}$, which should be considered directly at the level of the object to be calculated rather than at component level. The C_sensitivity factor is specified for each component family in the type-based counting tables provided below.

Part manufacturing factor Π_{PM}

Π_{PM} is simplified and used as part of a component selection policy, to be considered globally and not by component family.

$$\Pi_{PM} = e^{1.39 \times (1 - \text{Part_Grade}) - 0.69}$$

$$\text{Where: Part_Grade} = \left[\frac{(QA_{\text{manufacturer}} + QA_{\text{component}} + RA_{\text{component}}) \times \varepsilon}{36} \right]$$

$QA_{\text{manufacturer}}$

The quality assurance level of the selected manufacturers is most often (for example):	Position relative to state of the art	QA _{manufacturer}
Certified under ISO/TS16949 V2002	Higher	3
Certified under one of: QS9000, TL9000, ISO/TS 29001, EN9100, AS9100, JISQ 9100, AQAP 2110, AQAP 2120, AQAP 2130, IRIS, IEC TS 62239, ESA/SCC QPL, MIL-PRF-38535 QML, MIL-PRF-19500	Equivalent	2
Certified under ISO 9000 version 2000	Lower	1
No information	Much lower	0

$QA_{\text{component}}$

The quality assurance level of the selected components is most often (for example):	Position relative to state of the art	QA _{component}
Qualification according to AEC Q100, Q101, or JESD47	Higher	3
Qualification according to JESD22, JEP143 or QML	Equivalent	2
Manufacturer qualification program, unidentified production sites	Lower	1
No information	Much lower	0

$RA_{\text{component}}$

The level of testing on most of the selected components is most often (for example):	Position relative to state of the art	QA _{component}
Very severe tests are usually carried out	Very reliable – Level A	3
Severe tests are usually carried out	Very reliable – Level B	2
Tests are usually carried out	Reliable	1
No tests	Not reliable	0

Typical tests are given, for example for integrated circuits, in the detailed method.

Experience factor ε

The selected manufacturers are most often:	Position relative to state of the art	ε factor
Known, with mature processes	Very low risk	4
Known, with processes that are not yet analysed or are not mature	Low risk	3
Not known	Risk	2
Previous disqualifications, problems observed, etc.	High risk	1

General model associated with all families

$$\lambda = \lambda_{\text{Physical}} \times \Pi_{\text{PM}} \times \Pi_{\text{Process}}$$

Where:

$$\lambda_{\text{Physical}} = \sum_i^{\text{Phases}} \left(\frac{t_{\text{phase}}}{T_{\text{Total}}} \right) \times (\lambda_{\text{ECU}})_i + \sum_i^{\text{Phases}} \left(\frac{t_{\text{phase}}}{T_{\text{Total}}} \right)_i \times \left(\begin{array}{c} \lambda_{0\text{TH}} \times \Pi_{\text{Thermal}} \\ + \lambda_{0\text{TCyCase}} \times \Pi_{\text{TCyCase}} \\ + \lambda_{0\text{TCySolderJoints}} \times \Pi_{\text{TCySolderJoints}} \\ + \lambda_{0\text{RH}} \times \Pi_{\text{RH}} \\ + \lambda_{0\text{Mech}} \times \Pi_{\text{Mech}} \end{array} \right)_i \times (\Pi_{\text{Induced}})_i$$

The failure rate λ_{ECU} represents the Electrical (relays and switches), Chemical and Wear-out contributions, where the response to environmental conditions is null or is ignored when using count-based reliability methods. However, λ_{ECW} does differ as between operating mode (under power) and non-operating mode (switched off).

Mission profile data

t_{annual}	Time associated with each phase over one year (hours)
$\text{RH}_{\text{ambient}}$	Humidity level associated with a phase (%)
T_{ambient}	Mean temperature during a phase (°C)
$\Delta T_{\text{cycling}}$	Amplitude of temperature variation associated with a cycling phase (°C)
$T_{\text{max-cycling}}$	Maximum temperature on the board during a cycling phase (°C)
N_{cy}	Number of cycles associated with each cycling phase (cycles)
θ_{cy}	Cycle duration (hours)
GRMS	Vibration amplitudes random associated with each phase (GRMS)

Factors contributing to physical stresses

Π_{Thermal}	$e^{11604 \times E_a \times \left[\frac{1}{T_0 + 273} - \frac{1}{T_{\text{ambient}} + \Delta T \times e^{-\alpha \cdot T_{\text{ambient}} + 273}} \right]}$
Π_{TCyCase}	$\left(\frac{12 \times N_{\text{cy}}}{T_{\text{phase}}} \right) \times \left(\frac{\Delta T_{\text{cycling}}}{20} \right)^{m_B} \times e^{1414 \times \left[\frac{1}{313} - \frac{1}{T_{\text{max-cycling}} + 273} \right]}$
$\Pi_{\text{TCySolder_joints}}$	$\left(\frac{12 \times N_{\text{cy}}}{T_{\text{phase}}} \right) \times \left(\frac{\min(\theta_{\text{cy}}, 2)}{2} \right)^{\frac{1}{3}} \times \left(\frac{\Delta T_{\text{cycling}}}{20} \right)^{m_{JB}} \times e^{1414 \times \left[\frac{1}{313} - \frac{1}{T_{\text{max-cycling}} + 273} \right]}$
$\Pi_{\text{Mechanical}}$	$\left(\frac{G_{\text{RMS}}}{0.5} \right)^n$
Π_{RH}	$\left(\frac{\text{RH}_{\text{ambient}}}{70} \right)^{4.4} \times e^{11604 \times E_a \times \left[\frac{1}{293} - \frac{1}{T_{\text{ambient}} + 273} \right]}$

Parts count: Parameters

Parameters associated with operating (ON) phases for counting by type

			ON phases: Average parameters by default per type of item													Induced	
			Thermal					Humidity		Thermal cycling				Mechanical		El.Ch.Wr	Csensitivity
			λTh	Ea_Th	To	ΔT	α	λRh	Ea_Rh	λTcy_B	m_B	λTcy_JB	m_JB	λM	n	λECW	
Digital integrated circuit	Hermetic	≤ 24 p	0.021	0.7	20	3	0	0	0	0.002	4	0.012	1.9	0.00028	1.5	0	6.3
		24 to 48 p	0.021	0.7	20	5	0	0	0	0.0044	4	0.041	1.9	0.0012	1.5	0	6.3
		48 to 144 p	0.021	0.7	20	7	0	0	0	0.0084	4	0.11	1.9	0.0036	1.5	0	6.3
		144 to 288 p	0.021	0.7	20	10	0	0	0	0.016	4	0.30	1.9	0.011	1.5	0	6.3
		> 288 p	0.021	0.7	20	14	0	0	0	0.034	4	0.96	1.9	0.043	1.5	0	6.3
	Non-hermetic	≤ 24 p	0.021	0.7	20	3	0	0	0	0.002	4	0.012	1.9	0.00028	1.5	0	6.3
		24 to 48 p	0.021	0.7	20	5	0	0	0	0.0044	4	0.041	1.9	0.0012	1.5	0	6.3
		48 to 144 p	0.021	0.7	20	7	0	0	0	0.0084	4	0.11	1.9	0.0036	1.5	0	6.3
		144 to 288 p	0.021	0.7	20	10	0	0	0	0.016	4	0.30	1.9	0.011	1.5	0	6.3
		> 288 p	0.021	0.7	20	14	0	0	0	0.034	4	0.96	1.9	0.043	1.5	0	6.3
Memory integrated circuit	Hermetic	≤ 24 p	0.054	0.7	20	6	0	0	0	0.002	4	0.012	1.9	0.00028	1.5	0	6.3
		24 to 48 p	0.054	0.7	20	8	0	0	0	0.0044	4	0.041	1.9	0.0012	1.5	0	6.3
		48 to 144 p	0.054	0.7	20	12	0	0	0	0.0084	4	0.11	1.9	0.0036	1.5	0	6.3
		144 to 288 p	0.054	0.7	20	17	0	0	0	0.016	4	0.30	1.9	0.011	1.5	0	6.3
		> 288 p	0.054	0.7	20	24	0	0	0	0.034	4	0.96	1.9	0.043	1.5	0	6.3
	Non-hermetic	≤ 24 p	0.054	0.7	20	6	0	0	0	0.002	4	0.012	1.9	0.00028	1.5	0	6.3
		24 to 48 p	0.054	0.7	20	8	0	0	0	0.0044	4	0.041	1.9	0.0012	1.5	0	6.3
		48 to 144 p	0.054	0.7	20	12	0	0	0	0.0084	4	0.11	1.9	0.0036	1.5	0	6.3
		144 to 288 p	0.054	0.7	20	17	0	0	0	0.016	4	0.30	1.9	0.011	1.5	0	6.3
		> 288 p	0.054	0.7	20	24	0	0	0	0.034	4	0.96	1.9	0.043	1.5	0	6.3
Micro*, DSP, simple ASIC integrated circuit	Hermetic	≤ 24 p	0.075	0.7	20	8	0	0	0	0.002	4	0.012	1.9	0.00028	1.5	0	6.3
		24 to 48 p	0.075	0.7	20	11	0	0	0	0.0044	4	0.041	1.9	0.0012	1.5	0	6.3
		48 to 144 p	0.075	0.7	20	17	0	0	0	0.0084	4	0.11	1.9	0.0036	1.5	0	6.3
		144 to 288 p	0.075	0.7	20	23	0	0	0	0.016	4	0.30	1.9	0.011	1.5	0	6.3
		> 288 p	0.075	0.7	20	33	0	0	0	0.034	4	0.96	1.9	0.043	1.5	0	6.3

				ON phases: Average parameters by default per type of item													Induced				
				Thermal					Humidity		Thermal cycling				Mechanical		El.Ch.Wr	Csensitivity			
				λTh	Ea_Th	To	ΔT	α	λRh	Ea_Rh	λTcy_B	m_B	λTcy_JB	m_JB	λM	n	λECW				
Non-hermetic				≤ 24 p	0.075	0.7	20	8	0	0	0	0.002	4	0.012	1.9	0.00028	1.5	0	6.3		
				24 to 48 p	0.075	0.7	20	11	0	0	0	0.0044	4	0.041	1.9	0.0012	1.5	0	6.3		
				48 to 144 p	0.075	0.7	20	17	0	0	0	0.0084	4	0.11	1.9	0.0036	1.5	0	6.3		
				144 to 288 p	0.075	0.7	20	23	0	0	0	0.016	4	0.30	1.9	0.011	1.5	0	6.3		
				> 288 p	0.075	0.7	20	33	0	0	0	0.034	4	0.96	1.9	0.043	1.5	0	6.3		
Integrated circuit Analog, Mixed, FPGA, CPLD, complex ASIC				≤ 24 p	0.14	0.7	20	10	0	0	0	0.002	4	0.012	1.9	0.00028	1.5	0	6.3		
				24 to 48 p	0.14	0.7	20	15	0	0	0	0.0044	4	0.041	1.9	0.0012	1.5	0	6.3		
				48 to 144 p	0.14	0.7	20	22	0	0	0	0.0084	4	0.11	1.9	0.0036	1.5	0	6.3		
				144 to 288 p	0.14	0.7	20	30	0	0	0	0.016	4	0.30	1.9	0.011	1.5	0	6.3		
				> 288 p	0.14	0.7	20	43	0	0	0	0.034	4	0.96	1.9	0.043	1.5	0	6.3		
				≤ 24 p	0.14	0.7	20	10	0	0	0	0.002	4	0.012	1.9	0.00028	1.5	0	6.3		
				24 to 48 p	0.14	0.7	20	15	0	0	0	0.0044	4	0.041	1.9	0.0012	1.5	0	6.3		
				48 to 144 p	0.14	0.7	20	22	0	0	0	0.0084	4	0.11	1.9	0.0036	1.5	0	6.3		
				144 to 288 p	0.14	0.7	20	30	0	0	0	0.016	4	0.30	1.9	0.011	1.5	0	6.3		
				> 288 p	0.14	0.7	20	43	0	0	0	0.034	4	0.96	1.9	0.043	1.5	0	6.3		
Diode				Non-hermetic	SMD	low-power	0.0075	0.7	20	10	0	0	0	0.0008	4	0.0040	1.9	0.000080	1.5	0	5.2
						power	0.15	0.7	20	43	0	0	0	0.0041	4	0.021	1.9	0.00041	1.5	0	5.2
				TH	low-power	0.0075	0.7	20	10	0	0	0	0.0011	4	0.0055	1.9	0.00011	1.5	0	5.2	
					power	0.15	0.7	20	43	0	0	0	0.003	4	0.015	1.9	0.00030	1.5	0	5.2	
				Hermetic	SMD		0.079	0.7	20	27	0	0	0	0.0078	4	0.039	1.9	0.00078	1.5	0	5.2
					TH		0.079	0.7	20	27	0	0	0	0.01	4	0.051	1.9	0.0010	1.5	0	5.2
Transistor				Non-hermetic	SMD	low-power	0.014	0.7	20	10	0	0	0	0.0008	4	0.0040	1.9	0.00008	1.5	0	5.2
						power	0.034	0.7	20	43	0	0	0	0.0041	4	0.021	1.9	0.00041	1.5	0	5.2
					TH	low-power	0.014	0.7	20	10	0	0	0	0.0011	4	0.0055	1.9	0.00011	1.5	0	5.2
						power	0.034	0.7	20	43	0	0	0	0.003	4	0.015	1.9	0.0003	1.5	0	5.2
				Hermetic	SMD		0.024	0.7	20	27	0	0	0	0.0078	4	0.039	1.9	0.00078	1.5	0	5.2
					TH		0.024	0.7	20	27	0	0	0	0.01	4	0.051	1.9	0.001	1.5	0	5.2
Photodiode				Non-hermetic	SMD	low-power	0.05	0.4	20	7	0	0	0	0.0008	4	0.014	1.9	0.0051	1.5	0	5.2
						power	0.05	0.4	20	29	0	0	0	0.0041	4	0.031	1.9	0.0054	1.5	0	5.2
					TH	low-power	0.05	0.4	20	7	0	0	0	0.0011	4	0.016	1.9	0.0051	1.5	0	5.2

				ON phases: Average parameters by default per type of item														Induced
				Thermal					Humidity		Thermal cycling				Mechanical		El.Ch.Wr	Csensitivity
				λTh	Ea_Th	To	ΔT	α	λRh	Ea_Rh	λTcy_B	m_B	λTcy_JB	m_JB	λM	n	λECW	
	Hermetic		power	0.05	0.4	20	29	0	0	0	0.003	4	0.025	1.9	0.0053	1.5	0	5.2
			SMD	0.05	0.4	20	18	0	0	0	0.0078	4	0.049	1.9	0.0058	1.5	0	5.2
				TH	0.05	0.4	20	18	0	0	0	0.01	4	0.061	1.9	0.0060	1.5	0
Phototransistor	Non-hermetic	SMD	low-power	0.11	0.4	20	7	0	0	0	0.0008	4	0.025	1.9	0.011	1.5	0	5.2
			power	0.11	0.4	20	29	0	0	0	0.0041	4	0.042	1.9	0.011	1.5	0	5.2
		TH	low-power	0.11	0.4	20	7	0	0	0	0.0011	4	0.027	1.9	0.011	1.5	0	5.2
			power	0.11	0.4	20	29	0	0	0	0.003	4	0.036	1.9	0.011	1.5	0	5.2
	Hermetic	SMD	0.11	0.4	20	18	0	0	0	0.0078	4	0.060	1.9	0.012	1.5	0	5.2	
		TH	0.11	0.4	20	18	0	0	0	0.01	4	0.072	1.9	0.012	1.5	0	5.2	
Potentiometer				0.13	0.15	20	33	0	0	0	0	1	0.11	1.9	0.066	1.5	0	2.5
Resistor			high-power	0.010	0.15	20	78	0	0	0	0	1	0.37	1.9	0.0040	1.5	0	2.3
			low-power	0.0050	0.15	20	12	0	0	0	0	1	0.19	1.9	0.0020	1.5	0	2.8
Etched metal foil resistor				0.021	0.15	20	26	0	0	0	0	1	0.097	1.9	0.012	1.5	0	5.8
Resistor network, Resistor chip				0.00021	0.15	20	7	0	0	0	0	1	0.020	1.9	0.00021	1.5	0	4.5
Ceramic capacitor			moderate CV product	0.048	0.1	20	0	0	0	0	0	1	0.019	1.9	0.0014	1.5	0	6.1
			high CV product	0.12	0.1	20	0	0	0	0	0	1	0.075	1.9	0.0074	1.5	0	6.1
Aluminium capacitor				0.26	0.4	20	0	0	0	0	0	1	0.043	1.9	0.0031	1.5	0	6.4
Tantalum capacitor			gel	0.33	0.15	20	0	0	0	0	0	1	0.0043	1.9	0.052	1.5	0	7.0
			solid	0.54	0.15	20	0	0	0	0	0	1	0.072	1.9	0.013	1.5	0	7.0
Inductor				0.013	0.15	20	17	0	0	0	0	1	0.024	1.9	0.0043	1.5	0	5.5
Transformer			low-power	0.0013	0.15	20	10	0	0	0	0	1	0.091	1.9	0.033	1.5	0	6.9
			high-power	0.038	0.15	20	30	0	0	0	0	1	0.17	1.9	0.040	1.5	0	6.8
Quartz crystal resonator			through-hole	0.23	0	20	0	0	0	0	0	1	0.37	1.9	0.22	1.5	0	4.6
			SMD	0.13	0	20	0	0	0	0	0	1	0.47	1.9	0.12	1.5	0	4.6
Quartz crystal oscillator			through-hole	1.65	0	20	0	0	0	0	0	1	0.68	1.9	0.22	1.5	0	8.1
			SMD	1.65	0	20	0	0	0	0	0	1	0.94	1.9	0.12	1.5	0	8.1
Relay up to 10 contacts	Resistive load		< 3 actuations/h	2.6	0.25	40	57	0.024	0.13	0.9	0	1	0.10	1.9	0.76	1.5	0.12	7.55
			≥ 3 actuations/h	26	0.25	40	57	0.024	0.13	0.9	0	1	0.10	1.9	7.6	1.5	1.2	7.55
	Non-resistive load		< 3 actuations/h	2.6	0.25	40	57	0.024	0.13	0.9	0	1	0.10	1.9	0.76	1.5	2.4	7.55
			≥ 3 actuations/h	26	0.25	40	57	0.024	0.13	0.9	0	1	0.10	1.9	7.6	1.5	24	7.55

			ON phases: Average parameters by default per type of item															Induced	
			Thermal					Humidity		Thermal cycling				Mechanical		El.Ch.Wr	Csensitivity		
			λTh	Ea_Th	To	ΔT	α	λRh	Ea_Rh	λTcy_B	m_B	λTcy_JB	m_JB	λM	n	λECW			
Relay > 10 contacts	Resistive load	< 3 actuations/h		6.2	0.25	40	57	0.024	0.31	0.9	0	1	0.24	1.9	1.8	1.5	0.28	7.55	
		≥ 3 actuations/h		62	0.25	40	57	0.024	0.31	0.9	0	1	0.24	1.9	18	1.5	2.8	7.55	
	Non-resistive load	< 3 actuations/h		6.2	0.25	40	57	0.024	0.31	0.9	0	1	0.24	1.9	1.8	1.5	5.7	7.55	
		≥ 3 actuations/h		62	0.25	40	57	0.024	0.31	0.9	0	1	0.24	1.9	18	1.5	57	7.55	
Push button, switch, invertor (< 4 pins)			Resistive load		0.45	0.25	40	52	0.024	0.070	0.9	0	1	0.041	1.9	0.32	1.5	0.030	7.45
DIP, encoder (4 to 10 pins)					0.88	0.25	40	52	0.024	0.14	0.9	0	1	0.081	1.9	0.63	1.5	0.058	7.45
Switch (> 10 pins)					1.9	0.25	40	52	0.024	0.30	0.9	0	1	0.18	1.9	1.4	1.5	0.13	7.45
Push button, switch, invertor (< 4 pins)			Non-resistive load		0.45	0.25	40	52	0.024	0.070	0.9	0	1	0.041	1.9	0.32	1.5	0.59	7.45
DIP, encoder (4 to 10 pins)					0.88	0.25	40	52	0.024	0.14	0.9	0	1	0.081	1.9	0.63	1.5	1.2	7.45
Switch (> 10 pins)					1.9	0.25	40	52	0.024	0.30	0.9	0	1	0.18	1.9	1.4	1.5	2.6	7.45
Connector for printed circuits and IC supports	soldered (PTH or SMD)	up to 20 contacts		0.29	0.1	20	5	0	0.066	0.8	0	1	0.020	1.9	0.025	1.5	0.22	4.4	
		20 to 200 contacts		0.93	0.1	20	5	0	0.21	0.8	0	1	0.064	1.9	0.080	1.5	0.69	4.4	
		over 200 contacts		2.1	0.1	20	5	0	0.47	0.8	0	1	0.14	1.9	0.18	1.5	1.5	4.4	
	wrapping or insertion	up to 20 contacts		0.073	0.1	20	5	0	0.016	0.8	0	1	0.0051	1.9	0.0063	1.5	0.055	4.4	
		20 to 200 contacts		0.23	0.1	20	5	0	0.052	0.8	0	1	0.016	1.9	0.020	1.5	0.17	4.4	
		over 200 contacts		0.52	0.1	20	5	0	0.12	0.8	0	1	0.036	1.9	0.045	1.5	0.39	4.4	
Other connector	soldered (PTH or SMD)	up to 20 contacts		0.18	0.1	20	5	0	0.039	0.8	0	1	0.012	1.9	0.015	1.5	0.13	4.4	
		20 to 200 contacts		0.56	0.1	20	5	0	0.12	0.8	0	1	0.038	1.9	0.048	1.5	0.42	4.4	
		over 200 contacts		1.2	0.1	20	5	0	0.28	0.8	0	1	0.086	1.9	0.11	1.5	0.93	4.4	
	wrapping or insertion	up to 20 contacts		0.044	0.1	20	5	0	0.010	0.8	0	1	0.0030	1.9	0.0038	1.5	0.033	4.4	
		20 to 200 contacts		0.14	0.1	20	5	0	0.031	0.8	0	1	0.010	1.9	0.012	1.5	0.10	4.4	
		over 200 contacts		0.31	0.1	20	5	0	0.070	0.8	0	1	0.021	1.9	0.027	1.5	0.23	4.4	
Printed circuit, PTH		up to 2000 mounting points		0	0	40	0	0	0.090	0.8	0	1	0.30	1.9	0.10	1.5	0.022	6.5	
		> 2000 mounting points		0	0	40	0	0	0.36	0.8	0	1	1.2	1.9	0.40	1.5	0.087	6.5	
Printed circuit, SMD		up to 2000 mounting points		0	0	40	0	0	1.2	0.8	0	1	4.1	1.9	1.4	1.5	0.30	6.5	
		> 2000 mounting points		0	0	40	0	0	4.9	0.8	0	1	16	1.9	5.5	1.5	1.2	6.5	
Hard drive		EIDE		8.2	0.785	20	0	0	0	0	0	1	0	1	230	1.5	510	5.0	
		SCSI		2.9	0.785	20	0	0	0	0	0	1	0	1	110	1.5	510	5.0	
LCD screen (Class III)		STN	up to 14”		449	0.5	20	0	0	0	0	1	0	1	3520	1.5	1167	1.8	

			ON phases: Average parameters by default per type of item														Induced
			Thermal					Humidity		Thermal cycling				Mechanical		El.Ch.Wr	Csensitivity
			λTh	Ea_Th	To	ΔT	α	λRh	Ea_Rh	λTcy_B	m_B	λTcy_JB	m_JB	λM	n	λECW	
		14” to 20”	713	0.5	20	0	0	0	0	0	1	0	1	5697	1.5	1167	1.8
	TFT		137	0.6	20	0	0	0	0	0	1	0	1	1346	1.5	1167	2.4
			over 20”	252	0.6	20	0	0	0	0	0	1	0	1	2156	1.5	1167
CRT monitor	up to 20”		42	0.35	20	0	0	64	0.8	0	1	526	1.9	263	1.5	2858	3.2
	over 20”		89	0.35	20	0	0	269	0.8	0	1	796	1.9	401	1.5	2858	3.2
Switching power supply	up to 500W		77	0.44	20	0	0	0	0	0	1	112	2.5	23	1.5	0	5.9
	over 500W		142	0.44	20	0	0	0	0	0	1	207	2.5	26	1.5	0	5.9
DC/DC converter	up to 50W		19	0.44	20	0	0	0	0	0	1	27	2.5	6.4	1.5	0	5.9
	over 50W		47	0.44	20	0	0	0	0	0	1	68	2.5	12	1.5	0	5.9

SMD: Surface Mounted Device

TH: Through-Hole

ECW: Electrical, Chemical, Wear

Parameters associated with non-operating (OFF) phases for counting by type

			OFF phases: Average parameters by default per type of item													Induced	
			Thermal					Humidity		Thermal cycling				Mechanical		El.Ch.Wr	Csensitivity
			λTh	Ea_Th	To	ΔT	α	λRh	Ea_Rh	λTcy_B	m_B	λTcy_JB	m_JB	λM	n	λECW	
Digital integrated circuit	Hermetic	≤ 24 p	0	0	0	0	0	0	0.9	0.002	4	0.012	1.9	0.00028	1.5	0	6.3
		24 to 48 p	0	0	0	0	0	0	0.9	0.0044	4	0.041	1.9	0.0012	1.5	0	6.3
		48 to 144 p	0	0	0	0	0	0	0.9	0.0084	4	0.11	1.9	0.0036	1.5	0	6.3
		144 to 288 p	0	0	0	0	0	0	0.9	0.016	4	0.30	1.9	0.011	1.5	0	6.3
		> 288 p	0	0	0	0	0	0	0.9	0.034	4	0.96	1.9	0.043	1.5	0	6.3
	Non-hermetic	≤ 24 p	0	0	0	0	0	0.011	0.9	0.002	4	0.012	1.9	0.00028	1.5	0	6.3
		24 to 48 p	0	0	0	0	0	0.031	0.9	0.0044	4	0.041	1.9	0.0012	1.5	0	6.3
		48 to 144 p	0	0	0	0	0	0.072	0.9	0.0084	4	0.11	1.9	0.0036	1.5	0	6.3
		144 to 288 p	0	0	0	0	0	0.17	0.9	0.016	4	0.30	1.9	0.011	1.5	0	6.3
		> 288 p	0	0	0	0	0	0.45	0.9	0.034	4	0.96	1.9	0.043	1.5	0	6.3
Memory integrated circuit	Hermetic	≤ 24 p	0	0	0	0	0	0	0.9	0.002	4	0.012	1.9	0.00028	1.5	0	6.3
		24 to 48 p	0	0	0	0	0	0	0.9	0.0044	4	0.041	1.9	0.0012	1.5	0	6.3
		48 to 144 p	0	0	0	0	0	0	0.9	0.0084	4	0.11	1.9	0.0036	1.5	0	6.3
		144 to 288 p	0	0	0	0	0	0	0.9	0.016	4	0.30	1.9	0.011	1.5	0	6.3
		> 288 p	0	0	0	0	0	0	0.9	0.034	4	0.96	1.9	0.043	1.5	0	6.3
	Non-hermetic	≤ 24 p	0	0	0	0	0	0.011	0.9	0.002	4	0.012	1.9	0.00028	1.5	0	6.3
		24 to 48 p	0	0	0	0	0	0.031	0.9	0.0044	4	0.041	1.9	0.0012	1.5	0	6.3
		48 to 144 p	0	0	0	0	0	0.072	0.9	0.0084	4	0.11	1.9	0.0036	1.5	0	6.3
		144 to 288 p	0	0	0	0	0	0.17	0.9	0.016	4	0.30	1.9	0.011	1.5	0	6.3
		> 288 p	0	0	0	0	0	0.45	0.9	0.034	4	0.96	1.9	0.043	1.5	0	6.3
Micro*, DSP, simple ASIC integrated circuit	Hermetic	≤ 24 p	0	0	0	0	0	0	0.9	0.002	4	0.012	1.9	0.00028	1.5	0	6.3
		24 to 48 p	0	0	0	0	0	0	0.9	0.0044	4	0.041	1.9	0.0012	1.5	0	6.3
		48 to 144 p	0	0	0	0	0	0	0.9	0.0084	4	0.11	1.9	0.0036	1.5	0	6.3
		144 to 288 p	0	0	0	0	0	0	0.9	0.016	4	0.30	1.9	0.011	1.5	0	6.3
		> 288 p	0	0	0	0	0	0	0.9	0.034	4	0.96	1.9	0.043	1.5	0	6.3
	Non-hermetic	≤ 24 p	0	0	0	0	0	0.011	0.9	0.002	4	0.012	1.9	0.00028	1.5	0	6.3
		24 to 48 p	0	0	0	0	0	0.031	0.9	0.0044	4	0.041	1.9	0.0012	1.5	0	6.3
		48 to 144 p	0	0	0	0	0	0.072	0.9	0.0084	4	0.11	1.9	0.0036	1.5	0	6.3

				OFF phases: Average parameters by default per type of item													Induced							
				Thermal					Humidity		Thermal cycling				Mechanical		El.Ch.Wr	Csensitivity						
				λTh	Ea_Th	To	ΔT	α	λRh	Ea_Rh	λTcy_B	m_B	λTcy_JB	m_JB	λM	n	λECW							
144 to 288 p				0	0	0	0	0	0.17	0.9	0.016	4	0.30	1.9	0.011	1.5	0	6.3						
				> 288 p				0	0	0	0	0	0.45	0.9	0.034	4	0.96	1.9	0.043	1.5	0	6.3		
Analog, Mixed, FPGA, CPLD, complex ASIC integrated circuit				Hermetic		≤ 24 p		0	0	0	0	0	0	0	0.9	0.002	4	0.012	1.9	0.00028	1.5	0	6.3	
						24 to 48 p		0	0	0	0	0	0	0.9	0.0044	4	0.041	1.9	0.0012	1.5	0	6.3		
						48 to 144 p		0	0	0	0	0	0	0.9	0.0084	4	0.11	1.9	0.0036	1.5	0	6.3		
						144 to 288 p		0	0	0	0	0	0	0.9	0.016	4	0.30	1.9	0.011	1.5	0	6.3		
						> 288 p		0	0	0	0	0	0	0.9	0.034	4	0.96	1.9	0.043	1.5	0	6.3		
				Non-hermetic		≤ 24 p		0	0	0	0	0	0.011	0.9	0.002	4	0.012	1.9	0.00028	1.5	0	6.3		
						24 to 48 p		0	0	0	0	0	0.031	0.9	0.0044	4	0.041	1.9	0.0012	1.5	0	6.3		
						48 to 144 p		0	0	0	0	0	0.072	0.9	0.0084	4	0.11	1.9	0.0036	1.5	0	6.3		
						144 to 288 p		0	0	0	0	0	0.17	0.9	0.016	4	0.30	1.9	0.011	1.5	0	6.3		
						> 288 p		0	0	0	0	0	0.45	0.9	0.034	4	0.96	1.9	0.043	1.5	0	6.3		
Diode		Non-hermetic		SMD		low-power		0	0	0	0	0	0	0.9	0.0008	4	0.0040	1.9	0.000080	1.5	0	5.2		
						power		0	0	0	0	0	0.034	0.9	0.0041	4	0.021	1.9	0.00041	1.5	0	5.2		
				TH		low-power		0	0	0	0	0	0.031	0.9	0.0011	4	0.0055	1.9	0.00011	1.5	0	5.2		
						power		0	0	0	0	0	0.059	0.9	0.003	4	0.015	1.9	0.00030	1.5	0	5.2		
		Hermetic		SMD		0	0	0	0	0	0	0.9	0.0078	4	0.039	1.9	0.00078	1.5	0	5.2				
				TH		0	0	0	0	0	0	0.9	0.01	4	0.051	1.9	0.0010	1.5	0	5.2				
Transistor				Non-hermetic		SMD		low-power		0	0	0	0	0	0	0.9	0.0008	4	0.0040	1.9	0.00008	1.5	0	5.2
								power		0	0	0	0	0	0.034	0.9	0.0041	4	0.021	1.9	0.00041	1.5	0	5.2
						TH		low-power		0	0	0	0	0	0.031	0.9	0.0011	4	0.0055	1.9	0.00011	1.5	0	5.2
								power		0	0	0	0	0	0.059	0.9	0.003	4	0.015	1.9	0.0003	1.5	0	5.2
						Hermetic		SMD		0	0	0	0	0	0	0.9	0.0078	4	0.039	1.9	0.00078	1.5	0	5.2
								TH		0	0	0	0	0	0	0.9	0.01	4	0.051	1.9	0.001	1.5	0	5.2
Photodiode				Non-hermetic		SMD		low-power		0	0	0	0	0	0	0.9	0.0008	4	0.014	1.9	0.0051	1.5	0	5.2
								power		0	0	0	0	0	0.034	0.9	0.0041	4	0.031	1.9	0.0054	1.5	0	5.2
						TH		low-power		0	0	0	0	0	0.031	0.9	0.0011	4	0.016	1.9	0.0051	1.5	0	5.2
								power		0	0	0	0	0	0.059	0.9	0.003	4	0.025	1.9	0.0053	1.5	0	5.2
						Hermetic		SMD		0	0	0	0	0	0	0.9	0.0078	4	0.049	1.9	0.0058	1.5	0	5.2
								TH		0	0	0	0	0	0	0.9	0.01	4	0.061	1.9	0.0060	1.5	0	5.2

				OFF phases: Average parameters by default per type of item														Induced
				Thermal					Humidity		Thermal cycling				Mechanical		El.Ch.Wr	Csensitivity
				λTh	Ea_Th	To	ΔT	α	λRh	Ea_Rh	λTcy_B	m_B	λTcy_JB	m_JB	λM	n	λECW	
Phototransistor	Non-hermetic	SMD	low-power	0	0	0	0	0	0.01	0.9	0.0008	4	0.025	1.9	0.011	1.5	0	5.2
			power	0	0	0	0	0	0.034	0.9	0.0041	4	0.042	1.9	0.011	1.5	0	5.2
		TH	low-power	0	0	0	0	0	0.031	0.9	0.0011	4	0.027	1.9	0.011	1.5	0	5.2
			power	0	0	0	0	0	0.059	0.9	0.003	4	0.036	1.9	0.011	1.5	0	5.2
	Hermetic	SMD	0	0	0	0	0	0	0.9	0.0078	4	0.060	1.9	0.012	1.5	0	5.2	
		TH	0	0	0	0	0	0	0.9	0.01	4	0.072	1.9	0.012	1.5	0	5.2	
Potentiometer				0	0	0	0	0	0.003	0.9	0	1	0.11	1.9	0.066	1.5	0	2.5
Resistor		high-power		0	0	0	0	0	0.014	0.9	0	1	0.37	1.9	0.0040	1.5	0	2.3
		low-power		0	0	0	0	0	0.0045	0.9	0	1	0.19	1.9	0.0020	1.5	0	2.8
Etched metal foil resistor				0	0	0	0	0	0.064	0.9	0	1	0.097	1.9	0.012	1.5	0	5.8
Resistor network, Resistor chip				0	0	0	0	0	0.00021	0.9	0	1	0.020	1.9	0.00021	1.5	0	4.5
Ceramic capacitor		moderate CV product		0	0	0	0	0	0	0	0	1	0.019	1.9	0.0014	1.5	0	6.1
		high CV product		0	0	0	0	0	0	0	0	1	0.075	1.9	0.0074	1.5	0	6.1
Aluminium capacitor				0	0	0	0	0	0	0	0	1	0.043	1.9	0.0031	1.5	0	6.4
Tantalum capacitor		gel		0	0	0	0	0	0	0	0	1	0.0043	1.9	0.052	1.5	0	7.0
		solid		0	0	0	0	0	0	0	0	1	0.072	1.9	0.013	1.5	0	7.0
Inductor				0	0	0	0	0	0	0	0	1	0.024	1.9	0.0043	1.5	0	5.5
Transformer		low-power		0	0	0	0	0	0	0	0	1	0.091	1.9	0.033	1.5	0	6.9
		high-power		0	0	0	0	0	0	0	0	1	0.17	1.9	0.040	1.5	0	6.8
Quartz crystal resonator		through-hole		0	0	0	0	0	0.094	0.9	0	1	0.37	1.9	0.22	1.5	0	4.6
		SMD		0	0	0	0	0	0.077	0.9	0	1	0.47	1.9	0.12	1.5	0	4.6
Quartz crystal oscillator		through-hole		0	0	0	0	0	0.19	0.9	0	1	0.68	1.9	0.22	1.5	0	8.1
		SMD		0	0	0	0	0	0.15	0.9	0	1	0.94	1.9	0.12	1.5	0	8.1
Relay up to 10 contacts	Resistive load	< 3 actuations/h		0	0	0	0	0	0.13	0.9	0	1	0.10	1.9	0.76	1.5	0	7.55
		≥ 3 actuations/h		0	0	0	0	0	0.13	0.9	0	1	0.10	1.9	7.6	1.5	0	7.55
	Non-resistive load	< 3 actuations/h		0	0	0	0	0	0.13	0.9	0	1	0.10	1.9	0.76	1.5	0	7.55
		≥ 3 actuations/h		0	0	0	0	0	0.13	0.9	0	1	0.10	1.9	7.6	1.5	0	7.55
Relay > 10 contacts	Resistive load	< 3 actuations/h		0	0	0	0	0	0.31	0.9	0	1	0.24	1.9	1.8	1.5	0	7.55
		≥ 3 actuations/h		0	0	0	0	0	0.31	0.9	0	1	0.24	1.9	18	1.5	0	7.55
		< 3 actuations/h		0	0	0	0	0	0.31	0.9	0	1	0.24	1.9	1.8	1.5	0	7.55

			OFF phases: Average parameters by default per type of item													Induced		
			Thermal					Humidity		Thermal cycling				Mechanical		El.Ch.Wr	Csensitivity	
			λTh	Ea_Th	To	ΔT	α	λRh	Ea_Rh	λTcy_B	m_B	λTcy_JB	m_JB	λM	n	λECW		
	Non-resistive load	≥ 3 actuations/h		0	0	0	0	0	0.31	0.9	0	1	0.24	1.9	18	1.5	0	7.55
Push button, switch, invertor (< 4 pins)			Resistive load	0	0	0	0	0	0.070	0.9	0	1	0.041	1.9	0.32	1.5	0	7.45
DIP, encoder (4 to 10 pins)				0	0	0	0	0	0.14	0.9	0	1	0.081	1.9	0.63	1.5	0	7.45
Switch (> 10 pins)				0	0	0	0	0	0.30	0.9	0	1	0.18	1.9	1.4	1.5	0	7.45
Push button, switch, invertor (< 4 pins)			Non-resistive load	0	0	0	0	0	0.070	0.9	0	1	0.041	1.9	0.32	1.5	0	7.45
DIP, encoder (4 to 10 pins)				0	0	0	0	0	0.14	0.9	0	1	0.081	1.9	0.63	1.5	0	7.45
Switch (> 10 pins)				0	0	0	0	0	0.30	0.9	0	1	0.18	1.9	1.4	1.5	0	7.45
Connector for printed circuits and IC supports	soldered (PTH or SMD)	up to 20 contacts		0	0	0	0	0	0.066	0.8	0	1	0.020	1.9	0.025	1.5	0.22	4.4
		20 to 200 contacts		0	0	0	0	0	0.21	0.8	0	1	0.064	1.9	0.080	1.5	0.69	4.4
		over 200 contacts		0	0	0	0	0	0.47	0.8	0	1	0.14	1.9	0.18	1.5	1.5	4.4
	wrapping or insertion	up to 20 contacts		0	0	0	0	0	0.016	0.8	0	1	0.0051	1.9	0.0063	1.5	0.055	4.4
		20 to 200 contacts		0	0	0	0	0	0.052	0.8	0	1	0.016	1.9	0.020	1.5	0.17	4.4
		over 200 contacts		0	0	0	0	0	0.12	0.8	0	1	0.036	1.9	0.045	1.5	0.39	4.4
Other connector	soldered (PTH or SMD)	up to 20 contacts		0	0	0	0	0	0.039	0.8	0	1	0.012	1.9	0.015	1.5	0.13	4.4
		20 to 200 contacts		0	0	0	0	0	0.12	0.8	0	1	0.038	1.9	0.048	1.5	0.42	4.4
		over 200 contacts		0	0	0	0	0	0.28	0.8	0	1	0.086	1.9	0.11	1.5	0.93	4.4
	wrapping or insertion	up to 20 contacts		0	0	0	0	0	0.010	0.8	0	1	0.0030	1.9	0.0038	1.5	0.033	4.4
		20 to 200 contacts		0	0	0	0	0	0.031	0.8	0	1	0.010	1.9	0.012	1.5	0.10	4.4
		over 200 contacts		0	0	0	0	0	0.070	0.8	0	1	0.021	1.9	0.027	1.5	0.23	4.4
Printed circuit, PTH		up to 2000 mounting points		0	0	0	0	0	0.090	0.8	0	1	0.30	1.9	0.10	1.5	0.022	6.5
		> 2000 mounting points		0	0	0	0	0	0.36	0.8	0	1	1.2	1.9	0.40	1.5	0.087	6.5
Printed circuit, SMD		up to 2000 mounting points		0	0	0	0	0	1.2	0.8	0	1	4.1	1.9	1.4	1.5	0.30	6.5
		> 2000 mounting points		0	0	0	0	0	4.9	0.8	0	1	16	1.9	5.5	1.5	1.2	6.5
Hard drive		EIDE		0	0	0	0	0	0	0	0	1	0	1	230	1.5	0	5.0
		SCSI		0	0	0	0	0	0	0	0	1	0	1	110	1.5	0	5.0
LCD screen (Class III)		STN	up to 14"		0	0	0	0	0	0	0	1	0	1	3520	1.5	0	1.8
			14" to 20"		0	0	0	0	0	0	0	1	0	1	5697	1.5	0	1.8

		OFF phases: Average parameters by default per type of item														Induced Csensitivity
		Thermal					Humidity		Thermal cycling				Mechanical		El.Ch.Wr	
		λ_{Th}	Ea_Th	To	ΔT	α	λ_{Rh}	Ea_Rh	λ_{Tcy_B}	m_B	λ_{Tcy_JB}	m_JB	λ_M	n	λ_{ECW}	
	TFT		0	0	0	0	0	0	0	1	0	1	1346	1.5	0	2.4
		over 20"	0	0	0	0	0	0	0	1	0	1	2156	1.5	0	2.4
CRT monitor		up to 20"	0	0	0	0	0	64	0	1	526	1.9	263	1.5	0	3.2
		over 20"	0	0	0	0	0	269	0	1	796	1.9	401	1.5	0	3.2
Switching power supply		up to 500W	0	0	0	0	0	7.1	0	1	112	2.5	23	1.5	0	5.9
		over 500W	0	0	0	0	0	8.2	0	1	207	2.5	26	1.5	0	5.9
DC/DC converter		up to 50W	0	0	0	0	0	2.0	0	1	27	2.5	6.4	1.5	0	5.9
		over 50W	0	0	0	0	0	3.7	0	1	68	2.5	12	1.5	0	5.9

SMD: Surface Mounted Device

TH: Through-Hole

ECW: Electrical, Chemical, Wear

Managing the transition to lead-free

Warning:

At the time of publication of this guide, given the state-of-the-art of the physics of failure and the short period during which lead-free assemblies have been used industrially, it is not possible to construct rigorous reliability models based on the physics of failure. However, this chapter of the guide does propose an approach for the reliability assessment of electronic products with lead-free assemblies. These results should consequently be used with caution.

Consequences for reliability

The transition to lead-free assembly technologies may affect the reliability of systems due, on the one hand, to changes in manufacturing processes and materials and, on the other, to an increased acceleration of certain failure mechanisms.

Developments in the physics of failure for lead-free assemblies

In principle, the fatigue mechanisms affecting solders in lead-free assemblies and in tin-lead assemblies obey different laws of acceleration. The current models in this guide are based on the physics of failure for tin-lead assemblies. The laws used for calculating acceleration factors for the mechanical and thermo-mechanical environments of lead-free assemblies therefore need to be modified accordingly.

It is also possible that sensitivity to different environments may be modified for some component families, due to a probable change in the level of residual stress of lead-free solder assemblies, induced by a higher solidification temperature, a reduction in stress relaxation at ambient temperature, and increased stiffness of solder joints. The basic failure rates and the distributions of physical factors in assembled lead-free components could also evolve away from those of assembled tin-lead components.

All of these adaptations will become possible only when models validated by the scientific and industrial community become available. This is not the case at the time of publication of this guide. It has not yet been formally demonstrated that the models described in this guide are applicable to lead-free assemblies, and, to be rigorous, the models will eventually have to be modified.

With these caveats, this guide suggests that all of the proposed models should be applied in the same way, regardless of whether the assembly is leaded or lead-free.

Variation in risk levels due to changes in the component manufacturing process

Potential variations in the risk of failure, as a result of modifications to components (including PCBs), must be taken into account in the part manufacturing factor Π_{PM} or in the λ_0 values. In some cases, as with plastic encapsulated components supplied by major component manufacturers, the transition to lead-free has not entailed any significant change in the risk of failure. The most probable reason for this is that the modification in the resins (coating and adhesive) designed to facilitate mounting (by increasing the reflow profiles) contributes to greater reliability of the package (lower sensitivity to humidity, and probably better bonding of resins).

In other cases, where components are manufactured in small production runs, changes in processes or materials will, at best, be technologically validated by tests on a limited number of parts, which will not be sufficient to reveal any potential process dispersion.

Consequently, the risk of an increase in the failure rate, most likely due to process changes, is not considered more significant than for any other type of change, and the criteria taken into account in the evaluation of Π_{PM} are sufficient to characterize it.

Variation in risk levels due to changes in the board manufacturing process

Potential variations in the risk of failure, as a result of modifications to the design and manufacturing of circuit boards for the transition to lead-free, are likely to have a significant influence on the reliability of systems.

Changes in manufacturing processes could lead to a temporary spike in failures due to lack of expertise in new processes. Modifications in processes and materials, linked to the increase in the solder reflow temperature (an increase of about 35°C in the melting temperature of the solder alloy), have been validated to varying degrees, but it is highly probable that anomalies will occur despite these validations, due, for example, to unforeseen incompatibilities between processes and materials, poor knowledge of new equipment and processes by operators, the coexistence of different processes (backward-compatible and full lead-free), and so on.

The temporary increase in the risk of failure due to process changes is factored in by means of a Π_{LF} factor (LF for Lead Free), which takes account of experience in the design and in manufacturing of lead-free electronic assemblies.

This factor represents the increase in the system failure rate attributable to the lead-free process. It evolves over time as follows:

- The factor is equal to 1 as long as series manufacturing is performed using the “mastered” tin-lead process.
- The factor becomes greater than 1 at the time of transition, and then diminishes with time and experience.
- It returns to 1 once the lead-free process has reached maturity.

The model for estimating the Π_{LF} factor is given below.

Lead-free process factor

General form of the model

The failure rate for a traditional tin-lead solder process is calculated as follows:

$$\lambda = \lambda_{\text{Physical}} \times \Pi_{\text{PM}} \times \Pi_{\text{Process}}$$

The failure rate for a lead-free process is calculated as follows:

$$\lambda = \lambda_{\text{Physical}} \times \Pi_{\text{PM}} \times \Pi_{\text{Process}} \times \Pi_{\text{LF}}$$

Where Π_{LF} is the factor of transition to the lead-free process.

$$\Pi_{\text{LF}} = 1 + \left(1 + 12 \times \text{LF_grade} + C_Q \times Y \right)^{\frac{-1}{(3.1 - 2 \times \text{LF_grade})}}$$

Where:

- Y (Year) is the number of complete years of effective production with a lead-free process. The factor Y can only take integer values (therefore for the first year Y = 0).
- LF_grade is a process factor that measures the effort made to prevent defects related to the transition to lead-free processes. LF_grade ranges from 0 (no precautions) to 1 (maximum precautions).
- C_Q is a factor that depends on the quantity produced by means of lead-free processes.

The Π_{LF} factor ranges from 1 (for a mature process) to 2 (for a process where no precautions were taken). When the Π_{LF} factor goes below 1.1, it no longer needs to be taken into account.

Determination of the LF_grade factor

The LF_grade factor is determined by means of a questionnaire on lead-free processes.

$$LF_grade = \frac{\sum \text{Weight of satisfied criteria}}{\sum \text{Weight of applicable criteria}}$$

N°	Category	Criterion	Weight (if criterion met)
1	Development	The LF status is identified in the component database	8
2	Development	The compatibility of the component with the tin-lead assembly process or with the lead-free process is identified in the component database	8
3	Development	Compatibility between component finishes and lead-free or tin-lead processes is defined in specific design documents (this criterion is exclusive of the previous criterion)	4
4	Development	The LF status is clearly identified and taken into account in the design process	8
5	Supply chain, logistics, storage	Traceability: lead-free or ROHS components are identifiable by appropriate labelling throughout the supply chain	10
6	Supply chain, logistics, storage	Leaded or lead-free components are clearly identified in the production database	3
7	Supply chain, logistics, storage	The need for leaded or lead-free components is clearly indicated on the order, with a request for specific marking on the delivery	5
8	Supply chain, logistics, storage	Leaded or lead-free components are clearly identified on reception	5
9	Supply chain, logistics, storage	In storage, there is a physical separation between components used for the lead-free process and components used for the lead process	5
10	Supply chain, logistics, storage	Compatibility with lead-free process (solderability) is taken into account when defining component storage conditions	4
11	Assembly process	The lead-free assembly status is clearly identified in the production file	3
12	Assembly process	The lead-free serial production process is formally qualified (the integrity of components and PCBs is checked after mounting and metallurgical analysis)	20
13	Assembly process	The lead-free repair process is formally qualified (the integrity of components and PCBs is checked after repair and metallurgical analysis)	20
14	Assembly process	A line is dedicated to each serial process: leaded and lead-free	8
15	Assembly process	A line is dedicated to each repair process: leaded and lead-free	8
16	Assembly process	The lead-free or leaded technology is marked unambiguously on the board	8
17	Assembly process	Operators have received appropriate training on the lead-free issues (new processes, compatibility, solder joints, wiring, inspection, repair, etc.).	16
18	Reliability	The equipment manufacturer has commissioned internal or external studies about modelling the reliability or lifetime of lead-free assemblies.	8
19	Reliability	The equipment manufacturer has commissioned internal or external studies on qualification test conditions for lead-free boards or equipment.	8

N°	Category	Criterion	Weight (if criterion met)
20	Reliability	The equipment manufacturer has commissioned internal or external studies on the environmental stress screening of lead-free boards or equipment.	6
21	Reliability	The JEDEC JESD201 standard, or equivalent precautions, are taken into account to avoid the risk of tin whiskers	5

Note:

- In the general case, the sum of the applicable criteria is 166.
- Criterion 3 is included in Criterion 2, and only one of these two criteria can be counted as being met at any one time (the weight of Criterion 3 is therefore not counted in the total).
- The fact that a component changes its partnumber when it is modified to become lead-free provides sufficient identification (Criteria 7 and 8).
- When there is no cohabitation between lead-free and tin-lead processes, i.e. when there is no longer any tin-lead production line and all supplied components are lead-free, then Criteria 1, 2, 3, 4, 5, 6, 7, 8, 9, 14, 15, and 16 can be considered to be met.
- For non-repairable products, Criteria 13 and 15 are non-applicable. The sum of applicable criteria then becomes 138.

Determination of the C_Q factor

Cumulative number of components mounted per week in a lead-free process	Size of production series	C _Q
Weekly quantity <100000	Small series	4
100000 ≤ weekly quantity < 500000	Average series	12
500000 ≤ weekly quantity < 2000000	Large series	36
Weekly quantity ≥ 2000000	Very large series	108

IV

Reliability Process Control and Audit Guide

1. Life cycle

The table below details the complete life cycle of a product, through which its reliability is determined. The FIDES methodology covers the evaluation and control of reliability throughout the life cycle.

Phases		Main activities	
1	SPECIFICATION	1.1	Statement of need by the customer
		1.2	Formalization of system requirements
		1.3	Definition of the architecture
		1.4	Allocation of system requirements
		1.5	Formalization of subsystem, equipment, etc. requirements
2	DESIGN	2.1	Feasibility / Preliminary studies
		2.2	Detailed design
		2.3	Tests and fine-tuning
		2.4	Qualification
		2.5	Preparation for production / Product industrialization
		2.6	Preparation of logistics support
3	BOARD OR SUBASSEMBLY MANUFACTURING	3.1	Reception / incoming inspection
		3.2	Storage
		3.3	Assembly of board or subassemblies
		3.4	Testing (board or subassemblies)
		3.5	Equipment integration
		3.6	ESS (board or subassembly)
		3.7	Acceptance
		3.8	Delivery of board or subassembly
4	EQUIPMENT INTEGRATION	4.1	Reception / incoming inspection
		4.2	Storage
		4.3	Assembly of equipment
		4.4	Testing of equipment
		4.5	ESS (equipment)
		4.6	Acceptance of equipment
		4.7	Delivery of equipment
5	SYSTEM INTEGRATION	5.1	Reception / incoming inspection
		5.2	Storage
		5.3	Assembly of system
		5.4	Testing of system
		5.5	ESS (system)
		5.6	Acceptance of system
		5.7	Delivery of system
6	OPERATION & MAINTENANCE	6.1	Transfer to user
		6.2	Operational use
		6.3	In-service support
7	SUPPORT ACTIVITIES	7.1	Management of subcontractors
		7.2	Management of reliability, supplies, incidents
		7.3	Management of the quality system, resources

2. The process factor

The process factor is denoted by Π_{process} in the evaluation part of this guide.

The scores are quantified by answering questionnaires about the product development, manufacturing and operating processes. The procedure for replying to these questionnaires is described in the audit guide.

3. Function-specific recommendations: Reliability control

A set of reliability recommendations is presented for each phase or activity of the life cycle.

The recommendations are either global, potentially involving all of the phases (in which case they are associated with the “Support activities” phase), or specific and recognized as impacting reliability during particular activities in one or more phases of the life cycle.

Applying these recommendations makes it possible to implement reliability control actions (reliability engineering) and to assess the reliability assurance level for each phase of the process. The reliability control approach consists in using the results of a preliminary assessment to act upon activities that influence the outcome.

The recommendations of the Reliability Process Control and Audit Guide mainly concern procedures and organization throughout the life cycle. The guide is not intended to offer technological recommendations regarding the use of components, boards or subassemblies in electronic equipment.

4. Calculating the process factor Π_{Process}

Π_{process} is based on a score (**Process_Grade**) that reflects the quality of the process, determined following an audit of the different phases of the life cycle.

4.1. Relative influence of life cycle phases

The 7 life cycle phases are as follows:

- Specification
- Design
- Board or subassembly manufacture
- Equipment integration
- System integration
- Operation and maintenance
- Support activities.

Each phase has a specific impact on reliability. For quantification purposes, each phase is assigned a scale factor so as to determine the relative weight of each phase. If the value distribution specific to the audited manufacturer is already known, it can be taken into account.

The distribution by default is the following:

PHASE	Phase contribution %
Specification	8
Design	16
Board or subassembly manufacture	20
Equipment integration	10
System integration	10
Operation and maintenance	18
Support activities	18
Total:	100

4.2. Level of compliance with recommendations

The audit is conducted phase by phase, through questions (relating to the recommendations) that assess the way the activities are carried out.

The answers and evidence provided by the auditees are used to determine a **Compliance_Score** for each recommendation (level L1 to L4):

- L1 = the recommendation is not applied → clear reliability risks
- L2 = the recommendation is only partially applied → potential reliability risks
- L3 = the recommendation is largely applied in practice → few reliability risks
- L4 = the recommendation is fully applied → no significant reliability risk.

The score for each level is determined as follows:

Level	Score
L1	0
L2	1
L3	2
L4	3

Each recommendation is weighted by a specific **Recom_Weight**, for example:

- 1 → the recommendation associated with the question has little impact on reliability.
- 10 or more → the recommendation associated with the question has a strong impact on reliability.

The attached implementation tables list the recommendations for each phase (and the corresponding audit question) with their specific **Recom_Weight** factors. A datasheet is also provided for each recommendation, with a detailed description of the recommendation and of the criteria for meeting each of the four levels of compliance.

4.3. Calibration

The purpose of this step is to eliminate questions about activities that are not relevant to the product/process considered (these questions are said to be “non-applicable”).

The first step in the calculation is to establish the **Max_Audit_Score_j** for each phase *j*.

The **Max_Audit_Score_j** corresponds to a “perfect” audit, where the compliance level is L4 for all selected questions.

Thus, for each recommendation *i*:

$$\text{Max_Weighted_Points}_i = \text{Recom_Weight}_i \times 3$$

Max_Audit_Score_j is then calculated by summing the **Max_Weighted_Points_i** for all the recommendations applicable to the product/process considered (*i*=1 to *n*) for the complete phase *j*:

$$\text{Max_Audit_Score}_j = \sum_{i=1}^n \text{Max_Weighted_Points}_i$$

4.4. Calculating the audit score

This step consists in conducting the FIDES audit proper with the people involved in the different process phases, and defining the compliance level on the basis of the evidence provided. The approach to be followed is set out in the Audit Guide chapter.

Proceed in phases, answering each question i ; the compliance level of each question, rated **0**, **1**, **2** or **3**, multiplied by the weight of the recommendation, gives the **Weighted_Points_i** for question:

$$\text{Weighted_Points}_i = \text{Recom_Weight}_i \times \text{Compliance_Score } (0, 1, 2, 3)_{ij}$$

The **Audit_Score** for phase j is the sum of all **Weighted_Points** for the recommendations i selected for the phase concerned:

$$\text{Audit_Score}_j = \sum_{i=1}^n \text{Weighted_Points}_i$$

4.5. Calculating the process factor

The formula for the process factor is as follows:

$$\Pi_{\text{Process}} = e^{\delta_2(1-\text{Process_Grade})}$$

The factor δ_2 determines the range of variation of the process factor. It has been set at **2.079**, which entails a dynamic range of 1 to 8 for the process factor.

The **Process_Grade** is calculated from the previously calculated **Audit_Scores** for each phase, weighted by the **Phase_Contribution** of each phase such that:

$$\text{Process_Grade} = \sum_{j=1}^7 \left(\text{Phase_Contribution}_j \times \frac{\text{Audit_Score}_j}{\text{Max_Audit_Score}_j} \right)$$

Process_Grade takes a value between 0 and 1:

- **0** represents a process that fails to satisfy any of the audit questions
 $\rightarrow \Pi_{\text{Process}} = 8$
- **1** represents a “perfect” process that satisfies all the audit questions
 $\rightarrow \Pi_{\text{Process}} = 1$

Note: A Process_Grade_j can be evaluated for each phase j in order to assess its level:

$$\text{Process_Grade}_j = \frac{\text{Audit_Score}_j}{\text{Max_Audit_Score}_j}$$

5. Audit guide

The guide can be used to audit a company. The audit procedure is generic, in order to allow for a degree of freedom relative to the company.

The FIDES methodology identifies a list of recommendations which, if followed, will make it easier to establish the reliability of a product. This set of recommendations has been formulated as a set of questions.

The answers that a company gives to these questions provide:

- A measurement of its ability to make reliable products
- A quantification of the process factors used in the calculation models
- The possibility of identifying improvement actions.

5.1. Audit procedure

To conduct an audit, the auditor must:

- Identify the scope of the audit
- Prepare the audit
- Perform the audit
- Collect evidence
- Process the collected information
- Draw conclusions
- Write an audit report
- Present the audit findings.

5.2. Identifying the scope of the audit

5.2.1. Identifying the product being audited

Start by identifying exactly which product is to be audited.

The audit recommendations for the different life cycle phases must be oriented and adapted to suit the product considered.

5.2.2. Selecting the applicable phases

For certain products (such as standalone equipment, e.g. a radio set), the concepts of “equipment integration” (in the sense of integration into equipment) and “system integration” (in the sense of integration into a system) can be merged. In such cases, we select whichever of the two phases is the more representative and assign the weight of both phases to it.

For example:

PHASE	Phase contribution %
Specification	8
Design	16
Board or subassembly manufacture	20
Equipment integration	20 (<i>instead of 10</i>)
System integration	0 (<i>instead of 10</i>)
Operation and maintenance	18
Support activities	18
Total:	100

This contributions table will usually be adjusted to cancel out the contribution of a phase that would not be at all applicable for a given product.

Caution: Under this approach, a phase should not be considered “not applicable” simply because it is under the responsibility of someone other than manufacturer who initiated the audit. For example, even if a company has not given any commitment about the operation and maintenance of the products it sells, this life phase very probably still exists nonetheless.

5.2.3. Incorporating the manufacturing breakdown structure

Where a manufacturing breakdown structure applies, the life cycle will be spread across different organizations. For example, the same product can be specified by a project manager, designed by an equipment manufacturer, and made by a subcontractor before being integrated and operated by another manufacturer.

In such cases, the Π_{Process} factor must be evaluated by considering the process of each manufacturer involved, for the life cycle phases under their responsibility.

Where there is a manufacturing breakdown structure, it frequently happens that a single phase in the FIDES life cycle is shared between multiple manufacturers. For example, the specification activity may begin at system level and then continue at equipment level, and subsequently at subassembly level. This is generally also the case for the “Support activities” part of the FIDES life cycle. In these cases, the **Process_grade** factor of the

shared phases must be evaluated for each manufacturer involved, and the lowest score retained for the final calculation of Π_{Process} . The selected Π_{Process} will therefore be the “worst case” (lowest score). This is not about adopting a conservative approach: it is the weak link in the chain that matters most, and so it is vital to take account of the least effective practices observed, as in a quality audit.

The usual approach is to consider that the *Process_grade* is attached to the manufacturer with the greatest responsibility during the phase in question. It is therefore possible, where there is a contractual manufacturing breakdown structure, to pass on an adjusted failure rate requirement to the subcontractor, using a perfect *Process_grade* for the phases under customer responsibility and weighting the remainder (the complement of the customer phase value) by Π_{Process} .

If one of the phases in the life cycle cannot be evaluated, the Π_{Process} factor can be calculated using a default value for *Process_grade* for the phase in question. The default value of *Process_grade* is 0.33.

For manufacturers of COTS products (boards or equipment) who wish to publish reliability information on their products, it is recommended that they specify the Π_{Process} factor used in their evaluation, even if this means that a large part of the life cycle will have to be quantified by default.

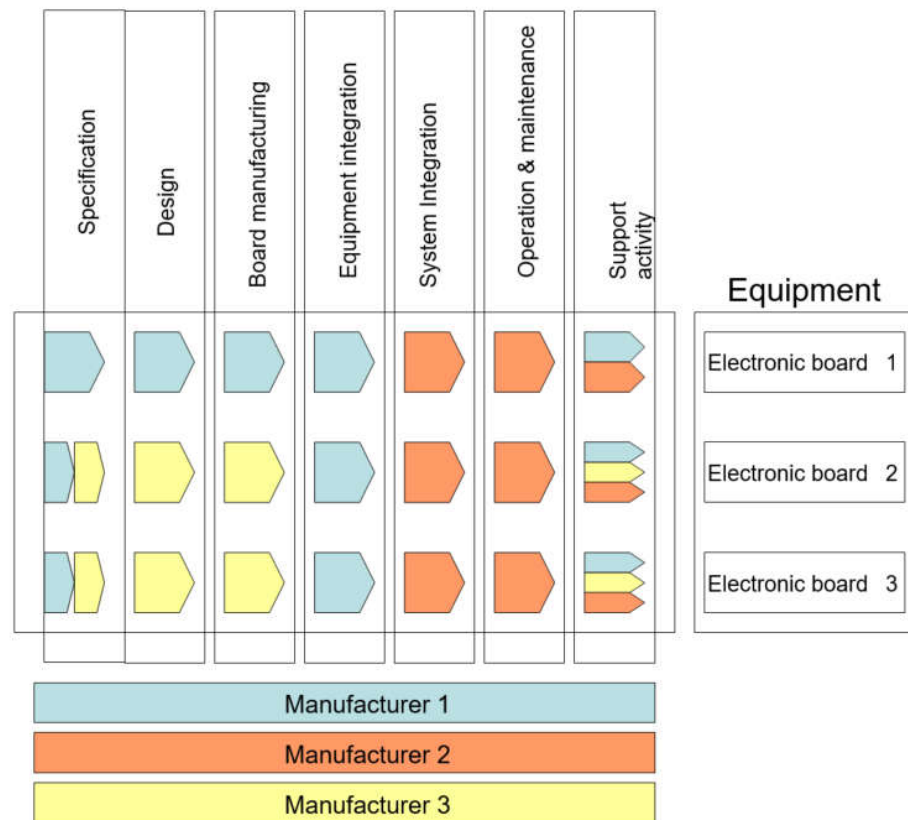
5.2.4. Multiple life cycles for the same product

If the product is an equipment item, it may be made up of boards or subassemblies that have not all followed the same life cycle. In this case the Π_{Process} factor must be evaluated for each board or subassembly.

5.2.5. Example of a manufacturing breakdown structure (MBS)

In the example represented by the following diagram, three manufacturers participate in the definition and production of an item composed of three circuit boards, which is integrated into a system.

In this breakdown, the first manufacturer is responsible for developing and manufacturing the equipment, the second for integrating it into the system and operating it. The first manufacturer outsources the design and manufacturing of two of the circuit boards to a third manufacturer.



For Circuit Board 1:

The process_grade of Manufacturer 1 must be used for the “Specification”, “Design”, “Board manufacture” and “Equipment integration” phases of the life cycle. The process_grade of Manufacturer 2 must be used for the “System integration” and “Operation and maintenance” phases. For the “Support activity” phase, the lower of the two process_grade values (as between Manufacturers 1 and 2) must be used.

For Circuit Boards 2 and 3:

The lower of the two process_grade values for Manufacturers 1 and 3 should be used for the "Specification" phase, which is assumed to be distributed between the two. The process_grade of Manufacturer 3 must be used for the “Design” and “Board manufacture” life cycle phases. The process_grade of Manufacturer 1 must be used for the "Equipment integration" phase. The process_grade of Manufacturer 2 must be used for the “System integration” and “Operation and maintenance” phases. For the “Support activity” phase, the lowest of the three process_grade values between Manufacturers 1, 2 and 3 must be used.

In this example, the reliability assessment of the complete equipment must be based on an audit of three manufacturers (none of whom is audited across the entire FIDES life cycle). The quantitative calculation must use two distinct $\Pi_{Process}$ factors: one for Circuit Board 1, the other for Circuit Boards 2 and 3.

5.3. ISO 9001 and EN 9100

The ISO 9001 and EN 9100 quality standards, when followed, guarantee a certain level of quality and process control in terms of both production and traceability. It is therefore possible to find parallels with the Π_{Process} factor audit, and a level of correspondence with similar or comparable recommendations.

Each recommendation sheet in the Audit Guide identifies, where applicable, the standard most similar to the recommendation, and the level corresponding to the application of the ISO 9001 or EN 9100 standard.

It is, however, highly recommended to conduct the audit; even if a company is ISO 9001 or EN 9100 certified, the certification process often identifies improvement areas or partial non-conformances, which may relate precisely to the recommendations in this Audit Guide. Care should therefore be taken when applying the corresponding levels directly.

5.4. Preparing the audit

The audit preparation consists of the following steps:

- Identify the nature and scope of the audit (full, partial, required information, duration, whether for a certification programme, etc.)
- Identify context elements of the audit
- Identify the right targets (FIDES targets listed in the following table)
- Draw up an audit plan (a schedule with a milestone calendar, convening notes, draft data collection documents, draft output document templates, involvement of the audit client and the organization to be audited), calculate the maximum score possible for the audit, define the rules, etc.
- Have the audit plan validated (by the external or internal audit client and by the audited company's representative)
- Initialize the implementation of the audit plan (send convening notes)
- Inform the auditee about the content of the audit in good time, making it clear that failure to supply evidence will be considered absence of evidence.

Obtaining buy-in for the audit is key to its success. To advocate the benefits of a FIDES audit, the auditor might mention the following points:

- The importance of the product reliability level as:
 - A cost generator, and a new key parameter in competition
 - A defined objective to be achieved.
- The audit is a risk control tool, focusing particularly on:
 - The robustness of the product definition
 - The environment of products in operation
 - The effective integration of reliability into the entire life cycle.
- The FIDES audit is complementary to the ISO 9001 V2000 audit, with FIDES being more specific and oriented towards dependability
- The specific objectives of carrying out a FIDES audit:
 - To establish a measurable quality indicator that reflects a company's ability to control the reliability of its products (process_grade or Π_{Process} factor)
 - To assess a key reliability factor: calculating Π_{Process}

- To identify the company's strengths and weaknesses and formulate targeted recommendations to improve the internal process.

5.5. Conducting the audit

To conduct the audit:

- Present the audit (summary of objectives, scope and rules). Ask the audit questions, (where applicable, ask the supplementary questions to determine the precise level for the criterion)
- Record the answers of the auditees to each question
- Collect the available evidence immediately so as to attach it to the report
- Classify the evidence collected during the audit
- Take into account any additional evidence.

During the audit, the auditors will identify any questions that are not relevant (i.e. where the process activities are not required) if they have not already done so in the preparation phase: this will lead to recalculation of the maximum expected audit score.

5.6. Processing the collected information

Processing the information involves evaluating, for each recommendation, the position of the audited entity relative to the criteria on the basis of the answers to the audit questions, the evidence supplied in support of these answers, and the weightings for each recommendation.

The result of this processing is used to:

- Determine the reliability level to be associated with the audited entity
- Quantify the process factor (Π_{Process}) to be used
- Where applicable, identify improvement areas for the audited entity.

We have to ask which risk we are trying to pin down, and how it is addressed. If the manufacturer does not identify with any of the defined levels, but has put in place a specific process which, in the manufacturer's opinion, is equivalent to a given level, then the manufacturer can reasonably assign this level to the recommendation.

5.7. Presenting the audit findings

After the audit is complete, the auditor will present the findings to the client and to the auditee. This presentation will address:

- The purpose of the audit
- The audit plan and its implementation
- The audit findings
- The identified improvement areas
- The conclusions.

The written audit report will subsequently be submitted to the client.

5.8. Ranking principle

Following the audit, the score for the audited organisation (process_grade for one or several phases, Π_{Process} for a complete life cycle) can be calculated by evaluating the answers to the questions against the criteria, and applying their weightings.

The minimum possible score corresponds to a process that meets none of the criteria. The FIDES methodology does not have a rule defining a minimum acceptable score for the FIDES methodology to be considered applicable. Such rules can only emerge from the practical use of the methodology in the industrial setting.

Depending on the position of the score (**process_grade**) relative to the maximum possible score, the audited entity's reliability level may be ranked as "very high", "high", "standard" or "low".

Level	Process	Π_{Process}	Process grade
Very high reliability	Process almost free of weaknesses	<1.7	> 75%
High reliability	Controlled process; reliability engineering	1.7 to 2.8	50% to 75%
Standard	Usual quality procedures e.g. ISO 9001 version 2000	2.8 to 4.8	25% to 50%
Low reliability	Reliability issues not taken into account	>4.8	<25%

5.9. Profile of audit participants

5.9.1. Profile of auditors

Auditors should:

- Be engineers, executives or technicians with at least 5 years' experience
- Be familiar with the ISO 9001 standard, version 2008 or 2000
- Have skills, as well as theoretical and practical experience, in reliability
- Have training in the conduct of audits.

These prerequisites will be supplemented by solid knowledge of the FIDES methodology.

These skills can also be obtained by an audit team (typically an experienced auditor and a reliability specialist).

5.9.2. Profile of auditees

Given the diversity of organizations liable to be audited, auditees may have a variety of profiles.

However, they will be representative of the eighteen target groups identified by FIDES. In a partial audit, the auditees may represent just one subset of these targets.

N°	Target population	Description
1	Procurement	- Responsible for the procurement process and document guidelines (creation, implementation) - Project buyer: responsible for negotiating technical clauses / costs and monitoring adherence to commitments
2	Engineering department / Design Office	- Analysis or production of requirement documents, technical specifications, and justification and traceability files - Setting up of design, evaluation, approval, and validation teams - Management of schedules, reviews, indicators (costs, quality, etc.)
3	Customer (client)	- Responsible for the specification of RAMS requirements and related specifications (mission profiles, analysis guidelines, etc.)
4	Site management	- Responsible for managing the overall site resources (design, production, product industrialization)
5	Document management	- Recording / archiving documentation (definition files, specifications, procurement files, etc.) and making it available for consultation - Managing project documentation
6	Operation	- Responsible for use-related matters: compliance with recommendations, user documentation, user training - End users
7	Product industrialization / Production / Integration: Methods and Quality management	- Product traceability and production / product industrialization / integration files (Product industrialization Manager) - Quality control of workplace services, fluids and environment - Inspection guidelines (creation, implementation), etc.

N°	Target population	Description
8	Product industrialization / Production / Integration: Site and Resources management	- Control of inspection resources (workshop manager) - Control of test and inspection procedures and reports
9	Product industrialization / Production / Integration: Operational Personnel	- Conduct of tests (resources, scheduling, etc.)
10	Maintenance	- Responsible for maintenance systems and procedures, compliance with recommendations, and anomaly processing (In-Service Support) - Maintenance technician
11	Handling / Logistics	- Responsible for transport / handling / packaging / storage clauses and procedures
12	Project	- Management and specification / creation of supplier or in-house clauses - Synthesis of activities including RAMS, logistics support, obsolescence, qualification, quality, handling/conditioning/storage, production, and after-sales - Risk management (technical, planning, nonconformities)
13	Quality	Description and implementation of processes: - Product traceability for design, production, delivery and customer care - Ensuring implementation of and compliance with professional guidelines and QA - Monitoring the processing of anomalies and nonconformities
14	Human Resources	- Balancing workload / qualification / human resources and leveraging knowledge and experience
15	After Sales Service, Customer Support	- Handling customer issues and anomalies or nonconformities
16	Components Service / Supplier Qualification / Technology Watch / Sourcing	- Inspection guidelines (creation, implementation) and qualification (technical and functional) of items purchased
17	Logistics Support	- Project actor for the implementation of support analyses - Logistics support process guidelines (creation, implementation) (process description, analysis and substantiation tests, qualification tests)
18	Dependability, RAMS (Reliability, Availability, Maintainability, Safety)	- Pilot driver for the implementation of RAMS / guidelines (resources, systems), project monitoring (indicators, specifications, risk management, RAMS feasibility, etc.) and promoting RAMS awareness in other entities - Guidelines (creation, implementation, realization)

V

Reliability Process Control and Audit Guide Recommendations

- 1. Weighted recommendation tables**
- 2. Detailed recommendation sheets**

Weighted recommendation tables

Specification

#	Recommendation	Question	Weight
1	Assign human and financial resources to dependability a.k.a. RAMST (Reliability, Availability, Maintainability, Safety, Testability) studies.	Is there a financing item for dependability studies? Have the necessary means and personnel been identified?	10.7
2	Allocate reliability requirements down to subassembly level.	Are the general reliability requirements allocated down to subassembly level? If so, which method of allocation was used?	10.4
26	Describe in full the environment in which the product is to be used and maintained.	Is there a description and characterization of the environment in which the product is to be stored, transported, used and maintained? If applicable, are there any recommendations on storage, transport, use and maintenance?	12.4
28	Define the criteria for product failure.	What counts as a product failure?	10.3
29	Define the method for demonstrating reliability during operation.	How is it planned to demonstrate the reliability of the product?	8.1
31	Define the product mission profile for which the reliability performance is expected.	Is there a clear definition of the product mission profile for which the reliability performance is expected?	9.9
53	Leverage experience feedback.	Is experience feedback used to maintain confidence in delivering the required reliability performance?	8.5
54	Involve the Dependability function in the functional and organic design of the product.	Are reliability criteria taken into account in the product architecture, and in decisions about design, product industrialization, and support?	12.6
57	Formalize the reliability requirements, and in particular the quantitative requirement.	Are the reliability requirements, in particular the quantitative requirement, formalized? Do they accurately describe the assumptions involved in assessing reliability performance?	13
62	Formally identify the technical risks that affect reliability.	Are the technical risks that affect reliability identified?	12.4
65	Identify the customer's reliability-related requirements.	Have the customer's reliability-related requirements been identified, documented and tracked?	7.3
103	Analyse the customer's reliability requirements.	Are the reliability requirements analysed? If applicable, does this analysis lead to the negotiation of requirements, or to the formalization of a risk management plan?	10.7

#	Recommendation	Question	Weight
106	Address reliability aspects during design reviews.	Do design reviews address reliability aspects?	10.3
107	Address reliability aspects during requirements reviews.	Do requirements reviews address reliability requirements?	10.3
117	Take account of the product maintenance policy, as defined by the customer.	How is the customer's product maintenance policy taken into account?	5.8
122	Write a reliability plan.	Has a reliability plan been written for the product?	7.6
181	Know and control the mission profile.	Is the product mission profile known and controlled?	13

Design

#	Recommendation	Question	Weight
6	Ensure the completeness of information on subassemblies in order to establish subassembly test manuals.	Is the technical data on subassemblies available for the development of production tests?	7.8
7	Implement corrective actions.	What process is used to collect data on technical events, to report anomalies, and to manage them?	6.7
8	Implement preventive actions.	Do the preventive action procedures include: - The use of diversified sources of information? - The definition of the steps to follow? - The criteria for triggering preventive actions and applying inspection procedures? - A preventive action follow-up review?	6.8
16	Make sure that a person responsible for logistics support, product industrialization, procurement, development and RAMS is involved at every step (concurrent engineering).	Is the viewpoint of the different disciplines involved in the engineering work taken into account?	16.7
28	Define the criteria for product failure.	What counts as a product failure?	14
34	Assign personnel qualified to work with the test equipment, measurements, and corresponding standards.	What training have the test personnel received to qualify them to work with the test equipment, measurements, and corresponding standards?	5.8
36	Conduct a preliminary technical reliability study.	Is a preliminary reliability study conducted during the preliminary development phases?	8
38	Apply specialist design procedures to capitalize on know-how.	Are specialist design procedures managed? Are these specialist design procedures updated?	13.8
39	Manage and maintain a design skills matrix.	Are design skills managed? Is there a regularly updated skills matrix? Is the relevance of training verified?	24.5

#	Recommendation	Question	Weight
44	Control handling and storage operations at the user site.	Is there a guideline document on customer-site handling and storage operations?	7.7
48	Improve product reliability by managing a preferred components list.	Are the component databases updated so as to capitalize on existing knowledge? Is there a regularly maintained preferred components list?	8
50	Maintain and operate a technical event database.	Is there a technical event database? How is it maintained and operated?	24.2
51	Maintain a database to capitalize on reliability assessment studies.	Is there a database to capitalize on reliability assessment studies?	10.6
52	Control design definitions and changes to products in development.	Is there a process for describing, tracking and justifying design definitions and changes to developed products? Is there a database on the design history and justifications?	7.8
54	Involve the Dependability function in the functional and organic design of the product.	Are reliability criteria taken into account in the product architecture, and in decisions about design, product industrialization, and support?	12.6
61	Identify and implement measures to protect subassemblies.	Have measures been identified and implemented to protect subassemblies during certain production activities?	7.3
62	Formally identify the technical risks that affect reliability.	Are the technical risks that affect reliability identified?	21
83	Maximize test coverage based on the specification.	Are checks performed to ensure that test coverage is maximal and is based on the specification? Is there a document describing the method of calculation and justifying the coverage rate?	6
86	Implement design reviews	Are design reviews held?	27.1
87	Implement a maintenance concept as part of logistics support.	Is there a maintenance concept?	5.4

#	Recommendation	Question	Weight
106	Address reliability aspects during design reviews.	Do design reviews address reliability aspects?	12.1
116	Control the quality of data in nonconformity reports.	What information is described in the nonconformity reports?	13.0
123	Plan dependability activities.	Is there a plan that integrates the dependability aspects? Are activities planned and key personnel identified?	17.7
124	Write a final acceptance procedure.	Is there a final acceptance procedure for production tests?	7.8
130	Make sure that the elements required for the reliability assessment are available.	Are the elements required for the reliability assessment available? Does the input data come from validated documents?	7.5
131	Make sure there are design rules for adapting the choice of component to the required reliability level.	Are there any design rules for adapting the choice of component to the required reliability level?	12.7
132	Make sure that production test points are defined and that test recommendations are applied by Production.	Are test points defined and are recommendations on production tests applied?	6
133	Make sure there is a qualification procedure for the product and the manufacturing process.	Is there a qualification procedure for the products and the manufacturing process?	7.2
134	Make sure there is a qualification procedure for Tier-1 suppliers/products.	Is there a qualification procedure for Tier-1 suppliers/products? On what criteria is this qualification based?	7.6
135	Make sure there is a qualification procedure for specific components or new technologies.	Are specific components and new technologies qualified before use? Is there a qualification procedure?	7.2
137	Use a proven methodology and a qualified tool for predictive reliability calculations.	Is the method used for predictive reliability assessment recognized and validated by field data? Is the calculation tool qualified?	7.7
147	Take account of the balance between reliability and the complexity of built-in tests.	Is the impact of the complexity of built-in tests on reliability evaluated?	10.2

#	Recommendation	Question	Weight
150	Use validated and recognized modelling techniques.	Are the modelling techniques (electrical, thermal, mechanical, etc.) validated and recognized?	13.8
181	Know and control the mission profile.	Is the product mission profile known and controlled?	15
182	Implement an improvement process to improve product robustness.	Is there an improvement process to improve product robustness?	20
183	Identify and reduce design risks.	Are analyses (such as Design FMECA) conducted in order to identify and reduce risks? Are the recommendations that emerge from these analyses formalized? Are the recommendations that emerge from these analyses followed?	15
184	Prepare the product industrialization phase.	Is the industrial strategy defined in a plan? Have the product industrialization requirements been detailed in an industrial specification?	16
185	Analyse the need for environmental stress screening (ESS) and define the ESS procedure.	Has the need for environmental stress screening (ESS) been analysed? If applicable, has the ESS procedure been defined?	18.0
186	Check that the environmental stress screening (ESS) test coverage is correctly defined and formalized.	Has the coverage rate of the ESS test been estimated?	11.0

Manufacturing of the board or subassembly

#	Recommendation	Question	Weight
5	Optimize the final test coverage with regard to the product specification, with the aid of the test result summary.	Is a test result summary produced in order to optimize the final test coverage?	6.6
7	Implement corrective actions.	What process is used to collect data on technical events, to report anomalies, and to manage them?	15.4
8	Implement preventive actions.	Do the preventive action procedures include: - The use of diversified sources of information? - The definition of the steps to follow? - The criteria for triggering preventive actions and applying inspection procedures? - A preventive action follow-up review?	15.6
17	Provide continuous environmental monitoring (cleanliness, temperature, humidity) for sensitive manufacturing operations	Is continuous environmental monitoring provided for sensitive manufacturing operations (cleanliness, temperature, humidity)?	9.9
18	Perform corrective maintenance following the occurrence of an anomaly in the production system.	Is corrective maintenance performed following the occurrence of an anomaly in the production system or in the subassemblies produced?	6.9
19	Control working time limits on transformable (limited lifetime) materials	Is there effective control over the working times of transformable materials?	4.0
21	Perform periodic checks on embedded software programming devices.	Are embedded software programming devices checked periodically to ensure that software is correctly loaded?	4.1
23	Automate handling in order to limit potential damage to components, boards and subassemblies.	Is the production and handling of components, boards and subassemblies automated (partially or totally)?	6.5
25	Control and maintain (through updates) the data loaded in programmable production systems.	Is the data loaded in programmable production systems recorded?	2.8
33	Delegate the final inspection of key operations in order to optimize filtering before moving on to the next step in the process.	Is quality control performed by someone other than the operator during the main manufacturing steps?	10.0

#	Recommendation	Question	Weight
37	Make sure that operators and technicians in charge of sensitive operations are trained and sufficiently experienced.	Are sensitive manufacturing processes entrusted to trained and experienced personnel?	12.0
42	Take periodic temperature readings for each soldering operation to make sure it is not damaging the component or subassembly.	Are periodic temperature checks or readings taken for soldering operations?	6.9
43	Ensure compatibility between the production systems and the subassemblies produced.	How is it ensured that the production systems are fully compatible with the items to be produced?	7.2
46	Record anomalies liable to lead to the application of corrective and/or preventive actions.	Are the anomalies encountered recorded so as to enable corrective and/or preventive actions?	7.6
60	Manage priorities that need to be met within process deadlines (Planning).	How are priorities managed with regard to process deadlines?	3.1
61	Identify and implement measures to protect subassemblies.	Have measures been identified and implemented to protect subassemblies during certain production activities?	7.3
75	Manage the workplace environment.	How is the workplace environment controlled? (Cleanliness, temperature and humidity)	13.0
77	Manage changes to manufacturing processes.	How are changes to manufacturing processes managed? Which changes are notified to the customer? Is there a re-qualification plan according to the nature of the change?	13.9
79	Inspect production machines, tools and programmable devices before use.	How are production machines, tools and programmable devices inspected before use?	12.0
84	Periodically check for bath contamination during wave soldering operations	Are solder baths periodically checked for contamination? Using which procedure?	5.8
88	Implement self-inspection by the operator at the end of each production process.	Is self-inspection performed by the operator at the end of each production process? Is it instruction-based? Is the inspection result recorded?	5.3

#	Recommendation	Question	Weight
89	Inspect solder paste deposition after screen-printing.	Is the quality of solder paste deposition inspected systematically after screen-printing?	6.0
90	Implement periodic inventories of component and material stocks.	Are periodic inventories carried out on component and material stocks?	5.5
91	Implement ESD protections for components and subassemblies during handling and storage.	Are specific ESD protections put in place for components and subassemblies during handling and storage?	26.0
92	Perform periodic checks on production system inspection tools.	Are periodic checks performed on production system inspection tools? Are the measurements checked for reproducibility?	4.9
93	Implement suitable protections to avoid damaging subassemblies during cleaning.	Are suitable protections applied to avoid damaging subassemblies during cleaning?	6.0
94	Implement testing tool self-tests to detect any anomalies.	Is there a testing tool self-test to detect any anomalies before use on the subassembly?	5.1
96	Manage the conformity of items and materials in stock.	Is the conformity of materials and items in stock identified?	6.0
97	Use Statistical Process Control (SPC) to manage the critical characteristics of the production process.	Is the production process managed by SPC (Statistical Process Control)? Have the critical characteristics identified by SPC been analysed? Do they feature clearly on the Control Plan? Are the upper and lower control limits updated?	4.5
98	Provide detailed instructions for each production process.	Are detailed instructions available for each production process? Are these instructions available online? Are help documents permanently visible to the operator?	12.0
99	Implement a procedure to prevent items/consumables that are expired or due to be scrapped being used in production	Is there a procedure for identifying expired items/consumables and withdrawing them from stock (or reforming them)?	6.4
100	Implement real-time processing of test monitoring indicators.	Are test monitoring indicators processed in real time in order to avoid damage to the subassembly if an anomaly occurs?	4.7

#	Recommendation	Question	Weight
102	Continuously monitor the temperature of the drying and polymerization ovens.	Is the temperature of the drying and polymerization ovens continuously monitored, using external probes?	12.7
112	Define upper and lower safety limits on inspection and monitoring system parameters.	Are upper and lower safety limits defined on the parameters of the inspection and monitoring systems? Do they trigger systematic shutdown of the cycle and analysis by a technician before restart?	5.7
116	Control the quality of data in nonconformity reports.	What information is described in the nonconformity reports?	12.0
120	Verify solder mask deposition before varnishing.	Is there an inspection step (even if only visual) to verify correct deposition of the solder mask before varnishing?	6.5
121	Check the machine parameters every time before using a production machine on a board or a subassembly (Setup).	Are the machine parameters checked before each use of a production machine? Is this check described in the corresponding machine instructions?	4.7
126	Revise and strengthen production system controls to eliminate any possibility of damaging component connections.	Is the inspection specification for the pick-and-place machine updated periodically? Are the board inspection results taken into account?	6.7
127	Control the preparation of transformable materials by means of a qualified procedure and inspection measures.	How is the preparation of transformable materials (solder paste, glues, varnish) controlled?	5.9
128	Maintain the skills of operators and technicians through continuous training and frequent assessment.	Are training, assessment and other actions performed to maintain the skills of operators and technicians? Are process changes accompanied by additional training? How is operator or technician turnover managed?	4.4
129	Verify the effectiveness of the final varnishing quality inspection by means of an appropriate procedure.	Is the effectiveness of the final varnishing quality inspection assured by strict application of the inspection procedure?	5.2
138	Make sure that the correct embedded software is loaded, and record its version details.	Are checks performed to ensure that the correct embedded software is loaded, and are its version details recorded?	6.7

#	Recommendation	Question	Weight
139	Ensure that preventive maintenance is performed on the production systems and that it is monitored.	Is preventive maintenance performed on the production systems? Is this maintenance monitored?	5.9
144	Sensitize personnel to the importance of visual checks on subassemblies after placement and before reflow.	Are personnel made aware of the importance of visual checks on subassemblies after placement and before reflow?	5.9
153	Check the conformity of purchased products.	Is the conformity of purchased products checked against clearly defined specifications?	8.6
154	Before starting the test, verify (by reading the barcode or S/N) that the test device and software are compatible with the product under test.	Before starting the test, is a check performed (by reading the barcode or S/N) to ensure that the test device and software are compatible with the product under test?	6.1
155	Check that the ESS procedure is correctly applied.	Have checks been done to ensure that the ESS procedure defined at the design stage is correctly applied?	5.2
172	Control the viscosity and thickness of the varnish.	Is the viscosity of the varnish inspected periodically? Is the thickness of the varnish inspected using test specimens?	6.0
173	Systematically inspect the quality of the solder joints on board-mounted components after solder reflow.	Is the quality of the solders controlled systematically, by inspecting the solder joints after reflow?	6.0
174	Implement conformity checks on finished products entering storage.	Are conformity checks performed on finished products entering storage, so that nonconforming items are excluded?	8.0
175	Control the environmental conditions for the stock of components and materials.	Are the temperature/humidity conditions of the component and material storage zones controlled?	9.0
176	Control the sensitivity of components to moisture (MSL) and to the reflow process.	Is the MSL of components taken into account during their industrial implementation?	12.0
177	Track all production and testing operations on the batch tracking sheet.	Is there a batch tracking sheet describing the required traceability information for all operations?	8.0

#	Recommendation	Question	Weight
178	Optimize the production line organization.	Are actions carried out to continuously improve the organization of the production line?	9.0
179	Guard against the purchase of fraudulent components.	Is the risk of buying fraudulent/counterfeit components covered by a procedure?	10.0
180	Control the bending of the board during electrical probe tests.	Is the bending of the board controlled during electrical probe tests? Is there a document specifying what is acceptable?	7.0

Equipment integration

#	Recommendation	Question	Weight
5	Optimize the final test coverage with regard to the product specification, with the aid of the test result summary.	Is a test result summary produced in order to optimize the final test coverage?	6.6
7	Implement corrective actions.	What process is used to collect data on technical events, to report anomalies, and to manage them?	15.4
8	Implement preventive actions.	Do the preventive action procedures include: - The use of appropriate sources of information? - The definition of appropriate steps? - The triggering of preventive actions and the application of control measures? - A management review of preventive actions?	15.6
9	Ensure product traceability.	How is product traceability ensured?	16.5
10	Manage the packaging of systems and equipment.	Does the supplier control wrapping, packaging and marking processes to ensure compliance with specified requirements? Is there a list of equipment that requires packaging?	12.3
11	Manage the storage conditions for subassemblies.	Are there any designated storage areas or premises? Are they adapted to prevent deterioration or damage to subassemblies? Are appropriate measures taken to enable reception and shipping in these areas?	10.8
12	Manage the delivery conditions for systems and equipment.	Does the supplier take steps to maintain product quality after final testing and inspection? When specified by contract, are these steps extended to include delivery for final use?	17.5
13	Validate the required tests and inspections before moving on to the next phase.	How are the planned tests and inspections for a phase validated before moving on to the next phase?	7.2
14	Perform final tests and inspections.	How is it ensured that all final tests and inspections are conducted in accordance with the control plan and the written procedures?	7.9

#	Recommendation	Question	Weight
15	Perform tests and inspections specific to the reception of materials and parts.	Are appropriate tests and inspections carried out on all incoming parts or materials before use, in accordance with a specification?	6.7
18	Perform corrective maintenance following the occurrence of an anomaly in the production system.	Is corrective maintenance performed following the occurrence of an anomaly in the production system or in the subassemblies produced?	6.9
19	Control working time limits on transformable (limited lifetime) materials.	Is there effective control over the working times of transformable materials?	4.0
21	Perform periodic checks on embedded software programming devices.	Are embedded software programming devices checked periodically to ensure that software is correctly loaded?	4.1
23	Automate handling in order to limit potential damage to components, boards and subassemblies.	Is the production and handling of components, boards and subassemblies automated (partially or totally)?	6.5
25	Control and maintain (through updates) the data loaded in programmable production systems.	Is the data loaded in programmable production systems managed?	2.8
30	Define the degree of nonconformity.	Is the description of accepted nonconformities, or of repairs performed, recorded to indicate the real condition of the product?	10.3
32	Define the necessary resources for the testing and inspection of the product.	Which procedure defines the necessary resources for the testing and inspection of the product?	11.6
35	Provide documents for performing incoming inspections on subassembly supplies.	Are there any documents for performing incoming inspections on supplies?	8.8
37	Make sure that operators and technicians in charge of sensitive operations are trained and sufficiently experienced.	Are sensitive manufacturing processes entrusted to trained and experienced personnel?	12.0
43	Ensure compatibility between the production systems and the subassemblies produced.	How is it ensured that the production systems are fully compatible with the items to be produced?	7.2

#	Recommendation	Question	Weight
46	Record any anomalies liable to lead to the application of corrective and/or preventive actions.	Are the anomalies encountered recorded so as to enable corrective and/or preventive actions?	7.6
47	Establish procedures to verify that purchased assemblies and subassemblies meet the specified requirements.	Are there any written procedures for checking that purchased assemblies and subassemblies meet the specified requirements?*	10.6
49	Manage the processing of nonconformities.	How are nonconformities analysed? Are there any procedures for processing nonconforming products?	13.6
60	Manage priorities that need to be met within process deadlines (Planning).	How are priorities managed with regard to process deadlines?	3.1
61	Identify and implement measures to protect subassemblies.	Have measures been identified and implemented to protect subassemblies during certain production activities?	7.3
66	Identify the necessary means for special processes.	Are the necessary means for special processes identified?	13.1
67	Identify the human resources for special processes.	Are the human resources for special processes identified?	11.7
71	Manage the equipment or system documentation.	Is the documentation associated with the equipment or system strictly managed? Does it take account of all equipment upgrades?	15
73	Make sure that test/inspection equipment and software are compatible with requirements.	What steps are taken to ensure that test/inspection equipment and software are compatible with requirements?	9.6
74	Control the environment of inspection, measurement and testing equipment.	How is the environment of inspection, measurement and testing equipment controlled?	7.9
75	Control the workplace environment.	How is the workplace environment controlled? (Cleanliness, temperature and humidity)	9.6

#	Recommendation	Question	Weight
77	Manage changes to manufacturing processes.	How are changes to manufacturing processes managed? Which changes are notified to the customer? Is there a re-qualification plan according to the nature of the change?	13.9
78	Control handling methods.	Are handling and inter-site transport methods defined?	8.8
79	Inspect production machines, tools and programmable devices before use.	How are production machines, tools and programmable devices inspected before use?	10.5
80	Control handling, storage, packaging and preservation operations.	How are handling, storage, packaging and preservation operations controlled?	6.5
81	Control special processes.	How are special processes controlled? Which documents describe these processes, and how are they updated? How are special processes qualified?	14.4
82	Control services and fluids in the work environment.	How are services and fluids controlled in the work environment?	10.1
88	Implement self-inspection by the operator at the end of each production process.	Is self-inspection performed by the operator at the end of each production process? Is it instruction-based? Is the inspection result recorded?	5.3
91	Implement ESD protections for components and subassemblies during handling and storage.	Are specific ESD protections put in place for components and subassemblies during handling and storage?	26.0
92	Perform periodic checks on production system inspection tools.	Are periodic checks performed on production system inspection tools? Are the measurements checked for reproducibility?	4.9
94	Implement testing tool self-tests to detect any anomalies.	Is there a testing tool self-test to detect any anomalies before use on the subassembly?	5.1
98	Provide detailed instructions for each production process.	Are detailed instructions available for each production process? Are these instructions available online? Are help documents permanently visible/accessible to the operator?	12.0

#	Recommendation	Question	Weight
99	Implement a procedure to prevent items/consumables that are expired or due to be withdrawn being used in production.	Is there a procedure for identifying expired items/consumables and withdrawing them from stock?	6.4
100	Implement real-time processing of test monitoring indicators.	Are test monitoring indicators processed in real time in order to avoid damage to the subassembly if an anomaly occurs?	4.7
113	Keep inspection and test records.	Are records produced and kept to prove that the product has been inspected and/or tested in accordance with defined criteria? Do the records identify the person who performed the inspections?	5.3
114	Maintain an inspection file.	Is there an inspection file containing the acceptance criteria, the inspection and testing sequence, the inspection records, and the list of specific and non-specific inspection instruments?	5.7
116	Control the quality of data in nonconformity reports.	What information is described in the nonconformity reports?	11.1
121	Check the machine parameters every time before using a production machine on a board or a subassembly (Setup).	Are the machine parameters checked before each use of a production machine? Is this check described in the corresponding machine instructions?	4.7
128	Maintain the skills of operators and technicians through continuous training and frequent assessment.	Are training, assessment and other actions performed to maintain operator skills? Are process changes accompanied by additional training?	10.0
138	Make sure that the correct embedded software is loaded, and record its version details.	Are checks performed to ensure that the correct embedded software is loaded, and are its version details recorded?	6.7
139	Ensure that preventive maintenance is performed on the production systems and that it is monitored.	Is preventive maintenance performed on the production systems? Is this maintenance monitored?	5.9
153	Check the conformity of purchased products.	Is the conformity of purchased products checked against clearly defined specifications?	8.6

#	Recommendation	Question	Weight
155	Check that the ESS procedure is correctly applied.	Have checks been done to ensure that the ESS procedure defined at the design stage is correctly applied?	9.0

System integration

#	Recommendation	Question	Weight
7	Implement corrective actions.	What process is used to collect data on technical events, to report anomalies, and to manage them?	15.4
8	Implement preventive actions.	Do the preventive action procedures include: - The use of appropriate sources of information? - The definition of appropriate steps? - The triggering of preventive actions and the application of control measures? - A management review of preventive actions?	15.6
9	Ensure product traceability.	How is product traceability ensured?	16.5
10	Manage the packaging of systems and equipment.	Does the supplier control wrapping, packaging and marking processes to ensure compliance with specified requirements? Is there a list of equipment that requires packaging?	12.3
11	Manage the storage conditions for subassemblies.	Are there any designated storage areas or premises? Are they adapted to prevent deterioration or damage to subassemblies? Are appropriate measures taken to enable reception and shipping in these areas?	10.8
12	Manage the delivery conditions for systems and equipment.	Does the supplier take steps to maintain product quality after final testing and inspection? When specified in the contract, are these steps extended to include delivery for final use?	17.5
13	Validate the required tests and inspections before moving on to the next phase.	How are the planned tests and inspections for a phase validated before moving on to the next phase?	7.2
14	Perform final tests and inspections.	How is it ensured that all final tests and inspections are conducted in accordance with the control plan and the written procedures?	7.9
15	Perform tests and inspections specific to the reception of materials and parts.	Are appropriate tests and inspections carried out on all incoming parts or materials before use, in accordance with a specification?	6.7

#	Recommendation	Question	Weight
30	Define the degree of nonconformity.	Is the description of accepted nonconformities, or of repairs performed, recorded to indicate the real condition of the product?	10.3
32	Define the necessary resources for the testing and inspection of the product.	Which procedure defines the necessary resources for the testing and inspection of the product?	11.6
35	Provide documents for performing incoming inspections on subassembly supplies.	Are there any documents for performing incoming inspections on supplies?	8.8
37	Make sure that operators and technicians in charge of sensitive operations are trained and sufficiently experienced.	Are sensitive manufacturing processes entrusted to trained and experienced personnel?	11.0
47	Establish procedures to verify that purchased assemblies and subassemblies meet the specified requirements.	Are there any written procedures for checking that purchased assemblies and subassemblies meet the specified requirements?	10.6
49	Manage the processing of nonconformities.	How are nonconformities analysed? Are there any procedures for processing nonconforming products?	13.6
66	Identify the necessary means for special processes.	Are the necessary means for special processes identified?	13.1
67	Identify the human resources for special processes.	Are the human resources for special processes identified?	11.7
71	Manage the equipment or system documentation.	Is the documentation associated with the equipment or system strictly managed? Does it take account of all equipment upgrades?	15
73	Make sure that test/inspection equipment and software are compatible with requirements.	What steps are taken to ensure that test/inspection equipment and software are compatible with requirements?	9.6
74	Control the environment of inspection, measurement and testing equipment.	How is the environment of inspection, measurement and testing equipment controlled?	7.9
75	Control the workplace environment.	How is the workplace environment controlled? (Cleanliness, temperature and humidity)	9.6

#	Recommendation	Question	Weight
77	Manage changes to manufacturing processes.	How are changes to manufacturing processes managed? Which changes are notified to the customer? Is there a re-qualification plan according to the nature of the change?	13.9
78	Control handling methods.	Are handling and inter-site transport methods defined?	8.8
79	Inspect production machines, tools and programmable devices before use.	How are production machines, tools and programmable devices inspected before use?	10.5
80	Control handling, storage, packaging and preservation operations.	How are handling, storage, packaging and preservation operations controlled?	6.5
81	Control special processes.	How are special processes controlled? Which documents describe these processes, and how are they updated? How are special processes qualified?	14.4
82	Control services and fluids in the work environment.	How are services and fluids controlled in the work environment?	10.1
88	Implement self-inspection by the operator at the end of each production process.	Is self-inspection performed by the operator at the end of each production process? Is it instruction-based? Is the inspection result recorded?	5.3
91	Implement ESD protections for components and subassemblies during handling and storage.	Are specific ESD protections put in place for components and subassemblies during handling and storage?	18.4
99	Implement a procedure to prevent items/consumables that are expired or due to be withdrawn being used in production.	Is there a procedure for identifying expired items/consumables and withdrawing them from stock?	6.4
113	Keep inspection and test records.	Are records produced and kept to prove that the product has been inspected and/or tested in accordance with defined criteria? Do the records identify the person who performed the inspections?	5.3

#	Recommendation	Question	Weight
114	Maintain an inspection file.	Is there an inspection file containing the acceptance criteria, the inspection and testing sequence, the inspection records, and the list of specific and non-specific inspection instruments?	5.7
116	Control the quality of data in nonconformity reports.	What information is described in the nonconformity reports?	11.1
138	Make sure that the correct embedded software is loaded, and record its version details.	Are checks performed to ensure that the correct embedded software is loaded, and are its version details recorded?	6.7
153	Check the conformity of purchased products.	Is the conformity of purchased products checked against clearly defined specifications?	8.6
155	Check that the ESS procedure is correctly applied.	Have checks been done to ensure that the ESS procedure defined at the design stage is correctly applied?	9.0

Operation and maintenance

#	Recommendation	Question	Weight
7	Implement corrective actions.	What process is used to collect data on technical events, to report anomalies, and to manage them?	17.5
8	Implement preventive actions.	Do the preventive action procedures include: - The use of diversified sources of information? - The definition of the steps to follow? - The criteria for triggering preventive actions and applying inspection procedures? - A preventive action follow-up review?	17.7
9	Ensure product traceability.	How is product traceability ensured?	9.2
10	Manage the packaging of systems and equipment	Does the supplier control wrapping, packaging and marking processes to ensure compliance with specified requirements? Is there a list of equipment that requires packaging?	13.8
11	Manage the storage conditions for subassemblies.	Are there any designated storage areas or premises? Are they adapted to prevent deterioration or damage to subassemblies? Are appropriate measures taken to enable reception and shipping in these areas?	15.6
13	Validate the required tests and inspections before moving on to the next phase.	How are the planned tests and inspections for a phase validated before moving on to the next phase?	11.2
14	Perform final tests and inspections.	How is it ensured that all final tests and inspections are conducted in accordance with the control plan and the written procedures?	10.4
20	Adopt a policy for managing nonconformity risks.	Is a policy applied to identify, evaluate and manage the potential risks associated with nonconformities, whether in the products or in the design, planning, manufacturing, assembly, inspection, etc. processes?	16.3
30	Define the degree of nonconformity.	Is the description of accepted nonconformities, or of repairs performed, recorded to indicate the real condition of the product?	12.8
32	Define the necessary resources for the testing and inspection of the product.	Which procedure defines the necessary resources for the testing and inspection of the product?	14.3
35	Provide documents for performing incoming inspections on subassembly supplies.	Are there any documents for performing incoming inspections on supplies?	9.9
47	Establish procedures to verify that purchased assemblies and subassemblies meet the specified requirements.	Are there any written procedures for checking that purchased assemblies and subassemblies meet the specified requirements?	6.8

#	Recommendation	Question	Weight
63	Identify documentation for special processes.	Is there documentation for special processes? Is this documentation updated?	12.2
66	Identify the necessary means for special processes.	Are the necessary means for special processes identified?	13.1
67	Identify the human resources for special processes.	Are the human resources for special processes identified?	13.7
71	Manage the equipment or system documentation.	Is the documentation associated with the equipment or system strictly managed? Does it take account of all equipment upgrades?	14
72	Control product testability and maintainability.	How are product testability and maintainability controlled?	17.6
73	Make sure that test/inspection equipment and software are compatible with requirements.	What steps are taken to ensure that test/inspection equipment and software are compatible with requirements?	11.3
74	Control the environment of inspection, measurement and testing equipment.	How is the environment of inspection, measurement and testing equipment controlled?	11.7
75	Control the workplace environment.	How is the workplace environment controlled? (Cleanliness, temperature and humidity)	10.8
77	Manage changes to manufacturing processes.	How are changes to manufacturing processes managed? Which changes are notified to the customer? Is there a re-qualification plan according to the nature of the change?	13.9
78	Control handling methods.	Are handling and inter-site transport methods defined?	9.9
79	Inspect production machines, tools and programmable devices before use.	How are production machines, tools and programmable devices inspected before use?	11.3
80	Control handling, storage, packaging and preservation operations.	How are handling, storage, packaging and preservation operations controlled?	11.3
81	Control special processes.	How are special processes controlled? Which documents describe these processes, and how are they updated? How are special processes qualified?	15.2
82	Control services and fluids in the work environment.	How are services and fluids controlled in the work environment?	12.2
91	Implement ESD protections for components and subassemblies during handling and storage.	Are specific ESD protections put in place for components and subassemblies during handling and storage?	17.4

#	Recommendation	Question	Weight
114	Maintain an inspection file.	Is there an inspection file containing the acceptance criteria, the inspection and testing sequence, the inspection results, and the list of specific and non-specific inspection instruments?	5.7
116	Control the quality of data in nonconformity reports.	What information is described in the nonconformity reports?	11.1
181	Know and control the mission profile.	Is the product mission profile known and controlled?	9

Support activities

#	Recommendation	Question	Weight
3	Allocate the necessary infrastructures to avoid deterioration of products during production and integration operations.	Have analyses (such as Process FMECA) been carried out in order to formulate recommendations on the necessary infrastructures for production and integration? Are the recommendations that emerge from these analyses formalized? Are the recommendations that emerge from these analyses followed?	7.4
4	Continuously improve the company's reliability engineering.	Have improvement targets been defined for reliability engineering? Have performance indicators been defined for these targets?	6.6
24	Collect customer comments about the reliability of the product in operation.	Is there a procedure to collect customer comments about the reliability of the product in operation?	7.9
45	Commit the company to a quality certification process.	Does the company have one or more quality certifications, such as ISO 9001 v2016 or EN9100?	6.5
53	Leverage experience feedback.	Is experience feedback used to maintain confidence in delivering the required reliability performance?	11
55	Involve the Dependability function in every phase of the project.	Does the Dependability function participate in every phase of the project?	8.8
56	Provide training for personnel in reliability studies or whose work has an influence on reliability, or employ qualified personnel.	Is the training of reliability personnel adapted to the criticality of the reliability performance expected from the product?	7.5
58	Supply the necessary resources for reliability studies.	Is the necessary technical data for reliability studies accessible? Are the necessary tools available for reliability studies? Are the necessary time and funding provided for the studies?	8.3

#	Recommendation	Question	Weight
59	Manage reliability studies documentation.	Are reliability studies documents managed?	5.4
68	Identify product reliability risks at subcontractors.	Have product reliability risks at subcontractors been identified?	7.2
69	Integrate reliability into the company's quality policy.	Does the subject of reliability feature in the company's quality policy?	7.4
76	Control monitoring and measurement devices, the metrology of measurement apparatus and industrial machinery.	What procedure is there to control monitoring and measurement devices, and the metrology of measurement apparatus and industrial machinery?	7.8
104	Appoint someone to be in charge of reliability studies.	Has someone been formally appointed to be in charge of reliability studies?	8.5
105	Manage the subcontracting of reliability studies.	How is the subcontracting of reliability studies managed?	10
110	Implement reliability growth actions.	Are reliability growth actions implemented?	9.1
111	Plan reliability studies.	Are reliability studies planned?	8
119	Plan for periodic customer consultations on reliability aspects.	Are periodic consultations planned with customers on reliability aspects?	7.3
142	Select the components used.	Are reliability criteria considered when selecting the components to be used?	12.9
143	Select component suppliers.	Are reliability criteria considered when selecting component suppliers?	10.8
146	Monitor and control corrective actions by subcontractors relating to product reliability.	Are reliability-related corrective actions by subcontractors monitored?	7.2
148	Address reliability in management reviews.	Is reliability addressed in management reviews?	5.6
149	Handle anomalies through an incident management and corrective action approach.	What process is used to collect data on technical events, report anomalies and measure reliability growth? How are equipment upgrades managed?	8.3
151	Use appropriate statistical methods to leverage experience feedback.	Are appropriate statistical methods used to leverage experience feedback?	6

#	Recommendation	Question	Weight
152	Validate the subcontractor's reliability management guidelines.	Are the subcontractor's reliability management guidelines validated?	7.7

Ruggedizing

#	Recommendation	Question	Weight
156	Ensure that environmental specifications are complete.	How is the completeness of environmental specifications ensured?	4
157	Provide training and maintain skills in product implementation and maintenance.	Have users (operation and maintenance) been trained in the product? Is this training renewed and updated as needed?	7
158	Ensure compliance with product-specific procedures and function-specific rules by means of an appropriate monitoring system.	Are there monitoring measures (processes, recording methods) that enable the supplier to ensure that users comply with the product instructions?	7
159	Design dependable electrical protection devices.	How are the electrical protection devices designed?	4
160	Analyse and manage the risk of the product under test being damaged due to failures in the test or maintenance equipment.	Is the risk of the product under test being damaged by test equipment failures taken into account?	4
161	Identify reasonably foreseeable weather-related hazards and take appropriate preventive measures.	Has allowance been made for reasonably foreseeable weather-related hazards?	4
162	Identify reasonably foreseeable abnormal uses and take appropriate preventive measures.	Has allowance been made for reasonably foreseeable abnormal uses?	4
163	Include the production, storage and maintenance environments in the product's environmental specifications.	How are the production, storage and maintenance environments taken into account in the product's environmental specifications?	4
164	Demonstrate compliance with environmental specifications.	How is compliance with environmental specifications substantiated?	4
165	Implement a product improvement process (e.g. accelerated stress tests) to limit the product's sensitivity to environmental stresses (disturbance, severe environments, overstress).	Is there a product improvement process to increase its robustness and accelerate its maturity?	7
166	Conduct an analysis of failures that could lead to failure propagation.	Is the risk of failure propagation analysed?	4
167	Conduct a process analysis of implementation and maintenance operations.	How is the risk of error in implementation and maintenance operations analysed?	4
168	Review maintenance operations performed by end users and address their recommendations.	Is there a review of maintenance operations carried out by the user?	4

#	Recommendation	Question	Weight
169	Write complete procedures for all product implementation and maintenance operations.	Is there documentation describing all of the product implementation and maintenance operations?	7
170	Comply with a standard on power supplies (such as EN2282, which defines possible disturbances and variations). The standard must be complied with both for electricity generation and for electricity consumption.	Is there a standard on electrical power supplies that applies to the product and the surrounding system? If so, how is it applied?	4
171	Comply with a standard on conducted and radiated electromagnetic interference; this applies both to the product and to the system into which it is integrated.	Is there a standard on conducted and radiated electromagnetic interference that applies to the product and the surrounding system? If so, how is it applied?	3

Detailed recommendation sheets

Recommendation Assign human and financial resources to dependability a.k.a. “RAMST” (Reliability, Availability, Maintainability, Safety, Testability) studies.	Correspondence ISO9001: L2 EN9100: L3	N° 1
Phase(s) in which the recommendation applies SPECIFICATION		Weight 10.7
Further description Financing is assigned to the project’s Dependability Manager(s). This is the subject of a separate project management item (at accounting level). The means and personnel required to carry out dependability studies are made available to the Product Dependability Manager.		
Audit question Is there a financing item for dependability studies? Have the necessary means and personnel been identified?		
Level 1	No specific resources are allocated to dependability studies: they are integrated into other studies or there is a specific non-formalized allocation.	
Level 2	Resources allocated to dependability studies are identified at project management level and formalized in a document.	
Level 3	Resources allocated to dependability studies are identified at project management level and formalized in a validated plan.	
Level 4	Resources allocated to dependability studies are identified at project management level and formalized in a validated plan. Proof of the actual availability of the resources is established.	

<u>Recommendation</u> Allocate reliability requirements down to subassembly level.		Correspondence	N° 2
Phase(s) in which the recommendation applies SPECIFICATION			Weight 10.4
Further description The reliability requirements must be applied at every level (system, equipment, board, etc.) by reliability specialists or under the control of reliability specialists. They can be grouped together in a dedicated document or integrated into the design documents.			
Audit question Are the general reliability requirements allocated down to subassembly level? If so, which method of allocation was used?			
Level 1	Reliability requirements are not or will not be assigned to the different subassembly levels.		
Level 2	People in charge of reliability engineering have defined (or participated in defining) the detailed allocation of reliability requirements to the different subassembly levels. The detailed allocations are not attested by any document.		
Level 3	People in charge of reliability engineering have defined (or participated in defining) the detailed allocation of reliability requirements to the different subassembly levels. The detailed allocations are attested by validated documents.		
Level 4	People in charge of reliability engineering have defined (or participated in defining) the detailed allocation of reliability requirements to the different subassembly levels. The detailed allocations are attested by validated documents. This allocation is based on prior data relating to similar equipment (technology, operating environment).		

Recommendation Allocate the necessary infrastructures to avoid deterioration of products during production and integration operations.	Correspondence ISO9001: L4 EN9100: L4	N° 3
Phase(s) in which the recommendation applies SUPPORT		Weight 7.4
Further description A Process FMECA analysis is conducted to ensure that production and integration operations do not impact reliability; this analysis serves to define the necessary infrastructures or to issue recommendations about existing infrastructures. Typical recommendations: apply 5S methods; provide appropriate power networks, clean rooms, ergonomic buildings; increase surface areas to facilitate handling; improve lighting; implement tidiness and cleanliness standards; improve tool quality; raise awareness of reliability issues.		
Audit question Have analyses (such as Process FMECA) been carried out in order to formulate recommendations on the necessary infrastructures for production and integration? Are the recommendations that emerge from these studies formalized? Are the recommendations that emerge from these analyses followed?		
Level 1	No impact analyses have been conducted on the infrastructures; no measures are planned.	
Level 2	Impact analyses are conducted informally; the infrastructures take account of these analyses.	
Level 3	Impact analyses are formalized. The recommendations that emerge from these analyses are formalized and taken into account when defining and/or improving the infrastructures.	
Level 4	Impact analyses are formalized. The recommendations that emerge from these analyses are formalized and taken into account when defining and/or improving the infrastructures. The infrastructures are subject to periodic inspection.	

Recommendation	Correspondence	N°
Continuously improve the company's reliability engineering.		4
Phase(s) in which the recommendation applies	Weight	
SUPPORT	6.6	
Further description		
Set reliability engineering improvement targets and tie them to performance indicators.		
Examples of targets:		
- Mandatory: maintaining reliability guidelines; achieving reliability performance levels, etc. - Nice to have: training; participation in working groups, conferences, etc.		
Audit the company's reliability engineering.		
Audit question		
Have improvement targets been defined for reliability engineering?		
Have performance indicators been defined for these targets?		
Level 1	No reliability engineering improvement targets are defined.	
Level 2	Reliability engineering improvement targets are defined informally, but no performance indicators are defined for these targets.	
Level 3	Reliability engineering improvement targets are formalized and performance indicators are defined for these targets.	
Level 4	Reliability engineering improvement targets are formalized, in particular those relating to reliability performance levels and maintaining the guidelines. Performance indicators are in place for these targets. Reliability engineering is evaluated by regular audits.	

Recommendation Optimize the final test coverage with regard to the product specification, with the aid of the test result summary.	Correspondence ISO9001: L3 EN9100: L3	N° 5
Phase(s) in which the recommendation applies MANUFACTURE EQUIPMENT_INTEGRATION		Weight 6.6 6.6
Further description The final product test, and in particular the level of coverage achieved by this test, must be analysed, and if necessary modified, even after the product has been specified and designed. A test result summary can optimize the effectiveness of this final test.		
Audit question Is a test result summary produced in order to optimize the final test coverage?		
Level 1	The predefined test coverage rate is not reviewed.	
Level 2	A product test summary can be compiled with the aim of reviewing and improving the predefined coverage rate. However, there is no document that formally describes any corresponding actions.	
Level 3	The final product tests are regularly reviewed, even after specification and design, with the aim of increasing the predefined test coverage. There are documents that describe the procedure to follow.	
Level 4	The final product tests are regularly reviewed, even after specification and design, with the aim of increasing the predefined test coverage. There are documents that describe the procedure to follow. These have been validated by an authority independent from the operating entity.	

Recommendation Ensure the completeness of information on subassemblies in order to establish subassembly test manuals.		Correspondence EN9100: L4	N° 6
Phase(s) in which the recommendation applies DESIGN			Weight 7.8
Further description Obtain technical data on subassemblies in order to develop production tests.			
Audit question Is the technical data on subassemblies available for the development of production tests?			
Level 1	Technical data on subassemblies is not available.		
Level 2	Technical data on subassemblies is available, but it is not validated and is only partially usable.		
Level 3	Technical data on subassemblies is available; it is validated, but incomplete or only partially usable.		
Level 4	Technical data on subassemblies is available; it is validated, complete, and usable for all of the subassemblies.		

Recommendation Implement corrective actions.	Correspondence ISO9001: L4 EN9100: L4	N° 7
Phase(s) in which the recommendation applies DESIGN MANUFACTURE EQUIPMENT_INTEGRATION SYSTEM_INTEGRATION OPERATION_&_MAINTENANCE		Weight 6.7 15.4 15.4 15.4 17.5
Further description This recommendation deals with the implementation of corrective actions (the documentation aspect is dealt with in Recommendation 116). The following functions must be covered by a quality assurance procedure: - Processing customer complaints and product nonconformity reports - Searching for causes of nonconformity relating to the product, the process and the quality system, and recording the results of this search - Determining corrective and preventive actions required to eliminate causes of nonconformity - Applying resources to measure the efficacy of these corrective and preventive actions.		
Audit question What process is used to collect data on technical events, to report anomalies, and to manage them?		
Level 1	There are no corrective action procedures.	
Level 2	Corrective actions are implemented following a customer complaint or nonconformity report, but they are not formalized.	
Level 3	Corrective action procedures include at least: - Effective processing of customer complaints and product nonconformity reports - The search for causes of nonconformity relating to the product, the process and the quality system, and recording the results of this search - The determination of corrective actions required to eliminate causes of nonconformity These procedures do not define the application of control measures to ensure that corrective actions are implemented and that they produce the desired effect.	
Level 4	Corrective action procedures include: - Effective processing of customer complaints and product nonconformity reports - The search for causes of nonconformity relating to the product, the process and the quality system, and recording the results of this search - The determination of corrective actions required to eliminate causes of nonconformity - The application of control measures to ensure that corrective actions are implemented and that they produce the desired effect.	

Recommendation Implement preventive actions.	Correspondence ISO9001: L2 EN9100: L3	N° 8
Phase(s) in which the recommendation applies DESIGN MANUFACTURE EQUIPMENT_INTEGRATION SYSTEM_INTEGRATION OPERATION_&_MAINTENANCE		Weight 6.8 15.6 15.6 15.6 17.7
Further description Preventive action procedures include: - Using appropriate sources of information – such as processes and operations affecting product quality, waivers, audit results, quality records, maintenance reports and customer complaints – to detect, analyse and eliminate potential causes of nonconformity - The definition of appropriate steps to deal with any problem that requires preventive action - Triggering preventive actions and applying inspection procedures to make sure they produce the desired effect - Ensuring that relevant information on the actions implemented is submitted for management review.		
Audit question Do the preventive action procedures include: - The use of diversified sources of information? - The definition of the steps to follow? - The criteria for triggering preventive actions and applying inspection procedures? - A preventive action follow-up review?		
Level 1	No preventive action procedures are implemented.	
Level 2	Preventive action procedures exist but are incomplete.	
Level 3	Preventive action procedures exist and are almost complete with regard to the listed criteria (there may be minor nonconformities in the way criteria are applied or met).	
Level 4	Preventive action procedures exist, are formalized, and are complete with regard to the listed criteria.	

Recommendation Ensure product traceability.	Correspondence ISO9001: L3 EN9100: L4	N° 9
Phase(s) in which the recommendation applies EQUIPMENT_INTEGRATION SYSTEM_INTEGRATION OPERATION_&_MAINTENANCE		Weight 16.5 16.5 9.2
Further description The system implemented must make it possible to: - Maintain the product identification throughout the life cycle - Know the history (definition file and changes) and final use (delivery, scrap) of all products made from the same batch of raw material or belonging to the same production batch - Identify the elements that make up an assembly, and identify the higher assembly - Find sequential documentation on the production (manufacture, assembly, inspection) of a given product (e.g. configuration tracking sheet listing operations performed and anomalies observed). The traceability system must be capable of determining the configuration of the product ready for delivery, including any variance between the actual condition and the agreed/specified condition.		
Audit question How is product traceability ensured?		
Level 1	There is no product traceability during the life cycle; the product is distinguished only by its markings.	
Level 2	There is sufficient traceability to identify the product, but not to determine its origin or history.	
Level 3	There is sufficient traceability to identify the product and determine its history (design definition and changes), but not to identify the documentation associated with its life cycle (e.g. there is no configuration tracking sheet listing operations performed and any anomalies observed).	
Level 4	There is sufficient traceability to identify the product and determine its history (design definition and changes), including the components e.g. date code, etc., as well as to identify the documentation associated with its life cycle (e.g. a configuration tracking sheet listing operations performed and any anomalies observed). The recommendation is applied in full.	

Recommendation Manage the packaging of systems and equipment.	Correspondence ISO9001: L3 EN9100: L3	N° 10
Phase(s) in which the recommendation applies EQUIPMENT_INTEGRATION SYSTEM_INTEGRATION OPERATION_&_MAINTENANCE		Weight 12.3 12.3 13.8
Further description The wrapping, packaging and marking processes must be controlled to ensure compliance with specified requirements. Define a list of equipment that requires packaging. Propose a way to manage packaging specific to particular products (dates, methods, duration). Periodically inspect packaging quality. Use appropriate product-specific packaging.		
Audit question Does the supplier control wrapping, packaging and marking processes to ensure compliance with specified requirements? Is there a list of equipment that requires packaging?		
Level 1	Product packaging is not defined; packaging materials are used on the basis of their availability. Information on the packaging dates, management methods and required inspections is not recorded.	
Level 2	Standard packaging is used for the products. Information is supplied about the packaging. There is no specific packaging inspection.	
Level 3	Product-specific packaging is used, and documentation is associated with it. There is no specific packaging inspection.	
Level 4	Product-specific packaging is used, and documentation is associated with it. A specific packaging inspection is performed periodically, following a formal procedure.	

Recommendation Manage the storage conditions for subassemblies.	Correspondence ISO9001: L3 EN9100: L3	N° 11
Phase(s) in which the recommendation applies EQUIPMENT_INTEGRATION SYSTEM_INTEGRATION OPERATION_&_MAINTENANCE		Weight 10.8 10.8 15.6
Further description Designated storage areas or premises must be adapted to prevent damage or deterioration to subassemblies. - Appropriate measures are taken to enable reception in these areas and shipment from them - The state of products in stock must be evaluated at appropriate intervals to detect any deterioration - The storage atmosphere must be controlled - Tailor positioning in storage - Organize periodic interventions (e.g. powering up) to maintain product characteristics in storage.		
Audit question Are there any designated storage areas or premises? Are they adapted to prevent deterioration or damage to subassemblies? Are appropriate measures taken to enable reception and shipping in these areas?		
Level 1 Product storage areas are not specific; the storage environment is not considered.		
Level 2 Product storage areas are not specific; the storage environment is controlled and adapted to the stored products.		
Level 3 Product storage areas are specific. The storage environment is controlled and adapted to the stored products. Storage positions are individually defined. Periodic interventions are performed to maintain product characteristics.		
Level 4 Product storage areas are specific. The storage environment is controlled and adapted to the stored products. Storage positions are individually defined. Periodic interventions are performed to maintain product characteristics. The condition of products in stock is regularly inspected; the stock is verified and storage conditions are regularly optimized.		

Recommendation		Correspondence	N°
Manage the delivery conditions for systems and equipment.		EN9100: L4	12
Phase(s) in which the recommendation applies			Weight
EQUIPMENT_INTEGRATION			17.5
SYSTEM_INTEGRATION			17.5
Further description			
The supplier must take measures to protect the quality of the product after the final tests and inspections. When specified in the contract, this protection is extended to include delivery for final use.			
The supplier makes sure that the accompanying product documentation, as specified on the order, is present at the time of delivery and that it is protected against loss and damage.			
Audit question			
Does the supplier take steps to maintain product quality after final testing and inspection? When specified in the contract, are these steps extended to include delivery for final use?			
Level 1	Normal protections for products during delivery are not used.		
Level 2	Product quality protections are used during delivery to the customer. The supplier does not check that the accompanying documents are present.		
Level 3	Product quality protections are used during delivery to the customer. The supplier checks that the accompanying documents are present, but does not protect them against loss and damage.		
Level 4	Product quality protections are used during delivery to the customer. The supplier checks that the accompanying product documentation, as specified on the order, is present, and that it is protected against loss and damage.		

Recommendation Validate the required tests and inspections before moving on to the next phase.	Correspondence EN9100: L4	N° 13
Phase(s) in which the recommendation applies EQUIPMENT_INTEGRATION SYSTEM_INTEGRATION OPERATION_&_MAINTENANCE		Weight 7.2 7.2 11.2
Further description During the phase, the product must be inspected and tests must be carried out in accordance with the quality plan and/or written procedures. The product must remain blocked until the required inspections and tests are terminated, or until the necessary reports have been received and verified. Corrective actions must be carried out if applicable.		
Audit question How are the planned tests and inspections for a phase validated before moving on to the next phase?		
Level 1 Test and inspection results are not validated before moving on to the next phase.		
Level 2 Test and inspection results are validated before moving on to the next phase, but this is not formalized by a written procedure or a quality plan.		
Level 3 Test and inspection results are validated before moving on to the next phase. This validation is described in a written procedure.		
Level 4 Test and inspection results are validated before moving on to the next phase. This validation is described in a written procedure, approved by Quality. The procedure is effectively applied, and, where applicable, corrective actions are based on these controls.		

Recommendation Perform final tests and inspections.	Correspondence EN9100: L4	N° 14
Phase(s) in which the recommendation applies		Weight
EQUIPMENT_INTEGRATION		7.9
SYSTEM_INTEGRATION		7.9
OPERATION_&_MAINTENANCE		10.4
Further description The control plan must contain a “Final Tests and Inspections” chapter, describing the equipment, the general conditions and the references to the associated procedures. The acceptance criteria and the specific product requirements must be described in detail. These inspections must be based on an analysis (e.g. FMECA) of the history of failures encountered and customer feedback. Additionally, it is important for an independent inspector to check, prior to shipping, that: - All the activities specified in the control plan have been completed satisfactorily - The associated data and documentation are available and accepted.		
Audit question How is it ensured that all final tests and inspections are conducted in accordance with the control plan and the written procedures?		
Level 1	No final tests or inspections.	
Level 2	Final tests and inspections are carried out, but these are not described in procedures adapted to the product under test.	
Level 3	Final tests and inspections are carried out in accordance with the control plan and the written procedures, and are adapted to the products under test. However, the application of these tests and inspections is neither verified nor validated.	
Level 4	Final tests and inspections are carried out in accordance with the control plan and the written procedures. The control plan is complete and is based on the experience and analysis of failures encountered (FMECA). Before shipping, the following points are controlled by an independent inspector: - All the activities specified in the control plan have been completed satisfactorily - The associated data and documentation are available and accepted (document such as a tracking sheet that logs the configuration, the operations performed and any anomalies observed).	

Recommendation Perform tests and inspections specific to the reception of materials and parts.	Correspondence EN9100: L4	N° 15
Phase(s) in which the recommendation applies EQUIPMENT_INTEGRATION SYSTEM_INTEGRATION		Weight 6.7 6.7
Further description Make sure that purchased parts and materials are not used or implemented until they have been inspected or until their conformity to the specified requirements has been verified in some other way. The verification of conformity to the specified requirements must be performed in accordance with the quality plan and/or the written procedures. Inspections carried out on subcontractor premises, and proofs of conformity supplied, must be taken into account in determining the extent and nature of the inspections to be performed on reception. When, for reasons of urgency, the incoming product is released before being verified, it must be identified and the decision to release recorded.		
Audit question Are appropriate tests and inspections carried out on all incoming parts or materials before use, in accordance with a specification?		
Level 1 No tests or inspections on reception.		
Level 2 Tests or inspections are performed on reception, but no specific procedure is described for these actions.		
Level 3 Conformity to specified requirements is verified in accordance with a quality plan (or control plan) and/or written procedures. Products entering without inspection, due to urgency, are not monitored.		
Level 4 Conformity to specified requirements is verified in accordance with a quality plan (or control plan) and/or written procedures. Inspections carried out on subcontractor premises, and proofs of conformity supplied are taken into account in determining the extent and nature of inspections to be performed on reception. When, for reasons of urgency, the incoming product is released before being verified, it is identified and the decision to release is recorded.		

Recommendation Make sure that a person responsible for logistics support, product industrialization, procurement, development and RAMS is involved at every step (concurrent engineering).	Correspondence	N° 16
Phase(s) in which the recommendation applies DESIGN	Weight 16.7	
Further description Make sure that a person responsible for logistics support, product industrialization, procurement, development and RAMS is involved at every step. Make sure that the guidelines used impose concurrent engineering: the company's organization relies on permanent function specialists.		
Audit question Is the viewpoint of the different disciplines involved in the engineering work taken into account?		
Level 1	The guidelines do not impose concurrent engineering.	
Level 2	There is an overall instruction that does not specify the precise modalities. No formal organization.	
Level 3	There is a procedure imposing concurrent engineering, but it is not adapted to the company's organization: positions of responsibility for logistics support, product industrialization, procurement, development and RAMS are assigned independently of the functions.	
Level 4	There is a procedure imposing concurrent engineering. The company's organization relies on permanent function specialists.	

Recommendation Provide continuous environmental monitoring (cleanliness, temperature, humidity) for sensitive manufacturing operations	Correspondence ISO9001: L2 EN9100: L3	N° 17
Phase(s) in which the recommendation applies MANUFACTURE		Weight 9.9
Further description Several steps in the board and subassembly manufacturing process are sensitive to particle rate, temperature and relative humidity factors in the manufacturing environment. The following processes (non-exhaustive list) are particularly concerned: - Soldering - Gluing - Underfill - Varnishing. These influence factors must be monitored and recorded continuously, or periodically, in order to ensure the reproducibility of product performance.		
Audit question Is continuous environmental monitoring provided for sensitive manufacturing operations (cleanliness, temperature, humidity)?		
Level 1	Some of the environment parameters of sensitive process steps are not monitored, or not sufficiently regularly.	
Level 2	The environment parameters of sensitive process steps are monitored continuously, but this monitoring is not covered by any formal specification, or there is no analysis of their criticality to the reliability of the board or subassembly.	
Level 3	The environment parameters of sensitive process steps are monitored continuously, and this monitoring is covered by a formal specification based on an analysis of their criticality to the reliability of the board or subassembly. However, the specification has not been validated by an independent authority.	
Level 4	The environment parameters of sensitive process steps are monitored continuously, and this monitoring is covered by a formal specification based on an analysis of its criticality to the reliability of the board or subassembly. The specification and the criticality analysis have been validated by an independent authority.	

Recommendation Perform corrective maintenance following the occurrence of an anomaly in the production system.	Correspondence ISO9001: L2 EN9100: L3	N° 18
Phase(s) in which the recommendation applies MANUFACTURE EQUIPMENT_INTEGRATION		Weight 6.9 6.9
Further description Applied to anomalies in production systems and subassemblies, corrective maintenance procedures involve: - Determining the root causes of the anomaly, by setting up a multidisciplinary team - Defining corrective actions, stipulating action owners and deadlines - Monitoring the implementation of corrective actions and their effectiveness - Updating the associated Process FMECA chapter and, where applicable, the Control Plan - Updating the machinery maintenance plan, if necessary. These actions are often standardized in a process called “8D”, overseen by the Quality department.		
Audit question Is corrective maintenance performed following the occurrence of an anomaly in the production system or in the subassemblies produced?		
Level 1 There is no corrective maintenance following the occurrence of an anomaly on a production machine or on a subassembly (non-systematic).		
Level 2 Corrective actions are applied directly where the anomaly was observed, without a corrective maintenance plan being put in place.		
Level 3 Real maintenance procedures are implemented for corrective actions, following anomalies in production. They are covered by a formalized corrective maintenance procedure, but this is not validated by an authority independent from the operating entity.		
Level 4 Real maintenance procedures are implemented for corrective actions, following anomalies in production. They are covered by a formalized corrective maintenance procedure, validated by an authority independent from the operating entity. This procedure describes the following steps: anomaly identification, root cause analysis, correction and preventive action, checking the effectiveness of corrective measures, and (where applicable) internal auditing.		

Recommendation	Correspondence	N°
Control working time limits on transformable (limited lifetime) materials		19
Phase(s) in which the recommendation applies	Weight	
MANUFACTURE		4.0
EQUIPMENT_INTEGRATION		4.0
Further description Several materials have a limited working time after withdrawal from stores, or after preparation (glue, solder paste, varnish, etc.). It is essential that these materials be withdrawn from production before this time limit is exceeded, and replaced with conforming batches. These working time durations are based on manufacturer recommendations, and can be reduced on the basis of experience in production. This control is performed by reading the expiry date on a dedicated label, and/or by the system alerting the operator, or even stopping the machine.		
Audit question Is there effective control over the working times of transformable materials?		
Level 1	The working times of transformable materials are not under clear control.	
Level 2	The working times of transformable materials are controlled, but they are not described in any clear chapter or document.	
Level 3	The working times of transformable materials are controlled. There are instruction documents and/or chapters that clearly define the procedure to follow for each material concerned. The expiry dates for each batch are indicated on dedicated labels, but there is no automatic alert to tell the operator to change batches.	
Level 4	The working times of transformable materials are controlled. There are instruction documents and/or chapters that clearly define the procedure to follow for each material concerned. The expiry dates for each batch are indicated on dedicated labels. Furthermore, these dates are input into the system in order to alert the operator before changing the batch.	

Recommendation Adopt a policy for managing nonconformity risks.	Correspondence ISO9001: L3 EN9100: L4	N° 20
Phase(s) in which the recommendation applies OPERATION_&_MAINTENANCE		Weight 16.3
Further description A policy is applied to identify, evaluate and manage the potential risks associated with nonconformities, whether in the products or in the design, planning, manufacturing, assembly, inspection, etc. processes This policy must take account of the potential risks associated with human factors.		
Audit question Is a policy applied to identify, evaluate and manage the potential risks associated with nonconformities, whether in the products or in the design, planning, manufacturing, assembly, inspection, etc. processes?		
Level 1	No policy is applied to evaluate nonconformity risks.	
Level 2	A policy to identify, evaluate and manage the potential risks associated with nonconformities is applied to products, but not to all of the design, planning, manufacturing, assembly, inspection, etc. processes.	
Level 3	A policy to identify, evaluate and manage the potential risks associated with nonconformities, is applied not only to products but also to the design, planning, manufacturing, assembly, inspection, etc. processes. This policy does not take account of the potential risks associated with human factors.	
Level 4	A policy to identify, evaluate and manage the potential risks associated with nonconformities, is applied not only to products but also to the design, planning, manufacturing, assembly, inspection, etc. processes. This policy takes account of the potential risks associated with human factors.	

Recommendation	Correspondence	N°
Perform periodic checks on embedded software programming devices.		21
Phase(s) in which the recommendation applies	Weight	
MANUFACTURE	4.1	
EQUIPMENT_INTEGRATION	4.1	
Further description This check, performed by the user of the programming device (examples: JTAG probe, programming boards, etc.), is designed to ensure that the right software is loaded or that the configuration is correct. The frequency of checks must be specified, based on a risk analysis.		
Audit question Are embedded software programming devices checked periodically to ensure that software is correctly loaded?		
Level 1	There is no periodic check on programming devices used to load software.	
Level 2	A number of checks are performed on programming devices. These checks are brief and do not necessarily take account of all software loading rules. There is no clear formal definition of the procedure or the limits of these checks.	
Level 3	A schedule of checks on programming devices has been designed; this schedule is adhered to, and all of the points verified (and the procedure to follow) are covered by a written document.	
Level 4	A strict schedule of checks on programming devices has been designed; this schedule is adhered to, and all of the points verified (and the procedure to follow) are covered by a written document. This document has been produced with reference to the entire loading software process and validated by an authority independent from the operating entity.	

Recommendation Automate handling in order to limit potential damage to components, boards and subassemblies.	Correspondence EN9100: L2	N° 23
Phase(s) in which the recommendation applies MANUFACTURE EQUIPMENT_INTEGRATION		Weight 6.5 6.5
Further description Efforts must be made to keep the handling of components, boards and subassemblies to a minimum during the production phase in order to limit the risk of mechanical shock, contamination, and other overstresses. Additionally, the automation of handling operations between activities throughout the production process eliminates a large number of failures caused by human intervention. This recommendation also applies to very small production series. Beyond using robots, this risk can be addressed by using stores, trolleys, vacuum lifters, etc.		
Audit question Is the production and handling of components, boards and subassemblies automated (partially or totally)?		
Level 1 None of the handling of components, boards and subassemblies is automated.		
Level 2 Some component, board and subassembly handling operations have been automated.		
Level 3 Component, board and subassembly handling operations are maximally automated. The level of automation has been established by a feasibility and benefit analysis. The whole procedure is formalized, but the study has not been validated by an authority independent from the operating entity.		
Level 4 Component, board and subassembly handling operations are maximally automated. The level of automation has been established by a feasibility and benefit analysis. The whole procedure is formalized, following a study validated by an authority independent from Production.		

<u>Recommendation</u> Collect customer comments about the reliability of the product in operation.		Correspondence ISO9001: L4 EN9100: L4	N° 24
Phase(s) in which the recommendation applies SUPPORT			Weight 7.9
Further description Survey customers and users of the product to gather information on its reliability in the operational environment, and conduct corresponding action plans.			
Audit question Is there a procedure to collect customer comments about the reliability of the product in operation?			
Level 1	No information is available about how the reliability of the product is perceived by the customer.		
Level 2	Some information is available about how the reliability of the product is perceived by the customer.		
Level 3	Customer satisfaction surveys, addressing the reliability aspect, have been conducted.		
Level 4	Customer satisfaction surveys, addressing the reliability aspect, have been conducted, action plans to improve reliability have been implemented, and the results have been observed by the customer.		

Recommendation Control and maintain (through updates) the data loaded in programmable production systems.	Correspondence	N° 25
Phase(s) in which the recommendation applies MANUFACTURE EQUIPMENT_INTEGRATION	Weight 2.8 2.8	
Further description For task automation and for reliable execution of activities, it is essential to specifically monitor and maintain (i.e. update) the references (coordinates, batch numbers, etc.) loaded in programmable production tools.		
Audit question Is the data loaded in programmable production systems managed?		
Level 1	There are no controls on the maintenance of programming data in programmable production systems.	
Level 2	The parameters loaded in programmable production systems are controlled or maintained, but there is no formal definition of actions to be carried out to guarantee this maintenance.	
Level 3	The data programmed into the production systems is controlled and maintained in accordance with an identified formal definition (document, inspection procedure, update procedure).	
Level 4	The data programmed into the production systems is controlled and maintained in accordance with an identified formal definition (document, inspection procedure, update procedure). All the documents have been validated by an authority independent from the operating entity.	

Recommendation Describe in full the environment in which the product is to be used and maintained.	Correspondence ISO9001: L4 EN9100: L4	N° 26
Phase(s) in which the recommendation applies SPECIFICATION		Weight 12.4
Further description The environment in which the product is to be stored, transported, used and maintained is described. The following parameters are characterized by their mean and maximum values: - Temperature - Humidity - Shocks - Vibrations - Pressure - Penetration/abrasion - Ambient light - Mount position - Climate (wind, rain, snow) - Operator qualification level. If applicable, recommendations are made on the storage, transport, use and maintenance of the product.		
Audit question Is there a description and characterization of the environment in which the product is to be stored, transported, used and maintained? If applicable, are there any recommendations on storage, transport, use and maintenance?		
Level 1	The product environment is not known (or is hardly known). No formal analysis has been conducted by the manufacturer.	
Level 2	The product environment is partially known (the parameters listed in the further description are partially known). The analysis of the product environment, and the resulting recommendations, have not been formalized.	
Level 3	The product environment is partially known (the parameters listed in the further description are partially known). The analysis of the product environment, and the resulting recommendations, have not been formalized.	
Level 4	The product environment is fully known (the parameters listed in the further description are known). The analysis and the resulting recommendations are formalized in a document.	

Recommendation Define the criteria for product failure.	Correspondence	N° 28
Phase(s) in which the recommendation applies SPECIFICATION DESIGN	Weight 10.3 14	
Further description The criteria for product failure are described in detail. Where applicable: - Tolerated degraded modes are described and validated - Redundancies are utilized. This work may lead, for example, to the calculation of a MTBCF (Mean Time Between Critical Failures).		
Audit question What counts as a product failure?		
Level 1	Product failure (along with acceptable degraded modes and usable redundancies) is not defined.	
Level 2	Product failure (along with acceptable degraded modes and usable redundancies) is defined by the manufacturer, but not validated by the customer.	
Level 3	Product failure (along with acceptable degraded modes and usable redundancies) is defined by the manufacturer and subsequently validated by the customer.	
Level 4	Product failure (along with acceptable degraded modes and usable redundancies) is defined by the manufacturer and subsequently validated by the customer. Product failure is specified in the appropriate documents: design documents and reliability plans.	

Recommendation Define the method for demonstrating reliability during operation.	Correspondence	N° 29
Phase(s) in which the recommendation applies SPECIFICATION		Weight 8.1
Further description Define, and validate with the customer, the method to be used for demonstrating compliance with the quantitative reliability requirement. In general, this method relies on the analysis of experience feedback or on a reliability demonstration test.		
Audit question How is it planned to demonstrate the reliability of the product?		
Level 1	The method for demonstrating reliability is not described in the call for tenders or contract.	
Level 2	The call for tenders or contract requires a demonstration of reliability, but does not specify the method.	
Level 3	The call for tenders or contract requires a demonstration of reliability but the method to be used is only partially described.	
Level 4	The reliability demonstration method is clearly defined in the call for tenders or contract.	

Recommendation Define the degree of nonconformity.	Correspondence ISO9001: L2 EN9100: L3	N° 30
Phase(s) in which the recommendation applies EQUIPMENT_INTEGRATION SYSTEM_INTEGRATION OPERATION_&_MAINTENANCE		Weight 10.3 10.3 12.8
Further description The description of the accepted nonconformity, or of repairs performed, is recorded to indicate the real condition of the product. Written procedures are kept up to date, defining, as a minimum: - How nonconformities are classified, and how the use of nonconforming components in finished products is monitored - The formal authorization process and scope of application for personnel authorizing the use of replacement materials and/or nonconforming products (waiver procedures) - The process for managing scrapped parts.		
Audit question Is the description of accepted nonconformities, or of repairs performed, recorded to indicate the real condition of the product?		
Level 1	There is no provision for indicating the degree of nonconformity of the product.	
Level 2	The degree of nonconformity of products is indicated for information only, and not to assist the decision on whether to use nonconforming equipment.	
Level 3	The description of the accepted nonconformity, or of repairs performed, is recorded to indicate the real condition of the product. The process for classifying nonconformities, and the monitoring of the use of nonconforming components in finished products, are defined in written procedures. The process for authorizing personnel to use replacement materials and/or nonconforming products is not formalized.	
Level 4	The description of the accepted nonconformity, or of repairs performed, is recorded to indicate the real condition of the product. There are written procedures defining: - The process for classifying nonconformities, and the monitoring of the use of nonconforming components in finished products - The process for authorizing personnel to use replacement materials and/or nonconforming products - The process for managing scrapped parts.	

Recommendation Define the product mission profile for which the reliability performance is expected.	Correspondence	N° 31
Phase(s) in which the recommendation applies SPECIFICATION		Weight 9.9
Further description The product mission profile (broken down into operational scenarios for which the reliability performance is expected) is fully defined. There is a description of the successive phases of use of the product (with the phase duration and environmental characteristics). This description must include, as a minimum: - The storage phases (non-operating, protected environment, low temperature variation, controlled humidity, etc.) - The non-operating phases (the product may be in its operational environment) - The operating phases (e.g. flight, taxiing, navigation, etc.). The mission profile can be refined during development.		
Audit question Is there a clear definition of the product mission profile for which the reliability performance is expected?		
Level 1 The mission profile is not defined.		
Level 2 The mission profile is not supplied in the specifications (contract), but has been defined fully or partially by the manufacturer, but not validated by the customer.		
Level 3 The mission profile is supplied in the specifications (contract) and partially complies with the recommendation, or has been partially defined by the manufacturer and formally validated by the customer.		
Level 4 The mission profile is supplied in the specifications (contract) and complies with the recommendation, or has been fully defined by the manufacturer and formally validated by the customer.		

Recommendation Define the necessary resources for the testing and inspection of the product.	Correspondence ISO9001: L4 EN9100: L4	N° 32
Phase(s) in which the recommendation applies EQUIPMENT_INTEGRATION SYSTEM_INTEGRATION OPERATION_&_MAINTENANCE	Weight 11.6 11.6 14.3	
Further description The product testing and inspection procedures must specify the resources required (human and technical), the methods to be implemented, the acceptance criteria, and the methods for recording the results. These procedures must also define training requirements and, if necessary, specify operator qualification requirements.		
Audit question Which procedure defines the necessary resources for the testing and inspection of the product?		
Level 1	No product testing or inspection procedure is specified, and there is no description of the acceptance criteria and methods.	
Level 2	The product testing or inspection procedures are specified. The acceptance criteria and methods are described. The results are not kept.	
Level 3	The product testing or inspection procedures are specified. The acceptance criteria and methods are described. The results are not recorded and used as experience feedback. The procedures also describe the training and qualification of operators.	
Level 4	The product testing or inspection procedures are specified. The acceptance criteria and methods are described. The results are recorded and used as experience feedback. The procedures also describe the training and qualification of operators.	

Recommendation Delegate the final inspection of key operations in order to optimize filtering before moving on to the next step in the process.	Correspondence	N° 33
Phase(s) in which the recommendation applies MANUFACTURE		Weight 10.0
Further description Cross-inspection filters out nonconformities before passing the subassembly on to the next step in the production process. It can be performed by a quality inspector or by a technician trained for the task. This inspection particularly concerns the component soldering, board varnishing, gluing, marking and electrical testing processes. These inspection measures generally involve updating the tracking sheet in order to track all operations, in accordance with the control plan.		
Audit question Is quality control performed by someone other than the operator, during the main manufacturing steps?		
Level 1	For some key processes, no quality control is performed by a person other than the operator.	
Level 2	Quality control of key processes is performed by a person other than the operator, but this inspection is not based on any formal document describing the procedure.	
Level 3	Quality control of key processes is performed by a person other than the operator. This inspection follows a formal procedure, but the corresponding document has not been validated by an independent authority.	
Level 4	Quality control of key processes is performed by a person other than the operator. This inspection follows a formal procedure set out in a document validated by an independent authority. These cross-inspections are recorded in the batch tracking sheet.	

<u>Recommendation</u> Assign personnel qualified to work with the test equipment, measurements, and corresponding standards.	Correspondence ISO9001: L4 EN9100: L4	N° 34
Phase(s) in which the recommendation applies DESIGN		Weight 5.8
Further description Provide training to ensure that personnel have full mastery of the test equipment, the standards, and the interpretation of the measurements: training planned and skills monitored.		
Audit question What training have the test personnel received to qualify them to work with the test equipment, measurements, and corresponding standards?		
Level 1	No training or skills monitoring on these points.	
Level 2	Training is available, but not tracked; courses are not individualized.	
Level 3	Training is followed on an individual basis.	
Level 4	Training is followed on an individual basis, with refresher courses.	

Recommendation Provide documents for performing incoming inspections on subassembly supplies.	Correspondence	N° 35
Phase(s) in which the recommendation applies EQUIPMENT_INTEGRATION SYSTEM_INTEGRATION OPERATION_&_MAINTENANCE		Weight 8.8 8.8 9.9
Further description Procurement documents must include, where applicable: - The type, the category, and any other precise identification - The title, or any other formal identification, and the applicable version of specifications, drawings, process requirements, inspection instructions and other relevant technical data - The title, identifier and version of the applicable quality system standard - The list of procurement documents reviewed and approved prior to distribution as conforming to the requirements. The documented procurement requirements must include, where applicable: - The tests, examinations, inspections and acceptance conditions imposed by the customer, and any corresponding instructions or requirements - The requirements relating to specimens (method of production, number, storage conditions) for inspections, investigations or audits - The requirements relating to the notification of anomalies, to changes in the definition, and to the approval of their processing. Customer requirements must be passed on to suppliers.		
Audit question Are there any documents for performing incoming inspections on supplies?		
Level 1	There is no specific documentation on the incoming inspection of supplies.	
Level 2	The only documents for performing incoming inspections on supplies are the product identification documents.	
Level 3	The procurement documents include: a precise identification; the applicable versions of specifications, plans, process requirements, inspection instructions and other relevant technical data; the title, identifier and version of the applicable quality system standard; and the list of procurement documents reviewed and approved prior to distribution as conforming to the requirements.	
Level 4	The procurement documents include: a precise identification; the applicable versions of specifications, plans, process requirements, inspection instructions and other relevant technical data; the title, identifier and version of the applicable quality system standard; and the list of procurement documents reviewed and approved prior to distribution as conforming to the requirements. The documented procurement requirements also include: the tests, examinations, inspections and acceptance conditions imposed by the customer, and any corresponding instructions or requirements; the requirements relating to specimens (method of production, number, storage conditions) for inspections, investigations or audits, the requirements relating to the notification of anomalies, to changes in the definition, and to the approval of their processing. Customer requirements are passed on to suppliers.	

Recommendation Conduct a preliminary technical reliability study.	Correspondence	N° 36
Phase(s) in which the recommendation applies DESIGN		Weight 8
Further description Perform a preliminary reliability study during the preliminary development phases. The preliminary study serves to allocate reliability requirements down to subassembly level. The requirements are formalized in the subassembly specifications. There is a directive that makes it mandatory to formalize the preliminary study and the requirements.		
Audit question Is a preliminary reliability study conducted during the preliminary development phases?		
Level 1	There is no preliminary reliability study.	
Level 2	A preliminary study is conducted, but not formalized.	
Level 3	A preliminary study is formalized. The reliability requirements are not allocated down to subassembly level.	
Level 4	A preliminary study is formalized. The reliability requirements are allocated down to subassembly level and are formalized in the subassembly specifications.	

Recommendation Make sure that operators and technicians in charge of sensitive operations are trained and sufficiently experienced.	Correspondence ISO9001: L3 EN9100: L3	N° 37
Phase(s) in which the recommendation applies MANUFACTURE EQUIPMENT_INTEGRATION SYSTEM_INTEGRATION		Weight 12.0 12.0 11.0
Further description Certain steps in the manufacturing process require a particular level of training and experience. Certain types of stress can impact equipment reliability in a way that cannot be detected by quality controls or final testing. This is the case with the polymerization or drying steps, for example, where the maximum temperature and duration must be kept under control Or with the soldering of components on the board (screen-printing in particular). Or with the X-ray inspection of boards Or with quality inspection (acceptance criteria) Or with electrical testing, where special care is needed to apply ESD protections, etc. This requirement must be covered by a training and certification plan. Records will be kept of these training courses.		
Audit question Are sensitive manufacturing processes entrusted to trained and experienced personnel?		
Level 1	There is no evidence of training for employees in charge of sensitive operations that can impact equipment reliability.	
Level 2	Operators performing sensitive operations are experienced, but have not followed specific training for these operations.	
Level 3	Operators performing sensitive operations are experienced and have followed specific training for these processes. The training in question is not independently validated and tracked.	
Level 4	Operators performing sensitive operations are experienced and have followed specific training for these processes. The training in question has been independently validated and is tracked.	

Recommendation Apply specialist design procedures to capitalize on know-how.	Correspondence	N° 38
Phase(s) in which the recommendation applies DESIGN		Weight 13.8
Further description Capitalize on know-how and on the application of technical standards by means of specialist design procedures (information leveraging designer know-how: guidelines, checklists, processes, operating procedures, etc.). Make sure that these procedures cover the reliability risks. Update these procedures in response to technical developments, and manage the resulting changes.		
Audit question Are specialist design procedures managed? Are these specialist design procedures updated?		
Level 1 There are no specialist design procedures.		
Level 2 There are specialist design procedures, but they are incomplete and/or not updated.		
Level 3 There are specialist design procedures, and they are updated.		
Level 4 There are specialist design procedures, and they are updated. Moreover: - They cover reliability risks - There is a plan to improve and capitalize on procedures - There are regular formalized reviews that define action plans to improve practices.		

Recommendation Manage and maintain a design skills matrix.	Correspondence ISO9001: L4 EN9100: L4	N° 39
Phase(s) in which the recommendation applies DESIGN		Weight 24.5
Further description Make sure that the skills required in design are attributed to named individuals in a regularly updated skills matrix. Perform periodic checks on the suitability of training.		
Audit question Are design skills managed? Is there a regularly updated skills matrix? Is the relevance of training verified?		
Level 1	Design skills are not monitored.	
Level 2	Design skills are monitored and are attributed to named individuals in a skills matrix.	
Level 3	Design skills are monitored and are attributed to named individuals in a regularly updated skills matrix.	
Level 4	Design skills are monitored and are attributed to named individuals in a regularly updated skills matrix. The skills matrix incorporates monitoring of training, in order to verify that courses are commensurate with the objectives.	

Recommendation Take periodic temperature readings for each soldering operation to make sure it is not damaging the component or subassembly.	Correspondence	N° 42
Phase(s) in which the recommendation applies MANUFACTURE		Weight 6.9
Further description The temperature (e.g. for soldering irons) or temperature profile (e.g. for reflow ovens) must be regularly monitored, and the results must fall within certain temperature and duration limits (as per the specifications). The periodicity of monitoring is determined by experience and specified in the process control instruction.		
Audit question Are periodic temperature checks or readings taken for soldering operations?		
Level 1 Temperature checks are not performed regularly for soldering operations.		
Level 2 Temperature checks are performed regularly for soldering operations. The readings do not follow any specific formal procedure.		
Level 3 Temperature checks are performed regularly for soldering operations, in line with a formal procedure pre-defined in documents that specify the protocol, frequency, etc., but which have not been validated by an authority independent from the operating entity.		
Level 4 Temperature checks are performed regularly for soldering operations, in line with a formal procedure pre-defined in documents that specify the protocol, frequency, etc., and which have been validated by an authority independent from the operating entity.		

Recommendation Ensure compatibility between the production systems and the subassemblies produced.	Correspondence EN9100: L2	N° 43
Phase(s) in which the recommendation applies MANUFACTURE EQUIPMENT_INTEGRATION	Weight 7.2 7.2	
Further description The description of the actions to be performed when using a production tool on a subassembly must be sufficiently explicit to rule out any interpretation by the operator that might result in the accidental use of unsuitable tools. Identification via barcodes on machines and tracking sheets is an effective error-proofing mechanism.		
Audit question How is it ensured that the production systems are fully compatible with the items to be produced?		
Level 1	There is no explicit description ensuring that there will be no mismatch between the production systems and the subassembly.	
Level 2	A number of criteria need to be verified in order to ensure compatibility between production systems and subassembly, but they are not formally identified in a document.	
Level 3	Compatibility between production machine and product is verified before use on a subassembly. This operation is formally identified in a document. However, this document has not been validated by an authority independent from the operating entity, or there is no double-check on compatibility.	
Level 4	Compatibility between production machine and product is verified before use on a subassembly. All the parameters that need to be checked are formalized in a document validated by an authority independent from the operating entity. Additionally, compatibility is double-checked, generally using a barcode system.	

Recommendation Control handling and storage operations at the user site.	Correspondence EN9100: L2	N° 44
Phase(s) in which the recommendation applies DESIGN		Weight 7.7
Further description Make sure there is a list of function-specific recommendations on handling and storage operations at the user site, and that it is applied. This list must be enriched by experience feedback. These recommendations will enable Integrated Logistics Support to write the user manual.		
Audit question Is there a guideline document on customer-site handling and storage operations?		
Level 1 There is no list of recommendations or procedures for processing feedback.		
Level 2 There is a list of recommendations, but it is neither formalized nor managed. Feedback is processed unsystematically.		
Level 3 There is a list of formalized recommendations, not necessarily applicable to the project (not referenced to the project) and not validated. Feedback is formalized in an unmanaged database and is rarely used in design.		
Level 4 The list of recommendations is formalized, validated, and referenced to the project. Feedback is formalized, validated, referenced to the project, and usable, and serves as design input data to improve reliability.		

<u>Recommendation</u>		Correspondence	N°
Commit the company to a quality certification process.			45
Phase(s) in which the recommendation applies			Weight
SUPPORT			6.5
Further description			
Certify the company's quality system, for example to ISO 9001 v2016 or EN9100 standard.			
Audit question			
Does the company have one or more quality certifications, such as ISO 9001 v2016 or EN9100?			
Level 1	The manufacturer has not set up a quality system.		
Level 2	The manufacturer has set up a quality system, but is not recognized by a quality certification standard, or the certification is more than one year old.		
Level 3	The manufacturer has set up a quality system and has obtained certification, e.g. ISO 9001 v2016 or EN9100.		
Level 4	The manufacturer has set up a quality system and has obtained certification, e.g. ISO 9001 v2016 or EN9100. The manufacturer conducts an internal audit on the reliability activity regularly (at least every two years) to define improvement actions.		

Recommendation Record any anomalies liable to lead to the application of corrective and/or preventive actions.	Correspondence ISO9001: L3 EN9100: L3	N° 46
Phase(s) in which the recommendation applies MANUFACTURE EQUIPMENT_INTEGRATION		Weight 7.6 7.6
Further description Any problem or dysfunction in production, regardless of its nature, must be identified and recorded in an “anomaly report”. Anomaly reports follow a set format: all the information to be recorded, and the way of recording it, is predetermined. The anomaly report is the element that triggers the implementation of preventive and/or corrective maintenance actions. Tracking anomaly reports provides traceability on the different steps taken to manage nonconformities. The potential sources of anomaly are the process, the production systems (test benches, test instruments, etc.) but also the products themselves: product failures must also be recorded in anomaly reports.		
Audit question Are the anomalies encountered recorded so as to enable corrective and/or preventive actions?		
Level 1	There is no recording or traceability of problems encountered during production.	
Level 2	The anomalies identified are recorded in order to undertake corrective actions, but nothing is formalized.	
Level 3	Every production problem, regardless of its nature, is identified and recorded in a document provided for this purpose, which can then be used for preventive and corrective maintenance. However, this method of leveraging information has not been validated.	
Level 4	Every production problem, regardless of its nature, is identified and recorded in a document provided for this purpose, following a set format. The whole procedure, and particularly the way in which information is recorded for reuse during preventive and corrective maintenance, has been validated by an authority independent from the operating entity.	

Recommendation Establish procedures to verify that purchased assemblies and subassemblies meet the specified requirements.		Correspondence ISO9001: L3 EN9100: L4	N° 47
Phase(s) in which the recommendation applies			Weight
EQUIPMENT_INTEGRATION			10.6
SYSTEM_INTEGRATION			10.6
OPERATION_&_MAINTENANCE			6.8
Further description Establish written procedures to ensure that all purchased assemblies and subassemblies conform to the specified requirements. Define the terms and conditions of supply and the responsibilities of all concerned. Check that procedures are applied.			
Audit question Are there any written procedures for checking that purchased assemblies and subassemblies meet the specified requirements?			
Level 1	There are no procedures to check conformity of products to specified requirements. Nothing is formalized.		
Level 2	Generic procedures (all products) are defined to ensure conformity of purchased products. Formal evidence exists: e.g. notes.		
Level 3	Product-specific procedures are defined in a validated plan to ensure conformity of purchased products. The conditions of supply and the responsibilities of those involved are not described.		
Level 4	Product-specific procedures are defined in a validated plan to ensure conformity of purchased products. The conditions of supply and the responsibilities of those involved are described. There is evidence to support the evaluation of these procedures.		

Recommendation Improve product reliability by managing a preferred components list.	Correspondence	N° 48
Phase(s) in which the recommendation applies DESIGN		Weight 8
Further description A preferred components list takes account of: - Qualification of manufacturers/suppliers - Reliability characteristics - Estimated end-of-life dates (prevention of obsolescence). This list must be updated regularly, if possible in real time, to take account of new solutions, permanent quality alerts, and information gathered from manufacturers (manufacturer status, lifetime estimate).		
Audit question Are the component databases updated so as to capitalize on existing knowledge? Is there a regularly maintained preferred components list?		
Level 1 There is no preferred components list.		
Level 2 There is a preferred components list but it is not formalized, not validated, and contains only technical characteristics.		
Level 3 There is a preferred components list, and it is managed and formalized, with standardization targets. It is validated by Procurement, Methods and Technical Services and contains only technical characteristics.		
Level 4 There is a preferred components list, and it is managed and formalized, with standardization targets. It is validated by Procurement, Methods and Technical Services and contains not only technical characteristics but also information on component reliability.		

Recommendation Manage the processing of nonconformities.	Correspondence ISO9001: L4 EN9100: L4	N° 49
Phase(s) in which the recommendation applies EQUIPMENT_INTEGRATION SYSTEM_INTEGRATION		Weight 13.6 13.6
Further description The processing of nonconformities must be described in written procedures. These procedures must provide for the possibility of nonconforming products being: - Reworked to satisfy the specified requirements - Accepted by waiver, with or without repair - Declassified for other applications - Rejected or scrapped. If specified by the contract, the proposal to use or repair a nonconforming product may be submitted to the customer. The repaired and/or reworked product is inspected again, in accordance with the requirements of the quality plan and/or written procedures. The responsibility for examining nonconforming products and for deciding on their processing must be defined.		
Audit question How are nonconformities analysed? Are there any procedures for processing nonconforming products?		
Level 1 The processing of nonconformities is not clearly defined.		
Level 2 Nonconforming products are examined and processed, but these actions are performed without written procedures.		
Level 3 Nonconforming products are examined and processed in accordance with written procedures, but these do not allow for product modifications or for acceptance without modification.		
Level 4 Nonconforming products are examined and processed in accordance with written procedures. These provide for the product to be: - Reworked to satisfy the specified requirements - Accepted by waiver, with or without repair - Declassified for other applications - Rejected or scrapped. If specified by the contract, the proposal to use or repair a nonconforming product is submitted to the customer. The repaired and/or reworked product is inspected again, in accordance with the requirements of the quality plan and/or written procedures.		

<u>Recommendation</u>	Correspondence	N°
Maintain and operate a technical event database.		50
Phase(s) in which the recommendation applies	Weight	
DESIGN	24.2	
Further description		
Make sure that there is a technical event database, and that it is maintained and operated, with the aim of capitalizing on experience feedback and improving the reliability of future designs.		
Audit question		
Is there a technical event database? How is it maintained and operated?		
Level 1	There is no technical event database.	
Level 2	There is a technical event database, but it is not up to date.	
Level 3	There is a technical event database; it is up to date, but it is not operated.	
Level 4	There is a technical event database; it is up to date and it is operated.	

<u>Recommendation</u>	Correspondence	N°
Maintain a database to capitalize on reliability assessment studies.		51
Phase(s) in which the recommendation applies	Weight	
DESIGN	10.6	
Further description Ensure centralized management of reliability assessment studies in order to reuse previous calculations with stresses: - The baseline assumptions are clearly identified - The data is extractible and reusable by the design functions.		
Audit question Is there a database to capitalize on reliability assessment studies?		
Level 1	There is no database.	
Level 2	There is a database, but it is not centralized.	
Level 3	There is a centralized database, but no formal process	
Level 4	There is a centralized database with a formal enrichment process.	

Recommendation Control design definitions and changes to products in development.	Correspondence ISO9001: L3 EN9100: L4	N° 52
Phase(s) in which the recommendation applies DESIGN		Weight 7.8
Further description Verify the existence of a design definition files and design justification files; make sure changes are tracked and configuration-managed.		
Audit question Is there a process for describing, tracking and justifying design definitions and changes to developed products? Is there a database on the design history and justifications?		
Level 1 There is no design definition file and no design justification file.		
Level 2 There is a design definition file, but there is no design justification file.		
Level 3 There is a design definition file and a design justification file. However, the traceability of changes is not managed.		
Level 4 There is a design definition file and a design justification file. There is a configuration management plan that enables changes to be tracked.		

Recommendation Leverage experience feedback.	Correspondence	N° 53
Phase(s) in which the recommendation applies		Weight
SPECIFICATION		8.5
SUPPORT		11
Further description Leverage experience feedback on products in operation. Experience feedback is exploited for three purposes: - To establish reasonable confidence in delivering the required reliability performance: field data analyses are performed on similar products used in similar environments, to consolidate predictive data - To improve product reliability: feedback analyses are shared with the development teams, in order to identify improvement options in terms of design - To readjust/calibrate/control predictive reliability methods. The methodology for exploiting feedback must be correctly defined. These studies require a large amount of time to collect operational data, and meticulous recording of the anomalies encountered. Input data: - Records of observed anomalies - The conditions of use of the product (mission profile, operational environment, length of use) - Analysis of the cause of failures (whether attributable to the manufacturer or not) Output data: - Operational reliability, which can be extrapolated for different environments and mission profiles by using system engineering models.		
Audit question Is experience feedback used to maintain confidence in delivering the required reliability performance?		
Level 1	No experience feedback (measuring operational reliability on previous projects) is available.	
Level 2	Experience feedback is available, but it is neither used nor formalized.	
Level 3	The manufacturer's experience feedback is used and formalized in a document. This field data does not correspond exactly to the technologies employed. There is a procedure to validate or readjust predictive reliability methods.	
Level 4	The manufacturer's experience feedback is used and formalized in a document. This field data corresponds to the current technologies employed, or similarity studies are conducted and formalized to evaluate the differences (document). There is a procedure to validate or readjust predictive reliability methods, and it is regularly updated.	

Recommendation Involve the Dependability function in the functional and organic design of the product.	Correspondence	N° 54
Phase(s) in which the recommendation applies SPECIFICATION DESIGN		Weight 12.6 12.6
Further description Use reliability engineering to optimize the product architecture and design choices. Give reliability engineering teams authority to act in the event of a reliability performance failure.		
Audit question Are reliability criteria taken into account in the product architecture, and in decisions about design, product industrialization, and support?		
Level 1 Reliability engineers do not participate in the design of the product.		
Level 2 The participation of reliability engineers in the design of the product is sporadic and/or partial. Reliability engineers intervene only for reliability assessments.		
Level 3 Reliability engineers contribute to design and to decision-making if a reliability target is missed; there are documents to this effect, but the company's reference system does not describe their participation.		
Level 4 Reliability engineers contribute to design and to decision-making if a reliability target is missed; there are documents to this effect and the company's reference system describes their participation.		

Recommendation Involve the Dependability function in every phase of the project.		Correspondence	N° 55
Phase(s) in which the recommendation applies SUPPORT			Weight 8.8
Further description The Dependability function participates in every phase of the project, from the upstream development phases through to the start of production. The Dependability function is also involved in production and in operational monitoring.			
Audit question Does the Dependability function participate in every phase of the project?			
Level 1	No reliability engineers participate in the project.		
Level 2	Reliability engineers participate partially in the project (incomplete assistance as defined in the recommendation); there are no documents attesting to their participation.		
Level 3	Reliability engineers participate fully in the project (full assistance as defined in the recommendation); this participation is not formalized by any plan or procedure.		
Level 4	Reliability engineers participate fully in the project (full assistance as defined in the recommendation); this participation is formalized and attested in documents.		

Recommendation Provide training for personnel in reliability studies or whose work has an influence on reliability, or employ qualified personnel.		Correspondence	N° 56
Phase(s) in which the recommendation applies SUPPORT			Weight 7.5
Further description Provide training for personnel in reliability studies, with training varying from awareness-raising to expert refresher courses for reliability managers, depending on the criticality of the reliability performance expected from the product. Make production personnel aware of the importance of avoiding deterioration of products.			
Audit question Is the training of reliability personnel adapted to the criticality of the reliability performance expected from the product?			
Level 1	No specific training (whether initial or further training) is given to personnel in charge of reliability studies. No awareness actions are carried out for production personnel.		
Level 2	Personnel in charge of reliability studies have been trained, but no awareness actions are carried out for production personnel.		
Level 3	Personnel in charge of reliability studies have been trained and awareness actions are carried out for production personnel.		
Level 4	Personnel in charge of reliability studies have been trained; they are experienced, and engage with reliability work. Awareness actions are carried out for production personnel.		

Recommendation Formalize the reliability requirements, and in particular the quantitative requirement.	Correspondence ISO9001: L3 EN9100: L3	N° 57
Phase(s) in which the recommendation applies SPECIFICATION		Weight 13
Further description This recommendation concerns the way in which high-level requirements, generally formulated by the customer, are broken down into measureable elements. All reliability-related requirements (whether derived or taken directly from the customer's requirements) must be formalized. Reliability performance must be specified quantitatively, for example by using one of the following metrics: - MTTF - MTBF or failure rate: this concept applies to repairable systems that have a long lifetime; it assumes a constant failure rate over time, which sometimes needs to be validated - Reliability or probability of survival for a given period of time; this concept can be used when a high level of reliability is required throughout the mission - The probability of success (independently of time) for single-use or cyclical-use devices. The assumptions involved in assessing reliability performance must be described: mission profile (Recommendation 31), type of time measurement (hours of operation, flight hours, cycles), etc. Quantitative values will be specified either as mean values (design objectives) or as minimum acceptable values. Minimum acceptable values are the values below which the customer judges the system to be unfit for purpose. The type of target value (design objective or minimum acceptable) will be defined explicitly.		
Audit question Are the reliability requirements, in particular the quantitative requirement, formalized? Do they accurately describe the assumptions involved in assessing reliability performance?		
Level 1	No quantitative reliability requirements are specified.	
Level 2	No quantitative reliability requirements are specified, but a quantitative evaluation of reliability is requested.	
Level 3	The quantitative requirement is specified. Some quantification elements involve non-formalized assumptions.	
Level 4	The quantitative requirement is specified. All quantification assumptions are formally specified.	

Recommendation Supply the necessary resources for reliability studies.	Correspondence ISO9001: L3 EN9100: L4	N° 58
Phase(s) in which the recommendation applies SUPPORT		Weight 8.3
Further description Allocate the necessary resources for reliability studies: - Access to technical data - Tools and equipment - Time and funding.		
Audit question Is the necessary technical data for reliability studies accessible? Are the necessary tools available for reliability studies? Are the necessary time and funding provided for the studies?		
Level 1	There is no clear allocation of resources to reliability activities.	
Level 2	Resources are allocated to reliability activities, but they are insufficient: one of the three resources – technical data, tools and equipment, or time and funding – is not available.	
Level 3	Sufficient resources are allocated to reliability activities.	
Level 4	Sufficient resources are allocated to reliability activities. These resources are described in a project management plan or in a reliability management plan.	

Recommendation Manage reliability studies documentation.		Correspondence EN9100: L4	N° 59
Phase(s) in which the recommendation applies SUPPORT			Weight 5.4
Further description Manage reliability studies documentation: validation, backup, archiving, configuration management of documents. Document management extends to any document that has an influence on reliability studies, including any document that formally defines the assumptions and input data.			
Audit question Are reliability studies documents managed?			
Level 1	Reliability documents are not managed.		
Level 2	Only certain documents (for example, customer deliverables) are managed.		
Level 3	Reliability studies documentation is not fully managed, as defined in the recommendation: for example, configuration management is not systematic, the assumptions underlying predictive calculations are not formalized, etc.		
Level 4	Reliability studies documentation is fully managed: with systematic configuration management of all documents, including those relating to assumptions and input data. Predictive reliability studies documents are available for the lifetime of the programmes (for comparative predictive/operational studies).		

Recommendation Manage priorities that need to be met within process deadlines (planning).	Correspondence	N° 60
Phase(s) in which the recommendation applies MANUFACTURE EQUIPMENT_INTEGRATION		Weight 3.1 3.1
Further description The various subassemblies are made and integrated on the basis of planned tasks that may involve simultaneous activities. Priorities have to be managed, so that only a minimum number of subassemblies need to be kept in stock (any delay in the production flow entails additional handling and storage of subassemblies) and so that the potential impacts on component reliability can also be kept to a minimum.		
Audit question How are priorities managed with regard to process deadlines?		
Level 1 There is no ranking of priorities in production.		
Level 2 Depending on the production schedule, some priority is given to subassemblies in order to minimize handling and storage. These priorities are not described in formal documents.		
Level 3 A fully-fledged priority management system is in place, based on process deadlines. This planning activity relies on formal documents, but these have not been validated by an authority independent from the operating entity.		
Level 4 A fully-fledged priority management system is in place, based on process deadlines. This planning activity relies on formal documents, which have been validated by an authority independent from the operating entity.		

Recommendation		Correspondence	N°
Identify and implement measures to protect subassemblies.			61
Phase(s) in which the recommendation applies			Weight
DESIGN			7.3
MANUFACTURE			7.3
EQUIPMENT_INTEGRATION			7.3
Further description Hazard risks must be identified and protective measures implemented to avoid impacting the reliability of subassemblies: - Mechanical protection - Thermal protection - Protection from contaminants.			
Audit question Have measures been identified and implemented to protect subassemblies during certain production activities?			
Level 1	No particular measures to protect subassemblies are identified.		
Level 2	Measures to protect subassemblies are identified, but only partially applied in the various activities.		
Level 3	Measures to protect subassemblies are specified, and their application is verified. However, this specification is not updated on the basis of past data.		
Level 4	Measures to protect subassemblies are specified and their application is verified. This specification is based on past data from previous products, and on periodic analysis of observed anomalies.		

Recommendation		Correspondence	N°
Formally identify the technical risks that affect reliability.			62
Phase(s) in which the recommendation applies			Weight
SPECIFICATION			12.4
DESIGN			21
Further description			
Formally identify any technical risks that might jeopardize delivery of the required reliability performance. This information will be used by the risk management procedure. Trace and manage these risks (in an action plan). Risk examples: components subject to SEEs (Single Event Effects), limited lifetime components, technologies that are new to the manufacturer, etc.			
Audit question			
Are the technical risks that affect reliability identified?			
Level 1	There is no risk management policy, or the risk management policy does not address reliability aspects.		
Level 2	The risk management policy is informal, or addresses reliability aspects only partially.		
Level 3	An initial analysis has been carried out of the risks to the delivery of reliability performance. This is formalized, but risk management is not sustained over time; collaboration is organized between the equipment manufacturer and the system integrator to evaluate the risks inherent in the product environment.		
Level 4	The risks to the delivery of reliability performance are clearly identified. There is a procedure for managing these risks at the manufacturer, and it is followed. Collaboration is organized between the equipment manufacturer and the system integrator to evaluate the risks inherent in the product environment. A risk form is drawn up and updated for each risk. The risk form takes a quantitative approach to the probability of the risk, its severity (in terms of cost, planning, and performance), the proposed risk mitigation solutions, and the cost of these solutions.		

<u>Recommendation</u> Identify documentation for special processes.		Correspondence EN9100: L4	N° 63
Phase(s) in which the recommendation applies OPERATION_&_MAINTENANCE			Weight 12.2
Further description Records concerning processes, products and personnel are kept up to date.			
Audit question Is there documentation for special processes? Is this documentation updated?			
Level 1	There are no records on special processes.		
Level 2	There are records, but only on processes; the products or human resources involved are not recorded.		
Level 3	There are records on processes, products and the personnel associated with special processes, but they are not updated.		
Level 4	Records on special processes, and the associated products and personnel, are updated.		

<u>Recommendation</u>		Correspondence	N°
Identify the customer's reliability-related requirements.		EN9100: L4	65
Phase(s) in which the recommendation applies			Weight
SPECIFICATION			7.3
Further description			
The customer's reliability-related requirements must be identified, documented and tracked.			
Audit question			
Have the customer's reliability-related requirements been identified, documented and tracked?			
Level 1	The customer's requirements are not identified.		
Level 2	The customer's requirements are identified informally.		
Level 3	The customer's requirements are identified, documented and listed in a document (e.g. reliability plan) that is not revision-controlled.		
Level 4	The customer's requirements are identified, documented and listed in a revision-controlled document (e.g. reliability plan). Revision control ensures that all changes to the requirements are traceable.		

<u>Recommendation</u>		Correspondence	N°
Identify the necessary means for special processes.		EN9100: L4	66
Phase(s) in which the recommendation applies			Weight
EQUIPMENT_INTEGRATION			13.1
SYSTEM_INTEGRATION			13.1
OPERATION_&_MAINTENANCE			13.1
Further description			
The requirements for the qualification of process operations, including the associated equipment and personnel, must be specified.			
Audit question			
Are the necessary means for special processes identified?			
Level 1	The resources for special processes are not formally identified.		
Level 2	There are documents identifying the technical resources dedicated to special processes. The equipment and personnel associated with these processes are not defined.		
Level 3	The requirements for the qualification of special process operations, including the associated equipment and personnel, are specified. These specifications are not updated regularly.		
Level 4	The requirements for the qualification of the process operations, including the associated equipment and personnel, are specified. The documents identifying these requirements are regularly updated.		

<u>Recommendation</u>		Correspondence	N°
Identify the human resources for special processes.		EN9100: L4	67
Phase(s) in which the recommendation applies			Weight
EQUIPMENT_INTEGRATION			11.7
SYSTEM_INTEGRATION			11.7
OPERATION_&_MAINTENANCE			13.7
Further description			
Special processes must be performed by qualified operators, and must be continuously monitored, with control over process parameters to ensure conformity to requirements.			
Audit question			
Identify the human resources for special processes.			
Level 1	Special processes are not associated with qualified human resources.		
Level 2	Special processes are performed by trained operators, but there is no regular monitoring of their skills.		
Level 3	Special processes are performed by qualified operators, or are continuously monitored.		
Level 4	Special processes are performed by qualified operators, and are continuously monitored.		

Recommendation Identify product reliability risks at subcontractors.		Correspondence	N° 68
Phase(s) in which the recommendation applies SUPPORT			Weight 7.2
Further description Identify reliability risks in subcontracted products: risk of not delivering reliability performance, use of new technologies, components used outside specifications, etc.			
Audit question Have product reliability risks at subcontractors been identified?			
Level 1	Reliability risks are not identified with the subcontractor before the signing of the contract (no specific provision).		
Level 2	Reliability risks are not identified with the subcontractor before the signing of the contract, but risks are identified during the project. However, they are not managed.		
Level 3	An analysis of reliability risks is carried out before the signing of the contract and is covered by a formal document. However, these risks are not managed.		
Level 4	An analysis of reliability risks is carried out before the signing of the contract and is covered by a formal document, which is updated and configuration-managed.		

Recommendation Integrate reliability into the company's quality policy.	Correspondence	N° 69
Phase(s) in which the recommendation applies SUPPORT		Weight 7.4
Further description Integrate reliability activities into the company's quality policy and adapt the policy to the levels concerned by reliability engineering. The company's quality manual deals specifically with reliability activities. An audit of the reliability process (as proposed by the FIDES Guide, for example) is integrated into the company's quality audits.		
Audit question Does the subject of reliability feature in the company's quality policy?		
Level 1	The quality policy does not take reliability into account.	
Level 2	Reliability is mentioned indirectly in the quality policy objectives. The company's quality manual does not specifically address reliability activities.	
Level 3	Reliability features in the company's quality policy. Reliability activities are specifically addressed in the company's quality manual.	
Level 4	Reliability is one of the key features of the quality policy. Reliability activities are specifically addressed in the company's quality manual. An audit of the reliability process is integrated into the company's quality audits.	

Recommendation Manage the equipment or system documentation.	Correspondence ISO9001: L3 EN9100: L4	N° 71
Phase(s) in which the recommendation applies EQUIPMENT_INTEGRATION SYSTEM_INTEGRATION OPERATION_&_MAINTENANCE		Weight 15 15 14
Further description Managing the documentation involves: - Recording and backing up product and process documents - Training a group of workshop personnel in the management of technical documents - Regularly updating the document inventory - Specifying technical documents for each process - Updating documents to take account of equipment upgrades - Submitting the documents, and their modifications, for validation - Making the process documents available, and usable, in the workshop.		
Audit question Is the documentation associated with the equipment or system strictly managed? Does it take account of all equipment upgrades?		
Level 1	There is no specific documentation on products or processes, and there is no provision for making specific documentation available.	
Level 2	There is specific documentation on products and processes, however, it is not always updated, and the validity of the documents is not analysed.	
Level 3	There is specific documentation on products and processes. It is periodically checked to verify that it is up to date, but the validity of the documents used is not analysed.	
Level 4	There is specific documentation on products and processes. It is periodically checked to verify that it is up to date, and the validity of the documents used is analysed. Strict procedures are followed for indexing and preserving the documentation, and for making it available to users.	

Recommendation		Correspondence	N°
Control product testability and maintainability.			72
Phase(s) in which the recommendation applies			Weight
OPERATION & MAINTENANCE			17.6
Further description			
Control the capability of products to detect failures; control failure detection systems; facilitate maintenance.			
Audit question			
How are product testability and maintainability controlled?			
Level 1	There are no built-in tests; maintenance is performed when a failure occurs.		
Level 2	Existence of minimum monitoring by lights or alarms.		
Level 3	Implementation on integrated tests such as PBIT, CBIT, IBIT (Power up Built In Test, Continuous Built In Test, Initiated Built In Test).		
Level 4	Implementation on integrated tests (PBIT, CBIT, IBIT) and complementary testability by one (or more) external test means.		

Recommendation		Correspondence	N°
Make sure that test/inspection equipment and software are compatible with requirements.		ISO9001: L4 EN9100: L4	73
Phase(s) in which the recommendation applies			Weight
EQUIPMENT_INTEGRATION			9.6
SYSTEM_INTEGRATION			9.6
OPERATION_&_MAINTENANCE			11.3
Further description			
Inspection, measurement and test equipment is used to ensure that the measurement uncertainty is known and is compatible with the required measurement accuracy. The test software, or the comparison references used for inspection purposes, are checked before being brought into service in order to demonstrate that they are capable of verifying the acceptability of the product. The equipment covered by this recommendation includes tools (e.g. drills), test equipment (e.g. harmonization benches), and test apparatus (e.g. ovens), etc.			
Audit question			
What steps are taken to ensure that test/inspection equipment and software are compatible with requirements?			
Level 1	There is no procedure that defines how to check that inspection, measurement and testing equipment is compatible with requirements.		
Level 2	There are procedures defining how to check that inspection, measurement and testing equipment is compatible with requirements, but no checks are performed to ensure that they are used.		
Level 3	The inspection, measurement and testing equipment is used to ensure that the measurement uncertainty is known and is compatible with the required measurement accuracy. However, inspection equipment is not checked before being brought into service.		
Level 4	The inspection, measurement and testing equipment is used to ensure that the measurement uncertainty is known and is compatible with the required measurement accuracy. The test software, or the comparison references used for inspection purposes, are checked before being brought into service in order to demonstrate that they are capable of verifying the acceptability of the product. If systematic verification before use is industrially impossible, a metrology procedure is used (with a validation period and definition of the equipment class).		

Recommendation		Correspondence	N°
Control the environment of inspection, measurement and testing equipment.		ISO9001: L4 EN9100: L4	74
Phase(s) in which the recommendation applies			Weight
EQUIPMENT_INTEGRATION			7.9
SYSTEM_INTEGRATION			7.9
OPERATION_&_MAINTENANCE			11.7
Further description			
The handling, preservation and storage of inspection and measurement equipment ensure that accuracy and usability are maintained. Inspection, measurement and testing equipment, including the test benches and test software, must be protected against handling operations that might invalidate the calibration settings.			
Audit question			
How is the environment of inspection, measurement and testing equipment controlled?			
Level 1	The environment of inspection, measurement and testing equipment is not taken into account.		
Level 2	Inspection, measurement and testing equipment is protected against potential hazards that might damage it.		
Level 3	Inspection, measurement and testing equipment is protected against potential hazards that might damage it, and against handling operations that might invalidate the calibration settings. However, the handling, preservation and storage of inspection equipment are not defined by strict procedures.		
Level 4	Inspection, measurement and testing equipment is protected against potential hazards that might damage it, and against handling operations that might invalidate the calibration settings. The handling, preservation and storage of inspection equipment are defined by strict procedures, which are applied by the operators.		

Recommendation Control the workplace environment.		Correspondence ISO9001: L3 EN9100: L3	N° 75
Phase(s) in which the recommendation applies			Weight
MANUFACTURE			13.0
EQUIPMENT_INTEGRATION			9.6
SYSTEM_INTEGRATION			9.6
OPERATION_&_MAINTENANCE			10.8
Further description The workplace environment is important for the quality of the product: appropriate limits must be specified for the ambient temperature and humidity. The level of cleanliness must be controlled. The layout of the workshop and the ergonomics of the workstation must be optimized.			
Audit question How is the workplace environment controlled? (Cleanliness, temperature and humidity)			
Level 1	The workshop layout and the workplace environment are not suited to the product.		
Level 2	The workstations are specific to the equipment. The work environment is controlled (cleanliness/temperature/humidity).		
Level 3	The workstations are specific to the equipment. The work environment is controlled and periodically verified (cleanliness/temperature/humidity).		
Level 4	The workstations are adapted to the specific needs of the product. The work environment is controlled and periodically verified (cleanliness/temperature/humidity). The workshop layout is optimized with the aim of improving product quality.		

Recommendation Control monitoring and measurement devices, the metrology of measurement apparatus and industrial machinery.	Correspondence ISO9001: L4 EN9100: L4	N° 76
Phase(s) in which the recommendation applies SUPPORT		Weight 7.8
Further description Control monitoring and measurement devices, the metrology of measurement apparatus and industrial machinery. Control the verification, calibration and rating of the measurement instruments and the test and trial equipment used by the company. The measurement instruments are linked to national measurement standards.		
Audit question What procedure is there to control monitoring and measurement devices, the metrology of measurement apparatus and industrial machinery?		
Level 1	There is no verification, calibration and rating procedure in the company for measurement instruments or for test and trial equipment.	
Level 2	There is a verification, calibration and rating procedure in the company for measurement instruments and for test and trial equipment, but is not followed.	
Level 3	There is a verification, calibration and rating procedure in the company for measurement instruments and for test and trial equipment, and it is applied.	
Level 4	The verification, calibration and rating of the measurement instruments and test and trial equipment used by the company is controlled (accreditation, certification, etc.). The measurement instruments are linked to national measurement standards.	

Recommendation Manage changes to manufacturing processes.	Correspondence ISO9001: L3 EN9100: L4	N° 77
Phase(s) in which the recommendation applies MANUFACTURE EQUIPMENT_INTEGRATION SYSTEM_INTEGRATION OPERATION_&_MAINTENANCE		Weight 13.9 13.9 13.9 13.9
Further description There must be a clear list of people authorized to approve process changes. Changes that require customer acceptance before being applied must be identified in a document. Any change relating to the processes, production equipment, tools or programs must be documented and must trigger a procedure to control its application. Ensure that the results of process changes produce the desired effect, and that the changes have not impacted the quality of the product. The actions and tests required to qualify a new process, material or tool must be specified in a document.		
Audit question How are changes to manufacturing processes managed? Which changes are notified to the customer? Is there a requalification plan according to the nature of the change?		
Level 1	Process changes are implemented without being recorded; these modifications are not submitted for authorization.	
Level 2	Process changes are recorded and submitted for authorization. These changes are not documented, and they do not trigger a procedure to control their implementation.	
Level 3	Process changes are recorded; the people authorized to approve changes to production processes, including quality/reliability managers, are clearly designated. Changes that require customer acceptance before being applied are identified. Any change relating to the processes, production equipment, tools or programs is documented and follows a procedure to control its implementation. However, there is no systematic check to ensure that the results of process changes produce the desired effect and that the changes have not impacted the quality of the product (qualification).	
Level 4	Process changes are recorded; the people authorized to approve changes to production processes, including quality/reliability managers, are clearly designated. Changes that require customer acceptance before being applied are identified. Any change relating to the processes, production equipment, tools or programs is documented and follows a procedure to control its implementation. Changes to processes, materials or production machinery give rise to requalification in accordance with a written procedure approved by Quality.	

Recommendation Control handling methods.		Correspondence ISO9001: L2 EN9100: L3	N° 78
Phase(s) in which the recommendation applies			Weight
EQUIPMENT_INTEGRATION			8.8
SYSTEM_INTEGRATION			8.8
OPERATION_&_MAINTENANCE			9.9
Further description Product handling methods and means are provided to prevent damage or deterioration to the product and include: - Intersite transport procedures - Handling modes specific to each product.			
Audit question Are handling and inter-site transport methods defined?			
Level 1	Handling methods are not defined; there are no specific means to avoid deterioration during handling.		
Level 2	Handling methods are defined, but they are not product-specific.		
Level 3	Product-specific handling methods are defined, and specific means are made available to avoid any deterioration during handling. However, no checks are performed to ensure their application.		
Level 4	Product-specific handling methods are defined, and the associated means prevent any deterioration to the product during handling. Checks are performed to ensure their application.		

<u>Recommendation</u> Inspect production machines, tools and programmable devices before use.	Correspondence EN9100: L2	N° 79
Phase(s) in which the recommendation applies MANUFACTURE EQUIPMENT_INTEGRATION SYSTEM_INTEGRATION OPERATION_&_MAINTENANCE	Weight 12.0 10.5 10.5 11.3	
Further description To avoid damaging the equipment, it is important to check, before use, that the production machines (test benches, ovens, vibrators, etc.), tools (torque screwdrivers, etc.) and programmable devices (numerical control machines, software programming devices used on test benches, etc.) are suited to the product, and that the parameters conform to the specifications. This prior inspection activity must be described in written procedures.		
Audit question How are production machines, tools and programmable devices inspected before use?		
Level 1	Production machines, tools and programmable devices are not inspected and validated before use.	
Level 2	Production machines, tools and programmable devices are inspected before use, but not all these inspections are formalized.	
Level 3	Production machines, tools and programmable devices are inspected before use. These inspections are covered by written procedures.	
Level 4	Production machines, tools and programmable devices are inspected before use. These inspections are covered by written procedures. The results of these inspections are recorded.	

Recommendation Control handling, storage, packaging and preservation operations.	Correspondence ISO9001: L2 EN9100: L3	N° 80
Phase(s) in which the recommendation applies EQUIPMENT_INTEGRATION SYSTEM_INTEGRATION OPERATION_&_MAINTENANCE		Weight 6.5 6.5 11.3
Further description For each production step, and in accordance with the manufacturer recommendations and/or the applicable regulations, there must be a procedure covering the specific requirements for: - Cleaning - Prevention, detection and removal of foreign bodies - Handling of sensitive products - Marking and labelling, including safety marking - Control of shelf life and stock rotation - Hazardous materials. Establish specific management procedures for perishables. Eliminate all expired or unidentified products. Suggest criteria for evaluating and analysing the quality of storage conditions. List and analyse failures due to lack of quality in storage.		
Audit question How are handling, storage, packaging and preservation operations controlled?		
Level 1	Handling, storage, packaging and preservation conditions are not codified; the execution of these operations is not perfectly controlled.	
Level 2	Handling, storage, packaging and preservation conditions are codified; they give rise to procedures that can be adapted to all products. The execution of these operations is not specific to any one product.	
Level 3	Handling, storage, packaging and preservation conditions are codified; they give rise to procedures that are specific to each product.	
Level 4	Handling, storage, packaging and preservation conditions are codified; they give rise to procedures that are specific to each product. Considerations such as expiry dates, sensitivity of products to stress, and the hazardousness of products are also codified and applied.	

Recommendation Control special processes.	Correspondence EN9100: L3	N° 81
Phase(s) in which the recommendation applies		Weight
EQUIPMENT_INTEGRATION		14.4
SYSTEM_INTEGRATION		14.4
OPERATION_&_MAINTENANCE		15.2
Further description Special processes are processes for which the results cannot be fully verified retrospectively by inspecting or testing the product, and for which the consequences of defective implementation will not appear until the product is used. Examples include gluing, soldering, brazing, heat treatment, and surface treatment. The following points must be taken into account: - The special processes to be implemented must be identified. - The supplier must check that all special process parameters (e.g. materials, personnel, procedures and software) produce satisfactory results. - The supplier must identify and document significant operations and process parameters to be controlled. - Any change to these operations and parameters must be described in a proposal justifying the change and guaranteeing that it does not produce any harmful effects for the outcome of the process. - The supplier must verify special processes by producing one or more typical parts under the conditions defined for the phase. - Special processes or subcontracted processes must be qualified before use. - The supplier must keep qualified special processes up to date.		
Audit question How are special processes controlled? Which documents describe these processes, and how are they updated? How are special processes qualified?		
Level 1	Special processes are not identified.	
Level 2	Special processes are identified. The parameters of these processes (materials, personnel, procedures and software) are evaluated. However, these processes are not documented, or not defined by strict procedures.	
Level 3	Special processes are identified. The parameters of these processes (materials, personnel, procedures and software) are evaluated. The significant operations and the parameters of the process to be controlled in production have been identified and documented. Any change to these operations and parameters is described in a proposal justifying the change and guaranteeing that it does not produce any harmful effects for the result of the process. However, the special processes have not been verified by producing one or more typical parts under controlled conditions.	
Level 4	Same criteria as Level 3, plus: Checks are performed to ensure that all the special processes parameters (e.g. materials, personnel, procedures and software) produce the expected results. Special processes are verified by producing one or more typical parts under controlled conditions. Special processes (whether subcontracted or otherwise) are qualified before use and kept up to date.	

Recommendation Control services and fluids in the work environment.		Correspondence ISO9001: L3 EN9100: L4	N° 82
Phase(s) in which the recommendation applies			Weight
EQUIPMENT_INTEGRATION			10.1
SYSTEM_INTEGRATION			10.1
OPERATION_&_MAINTENANCE			12.2
Further description Services and supplies such as water, compressed air, electricity and chemicals used must be controlled and verified regularly to ensure that their effect on the process remains constant.			
Audit question How are services and fluids controlled in the work environment?			
Level 1	Services and supplies such as water, compressed air, electricity and chemicals are not verified.		
Level 2	Services and supplies such as water, compressed air, electricity and chemicals are verified occasionally and when a failure occurs (see ISO 14000).		
Level 3	Services and supplies such as water, compressed air, electricity and chemicals are controlled and verified periodically to ensure that their effect on the process remains constant.		
Level 4	Services and supplies such as water, compressed air, electricity and chemicals are controlled and verified continuously to ensure that their effect on the process remains constant.		

Recommendation Maximize test coverage based on the specification.	Correspondence	N° 83
Phase(s) in which the recommendation applies DESIGN		Weight 6
Further description Make sure that test coverage is maximal and is based on the specification. Produce a document justifying the coverage rate.		
Audit question Are checks performed to ensure that test coverage is maximal and is based on the specification? Is there a document describing the method of calculation and justifying the coverage rate?		
Level 1 The coverage rate is not evaluated.		
Level 2 Test coverage is evaluated, but no action is taken to achieve the specification target or to maximize the coverage rate.		
Level 3 Test coverage is evaluated and actions are taken to maximize the coverage rate. The justification of the coverage rate is not formalized.		
Level 4 Test coverage is evaluated and actions are taken to maximize the coverage rate. The justification of the coverage rate is formalized. Any non-coverage of certain functions must be justified (functions not used in operation, no impact, etc.).		

Recommendation Periodically check for bath contamination during wave soldering operations.	Correspondence	N° 84
Phase(s) in which the recommendation applies MANUFACTURE		Weight 5.8
Further description Solder baths, for wave soldering operations, must be checked regularly for contamination in order to avoid degrading the final solder quality. The contamination level and the checking frequency must be specified. The record of results must be taken into account when adjusting the periodicity of these checks.		
Audit question Are solder baths periodically checked for contamination? Using which procedure?		
Level 1 The solder bath contamination level is not measured.		
Level 2 Solder bath contamination levels are measured, but measurements are sporadic and are not formalized.		
Level 3 Solder bath contamination levels are measured. Measurements are taken in accordance with a protocol and periodicity identified and described in a document, but this document has not been validated by an authority independent from the operating entity.		
Level 4 Solder bath contamination levels are measured. Measurements are taken in accordance with a protocol and periodicity identified and described in a document that has been validated by an authority independent from the operating entity.		

Recommendation Implement design reviews.	Correspondence ISO9001: L3 EN9100: L4	N° 86
Phase(s) in which the recommendation applies DESIGN		Weight 27.1
Further description Design reviews must be organized throughout development to ensure that the project is moving in the right direction. These reviews will include, at a minimum: - A preliminary design review - A detailed design review - A final design review. These reviews must be attended by a representative from each function.		
Audit question Are design reviews held?		
Level 1 There is no design review.		
Level 2 There are design reviews at each stage of development, but they are informal, and not described in a procedure validated by the company.		
Level 3 There are design reviews at each stage of development. They are formal, and described in a procedure validated by the company. However, they do not involve all of the functions.		
Level 4 There are design reviews at each stage of development. These reviews are organized following a procedure validated by the company. These reviews involve all of the functions.		

Recommendation Implement a maintenance concept as part of logistics support.		Correspondence	N° 87
Phase(s) in which the recommendation applies DESIGN			Weight 5.4
Further description Make sure that the maintenance concept is formalized and validated by the customer. Example of documents to be presented to meet the concept requirements: - Integrated logistics support plan - Logistics supportability analysis (LSA).			
Audit question Is there a maintenance concept?			
Level 1	There are no support requirements. The customer's organization is not taken into account.		
Level 2	There are support requirements, but they are only partially formalized, and ad-hoc or even disorganized, and they are not defined down to subassembly level. The customer's organization is not taken into account.		
Level 3	Support requirements are formalized. The response to the requirements is formalized, but not validated, and is seen as secondary: the requirements are only partially justified, or are not met.		
Level 4	Support requirements are formalized (maintenance concept). A project organization is implemented at the manufacturer to respond to the customer requirements, in the form of a logistics support plan. Support requirements are taken into account from the design stage; they are defined down to subassembly level, justified and validated in a supportability analysis. The support system elements (documents, training, spare parts batches, tools and test equipment) are available, and are consistent and validated.		

Recommendation Implement self-inspection by the operator at the end of each production process.	Correspondence	N° 88
Phase(s) in which the recommendation applies MANUFACTURE EQUIPMENT_INTEGRATION SYSTEM_INTEGRATION		Weight 5.3 5.3 5.3
Further description Self-inspection by the operator, at the end of each process step, helps to filter out human errors that might jeopardize the reliability of the subassembly. The self-inspection procedure, usually based on a visual inspection of parts in production, is described in the operation instructions and in the Control Plan. The result of this inspection is recorded for batch traceability.		
Audit question Is there a self-inspection by the operator at the end of each production process? Is it instruction-based? Is the inspection result recorded?		
Level 1	There is no systematic self-inspection by the operator at the end of each production process.	
Level 2	Self-inspection is performed by the operator at the end of each production process. However, it is not described in any formal document.	
Level 3	Self-inspection is performed by the operator at the end of each production process. This is done in accordance with a predefined protocol formalized in a document. However, this document has not been validated by an independent authority.	
Level 4	Self-inspection is performed by the operator at the end of each production process. This is done in accordance with a predefined protocol formalized in a document. This document has been validated by an independent authority. The inspection result is recorded.	

Recommendation Inspect solder paste deposition after screen-printing.	Correspondence ISO9001: L2 EN9100: L3	N° 89
Phase(s) in which the recommendation applies MANUFACTURE		Weight 6.0
Further description Before mounting components, it is important to inspect the quality of screen-printing depositions, to minimize subsequent repairs and to supplement post-reflow inspections. Solder paste inspection (SPI) (of thickness/volume, quantity, etc.) must be specified, and the measurements recorded.		
Audit question Is the quality of solder paste deposition inspected systematically after screen-printing?		
Level 1 Solder paste deposition is not systematically inspected.		
Level 2 The thickness of the solder paste deposition is measured regularly, but this inspection is not based on a clearly identified document.		
Level 3 The thickness of the solder paste deposition is measured regularly. This inspection is based on a clearly identified document. However, this document does not refer to the IPC-A-610 standard, or there is no independent validation of the document.		
Level 4 The thickness of the solder paste deposition is measured regularly. This inspection is based on a clearly identified document. The IPC-A-610 standard is cited as a reference in this document. Additionally, the document has been validated by an authority independent from the operating entity.		

Recommendation Implement periodic inventories of component and material stocks.		Correspondence	N° 90
Phase(s) in which the recommendation applies MANUFACTURE			Weight 5.5
Further description Setting up periodic store inventories is useful for checking the accuracy of the recorded data: names or identifiers, batch quantities, geographic locations, expiry dates.			
Audit question Are periodic inventories carried out on component and material stocks?			
Level 1	No periodic stock inventories are carried out.		
Level 2	A number of inventories are carried out, but there is no formal plan for the frequency of these inventories.		
Level 3	Periodic inventories are carried out. If an inventory date is missed, a reminder is systematically issued until a new inventory has been validated. The actions to be performed, and the different tracking forms to be updated, are defined in formal documents. However, these documents have not been independently validated.		
Level 4	Periodic inventories are carried out. If an inventory date is missed, a reminder is systematically issued until a new inventory has been validated. The actions to be performed, and the different tracking forms to be updated, are defined in formal documents. These documents have been validated by an authority independent from the operating entity.		

Recommendation Implement ESD protections for components and subassemblies during handling and storage.	Correspondence	N° 91
Phase(s) in which the recommendation applies		Weight
MANUFACTURE		26.0
EQUIPMENT_INTEGRATION		26.0
SYSTEM_INTEGRATION		18.4
OPERATION_&_MAINTENANCE		17.4
Further description ESD protections for components and subassemblies include: - The grounding (earthing) of machines, tables, shelving, chairs and trolleys - The incoming inspection of conductive shoes and wristbands for the operators concerned - The connection of anti-ESD wristbands for operators - The use of ionizers, if necessary - The use of antistatic bags for board storage.		
Audit question Are specific ESD protections put in place for components and subassemblies during handling and storage?		
Level 1	Protection against ESD is not addressed.	
Level 2	Protection against ESD is handled by non-formalized rules and practices.	
Level 3	Protection against ESD is handled by validated procedures/instructions that define practices known to protect subassemblies. However, there is no external verification of the application of these rules (dedicated quality audits).	
Level 4	Protection of components and subassemblies against ESD is handled by validated procedures/instructions, which are monitored through dedicated internal audits.	

Recommendation		Correspondence	N°
Perform periodic checks on production system inspection tools.		EN9100: L3	92
Phase(s) in which the recommendation applies			Weight
MANUFACTURE			4.9
EQUIPMENT_INTEGRATION			4.9
Further description A number of production system parameters (temperature, dimensions, time, etc.) need to be controlled. Inspection tools (probes, sensors, detectors, micrometres, confocal microscopes, etc.) must be calibrated periodically in accordance with a set procedure, and, for certain critical tools, measurement repeatability and reproducibility tests are required, incorporating the operator factor.			
Audit question Are periodic checks performed on production system inspection tools? Are the measurements checked for reproducibility?			
Level 1	There is no periodic verification of production system inspection tools.		
Level 2	Tools and instruments used to inspect production systems are verified occasionally, but without following a formal verification plan.		
Level 3	Tools and instruments used to inspect production systems are periodically calibrated. There is evidence of these calibrations (records, labels). The calibration procedure is formalized in documents. However, the reproducibility and repeatability of critical measurements are not verified.		
Level 4	Tools and instruments used to inspect production systems are periodically verified. The calibration procedure is formalized in documents, which have been validated by an authority independent from the operating entity. The reproducibility and repeatability of critical measurements are verified periodically.		

Recommendation Implement suitable protections to avoid damaging subassemblies during cleaning.	Correspondence EN9100: L2	N° 93
Phase(s) in which the recommendation applies MANUFACTURE		Weight 6.0
Further description Suitable protections must be put in place to avoid damage to subassemblies during board cleaning operations. As the purpose of these protections is to isolate part of the subassembly, their effectiveness after completion of this activity must be verifiable (by inspection and measurement).		
Audit question Are suitable protections applied to avoid damaging subassemblies during cleaning?		
Level 1 No specific protections are used while cleaning subassemblies.		
Level 2 A number of protections are put in place during subassembly cleaning. These may be specific to certain subassemblies, but they are not covered by formal documents.		
Level 3 A number of protections are put in place during subassembly cleaning. The effectiveness of the protections is verified during subsequent inspections. The protections for each type of subassembly, and the correct procedures to follow, are identified and formalized in one or more documents. However, these documents are not validated by an independent authority.		
Level 4 A number of protections are put in place during subassembly cleaning. The effectiveness of the protections is verified during subsequent inspections. The protections for each type of subassembly, and the correct procedures to follow, are identified and formalized in one or more documents, which have been validated by an authority independent from the operating entity.		

Recommendation	Correspondence	N°
Implement testing tool self-tests to detect any anomalies.		94
Phase(s) in which the recommendation applies	Weight	
MANUFACTURE	5.1	
EQUIPMENT_INTEGRATION	5.1	
Further description Implement testing tool self-tests (built-in tests performed by the test equipment itself) to detect any anomalies. Reference units of known validity can be used for this self-test. The self-test is generally run during tester setup (with verifications during changes of operator or product, etc.) No test can be done if the self-test is not conclusive, unless there is a tracked waiver for the subassembly under test (i.e. authorization to conduct the test, but using markings and a signed tracking sheet that must constantly accompany the subassembly thereafter).		
Audit question Is there a testing tool self-test to detect any anomalies before use on the subassembly?		
Level 1	No self-test is implemented on testing tools.	
Level 2	A self-test is implemented on testing tools. It runs automatically, but there is no formal document or study to determine its effectiveness and limitations.	
Level 3	A self-test is implemented on testing tools. This self-test is described in documents that are used to determine the procedure and the degree of effectiveness. But these documents have not been validated by an authority independent from the operating entity.	
Level 4	A self-test is implemented on testing tools. This self-test is described in documents that are used to determine the procedure and the degree of effectiveness. These documents have been validated by an authority independent from the operating entity.	

Recommendation Manage the conformity of items and materials in stock.	Correspondence ISO9001: L3 EN9100: L3	N° 96
Phase(s) in which the recommendation applies MANUFACTURE		Weight 6.0
Further description The conformity of items and materials placed in stock, and available for production, must be clearly identified: - Stocks of nonconforming products must be placed in distinct areas, if possible inaccessible to production - Wrappings on items placed in production stock are usually marked (quality stamp or label) - The stock database records acceptance by the incoming quality control department, along with data on quantity, location, expiry dates, etc.		
Audit question Is the conformity of materials and items in stock identified?		
Level 1 There is no clear identification of the conformity of items and materials held in stock.		
Level 2 The conformity of items and materials held in stock is identified, but this is not formalized in writing.		
Level 3 The conformity of items and materials held in stock is clearly identified, in accordance with a written procedure. Nonconforming products are placed in dedicated areas. However, there is no stock database identifying conforming batches and their location.		
Level 4 The conformity of items and materials held in stock is clearly identified, in accordance with a written procedure. Nonconforming products are placed in dedicated areas. Conforming batches, and their location, are identified in a stock database.		

Recommendation Use Statistical Process Control (SPC) to manage the critical characteristics of the production process.	Correspondence	N° 97
Phase(s) in which the recommendation applies MANUFACTURE		Weight 4.5
Further description Statistical Process Control (SPC) charts can be used to monitor variability in critical processes with regard to the specification limits, and to take corrective or preventive action on the machine parameters. The process performance of SPC is measured by comparing the interval between the mean and a specified limit with the standard deviation of the measurement (3 Sigma). This process performance ratio (denoted Cpk) is used to calculate a quality target (lower limit). The SPC characteristics feature in the Process Control Plan, and are based on a failure mode effects and criticality analysis (FMECA). The control limits (upper and lower) are generally updated as a function of the measurement history.		
Audit question Is the production process managed by SPC (Statistical Process Control)? Have the critical characteristics identified by SPC been analysed? Do they feature clearly on the Control Plan? Are the upper and lower control limits updated?		
Level 1 The production process is not controlled by statistical methods (SPC).		
Level 2 There are statistical analyses of the critical production process characteristics, but these are not formally described in a document.		
Level 3 There are statistical analyses of the critical production process characteristics. This statistical control is formalized in documents, and control limits have been established. The result of the analyses is monitored by Quality, and corrective actions are initiated when the limits are exceeded. However, there is no evidence of a link between the FMECA study and the SPC characteristics, or there is no evidence that the reproducibility and repeatability of the measurements of these characteristics are regularly monitored.		
Level 4 There are statistical analyses of the critical production process characteristics. This statistical control is formalized in documents, and control limits have been established. The result of the analyses is monitored by Quality, and corrective actions are initiated when the limits are exceeded. The SPC characteristics are based on a FMECA conducted by a multidisciplinary team. The repeatability and reproducibility of the SPC tests are verified periodically.		

Recommendation	Correspondence	N°
Provide detailed instructions for each production process.	EN9100: L3	98
Phase(s) in which the recommendation applies	Weight	
MANUFACTURE	12.0	
EQUIPMENT_INTEGRATION	12.0	
Further description		
Reliability depends on the reproducibility of production operations (screen-printing, component mounting, reflow, cleaning, repair, masking, varnishing, etc.). Each of these operations must be clearly described in an instruction that defines the actions to be performed by the operator step-by-step. These instructions must be easily accessible on the production line, and should even, where appropriate, be printed on help documents permanently visible to the operator. These instructions must be constantly updated on the basis of internal or external quality audits.		
Audit question		
Are detailed instructions available for each production process?		
Are these instructions available online?		
Are help documents permanently visible to the operator?		
Level 1	There are no instructions describing the actions to be performed by each operator.	
Level 2	There is an instruction for each production operation, but its update procedure is not quality controlled, or it has not been validated by an independent authority, or is not easily accessible to the operator.	
Level 3	There is an instruction for each production operation. It is easily accessible to the operator and has been validated by an independent authority, but its update procedure is not quality controlled (internal audits).	
Level 4	There is an instruction for each production operation. It is easily accessible to the operator and has been validated by an independent authority. Its update procedure is quality controlled (internal audits) and only the latest version is available.	

Recommendation Implement a procedure to prevent items/consumables that are expired or due to be scrapped being used in production.	Correspondence	N° 99
Phase(s) in which the recommendation applies MANUFACTURE EQUIPMENT_INTEGRATION SYSTEM_INTEGRATION		Weight 6.4 6.4 6.4
Further description The use of expired consumables must be rendered impossible. The method of prevention is based on labelling, to identify the expiry date of the material, or on recording the date in the system, or on both. It is the responsibility of the storekeeper to check the date, and there must be a mandatory instruction to this effect. Preferably, the system will alert the storekeeper to withdraw the material/item before the date is exceeded, and will block the use of the batch. For the particular case of components that need to be reconditioned or reformed after a given storage time (e.g. electrolytic capacitors), there must be a corresponding procedure, and this action must be tracked in the stock database.		
Audit question Is there a procedure for identifying expired items/consumables and withdrawing them from stock (or reforming them)?		
Level 1	There is no procedure describing how to manage consumables that are out of date, or need to be reconditioned or reformed.	
Level 2	There is a procedure describing how to manage consumables that are out of date, or need to be reconditioned or reformed, but the labelling or signalling of expiry dates is either non-existent or very sporadic.	
Level 3	The expiry dates of consumables – and, where applicable, the reconditioning or reforming date – are correctly managed and identified by labelling, and/or via the system. The usage instructions clearly describe how to manage these dates, but these documents have not been validated by an authority independent from the operating entity.	
Level 4	The expiry dates of consumables – and, where applicable, the reconditioning or reforming dates – are correctly managed and identified, both by labelling and via the system. The usage instructions clearly describe how to manage these dates, and these documents are validated by an authority independent from the operating entity.	

Recommendation	Correspondence	N°
Implement real-time processing of test monitoring indicators.		100
Phase(s) in which the recommendation applies	Weight	
MANUFACTURE	4.7	
EQUIPMENT_INTEGRATION	4.7	
Further description		
The use of test monitoring indicators makes it possible to intervene as soon as an anomaly occurs:		
- An alarm is initiated on detection of an anomaly - The current activity is suspended if the anomaly is considered serious (to avoid stress to the subassembly) - The anomaly is addressed and corrected before restarting or resuming the activity - If possible (if volumes are sufficient), a minimum test rate is defined, below which the result is considered abnormal.		
Audit question		
Are test monitoring indicators processed in real time in order to avoid damage to the subassembly if an anomaly occurs?		
Level 1	There are no test monitoring indicators.	
Level 2	There are a number of indicators used to identify any anomaly occurring during the test. However, indicators are not part of a formal plan and/or they are not processed until later.	
Level 3	There are a number of indicators used to identify any anomaly occurring during the test, and the processing of these indicators makes it possible to intervene as soon as an anomaly occurs. There are documents that formally describe how these indicators, and the resulting actions, are managed. However, the relevant data has not been validated by an authority independent from the operating entity.	
Level 4	There are a number of indicators used to identify any anomaly occurring during the test, and the processing of these indicators makes it possible to intervene as soon as an anomaly occurs. There are documents that formally describe how these indicators, and the resulting actions, are managed. Additionally, these documents have been validated by an authority independent from the operating entity.	

Recommendation Continuously monitor the temperature of the drying and polymerization ovens.	Correspondence	N° 102
Phase(s) in which the recommendation applies MANUFACTURE		Weight 12.7
Further description It is important that the equipment and materials are not subject to unnecessary thermal stress during the drying and polymerization operations. The temperature of the ovens must be kept under control. This control must be exercised by continuously measuring the temperature, or by frequent periodic checks, but also by verifying the set points or program at the start of the operation. All such checks and measurements must be recorded by the operator and tracked for each production batch. Corrective actions (program shutdown, decisions about the production batch) must be clearly defined in an instruction.		
Audit question Is the temperature of the drying and polymerization ovens continuously monitored, using external probes?		
Level 1	The temperature of the ovens is not monitored continuously or even frequently, nor is it recorded.	
Level 2	The temperature of the ovens is monitored frequently or continuously, and the results are recorded, but this monitoring task is not covered by a document describing the actions to be performed.	
Level 3	The temperature of the ovens is monitored frequently or continuously; the results are recorded, and there are limits that trigger corrective action. This monitoring task is covered by a document describing the actions to be performed. However, this document has not been validated by an authority independent from the operating entity.	
Level 4	The temperature of the ovens is monitored frequently or continuously; the results are recorded, and there are limits that trigger corrective action. The result is recorded for each batch. This monitoring task is covered by a document describing the actions to be performed. This document has been validated by an authority independent from the operating entity.	

Recommendation Analyse the customer's reliability requirements.	Correspondence EN9100: L2	N° 103
Phase(s) in which the recommendation applies SPECIFICATION		Weight 10.7
Further description The reliability requirements must be analysed, with two aims in mind: - To assess the risk of nonconformity - To optimize reliability performance versus cost (cost of designing product, cost of conducting reliability studies). Analysis of the requirements may lead to: - Negotiation of requirements with the customer - The formalization of a plan to manage the risk of nonconformity to the requirements. In every case, analysis and negotiation must be carried out with reference to the state of the art in the technology.		
Audit question Are the reliability requirements analysed? If applicable, does this analysis lead to the negotiation of requirements, or to the formalization of a risk management plan?		
Level 1	The requirements are not analysed.	
Level 2	The requirements are analysed informally. However, there is no risk management plan and negotiations, if any, remain informal.	
Level 3	The requirements are analysed formally. The risk management plan or negotiations with the customer remain informal.	
Level 4	The requirements are analysed. Analysis of the requirements leads, where necessary, to the formalization of a plan to manage nonconformity risks, or to the negotiation of requirements with the customer. In the latter case, the result of the negotiation is integrated into the final bid submitted to the customer.	

<u>Recommendation</u> Appoint someone to be in charge of reliability studies.		Correspondence	N° 104
Phase(s) in which the recommendation applies SUPPORT			Weight 8.5
Further description Appoint a reliability manager for each project, with responsibility for ensuring that the product reliability objectives are achieved. This person will report on the progress of studies and on problems encountered.			
Audit question Has someone been formally appointed to be in charge of reliability studies?			
Level 1	Nobody is identified as being in charge of reliability studies.		
Level 2	There is somebody in charge of reliability studies, but there is no record of their appointment.		
Level 3	Somebody has been formally appointed to take charge of reliability studies. They are named in a programme management plan or in a function-specific plan (reliability or RAMS plan). They are trained and experienced in the field of reliability.		
Level 4	Somebody has been formally appointed to take charge of reliability studies. They are named in a programme management plan or in a function-specific plan (reliability or RAMS plan). They are trained and experienced in the field of reliability. They are integrated into the project and provide regular updates on the progress of studies at meetings or in reports.		

Recommendation Manage the subcontracting of reliability studies.	Correspondence ISO9001: L3 EN9100: L3	N° 105
Phase(s) in which the recommendation applies SUPPORT		Weight 10
Further description The subcontracting of reliability studies must be organized and formalized: - The process of communication and data exchange with the subcontractor must be described - Periodic meetings must be organized with the subcontractor.		
Audit question How is the subcontracting of reliability studies managed?		
Level 1 The management of subcontracting is neither organized nor formalized.		
Level 2 The management of subcontracting is incomplete (at least one of the two criteria in the further description is not met).		
Level 3 The management of subcontracting is complete, with regard to both the criteria in the further description, but it is not formalized in a project management plan.		
Level 4 The management of subcontracting is described in a project management plan, covering, at least, both of the criteria in the further description.		

Recommendation Address reliability aspects during design reviews.		Correspondence ISO9001: L3 EN9100: L4	N° 106
Phase(s) in which the recommendation applies			Weight
SPECIFICATION			10.3
DESIGN			12.1
Further description Design reviews must address reliability aspects. Conformity to reliability requirements must be addressed in these reviews. The reviews must define: - Reliability allocations - Conditions of use (mission profile).			
Audit question Do design reviews address reliability aspects?			
Level 1	There are no design reviews, or the design reviews do not address reliability aspects.		
Level 2	Design reviews address reliability aspects only in part, or they are held in the absence of reliability specialists.		
Level 3	Design reviews address reliability aspects in full and are held in the presence of reliability specialists.		
Level 4	Design reviews address reliability aspects in full and are held in the presence of reliability specialists. Design reviews, and the presence of reliability specialists, are imposed by a procedure or plan.		

Recommendation Address reliability aspects during requirements reviews.	Correspondence ISO9001: L3 EN9100: L4	N° 107
Phase(s) in which the recommendation applies SPECIFICATION		Weight 10.3
Further description Requirements reviews must address reliability requirements and verify that: - All reliability requirements are identified and tracked - The reliability requirements correspond to the customer's need (contract) - The reliability requirements are attainable, verifiable and validatable (conformity metrics).		
Audit question Do requirements reviews address reliability requirements?		
Level 1	There are no requirements reviews, or if there are, they do not address the reliability requirements.	
Level 2	Requirements reviews address reliability requirements only in part, or they are held in the absence of reliability specialists.	
Level 3	Requirements reviews address reliability requirements in full and are held in the presence of reliability specialists.	
Level 4	Requirements reviews address reliability requirements in full and are held in the presence of reliability specialists. Requirements reviews, and the presence of reliability specialists, are imposed by a procedure or plan.	

<u>Recommendation</u> Implement reliability growth actions.		Correspondence	N° 110
Phase(s) in which the recommendation applies SUPPORT			Weight 9.1
Further description Describe reliability growth activities in the reliability plan, and document their implementation.			
Audit question Are reliability growth actions implemented?			
Level 1	No product reliability growth activities are planned or implemented.		
Level 2	Product reliability growth activities do exist, but are not described in any plan.		
Level 3	Product reliability growth activities are described in a plan and are partially implemented.		
Level 4	Product reliability growth activities are described in a plan and are fully implemented.		

Recommendation Plan reliability studies.	Correspondence ISO9001: L3 EN9100: L3	N° 111
Phase(s) in which the recommendation applies SUPPORT		Weight 8
Further description Plan reliability studies and synchronize them with the design of the product. Integrate reliability activities into project schedules.		
Audit question Are reliability studies planned?		
Level 1 Reliability studies are not planned.		
Level 2 Reliability studies are planned informally (schedule defined and updated by the reliability team).		
Level 3 Reliability studies are identified in a project schedule.		
Level 4 Reliability studies are identified in a project schedule, which is updated. The synchronization with the design of the product is formalized through milestones, freezes, etc.		

Recommendation Define upper and lower safety limits on inspection and monitoring system parameters.	Correspondence	N° 112
Phase(s) in which the recommendation applies MANUFACTURE		Weight 5.7
Further description Upper and lower safety limits are defined on all inspection and monitoring system parameters (e.g. temperature measured by probes). In the event of excursion beyond the safety limits, the cycle is automatically shut down and a technician analyses the situation before anything can restart.		
Audit question Are upper and lower safety limits defined on the parameters of the inspection and monitoring systems? Do they trigger systematic shutdown of the cycle and analysis by a technician before restart?		
Level 1	The upper and lower safety limits, if any, on the inspection and monitoring system parameters do not trigger automatic shutdown.	
Level 2	There are upper and lower safety limits on the inspection and monitoring system parameters, and any excursion beyond the safety limits triggers an automatic shutdown. However, there is no document that formally identifies the safety limits and the shutdown procedures.	
Level 3	There are upper and lower safety limits on the inspection and monitoring system parameters, and any excursion beyond the safety limits triggers an automatic shutdown. The safety limits are formally identified in a document specific to each system, which also identifies the shutdown procedures.	
Level 4	There are upper and lower safety limits on the inspection and monitoring system parameters, and any excursion beyond the safety limits triggers an automatic shutdown. The safety limits are formally identified in a document specific to each system, which also identifies the shutdown procedures. Additionally, these documents have been validated by an authority independent from the operating entity.	

Recommendation Keep inspection and test records.		Correspondence EN9100: L2	N° 113
Phase(s) in which the recommendation applies EQUIPMENT_INTEGRATION SYSTEM_INTEGRATION			Weight 5.3 5.3
Further description Records must be established and kept proving that the product has been inspected and/or tested in accordance with the defined criteria. The records must identify the person who performed the inspections and released the product. Test records must indicate the measured values when they are required by the specification or by the acceptance plan. If specified, the supplier must demonstrate the qualification of the product.			
Audit question Are records produced and kept to prove that the product has been inspected and/or tested in accordance with defined criteria? Do the records identify the person who performed the inspections?			
Level 1	Records of tests and inspections are established but not kept.		
Level 2	Records of tests and inspections are established and kept, but do not identify the source of the inspection (persons, machines).		
Level 3	Records of tests and inspections are established and kept and identify the source of the inspection (persons, machines).		
Level 4	Records prove that the product has been inspected and/or tested in accordance with the defined criteria. The records identify the person who performed the inspections and authorized the release of the product. The test records indicate the measured values when they are required by the specification or by the acceptance plan.		

Recommendation Maintain an inspection file.	Correspondence EN9100: L4	N° 114
Phase(s) in which the recommendation applies		Weight
EQUIPMENT_INTEGRATION		5.7
SYSTEM_INTEGRATION		5.7
OPERATION_&_MAINTENANCE		5.7
Further description The inspection file must contain: - The acceptance and rejection criteria - A sequential list of test and inspection operations to be performed - The inspection results - A list of specific and non-specific inspection instruments - The documents associated with the specific inspection instruments, pertaining to their design, production, validation, management, use and maintenance.		
Audit question Is there an inspection file containing the acceptance criteria, the inspection and testing sequence, the inspection records, and the list of specific and non-specific inspection instruments?		
Level 1	There is no inspection file.	
Level 2	The inspection file is limited to the definition of the acceptance or rejection criteria.	
Level 3	The inspection file defines acceptance or rejection criteria, as well as the list of operations to be performed. It proposes documents for recording inspection results.	
Level 4	The inspection file contains: - The definition of the acceptance or rejection criteria, supplemented as needed by photographs of acceptable/unacceptable defects for visual inspection - The inspection and testing sequence to be performed - The inspection results - The list of specific and non-specific inspection instruments - The documents associated with the specific inspection instruments, pertaining to their design, production, validation, management, use and maintenance.	

Recommendation Control the quality of data in nonconformity reports.	Correspondence ISO9001: L4 EN9100: L4	N° 116
Phase(s) in which the recommendation applies DESIGN MANUFACTURE EQUIPMENT_INTEGRATION SYSTEM_INTEGRATION OPERATION_&_MAINTENANCE	Weight 13.0 12.0 11.1 11.1 11.1	
Further description Nonconformity reports must specify: - The product identification - The description of the nonconformity - The cause of the nonconformity - The actions taken to avoid recurrence of the nonconformity - The rework or repairs, if necessary - The inspection of the characteristics affected by the rework or repairs - The final decision.		
Audit question What information is described in the nonconformity reports?		
Level 1	There are no specific nonconformity documents.	
Level 2	The nonconformity documents serve only to identify nonconforming products.	
Level 3	The nonconformity documents specify the product identification, the description of the nonconformity, and the cause of the nonconformity. However, no formal actions are defined to avoid recurrence of the nonconformity, specify any rework or repairs required, and inspect the characteristics affected by the rework or repairs.	
Level 4	The nonconformity documents specify the product identification, the description of the nonconformity, and the cause of the nonconformity. Formal actions are defined to avoid recurrence of the nonconformity, specify any rework or repairs required, and inspect the characteristics affected by the rework or repairs.	

Recommendation Take account of the product maintenance policy, as defined by the customer.	Correspondence	N° 117
Phase(s) in which the recommendation applies SPECIFICATION		Weight 5.8
Further description The customer's maintenance policy must be taken into account in order to preserve the reliability of the product over time.		
Audit question How is the customer's product maintenance policy taken into account?		
Level 1 The product maintenance policy is unknown.		
Level 2 The maintenance policy is known, but was defined without taking account of reliability aspects.		
Level 3 The maintenance policy is known and was defined by taking account of reliability aspects (identification and monitoring of critical elements).		
Level 4 The maintenance policy is known and was defined by taking account of reliability aspects (identification and monitoring of critical elements). The maintenance policy was defined with input from reliability managers. The maintenance policy is formalized in a document.		

Recommendation		Correspondence	N°
Plan for periodic customer consultations on reliability aspects.		ISO9001: L3 EN9100: L3	119
Phase(s) in which the recommendation applies			Weight
SUPPORT			7.3
Further description			
Consult customers regularly on operational reliability aspects and take their comments into account for the design of new products. This consultation can be conducted through customer satisfaction surveys, which must take account of reliability aspects.			
Audit question			
Are periodic consultations planned with customers on reliability aspects?			
Level 1	There is no available customer feedback about the perception of product reliability.		
Level 2	There is some available customer feedback on reliability, but it is rarely used.		
Level 3	There is some available customer feedback on reliability, and it is used to improve product design, development and manufacture.		
Level 4	The company regularly consults its customers about the reliability of its products (formal interviews or questionnaire surveys). This feedback is used and give rise to action plans, the results of which are communicated to the customer. The effectiveness of this process can be demonstrated by customer satisfaction levels.		

<u>Recommendation</u>		Correspondence	N°
Verify solder mask deposition before varnishing.			120
Phase(s) in which the recommendation applies			Weight
MANUFACTURE			6.5
Further description			
The quality of solder mask applied before varnishing is important for avoiding potential reliability issues with connectors or heat sinks in operation. The inspection of this operation must feature on the Control Plan of the manufacturing process.			
Audit question			
Is there an inspection step (even if only visual) to verify correct deposition of the solder mask before varnishing?			
Level 1	No particular visual inspection is carried out when placing solder mask before varnishing.		
Level 2	A specific solder mask inspection is carried out before varnishing, but there is no document that describes the procedure to follow for this inspection.		
Level 3	A specific solder mask inspection is carried out before varnishing. This dedicated inspection is covered by a formal instruction chapter. However, this document has not been validated by an independent authority, or does not feature in the Control Plan.		
Level 4	A specific solder mask inspection is carried out before varnishing. This dedicated inspection is covered by a formal instruction chapter. This document is validated by an independent authority, and features in the Control Plan.		

Recommendation Check the machine parameters every time before using a production machine on a board or a subassembly (Setup).	Correspondence ISO9001: L2 EN9100: L2	N° 121
Phase(s) in which the recommendation applies MANUFACTURE EQUIPMENT_INTEGRATION		Weight 4.7 4.7
Further description It is important to detect any anomalies before using a machine on a board or subassembly. This is done through a series of “machine setup” checks, performed by the operator at key production steps: - Machine startup - Product changeover - Production batch changeover - Operator team changeover - Replacement of certain consumables. These checks are clearly described in the machine instructions, and are recorded.		
Audit question Are the machine parameters checked before each use of a production machine? Is this check described in the corresponding machine instructions?		
Level 1	There is no evidence that machine parameters are checked methodically during critical changeovers.	
Level 2	Some machine parameters are checked during critical changeovers, but these checks are not exhaustive and are not covered by a formal document.	
Level 3	Machine parameters are systematically checked during critical changeovers. These checks are described in a machine instruction. However, this document is not duly validated, or the checks are not recorded.	
Level 4	Machine parameters are systematically checked during critical changeovers. These controls are described in a machine instruction, which is validated by an independent authority. The parameter checks are recorded, and the subsequent anomaly analysis can lead to corrective or preventive actions.	

Recommendation Write a reliability plan.	Correspondence	N° 122
Phase(s) in which the recommendation applies SPECIFICATION		Weight 7.6
Further description A reliability plan is written, including, as a minimum: - A description of the project organization (human and hardware) - A reminder of the reliability-related requirements - A description of the calculation methodology and underlying assumptions (level of detail of the study, mission profile, etc.) - A description of the requirement demonstration method - A description of the method for exploiting feedback - A description of the activities to be performed in the event of nonconformity. In the absence of a reliability plan, the above topics must be fully covered in a RAMS/RAMT plan or a development plan.		
Audit question Has a reliability plan been written for the product?		
Level 1	No reliability plan has been written.	
Level 2	A reliability plan is written informally, but is not validated by the Project.	
Level 3	The reliability plan is written and validated by the Project. However, this document, written at the outset, is not updated during the project.	
Level 4	The reliability plan is written and validated by the Project. This document is updated throughout the project, to take stock of events that might cause it to be modified.	

Recommendation Plan dependability activities.		Correspondence ISO9001: L3 EN9100: L3	N° 123
Phase(s) in which the recommendation applies DESIGN			Weight 17.7
Further description Make sure that there is a plan (e.g. a project management plan or dependability management plan) that clearly describes the activities to be performed, their interactions (inputs/outputs) and how they fit in with the other activities, and any adjustments made to the guidelines.			
Audit question Is there a plan that integrates the dependability aspects? Are activities planned and key personnel identified?			
Level 1	There is no management plan, and no schedule describing the tasks to be performed. There is no organizational structure in place.		
Level 2	The management plan is incomplete: - It does not specify any adjustments made to the guidelines. - The schedule describing the tasks to be performed and the organization put in place is imprecise and/or incompatible with the available resources.		
Level 3	The management plan is incomplete: - It does not specify any adjustments made to the guidelines. - The schedule describing the tasks to be performed and the organization put in place is precise, but not peer-validated (by key function personnel).		
Level 4	The management plan is complete: - It specifies the adjustments made to the guidelines, if any. - The schedule describing the tasks to be performed and the organization put in place is precise and is validated. - The plan is peer-validated (by key function personnel).		

Recommendation Write a final acceptance procedure.	Correspondence ISO9001: L3 EN9100: L4	N° 124
Phase(s) in which the recommendation applies DESIGN		Weight 7.8
Further description Make sure that there is an acceptance procedure and that it is relevant. The acceptance procedure is drawn up on the basis of a definition file and a manufacturing file which focuses on testing and final inspection.		
Audit question Is there a final acceptance procedure for production tests?		
Level 1 There is no acceptance procedure.		
Level 2 There is an acceptance procedure, but it is established in production, independently from the development teams.		
Level 3 There is an acceptance procedure, which is established during development; it integrates configuration management, but is neither validated nor tracked.		
Level 4 The acceptance procedure is adapted to the product (with proof of the traceability of its application to the product and its configuration) and is validated.		

Recommendation Revise and strengthen production system controls to eliminate any possibility of damaging component connections.	Correspondence	N° 126
Phase(s) in which the recommendation applies MANUFACTURE		Weight 6.7
Further description To avoid damaging component connections during production operations, especially when placing components, the machine parameters and the tools must be covered by an inspection specification. This specification, and the associated instructions, must be periodically updated to take account of machine incidents, the results of board inspections (as per the IPC standard) and changes in the production process. Machine setup and calibration also contribute to this prevention role.		
Audit question Is the inspection specification for the pick-and-place machine updated periodically? Are the board inspection results taken into account?		
Level 1	There is no production system specification that covers component handling risks.	
Level 2	There is a production system specification that covers component handling risks, but it is not regularly updated.	
Level 3	There is a production system specification that covers component handling risks. The specification is regularly updated, but this document is not validated by an independent authority, or is not covered by a dedicated FMECA analysis.	
Level 4	There is a production system specification that covers component handling risks. This specification is regularly updated, and is validated by an independent authority. The risk of damaging component connections is the subject of a FMECA analysis.	

Recommendation Control the preparation of transformable materials by means of a qualified procedure and inspection measures.		Correspondence	N° 127
Phase(s) in which the recommendation applies MANUFACTURE			Weight 5.9
Further description The preparation of transformable materials (solder paste, glues, resin, varnish) must be qualified and its reproducibility guaranteed. The instructions need to have been analysed and the Control Plan must include the associated inspection specifications. The observations and inspections must be recorded in the batch tracking sheet.			
Audit question How is the preparation of transformable materials (solder paste, glues, varnish) controlled?			
Level 1	Not all transformable materials (solder paste, glues, varnish) are covered by a qualified preparation instruction and a quality control check.		
Level 2	All transformable materials (solder paste, glues, varnish) are covered by a preparation instruction. However, these documents are not based on process qualification studies, or do not mention associated quality controls.		
Level 3	All transformable materials (solder paste, glues, varnish) are covered by a preparation instruction. These documents systematically mention the associated quality controls, which are recorded on the batch tracking sheet, but not all of them are based on process qualification studies.		
Level 4	All transformable materials (solder paste, glues, varnish) are covered by a preparation instruction. These documents systematically mention the associated quality controls, which are recorded on the batch tracking sheet, and all of them are based on process qualification studies. Any change of material leads to requalification. These documents have been validated by an independent authority.		

Recommendation	Correspondence	N°
Maintain the skills of operators and technicians through continuous training and frequent assessment.	ISO9001: L3 EN9100: L3	128
Phase(s) in which the recommendation applies	Weight	
MANUFACTURE	4.4	
EQUIPMENT_INTEGRATION	10.0	
Further description Beyond initial training, and the certification of the operator/technician for the tasks entrusted to them, it is important to make sure that these skills are firmly rooted, and that they are updated in response to process changes. This can be monitored, and any necessary corrections made, by drawing on a range of actions: - Re-certification of the operator after a period of time that depends on the criticality of the operation in terms of quality and reliability - Training in new methods or processes - Internal audits conducted by Quality, and external audits conducted by independent services - Logging and analysis of nonconformities or customer feedback - Skills management within teams (e.g. preventing turnover).		
Audit question Are training, assessment and other actions performed to maintain the skills of operators and technicians? Are process changes accompanied by additional training? How is operator or technician turnover managed?		
Level 1	There is no plan in place to upgrade operator knowledge.	
Level 2	Operator skills are updated. However, this update is not described formally in documents.	
Level 3	Operator skills are updated. This update is covered by documents describing training plans and events that give rise to updates. However, these documents have not been validated, or there is no knowledge monitoring on the production line (quality audits).	
Level 4	Operator skills are updated. This update is covered by documents describing training plans and events that give rise to updates. These documents have been validated by an authority independent from the operating entity. Knowledge is monitored through internal quality audits. Skills management for each position is constantly analysed.	

Recommendation Verify the effectiveness of the final varnishing quality inspection by means of an appropriate procedure.	Correspondence	N° 129
Phase(s) in which the recommendation applies MANUFACTURE		Weight 5.2
Further description The inspection procedure must be strictly applied to the varnishing. This final check includes visual inspection and a verification of all the process steps. The reference standard is IPC-A-610.		
Audit question Is the effectiveness of the final varnishing quality inspection assured by strict application of the inspection procedure?		
Level 1 It is not clear that the final varnishing inspection is performed systematically.		
Level 2 The final varnishing inspection is performed systematically, but there is no document describing the control points and requirements in detail.		
Level 3 The final varnishing inspection is performed systematically. There is a document describing the control points and requirements in detail. However, this document does not refer to a recognized standard or has not been validated by an authority independent from the operating entity.		
Level 4 The final varnishing inspection is performed systematically. There is a document describing the control points and requirements in detail. This document refers to the IPC-A-610 standard, and has been validated by an authority independent from the operating entity.		

Recommendation Make sure that the elements required for the reliability assessment are available.	Correspondence ISO9001: L4 EN9100: L4	N° 130
Phase(s) in which the recommendation applies DESIGN		Weight 7.5
Further description Make sure that the necessary input data for reliability assessment is available. This data includes: - The definition file (including the functional and physical architectures, electrical diagrams, mission profile and reliability requirements) - The parts lists - The thermal analysis reports - The derating analysis. These input data have been validated.		
Audit question Are the elements required for the reliability assessment available? Does the input data come from validated documents?		
Level 1	The data required for reliability assessment is not validated.	
Level 2	The data required for reliability assessment is incomplete or only partially validated, or is not up to date.	
Level 3	The data required for reliability assessment is available, up to date and validated, but incomplete (see further description list).	
Level 4	All the data required for reliability assessment (see further description list) is available, up to date and validated.	

Recommendation Make sure there are design rules for adapting the choice of component to the required reliability level.	Correspondence ISO9001: L4 EN9100: L4	N° 131
Phase(s) in which the recommendation applies DESIGN		Weight 12.7
Further description Make sure there is a design methodology that imposes rules appropriate to the required reliability level: - Choice of packages (dimensions, materials) appropriate to the conditions of use - Choice of components with a lifetime commensurable with the lifetime of the equipment - Mechanical rules (stiffeners, etc.) - Etc. Make sure that the application of the rules is verified.		
Audit question Are there any design rules for adapting the choice of component to the required reliability level?		
Level 1 There are no reliability-oriented design rules.		
Level 2 There are rules, but they are not formalized (not updated, not validated).		
Level 3 There are formalized rules; they are updated, but not validated.		
Level 4 There are formalized rules; they are updated, validated and applied.		

Recommendation Make sure that production test points are defined and that test recommendations are applied by Production.	Correspondence	N° 132
Phase(s) in which the recommendation applies DESIGN		Weight 6
Further description Make sure test points are defined. Make sure that the test operation constraints (as specified by the test manager) are integrated into the product definition and communicated to the production manager. Remark: The definition of test points can sometimes be deduced from the key characteristics identified in the Design FMECA.		
Audit question Are test points defined and are recommendations on production tests applied?		
Level 1	The test points have not been defined, or the production manager has no information on the method for implementing tests on the product.	
Level 2	The production manager knows about the test implementation method, and contributed to the test recommendations.	
Level 3	The production manager knows about the test implementation method and contributed to the test recommendations. There is a validated set of recommendations explaining how to conduct the tests. There is no traceability (no guarantee that the recommendations are applied).	
Level 4	The production manager knows about the test implementation method and contributed to the test recommendations. There is a validated set of recommendations explaining how to conduct the tests. There are documents attesting that the recommendations are applied.	

Recommendation Make sure there is a qualification procedure for the product and the manufacturing process.	Correspondence ISO9001: L4 EN9100: L4	N° 133
Phase(s) in which the recommendation applies DESIGN		Weight 7.2
Further description Make sure there is a product/process qualification procedure, including the description of the qualification test. This procedure is validated and updated in accordance with the entity's practices.		
Audit question Is there a qualification procedure for the products and the manufacturing process?		
Level 1	There is no product/process qualification procedure.	
Level 2	There is a qualification procedure for the product but there is no qualification procedure for the process.	
Level 3	There is a qualification procedure for the product and a qualification procedure for the process, but the compatibility between the two procedures is not verified.	
Level 4	There is a qualification procedure for the product and a qualification procedure for the process; the compatibility between the two procedures is verified.	

Recommendation Make sure there is a qualification procedure for Tier-1 suppliers/products.	Correspondence EN9100: L2	N° 134
Phase(s) in which the recommendation applies DESIGN		Weight 7.6
Further description Make sure that Tier-1 suppliers are qualified and that the following aspects are covered: - Quality - Reliability - Long-term sustainability. Make sure that the initial qualification level is maintained over time (audits, analyses, tests, etc.).		
Audit question Is there a qualification procedure for Tier-1 suppliers/products? On what criteria is this qualification based?		
Level 1 There is no qualification of suppliers.		
Level 2 Supplier qualification is performed partially and informally.		
Level 3 The company guidelines call for the qualification of suppliers according to criteria of quality, reliability and long-term sustainability. This is implemented and is based on formal activities: analysis of supplier data, interviews with suppliers, track record analysis, audits, ISO certification, etc.).		
Level 4 The company guidelines call for the qualification of suppliers according to criteria of quality, reliability and long-term sustainability. This is implemented and is based on formal activities: analysis of supplier data, interviews with suppliers, track record analysis, audits, ISO certification, etc.). There is a procedure for maintaining qualification over time.		

Recommendation Make sure there is a qualification procedure for specific components or new technologies.	Correspondence	N° 135
Phase(s) in which the recommendation applies DESIGN		Weight 7.2
Further description This recommendation concerns specific components (e.g. ASICs or hybrids). Make sure there is a qualification procedure for evaluating the risks inherent in the use of specific components or new technologies.		
Audit question Are specific components and new technologies qualified before use? Is there a qualification procedure?		
Level 1 There is no procedure.		
Level 2 There are informal rules.		
Level 3 There is a procedure.		
Level 4 There is a managed procedure that follows developments in technologies; it is validated by the competent technical services. There is proof of its application.		

Recommendation Use a proven methodology and a qualified tool for predictive reliability calculations.	Correspondence	N° 137
Phase(s) in which the recommendation applies DESIGN		Weight 7.7
Further description Make sure that the predictive reliability calculation follows a recognized methodology and uses a qualified tool.		
Audit question Is the method used for predictive reliability assessment recognized and validated by experience? Is the calculation tool qualified?		
Level 1	There is no identified methodology: no standard, guide or instruction was followed. The tool is neither recognized nor validated by evaluation.	
Level 2	There is an identified methodology, but it is neither recognized nor validated. The tool is recognized, but not validated by evaluation.	
Level 3	The methodology and the tool are recognized, but not validated by evaluation.	
Level 4	The methodology and the tool are recognized and are validated by evaluation.	

Recommendation Make sure that the correct embedded software is loaded, and record its version details.	Correspondence	N° 138
Phase(s) in which the recommendation applies MANUFACTURE EQUIPMENT_INTEGRATION SYSTEM_INTEGRATION		Weight 6.7 6.7 6.7
Further description Make sure that the correct embedded software is loaded and, in particular, that it corresponds to the latest version for use on the subassembly. This identification information must be traceable for the rest of the process.		
Audit question Are checks performed to ensure that the correct embedded software is loaded, and are its version details recorded?		
Level 1 The embedded software that has been loaded is not identified.		
Level 2 After embedded software has been loaded into a hardware subassembly, an identifier is supplied for the loaded software, guaranteeing that it conforms to the subassembly. However, there is no document that precisely describes the format or recording of this identifier.		
Level 3 Every time embedded software is loaded, the operator is informed of the software version to be used. An identifier of the version to be used is supplied after the operation.		
Level 4 Every time embedded software is loaded, the operator is informed of the software version to be used. An identifier of the version to be used is supplied after the operation. Cross-checking is formalized.		

Recommendation Ensure that preventive maintenance is performed on the production systems and that it is monitored.	Correspondence	N° 139
Phase(s) in which the recommendation applies MANUFACTURE EQUIPMENT_INTEGRATION		Weight 5.9 5.9
Further description Preventive maintenance on production systems must take account of specified wear/usage criteria (duration, number of operations or drifts) and the recent history of nonconformities and characteristic measurement drifts. The maintenance of equipment involves the repair and recalibration of tools, and the replacement of worn parts or those that have reached their maximum authorized lifetime. This function can be entrusted to a dedicated team, which will also manage the need for inspection and repair by outside firms (which may depend on the machine manufacturer).		
Audit question Is preventive maintenance performed on the production systems? Is this maintenance monitored?		
Level 1 Preventive maintenance of production systems is not monitored.		
Level 2 Preventive maintenance is performed on the production systems and is monitored. However there is no documented formal maintenance plan indicating the periodicity and the mandatory inspection points for this maintenance.		
Level 3 There is an effective preventive maintenance procedure in place for the production systems. It is monitored on the basis of a plan that lists all the mandatory inspection points and the periodicities of each action. There is no evidence of recent nonconformities being taken into account in the maintenance plan.		
Level 4 There is an effective maintenance procedure in place for the production systems. It is monitored on the basis of a plan that lists all the mandatory inspection points and the periodicities of each action. Additionally, recent nonconformities are taken into account in the preventive maintenance plan.		

Recommendation Select the components used.		Correspondence	N° 142
Phase(s) in which the recommendation applies SUPPORT			Weight 12.9
Further description Select the components used on the basis of the expected reliability level: analyse the market; evaluate the reliability of the components.			
Audit question Are reliability criteria considered when selecting the components to be used?			
Level 1	There is no selection of components.		
Level 2	Components are selected on the basis of reliability (and/or manufacturing quality), but only informally.		
Level 3	The company guidelines call for components to be selected on the basis of reliability (and/or manufacturing quality). This is followed, but is based only on manufacturer data.		
Level 4	The company guidelines call for components to be selected on the basis of reliability (and/or manufacturing quality). This is followed, and is based on in-depth analyses (using manufacturer data, manufacturer audits, and evaluations of the technologies employed).		

<u>Recommendation</u> Select component suppliers.		Correspondence	N° 143
Phase(s) in which the recommendation applies SUPPORT			Weight 10.8
Further description Select component suppliers: analyse the market; evaluate how component reliability is taken into account by suppliers.			
Audit question Are reliability criteria considered when selecting component suppliers?			
Level 1	Component suppliers are not selected.		
Level 2	Component suppliers are selected informally.		
Level 3	The company guidelines call for suppliers to be selected on the basis of reliability (and/or manufacturing quality). This is followed, but is based only on supplier data.		
Level 4	The company guidelines call for suppliers to be selected on the basis of reliability (and/or manufacturing quality). This is followed, and is based on formal activities: interviews with suppliers, track record analysis, audits, ISO certifications, etc.		

Recommendation Sensitize personnel to the importance of visual checks on subassemblies after placement and before reflow.		Correspondence	N° 144
Phase(s) in which the recommendation applies MANUFACTURE			Weight 5.9
Further description In order to reduce the number of anomalies that go undetected during inspections that rely on the human factor (in this case visual checks), it is important to sensitize the personnel responsible for this task so as to minimize risks due to human error or to failure to detect anomalies.			
Audit question Are personnel made aware of the importance of visual checks on subassemblies after placement and before reflow?			
Level 1	No particular verification is performed after placing the components and before reflow.		
Level 2	The operator may check that placement is correct before reflow, but this verification is not described formally.		
Level 3	The operator checks that placement is correct before reflow. This verification conforms to a procedure (which may simply mention a visual check or may list the points to be verified systematically).		
Level 4	The operator checks that placement is correct before reflow. This verification conforms to a procedure (which may simply mention a visual check or may list the points to be verified systematically). Additionally, this procedure has been validated by an authority independent from the operating entity.		

Recommendation Monitor and control corrective actions by subcontractors relating to product reliability.		Correspondence	N° 146
Phase(s) in which the recommendation applies SUPPORT			Weight 7.2
Further description Monitor and control (through plans and records) corrective actions by subcontractors relating to product reliability.			
Audit question Are reliability-related corrective actions by subcontractors monitored?			
Level 1	There is no system for monitoring corrective actions that the subcontractor has been asked to perform.		
Level 2	Corrective actions required from subcontractors are monitored partially during meetings with the subcontractor.		
Level 3	A system has been set up to periodically monitor corrective actions required from subcontractors, but it is not fully or sufficiently controlled.		
Level 4	A system has been set up to periodically monitor corrective actions required from subcontractors, and there are documents attesting that they are monitored.		

Recommendation Take account of the balance between reliability and the complexity of built-in tests.	Correspondence	N° 147
Phase(s) in which the recommendation applies DESIGN		Weight 10.2
Further description To achieve effective test coverage, there is a trade-off to be made between the complexity of built-in tests and the reliability of the components used for operational functions. In particular, the coverage rate should be weighed against the false alarm rate.		
Audit question Is the impact of the complexity of built-in tests on reliability evaluated?		
Level 1	The influence of built-in tests on reliability is not evaluated.	
Level 2	The influence of built-in tests on reliability is taken into account informally.	
Level 3	There is a specified objective limiting the impact of built-in tests on reliability, but this objective is not verified.	
Level 4	There is a specified objective limiting the impact of built-in tests on reliability. This objective is verified and taken into account in the product design in order to optimize the design of built-in tests. The entire procedure is formalized.	

Recommendation Address reliability in management reviews.	Correspondence ISO9001: L2 EN9100: L2	N° 148
Phase(s) in which the recommendation applies SUPPORT		Weight 5.6
Further description Put reliability on the agenda of management reviews (progress targets, action plans, monitoring of target attainment, reliability assessment of products at customers).		
Audit question Is reliability addressed in management reviews?		
Level 1 Product reliability is not addressed in management reviews.		
Level 2 Product reliability is addressed sporadically in management reviews.		
Level 3 Product reliability is addressed systematically in management reviews.		
Level 4 Product reliability is addressed systematically in management reviews; progress targets are defined and their attainment is evaluated.		

Recommendation Handle anomalies through an incident management and corrective action approach.	Correspondence	N° 149
Phase(s) in which the recommendation applies SUPPORT		Weight 8,3
Further description Set up an anomaly processing system that can cover the entire FIDES life cycle. This system is intended to: - record the circumstances of the anomaly, - record references of the defective product, - propose remedial action, - analyse the causes of the anomaly, - propose corrective/preventive actions, - check the efficiency of the corrective/preventive actions. Processing in this system is suitable for: - quickly finding identical anomalies observed previously, - producing statistics, - use as feedback from operations.		
Audit question What process is set up to collect technical events, produce anomaly reports and measure improved reliability? How are hardware upgrades managed?		
Level 1 There is no system for processing anomalies.		
Level 2 The manufacturer has set up a system for processing anomalies that partially satisfies the requirements of the recommendation. It is not fully applied to the project.		
Level 3 The manufacturer has set up a system for processing anomalies that partially satisfies the requirements of the recommendation. It is fully applied to the project.		
Level 4 The manufacturer has set up a system for processing anomalies that fully satisfies the requirements of the recommendation. It is fully applied to the project. Furthermore: - Indicators are available, - They are regularly interpreted to produce conclusions, - Benefits of the system set up are measured.		

<u>Recommendation</u> Use validated and recognized modelling techniques.		Correspondence	N° 150
Phase(s) in which the recommendation applies DESIGN			Weight 13.8
Further description Use validated and recognized modelling techniques (electrical, thermal and mechanical, etc. models). Demonstrate that tools are monitored and updated.			
Audit question Are the modelling techniques (electrical, thermal, mechanical, etc.) validated and recognized?			
Level 1	Modelling tools and techniques are neither validated nor recognized.		
Level 2	Modelling techniques are validated and recognized, but not monitored.		
Level 3	Modelling techniques are validated, recognized and monitored, but responsibility for monitoring the tools is not clearly identified in the organization.		
Level 4	Modelling tools and techniques are validated, recognized and monitored. Responsibility for monitoring the tools is clearly identified in the organization.		

Recommendation Use appropriate statistical methods to leverage experience feedback.		Correspondence ISO9001: L2 EN9100: L2	N° 151
Phase(s) in which the recommendation applies SUPPORT			Weight 6
Further description Use appropriate statistical methods to leverage experience feedback.			
Audit question Are appropriate statistical methods used to leverage experience feedback?			
Level 1	Feedback is neither observed nor documented.		
Level 2	Feedback is documented, but it is not used, or it is used with unsuitable and non-formalized statistical methods.		
Level 3	Feedback is documented, and is exploited with suitable methods, but they are not formalized (no generalized methods).		
Level 4	Feedback is documented; it is processed using relevant statistical methods and is communicated to users.		

Recommendation		Correspondence	N°
Validate the subcontractor's reliability management guidelines.			152
Phase(s) in which the recommendation applies			Weight
SUPPORT			7.7
Further description			
Validate the subcontractor's reliability management guidelines. In particular, make sure that the contractual requirements are correctly incorporated by the subcontractor.			
Audit question			
Are the subcontractor's reliability management guidelines validated?			
Level 1	The contractual reliability requirements, though applicable, are not transmitted to the subcontractor.		
Level 2	The manufacturer transmits the contractual or internal reliability requirements to the subcontractor, but there is no document written by the subcontractor that guarantees the implementation of these requirements.		
Level 3	The subcontractor has produced reliability management guidelines (a management plan or reliability plan) that mirror the manufacturer's original requirements. However, the implementation of these guidelines is not verified by the manufacturer.		
Level 4	The subcontractor has produced reliability management guidelines (a management plan or reliability plan) that mirror the manufacturer's original requirements. The implementation of these guidelines is validated by the manufacturer (through progress meetings, audits, etc.).		

Recommendation	Correspondence	N°
Check the conformity of purchased products.		153
Phase(s) in which the recommendation applies	Weight	
MANUFACTURE	8.6	
EQUIPMENT_INTEGRATION	8.6	
SYSTEM_INTEGRATION	8.6	
Further description		
Implement measures to verify purchased products, e.g.:		
- Examination of incoming products (components, PCBs, subassemblies, accessories) - Examination of the required documentation, including the supplier's Certificate of Conformity, if specified - Inspection of incoming batches (critical characteristics).		
This incoming inspection is covered by dedicated specifications for each type of material or item purchased.		
Audit question		
Is the conformity of purchased products checked against clearly defined specifications?		
Level 1	The conformity of purchased products is not verified.	
Level 2	The conformity of purchased products is verified only partially, by examining the batch documents.	
Level 3	The conformity of purchased products is verified by inspecting incoming products and examining the required documentation. These inspections are recorded. However, this incoming inspection is not covered by dedicated specifications for each type of material or item purchased.	
Level 4	The conformity of purchased products is verified by inspecting incoming batches and examining the required documentation, in accordance with clearly defined and validated specifications (dedicated to each type of item purchased). These inspections are recorded.	

Recommendation		Correspondence	N°
Before starting the test, verify (by reading the barcode or S/N) that the test device and software are compatible with the product under test.		ISO9001: L3 EN9100: L3	154
Phase(s) in which the recommendation applies			Weight
MANUFACTURE			6.1
Further description			
The compatibility of the test device and software with the product under test is generally verified, before starting the test, by comparing the information on the batch tracking sheet with the test programme identifier on the test machine. This operator task is often confirmed by scanning and loading the barcodes that are also found on the batch tracking sheet (with a handheld scanner).			
Audit question			
Is a check performed (by reading the barcode or S/N) to ensure that the test device and software are compatible with the product under test before starting the test?			
Level 1	No checks are performed to ensure that the test device and program are compatible with the product under test.		
Level 2	The test device and test program are verified to ensure that they are compatible with the product under test, but this verification is not formally described.		
Level 3	The test device and test program are verified to ensure that they are compatible with the product under test. This is based on a documented procedure indicating how the verification is done (by scanning an identifier, etc.).		
Level 4	The test device and test program are verified to ensure that they are compatible with the product under test. This is based on a documented procedure indicating how the verification is done (by scanning an identifier, etc.). This verification procedure has been validated by an authority independent from the operating entity.		

Recommendation	Correspondence	N°
Check that the ESS procedure is correctly applied.		155
Phase(s) in which the recommendation applies		Weight
MANUFACTURE		5.2
EQUIPMENT_INTEGRATION		9.0
SYSTEM_INTEGRATION		9.0
Further description		
It is important to ensure that the environmental stress screening (ESS) procedure defined at the design stage is correctly applied, and that the results are correctly recorded and are traceable. It is also important for corrective or preventive actions to be planned, in order to minimize the ESS failure rate. Finally, the team in charge of defining the ESS procedure must be given the necessary information to optimize this procedure.		
Audit question		
Have checks been done to ensure that the ESS procedure defined at the design stage is correctly applied?		
Level 1	The ESS procedure defined at the design stage is not applied.	
Level 2	The ESS procedure defined at the design stage is applied, but there is no traceable record of the test results.	
Level 3	The ESS procedure defined at the design stage is applied; the results of the ESS test are recorded and traceable. ESS failures are analysed to assist possible corrective or preventive actions in production.	
Level 4	The ESS procedure defined at the design stage is applied; the results of the ESS test are recorded and traceable. ESS failures are analysed to assist possible corrective or preventive actions in production. Additionally, all of this information is communicated to the design team in charge of optimizing the ESS procedure.	

Recommendation		N°
Ensure that environmental specifications are complete.		156
Phase(s) in which the recommendation applies		Weight
RUGGEDIZING		4
Further description		
Ensure the completeness of specifications by using validation criteria: analysis, tests, experience feedback, compliance with standards.		
Audit question		
How is the completeness of environmental specifications ensured?		
Level 1	The completeness check is insufficient or non-existent.	
Level 2	The completeness of the environmental specifications is verified by at least 2 of the 4 methods: Analysis, Experience, Testing, Standards.	
Level 3	The completeness of the environmental specifications is verified by at least 3 of the 4 methods: Analysis, Experience, Testing, Standards.	
Level 4	The completeness of the environmental specifications is verified by all 4 methods: Analysis, Experience, Testing, Standards. The coverage approach is described in a formal procedure.	

<u>Recommendation</u>		N°
Provide training and maintain skills in product implementation and maintenance.		157
Phase(s) in which the recommendation applies		Weight
RUGGEDIZING		7
Further description		
Train users to make sure that the product is always used and repaired correctly.		
Audit question		
Have users (operation and maintenance) been trained in the product? Is this training renewed and updated as needed?		
Level 1	There is no training associated with the product.	
Level 2	There is some initial training, or training of only some users.	
Level 3	There is a complete training programme, but with no skills management.	
Level 4	There is a complete training programme. Skills management ensures that all users have been trained, and that their training is up to date.	

<u>Recommendation</u> Ensure compliance with product-specific procedures and function-specific rules by means of an appropriate monitoring system.	N° 158
Phase(s) in which the recommendation applies RUGGEDIZING	Weight 7
Further description Implement ways of supervising and monitoring how users operate and maintain the product, in order to identify and address deviations.	
Audit question Are there monitoring measures (processes, recording methods) that enable the supplier to ensure that users comply with the product instructions?	
Level 1 There are no monitoring measures.	
Level 2 There are some checking and monitoring measures.	
Level 3 Monitoring measures are non-exhaustive or informally applied. Deviations are not systematically addressed.	
Level 4 Monitoring measures are complete and formalized. Deviations are addressed.	

Recommendation Design dependable electrical protection devices.	N° 159
Phase(s) in which the recommendation applies RUGGEDIZING	Weight 4
Further description Identify protection needs. Design electrical protection devices. Make sure they are testable and maintainable. Integrate the use cases for these devices into the definition of the maintenance policy.	
Audit question How are the electrical protection devices designed?	
Level 1 Dependability principles are not applied to electrical protection devices.	
Level 2 Dependability principles are applied to electrical protection devices in some cases.	
Level 3 Dependability principles are applied to electrical protection devices. Provision is made to check the operation of these devices during the life of the product.	
Level 4 Dependability principles are applied to electrical protection devices. Provision is made to check the operation of these devices during the life of the product. This approach is described in a procedure.	

<u>Recommendation</u> Analyse and manage the risk of the product under test being damaged due to failures in the test or maintenance equipment.	N° 160
Phase(s) in which the recommendation applies RUGGEDIZING	Weight 4
Further description Minimize the risk of deterioration by factoring it into the design of the test equipment and the unit under test; develop suitable preventive measures.	
Audit question Is the risk of the product under test being damaged by test equipment failures taken into account?	
Level 1 Test and maintenance equipment failures are not analysed.	
Level 2 Certain known failures are taken into account.	
Level 3 These risks are analysed during the design of the test equipment and of the unit under test. Suitable preventive measures are in place.	
Level 4 These risks are analysed during the design of the test equipment and of the unit under test. Suitable preventive measures are in place. This approach is described in a procedure.	

Recommendation Identify reasonably foreseeable weather-related hazards and take appropriate preventive measures.	N° 161
Phase(s) in which the recommendation applies RUGGEDIZING	Weight 4
Further description Check for reasonably foreseeable weather-related hazards (UV, hail, condensation, etc.) and mitigate their effects.	
Audit question Has allowance been made for reasonably foreseeable weather-related hazards?	
Level 1 Weather-related hazards are not taken into account.	
Level 2 Well-known weather-related hazards are taken into account.	
Level 3 Instances of weather-related hazards are checked for and taken into account.	
Level 4 Instances of weather-related hazards are systematically checked for and taken into account. This approach is described in a procedure or a standard.	

Recommendation Identify reasonably foreseeable abnormal uses and take appropriate preventive measures.	N° 162
Phase(s) in which the recommendation applies RUGGEDIZING	Weight 4
Further description Check for reasonably foreseeable abnormal uses – product being used for another purpose, mounted upside-down, etc. – and mitigate their effects.	
Audit question Has allowance been made for reasonably foreseeable abnormal uses?	
Level 1 Abnormal uses are not taken into account.	
Level 2 Well-known abnormal uses are taken into account.	
Level 3 Instances of abnormal use are checked for and taken into account.	
Level 4 Instances of abnormal use are systematically checked for and taken into account. This approach is described in a procedure.	

Recommendation Include the production, storage and maintenance environments in the product's environmental specifications.	N° 163
Phase(s) in which the recommendation applies RUGGEDIZING	Weight4
Further description Extend the environmental specifications to cover production, storage and maintenance situations (and not only use cases). For storage (e.g. spares) duration may be the defining parameter.	
Audit question How are the production, storage and maintenance environments taken into account in the product's environmental specifications?	
Level 1 The production, storage and maintenance environments are not specified.	
Level 2 The production, storage and maintenance environments are taken into account, if known.	
Level 3 The production, storage and maintenance environments are taken into account in the environmental specifications.	
Level 4 The production, storage and maintenance environments are systematically taken into account. These environments are described in formal documentation.	

Recommendation		N°
Demonstrate compliance with environmental specifications.		164
Phase(s) in which the recommendation applies		Weight
RUGGEDIZING		4
Further description		
Implement a justification procedure for compliance with environmental specifications that guarantees completeness.		
Audit question		
How is compliance with environmental specifications substantiated?		
Level 1	There is no formal demonstration of compliance with environmental specifications.	
Level 2	Compliance with key environmental specifications is formally justified.	
Level 3	Compliance with all environmental specifications is formally justified.	
Level 4	Compliance with all environmental specifications is formally justified. The justification approach identifies the margins relative to the requirements. The justification process is described in a formal procedure.	

<u>Recommendation</u> Implement a product improvement process (e.g. accelerated stress tests) to limit the product's sensitivity to environmental stresses (disturbance, severe environments, overstress).	N° 165
Phase(s) in which the recommendation applies RUGGEDIZING	Weight 7
Further description Implement a product improvement process in development and production so as not to deliver a product that is not mature (presence of early-life defects) or which presents weaknesses (in environmental stress screening). Highly accelerated life tests (HALT) are cited as an important tool for this objective.	
Audit question Is there a product improvement process to increase its robustness and accelerate its maturity?	
Level 1 There is no process to improve robustness and maturity.	
Level 2 Some measures are taken to improve robustness and maturity.	
Level 3 A process is implemented to improve robustness and maturity.	
Level 4 A process is implemented to improve robustness and maturity. This approach is systematic and is described in a formal procedure.	

<u>Recommendation</u>		N°
Conduct an analysis of failures that could lead to failure propagation.		166
Phase(s) in which the recommendation applies		Weight
RUGGEDIZING		4
Further description		
Analyse the risk of failure propagation in order to contain the effects and limit unnecessary removals, in particular by using FMECA (Failure Modes, Effects and Criticality Analysis).		
Audit question		
Is the risk of failure propagation analysed?		
Level 1	There is no analysis of failure propagation.	
Level 2	Certain known instances of failure propagation are taken into account.	
Level 3	Risks of failure propagation are taken into account in the design, and when conducting FMECA analyses. These cases are addressed by means of protection or architecture solutions.	
Level 4	Risks of failure propagation are taken into account in the design, and when conducting FMECA analyses. These cases are addressed by means of protection or architecture solutions. This approach is described in a procedure.	

<u>Recommendation</u>		N°
Conduct a process analysis of implementation and maintenance operations.		167
Phase(s) in which the recommendation applies		Weight
RUGGEDIZING		4
Further description		
Conduct a Process FMECA (Failure Modes, Effects and Criticality Analysis) on implementation and maintenance operations.		
Audit question		
How is the risk of error in implementation and maintenance operations analysed?		
Level 1	There is no analysis of the risk of error in carrying out operations.	
Level 2	Data on known anomalies and drifts is collected and used in the design or maintenance of the product.	
Level 3	A Process FMECA is conducted in some cases on implementation and maintenance operations and is used in the design or maintenance of the product.	
Level 4	A Process FMECA is conducted systematically on implementation and maintenance operations and is used in the design or maintenance of the product.	

<u>Recommendation</u> Review maintenance operations performed by end users and address their recommendations.		N° 168
Phase(s) in which the recommendation applies RUGGEDIZING		Weight 4
Further description Take user recommendations into account in the design of product maintenance, after review.		
Audit question Is there a review of maintenance operations carried out by the user?		
Level 1	There is no review with the user.	
Level 2	User recommendations are taken into account informally, or there is a paper review without performing any operations on a product.	
Level 3	A maintenance procedure review is organized with the user, involving operations on the equipment.	
Level 4	A maintenance procedure review is organized with the user, involving operations on the equipment. The review process is described in a plan or a formalized procedure.	

<u>Recommendation</u>	N°
Write complete procedures for all product implementation and maintenance operations.	169
Phase(s) in which the recommendation applies	Weight
RUGGEDIZING	7
Further description	
Provide users and Maintenance with documentation describing the procedures to be applied for each situation.	
Audit question	
Is there documentation describing all of the product implementation and maintenance operations?	
Level 1 The documentation is superficial or non-existent.	
Level 2 There is documentation covering some of the needs.	
Level 3 There is complete documentation, but its completeness cannot be demonstrated.	
Level 4 There is complete documentation. The compilation of the documentation is described in a procedure that guarantees its completeness.	

<u>Recommendation</u> Comply with a standard on power supplies (such as EN2282, which defines possible disturbances and variations). The standard must be complied with both for electricity generation and for electricity consumption.	N° 170
Phase(s) in which the recommendation applies RUGGEDIZING	Weight4
Further description Apply a standard to all electrical interfaces in order to guarantee their operating conditions and define foreseeable extreme situations. It is also recommended that the person responsible for the system into which the product is integrated should guarantee compliance with the standard across the system. Compliance with the standard should also be assured for the generation of the electricity that powers the product.	
Audit question Is there a standard on electrical power supplies that applies to the product and the surrounding system? If so, how is it applied?	
Level 1 There is no standard for electrical power supplies.	
Level 2 A standard is used as a guide for defining electrical interfaces.	
Level 3 A standard is applied to all of the product's electrical interfaces.	
Level 4 A standard is applied to all of the product's electrical interfaces. Formal justification is provided to demonstrate compliance with the standard.	

Recommendation Comply with a standard on conducted and radiated electromagnetic interference; this applies both to the product and to the system into which it is integrated.	N° 171
Phase(s) in which the recommendation applies RUGGEDIZING	Weight 3
Further description Apply a standard on conducted and radiated electromagnetic interference.	
Audit question Is there a standard on conducted and radiated electromagnetic interference that applies to the product and the surrounding system? If so, how is it applied?	
Level 1 There is no standard for conducted and radiated electromagnetic interference.	
Level 2 A standard is used as a guide for defining tolerances to conducted and radiated electromagnetic interference.	
Level 3 A standard is applied for defining the tolerance of the product to conducted and radiated electromagnetic interference.	
Level 4 A standard is applied for defining the tolerance of the product to conducted and radiated electromagnetic interference. Formal justification is provided to demonstrate compliance with the standard.	

Recommendation Control the viscosity and thickness of the varnish.	Correspondence ISO9001: L2 EN9100: L3	N° 172
Phase(s) in which the recommendation applies MANUFACTURE		Weight 6.0
Further description Given the impact on final quality, the viscosity of the varnish must be inspected at least daily. Remark: For glues and solder paste, the viscosity is generally controlled by managing the working time under production conditions (Recommendation 19). To control the thickness of the varnish deposited, test specimens are generally produced for each new batch of boards, with tolerance limits that are specified in a document.		
Audit question Is the viscosity of the varnish inspected periodically? Is the thickness of the varnish inspected using test specimens?		
Level 1	The viscosity of the varnish is not controlled.	
Level 2	The viscosity of the varnish is measured periodically. However, there is no specification describing the criteria for rejection and/or corrective action.	
Level 3	The viscosity of the varnish is measured every day. The criteria for rejection and/or corrective action are clearly described in specifications and/or instructions. However, these documents have not been validated by an independent authority.	
Level 4	The viscosity of the varnish is measured every day. The criteria for rejection and/or corrective action are clearly described in specifications and/or instructions. These documents have been validated by an independent authority.	

Recommendation Systematically inspect the quality of the solder joints on board-mounted components after solder reflow.	Correspondence ISO9001: L2 EN9100: L3	N° 173
Phase(s) in which the recommendation applies MANUFACTURE		Weight 6.0
Further description Solder joints between boards and components can present the following defects: - Lack of solder, or overly large gaps - Excess of solder, and/or solder balling - Mispositioned components (decentring, tilt, etc.) - Incomplete mixing in the case of ball grid arrays. Solder joint inspections must be specified and recorded, in order to prevent all of these reliability issues in subassemblies. Optical and X-ray techniques are generally used. The reference standard is IPC-A-610.		
Audit question Is the quality of the solders controlled systematically, by inspecting the solder joints after reflow?		
Level 1	The quality of solder joints between boards and components is not systematically inspected.	
Level 2	The quality of solder joints between boards and components is systematically inspected, but this inspection is not based on a clearly identified document.	
Level 3	The quality of solder joints between boards and components is systematically inspected. This inspection is based on a clearly identified document. However, this document does not refer to IPC-A-610, or is not independently validated.	
Level 4	The quality of solder joints between boards and components is systematically inspected. This inspection is based on a clearly identified document. IPC-A-610 is cited as the reference standard in this document. Additionally, this document has been validated by an authority independent from the operating entity.	

Recommendation Implement conformity checks on finished products entering storage.	Correspondence ISO9001: L3 EN9100: L3	N° 174
Phase(s) in which the recommendation applies MANUFACTURE		Weight 8.0
Further description A “conformity check” step on finished products must be identified, and must be performed systematically before items definitively enter storage. The aim is to ensure that the accompanying data is complete, and is consistent with the items received.		
Audit question Are conformity checks performed on finished products entering storage, so that nonconforming items are excluded?		
Level 1 There is no conformity check on finished products before they enter storage.		
Level 2 Certain parameters of finished products are monitored on entering storage, but these are not formally defined.		
Level 3 A full conformity check is performed on finished products before they enter storage. This conformity check is documented formally (parameters, particular points, etc.). However, no independent authority has validated these documents.		
Level 4 A full conformity check is performed on finished products before they enter storage. This conformity check is documented formally (parameters, particular points, etc.). The relevance of the information in these documents has been validated by an authority independent from the operating entity.		

Recommendation Control the environmental conditions for the stock of components and materials.		Correspondence ISO9001: L2 EN9100: L3	N° 175
Phase(s) in which the recommendation applies MANUFACTURE			Weight 9.0
Further description Electronic components are sensitive to environmental conditions in storage (corrosion, moisture absorption, ESD). Certain materials require cold storage (cold room or freezer). It is important that the limits (specifications and control limits) are defined in an official document and that inspection systems (probes, recordings, alarms) are in place.			
Audit question Are the temperature/humidity conditions of the component and material storage zones controlled?			
Level 1	The storage conditions of items and materials are not measured, or are not monitored periodically.		
Level 2	The storage conditions of items and materials are measured, and are monitored continuously or periodically, but there are no specifications describing the acceptable limits.		
Level 3	The storage conditions of items and materials are measured and monitored continuously or periodically. There are specifications describing the acceptable limits. However, these specifications are not described in a document validated by an independent authority, or the specification limits are not backed up by control limits to avoid the risk of excursion.		
Level 4	The storage conditions of items and materials are measured and monitored continuously or periodically. There are specifications describing the acceptable limits. These specifications are described in a document validated by an independent authority. The specification limits are backed up by control limits to avoid the risk of excursion.		

Recommendation Control the sensitivity of components to moisture (MSL) and to the reflow process.	Correspondence	N° 176
Phase(s) in which the recommendation applies MANUFACTURE		Weight 12.0
Further description Manufacturers assign a moisture sensitivity level (MSL) to SMD plastic components; sensitive components are wrapped in sealed bags, and working time limits are specified. The reference document for production is the J-STD-033 standard. It is important for operators tasked with mounting components on boards to manage this risk, and for there to be an accompanying instruction. Operators must have the means to implement the necessary actions (an oven for drying components, etc.)		
Audit question Is the MSL of components taken into account during their industrial implementation?		
Level 1 The MSL of sensitive components is not managed by the operator.		
Level 2 The MSL of sensitive components is known and managed by the operator, but there is no internal document describing the actions associated with this risk.		
Level 3 The MSL of sensitive components is known and managed by the operator. There is an internal document describing the actions associated with this risk. However, this document is not validated by an independent authority.		
Level 4 The MSL of sensitive components is known and managed by the operator. There is an internal document describing the actions associated with this risk. This document is up to date and is validated by an independent authority.		

Recommendation Track all production and testing operations on the batch tracking sheet.	Correspondence ISO9001: L3 EN9100: L4	N° 177
Phase(s) in which the recommendation applies MANUFACTURE		Weight 8.0
Further description At each step of production and testing, it is important to know certain information in order to minimize quality and reliability risks: partnumber, batch number, incoming quantity, references of the required consumables, reference of the test to be performed, etc. Likewise, the operator must enter a certain number of traceability details on the tracking sheet: batch numbers of consumable materials, tools, outgoing quantity, reasons for rejecting or repairing parts, number of the machine used, name of the operator, etc. The same sheet is often used by Quality Control to track inspections (e.g. the Quality Assurance stamp).		
Audit question Is there a batch tracking sheet describing the required traceability information for all operations?		
Level 1	There is no batch tracking sheet recording the materials used and/or the test applied.	
Level 2	A tracking sheet is filled in for each production batch. However, it does not contain all the information necessary to perform every operation, or does not track the factors that influence quality/reliability.	
Level 3	A tracking sheet is filled in for each production batch. It contains all the information necessary to perform every operation, and tracks the factors that influence quality/reliability, but it is not validated by an independent authority.	
Level 4	A tracking sheet is filled in for each production batch. It contains all the information necessary to perform every operation, tracks the factors that influence quality/reliability, and is validated and updated by an independent authority.	

<u>Recommendation</u>		Correspondence	N°
Optimize the production line organization.			178
Phase(s) in which the recommendation applies			Weight
MANUFACTURE			9.0
Further description			
The organization of the production line, the flows of products and persons, and the clarity and ease of use of workstations all help to minimize reliability risks to the products on the line. Organizational methodologies are defined in a number of standards (5S, Lean, etc.)			
Audit question			
Are actions carried out to continuously improve the organization of the production line?			
Level 1	There are no concrete actions in place to improve the organization of the production line.		
Level 2	Improvements have been made to the organization of the production line, but there is no methodology for its continuous improvement.		
Level 3	Concrete actions are in place for the continuous improvement of the production line. These actions are not monitored by a quality review.		
Level 4	Concrete actions are in place for the continuous improvement of the production line. These actions are monitored by a quality review.		

Recommendation		Correspondence	N°
Guard against the purchase of fraudulent components.		EN9100: L3	179
Phase(s) in which the recommendation applies			Weight
MANUFACTURE			10.0
Further description			
The potential use of fraudulent or counterfeit components, generally purchased from brokers (independent suppliers not under franchise to the manufacturers), represents a reliability risk. It is important that there is a clear procurement procedure that authorizes the purchase of components from brokers only under particular circumstances, and in accordance with a specific protocol (waiver request, authorized broker, batch inspections, proofs of conformity, notification of the customer when specified).			
Audit question			
Is the risk of buying fraudulent/counterfeit components covered by a procedure?			
Level 1	The potential procurement of components from brokers is not subject to any particular procedure.		
Level 2	There is a broker procurement procedure, but no authorized list, or no reinforced control of the batch inspections.		
Level 3	There is a broker procurement procedure, a list of authorized brokers, and a specific incoming inspection procedure. However, these documents are not validated and updated periodically by an authority independent from the Procurement department.		
Level 4	There is a broker procurement procedure, a list of authorized brokers, and a specific incoming inspection procedure. These documents have been validated by an authority independent from Procurement, and are periodically updated.		

Recommendation		Correspondence	N°
Monitor the bending of the board during electrical probe tests.			180
Phase(s) in which the recommendation applies			Weight
MANUFACTURE			7.0
Further description			
During electrical probe tests, deformation of the board can damage the item under test, impacting its reliability. It is important to monitor the degree of bending by taking measurements at the start of each new batch. The rules to be applied must be set out in a specification.			
Audit question			
Is the bending of the board controlled during electrical probe tests? Is there a document specifying what is acceptable?			
Level 1	There is no evidence that the bending of the board is monitored during electrical probe tests.		
Level 2	The bending of the board is monitored during electrical probe tests at the start of the batch, but this is not specified in any document.		
Level 3	The bending of the board is monitored during electrical probe tests at the start of the batch. The measurement conditions and acceptance criteria are specified in a document. However, this document has not been validated by an independent authority.		
Level 4	The bending of the board is monitored during electrical probe tests at the start of the batch. The measurement conditions and acceptance criteria are specified in a document. This document has been validated by an authority independent from the operating entity.		

Recommendation	Correspondence	N°
Know and control the mission profile.		181
Phase(s) in which the recommendation applies	Weight	
SPECIFICATION		13
DESIGN		15
OPERATION_&_MAINTENANCE		9
Further description		
The product mission profile is fully known and controlled:		
- The potential variations in the different environmental stresses are known - The impact of environmental stress variation is controlled and taken into account in dependability analyses (e.g. MTBF for safety analyses).		
Audit question		
Is the product mission profile known and controlled?		
Level 1	The mission profile is not controlled; a “standard” mission profile is assumed.	
Level 2	The mission profile is not controlled. However, reliability calculations are performed using multiple mission profiles corresponding to typical use cases.	
Level 3	Sensitivity analyses are conducted to evaluate the impact of the different environmental stresses on product reliability. These analyses are taken into account to adjust reliability calculations to the circumstances (e.g. “average” mission profile for an ISS contract, or “worst case” mission profile for a safety analysis).	
Level 4	Integrated sensors (HUMS) are implemented; these measure environmental stresses in real time and process them to gain better knowledge of the product’s conditions of operation. This data is taken into account to update reliability analyses and adjust them to the circumstances (e.g. “average” mission profile for an ISS contract, or “worst case” mission profile for a safety analysis).	

Recommendation Implement an improvement process to improve product robustness.	Correspondence ISO9001: L2 EN9100: L2	N° 182
Phase(s) in which the recommendation applies DESIGN		Weight 20
Further description It is important to conduct a process to improve the robustness of the product in development and in production, so as to limit its sensitivity to environmental stresses (disturbance, severe environments, overstress); this improvement process must be formalized in a document (e.g. a robustness improvement plan). Robustness is characterized by increased operational stress margins, generally measured by accelerated stress tests, a combination of traditional vibration and thermo-mechanical tests, or highly accelerated life testing (HALT). The operational stress margin must be estimated and, if necessary, improved through successive iterations (with additional tests to verify the improvement). This margin will be necessary in order to conduct accelerated ESS testing, beyond the maximum stress levels.		
Audit question Is there an improvement process to improve product robustness?		
Level 1 There is no robustness improvement approach.		
Level 2 There is a robustness improvement approach, but it is not formalized through testing.		
Level 3 There is a robustness improvement approach; it is based on performing accelerated stress tests in order to estimate operational stress margins.		
Level 4 Margins are evaluated and demonstrated to be sufficient with regard to operational stress and the implementation of an accelerated ESS test such as the HASS test, based on HALT test results. If necessary, improvements are made to the product in order to guarantee the desired margins.		

Recommendation	Correspondence	N°
Identify and reduce design risks.	EN9100: L1	183
Phase(s) in which the recommendation applies	Weight	
DESIGN	15	
Further description A Design FMECA identifies and reduces risks, for example following design changes or when new technologies are brought in. Criticality analysis of the identified risks leads to formalized recommendations and/or corrective actions. Design FMECAs must be an iterative process: the efficacy of corrective actions must be measured.		
Audit question Are analyses (such as Design FMECA) conducted in order to identify and reduce risks? Are the recommendations that emerge from these analyses formalized? Are the recommendations that emerge from these analyses followed?		
Level 1	There is no Design FMECA.	
Level 2	There is a Design FMECA, but the risk criticality analysis does not lead to formalized recommendations and/or corrective actions.	
Level 3	Design FMECA is conducted; the risk criticality analysis leads to formalized recommendations and/or corrective actions, but these are not systematically tracked over time.	
Level 4	The Design FMECA is realized; the risk criticality analysis leads to formalized recommendations and/or corrective actions, which are tracked over time.	

<u>Recommendation</u>		Correspondence	N°
Prepare the product industrialization phase.		EN9100: L4	184
Phase(s) in which the recommendation applies			Weight
DESIGN			16
Further description			
Preparations for product industrialization must begin during the design phase. The industrial strategy must be formalized in a plan; the product industrialization requirements must be detailed in an industrial specification and taken into account by designers.			
Audit question			
Is the industrial strategy defined in a plan? Have the product industrialization requirements been detailed in an industrial specification?			
Level 1	Industrial activity is not taken into account in design: no plan, no specification.		
Level 2	Industrial activity is taken into account in design, without being formalized.		
Level 3	Industrial activity is taken into account and formalized in a plan. There is no specification (there are no requirements defined for designers).		
Level 4	Industrial activity is taken into account and formalized by a plan. The requirements are formalized in a specification and are taken into account by designers.		

Recommendation Analyse the need for environmental stress screening (ESS) and define the ESS procedure.	Correspondence EN9100: L3	N° 185
Phase(s) in which the recommendation applies DESIGN		Weight 18.0
Further description The need for environmental stress screening must be analysed and, if applicable, an ESS procedure established. Several criteria can be considered: reliability test results (e.g. HALT, qualification trials); experience feedback on other products; customer requirements; efficiency targets (e.g. maximum scrap rate after first use, after one year); product quantity, etc. Experience feedback on the product itself (ESS results and field data) must be used to optimize the ESS procedure for the duration of the programme.		
Audit question Has the need for ESS been analysed? If applicable, has the ESS procedure been defined?		
Level 1	The need for ESS is not analysed.	
Level 2	An analysis is conducted to justify ESS and, if applicable, define an appropriate ESS procedure for the product, but this procedure is not covered by a validated document.	
Level 3	An analysis is conducted to justify ESS and, if applicable, define an appropriate ESS procedure for the product, and this procedure is covered by a validated and configuration-managed document.	
Level 4	An analysis is conducted to justify ESS and, if applicable, define an appropriate ESS procedure for the product; this procedure is covered by a validated document. Additionally, experience feedback on the product itself (ESS results and field data) is used to optimize the ESS procedure for the duration of the programme.	

Recommendation Check that the environmental stress screening (ESS) test coverage is correctly defined and formalized.	Correspondence EN9100: L3	N° 186
Phase(s) in which the recommendation applies DESIGN		Weight 11.0
Further description If ESS is necessary, it is important to verify the testability of the product before, during and after applying stress. The ESS test coverage rate must be estimated. The built-in functional tests on the board, subassembly or system may be sufficient if their coverage rate is satisfactory. If so, the test coverage is formalized in the FMECA or the testability report.		
Audit question Has the coverage rate of the ESS test been estimated?		
Level 1 The ESS testability of the product is not taken into account.		
Level 2 The ESS testability of the product is taken into account, but the ESS test coverage rate has not been estimated.		
Level 3 The ESS test coverage rate is estimated (e.g. by a FMECA).		
Level 4 The ESS test coverage rate is estimated ((e.g. by a FMECA); experience feedback (ESS results and field data) is used to optimize the ESS test.		

Appendix

Recommendations for using DSM components

Calculating ageing for Deep Sub Micron (DSM) components

The ageing model presented below can be used to verify the impact of degradation mechanisms on the failure rate of a DSM component.

It is based on accelerated tests performed on FPGAs that use planar 28 nm technology. The recommendations below therefore apply to planar DSM technologies. They do not apply to 3D technologies (e.g. FinFET), for which a methodology is described in the next section.

The BTI (Bias Temperature Instability) mechanism was identified as the main degradation mechanism, significantly more prevalent than other mechanisms.

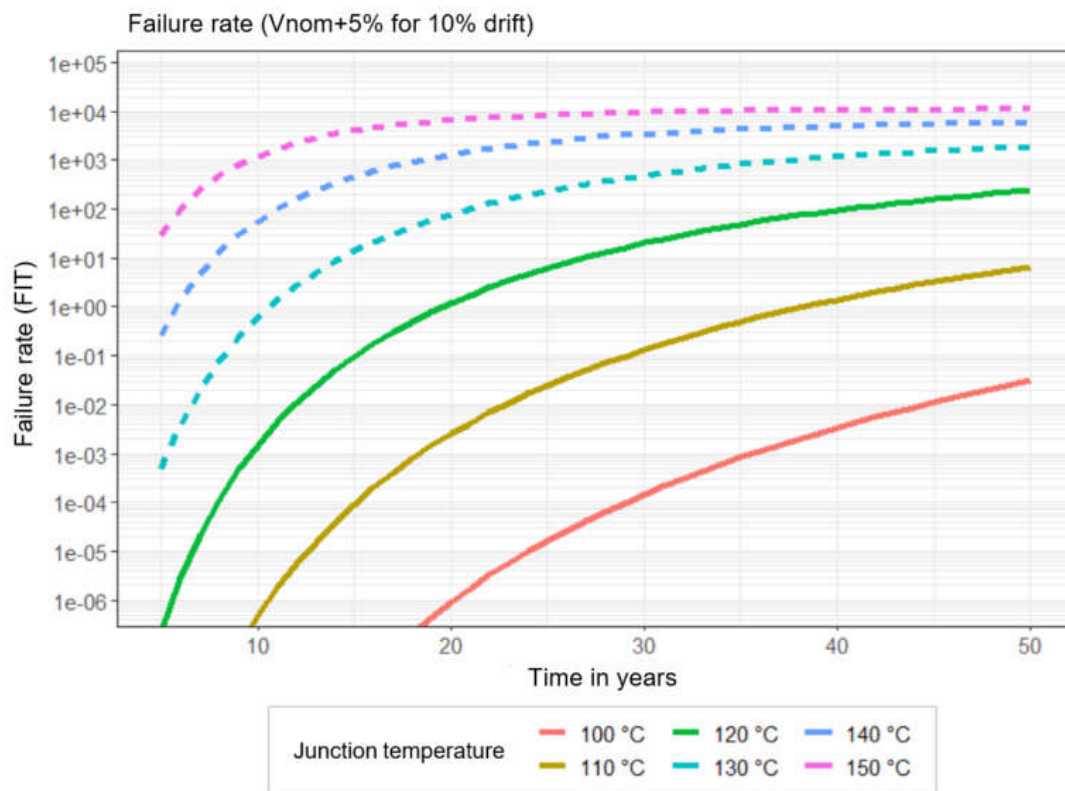
Up to a junction temperature of 100°C and for 24-hour operation, the failure rate associated with BTI is negligible: it remains strictly lower than 0.1 FIT after 50 years of use.

Above a junction temperature of 100°C, the failure BTI rate becomes significant: at 125°C (the maximum working temperature of most FPGAs), the failure rate exceeds 1 FIT shortly before 15 years of use, rising to 10³ FIT after 50 years of use.

The recommendations to users are consequently as follows:

- Below a junction temperature of 100°C, and for a $V_{\text{applied}}/V_{\text{rated}}$ ratio lower than or equal to 1.05, the BTI failure rate can be ignored; users can assume a constant failure rate, associated with random failures.
- Above a junction temperature of 100°C, and for a $V_{\text{applied}}/V_{\text{rated}}$ ratio lower than or equal to 1.05, users can initially use the figure below to evaluate the impact of the BTI failure rate as a function of the planned duration of use.

The figure below shows how the BTO failure rate evolves over time, at different junction temperatures and for a $V_{\text{applied}}/V_{\text{rated}}$ ratio of 1.05 (i.e. a voltage stress ratio of 105%).



Note:

- The failure criterion applied in testing is a drift of 10% in signal propagation time.
- The dotted-line curves are derived from modelling rather than testing.

Thereafter, if necessary, users can apply the ageing law given below to evaluate the BTI failure rate at any given moment; the overall failure rate – now time-dependent – will be:

$$\lambda(t) = \lambda + \lambda_{BTI}(t, T_j, V).$$

(where λ is the constant random failure rate as evaluated in the FIDES Guide).

It is then up to the user to choose between a failure rate that is not constant over time or – more conservatively – a constant failure rate evaluated at end of life:

$$\lambda(D) = \lambda + \lambda_{BTI}(D, T_j, V),$$

(where D is the planned duration of use).

- For a $V_{\text{applied}}/V_{\text{rated}}$ ratio lower than or equal to 1.1, and a junction temperature lower than or equal to 125°C, users can apply the following ageing law to evaluate the BTI failure rate at any given moment:

The associated ageing model follows a log-normal distribution, and the failure rate is expressed as follows:

$$\lambda_{BTI}(t, T_j, V) = \frac{1}{t \cdot \sigma \cdot \sqrt{2\pi}} \cdot e^{-\frac{1}{2} \left(\frac{\ln(t) - \mu_0 + \gamma \cdot V - \frac{E_a}{k_B \cdot T_j}}{\sigma} \right)^2} \cdot \frac{1}{1 - \Phi \left(\frac{\ln(t) - \mu_0 + \gamma \cdot V - \frac{E_a}{k_B \cdot T_j}}{\sigma} \right)}$$

Where:

- Φ is the cumulative distribution function of the standard normal distribution $\mathcal{N}(0,1)$
- T_j is the component junction temperature, in Kelvin
- V is the $V_{\text{applied}}/V_{\text{rated}}$ ratio
- μ_0 , σ and γ are the distribution parameters (values given below)
- E_a is the activation energy in eV (value given below)
- k_B is the Boltzmann constant.

The values to be considered for the different parameters are as follows:

Parameter	Estimated value
μ_0	2.71
σ	0.44
E_a	0.76
γ	10.52

Methodology for 3D technologies

The ageing model proposed in the previous paragraph applies only to planar technologies. Strictly, it cannot be used on 3D technologies (3D transistors such as FinFET), on which it has not been validated by testing.

Typically, these 3D technologies are used for technology nodes lower than (or equal to) 22 nm for Intel processors and 16 nm for FPGAs.

To evaluate the long-term reliability risk, the following 2-step methodology is proposed:

- 1/ As a first step, it is essential to obtain reliability data on the chip technology from the manufacturer/supplier, in particular data from WLR (Wafer Level Reliability) tests, performed on test structures. Such data is never public, but it can be obtained under non-disclosure agreements (NDAs).

Manufacturers provide this data to demonstrate a level of reliability, typically for 10 years' continuous use at a junction temperature of 85°C and a rated power supply voltage (the conditions given here are for illustration only). Comparing this data with the conditions of the mission and usage profile in application provides an initial first estimate of the level of risk.

By way of comparison with the typical values indicated previously, the long term reliability risk can be deemed negligible if the components involved are only used for a cumulative duration of 3 years, at a junction temperature of 100°C and at a rated voltage.

If doubt remains as to the level of reliability demonstrated by the manufacturer, or the effective long-term resilience, additional information is required.

- 2/ The mission and usage profile may be submitted to the manufacturer, who will analyse the risk in the light of their own internal reliability data,¹ supplied by the Wafer Fab production and engineering teams. Most manufacturers will agree to perform these analyses.

Note: For manufacturers who have marketed their products – FPGAs, for example – through several different processes and technology nodes, it is important to compare data from one generation to the next, so as to assess whether the reliability “margins” have changed. This will be all the more valuable if it is possible to compare two generations of product where there is a migration from planar to FinFET technology (as with FPGAs on the current market).

¹ Even under NDAs, manufacturers will not supply all of their internal data. This second part of the methodology can be used to obtain a more detailed risk estimate from the manufacturer, but, for liability reasons, this estimate cannot be treated as a “manufacturer’s warranty”.